



Research

The House View: Snapshot

A broadening tightening cycle

Deutsche Bank

Macro views

World

- Global growth of 3.3% in 2026 and 3.4% in 2027.
- **Rationale:** AI investment supports activity despite the energy shock. Brent is forecast to average \$95/bbl in Q4 2026, with supply normalisation expected from Q1 2027.
- **Key risks:** Prolonged Middle East conflict, higher energy costs and tighter financial conditions. Upside risk from stronger AI investment and productivity.

Europe

- Euro-area GDP upgraded to 0.8% (prev. 0.5%) in 2026; 1.2% (prev. 1.1%) in 2027. Growth has been surprisingly resilient in the face of the energy shock.
- **Rationale:** Consumption, defence and AI spending, modest fiscal easing and bank lending have cushioned the energy shock.
- **Key risks:** A worsening energy shock, higher borrowing costs and vulnerable public finances around elections could lift risk premia.

China

- GDP growth lowered to 4.6% in 2026; 2027 remains 4.5%.
- **Rationale:** Exports and AI support growth, but consumption, investment and property remain weak. Fiscal spending is picking up; PBoC rates remain unchanged.
- **Key risks:** Delayed fiscal support, further property weakness and weaker external or AI-related demand.

United States

- 2026 & 2027 GDP growth forecast at 2.2%.
- **Rationale:** Easy financial conditions, fiscal policy and AI-driven capex support growth. Unemployment is expected near 4.1% into year-end; PCE inflation stays above 3% into 2027.
- **Key risks:** A larger oil shock weakens real income; financial conditions tighten; or tighter labour market and AI productivity surprise on the upside.

Germany

- GDP growth forecast at 1.0% in 2026 and 1.3% in 2027.
- **Rationale:** Strong H1 offset by loss of purchasing power caused by the energy shock, although expansionary fiscal policy continues to provide tailwinds.
- **Key risks:** Energy exposure, higher financing costs and political uncertainty could delay the improvement.

India

- 7.3% y/y GDP for FY27 (prev. 6.7%). FY28 growth outlook unchanged at 7.2%.
- **Rationale:** Growth remains strong even as higher oil prices raise inflation and external-balance sensitivity. Current policy looks too accommodative, with strong growth and rising headline inflation (above 6% through Oct-Dec 2026); 100bps of rate hikes starting in Oct 2026.
- **Key risks:** Persistently high oil prices, weaker global demand and tighter domestic financial conditions.

Key downside risks

- H** **Middle East conflict.** Possibility that the war can extend into 2028 pushes Brent forecast to \$130-150/bbl, leading to higher stagflation concerns.
- H** **Inflation:** Higher energy costs, AI investment, fiscal expansion, and higher defence spending could require more monetary tightening.
- M** **Political and fiscal risks.** European elections and fragile public finances could increase risk premia. US cost-of-living concerns remain important ahead of the midterms.

Marion Laboure
thehouseview@list.db.com

Camilla Siazon
<http://houseview.research.db.com>

Jim Reid

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Deutsche Bank AG/London

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Market views

	Market Sentiment	– Risk assets remain resilient despite rate hikes and higher yields. – The global economy resilient despite geopolitical conflict and energy risks, supported partly by the AI investment boom. – Key question is how long activity can withstand high energy costs, rising rates and limited fiscal space.
	Equities	– S&P 500 YE-2026 target: 8000; STOXX 600 YE-2026 target: 670. – Earnings recovery has broadened beyond technology. – Overweight mega-cap growth & tech, financials, industrials, and metals & mining.
%	Rates	– YE-2026: 10Y UST at 4.80%; 10Y Bund at 3.20%. – Neutral duration after a long-standing bearish bias. Equilibrium rates and term premia remain structurally higher.
	Credit	– YE 2026: \$IG 85bp; \$HY 315bp; €IG 95bp; €HY 345bp. – Spreads expected to widen modestly. – Prefer US over European credit amid stronger US growth and greater ECB risk.
	FX	– YE: EUR/USD: 1.17; USD/JPY: 150. – Neutral EUR/USD with a mild upside bias.
	Oil & Gold	– Q4-2026: Brent: \$95/bbl; WTI: \$90/bbl. – Supply normalisation expected from Q1 2027, leading Brent to average \$74/bbl in 2027. – Gold: YE-2026 target \$4,600/oz. Higher real yields and a more hawkish Fed weigh near term; 2027 average forecast is \$5,100/oz.
	Monetary Policy	– Fed: Two 25bp hikes in Dec and Mar, taking policy rate to 4.375%. – ECB: 25bp hikes in Dec, taking the deposit rate to 2.75%. – BoJ: Two 25bp hikes in Jan and Apr, bringing policy rate to 1.75%. – BoE: Two 25bp hikes in Nov and Feb, taking Bank Rate to 4.25%. – PBoC: Unchanged at 1.40% through 2026.

Key macro & markets forecasts

GDP growth (%)			Central bank policy rate (%)				Key market forecasts			
	2026F	2027F		Current	Dec 26F	Jun 27F		Current	Dec 26F	Jun 27F
Global	3.3	3.4	US: Federal Funds Rate	3.875	4.125	4.375	US 10Y yield (%)	5.11	4.80	4.75
US	2.2	2.2	Eurozone: Deposit Facility Rate	2.50	2.75	2.75	EUR 10Y yield (%)	3.55	3.20	3.20
Eurozone	0.8	1.2	UK: Bank Rate	3.75	4.00	4.25	S&P 500	7,706	8,000	-
Germany	1.0	1.3	Japan: Policy Balance Rate	1.25	1.25	1.75	Gold	4,288	4,600	5,000
Japan	0.7	0.9	China: 7d OMO Rate	1.40	1.40	1.40	EUR/USD	1.14	1.17	-
UK	1.3	1.4					USD/JPY	158	150	-
China	4.6	4.5					Oil WTI (USD/bbl)	91	90	67
							Oil Brent (USD/bbl)	102	95	70

2026 Macro events calendar

October 2026			November 2026			December 2026		
28	CA	Bank of Canada Decision	3	US	Midterm Elections	9	CA	Bank of Canada Decision
28	US	Fed Decision	5	UK	Bank of England Decision	9	US	Fed Decision
29	EZ	ECB Decision				17	UK	Bank of England Decision
29-30	JN	Bank of Japan Decision				17	EZ	ECB Decision
						18	JN	Bank of Japan Decision



Recent editions

- [A broadening tightening cycle](#), 23 Sep 2026
- [1999 meets 1990](#), 9 Jun 2026
- [The Impact of the Middle East](#), 10 Mar 2026

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Appendix 1

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M&A Spotlight:

When rates rise ...

17 September 2026

Luke Templeman | Thematic Strategist | Luke.Templeman@db.com | +44 207 541 0130
Galina Pozdnyakova | Research Analyst | Galina.Pozdnyakova@db.com | +44 207 547 4994

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

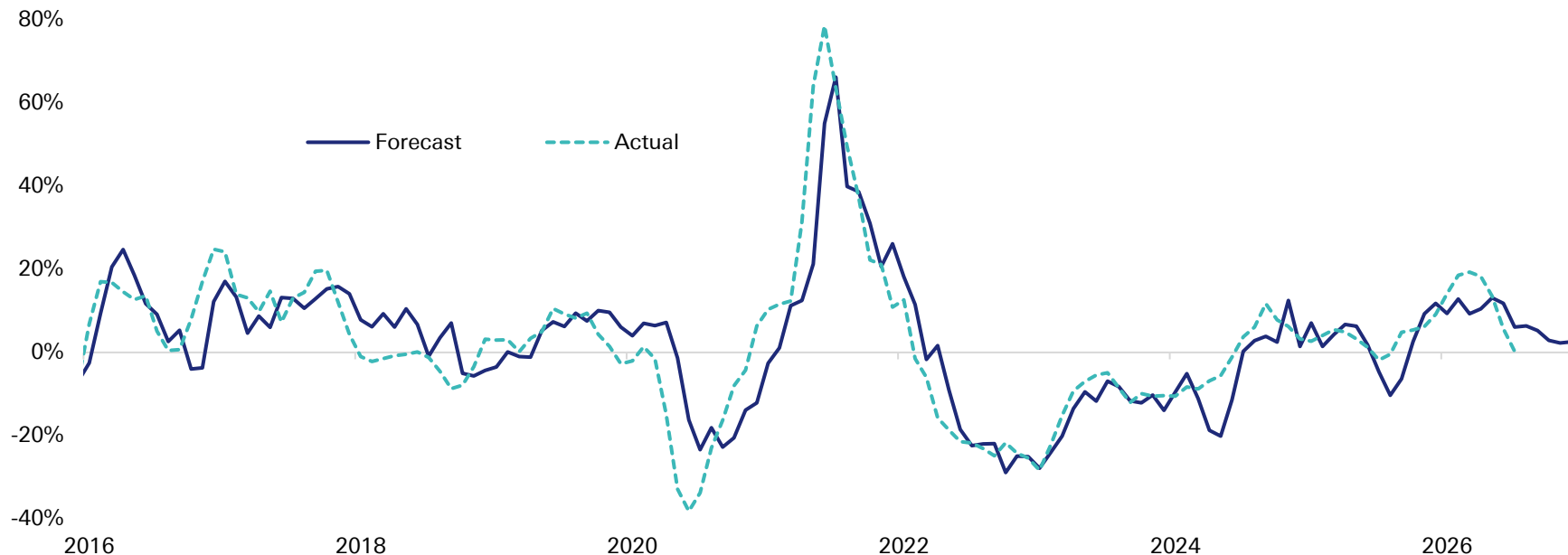
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GEOPOLITICS

Our M&A forecast model: In the US we see low but positive growth over the coming 3 months
Mid-term elections are incentivising some CEOs to wait for greater legislative certainty. That is combining with the reemergence of greater geopolitical uncertainty and the second round of rising oil prices to dampen near-term forecasts



Our 3m-forward forecast for growth in US M&A deal numbers

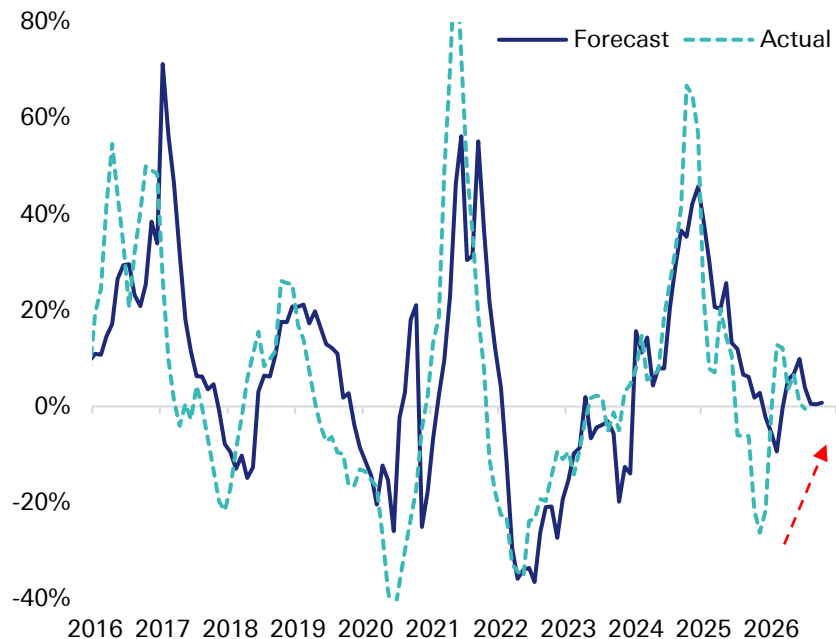


Source: Bloomberg Finance L.P., Dealogic, Deutsche Bank

Our M&A forecast model: In the UK and Europe, our model shows similar low but positive growth
In the UK, this will be encouraging given the market is recovering from a significant drop over the last 24 months

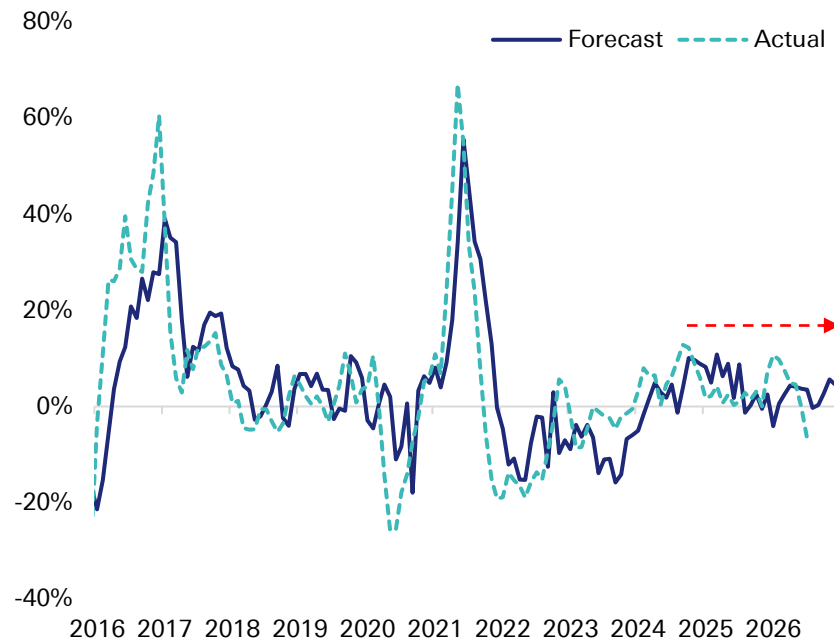


Our 3m-forward forecast for growth in UK M&A deal numbers



Source: Bloomberg Finance L.P., Dealogic, Deutsche Bank

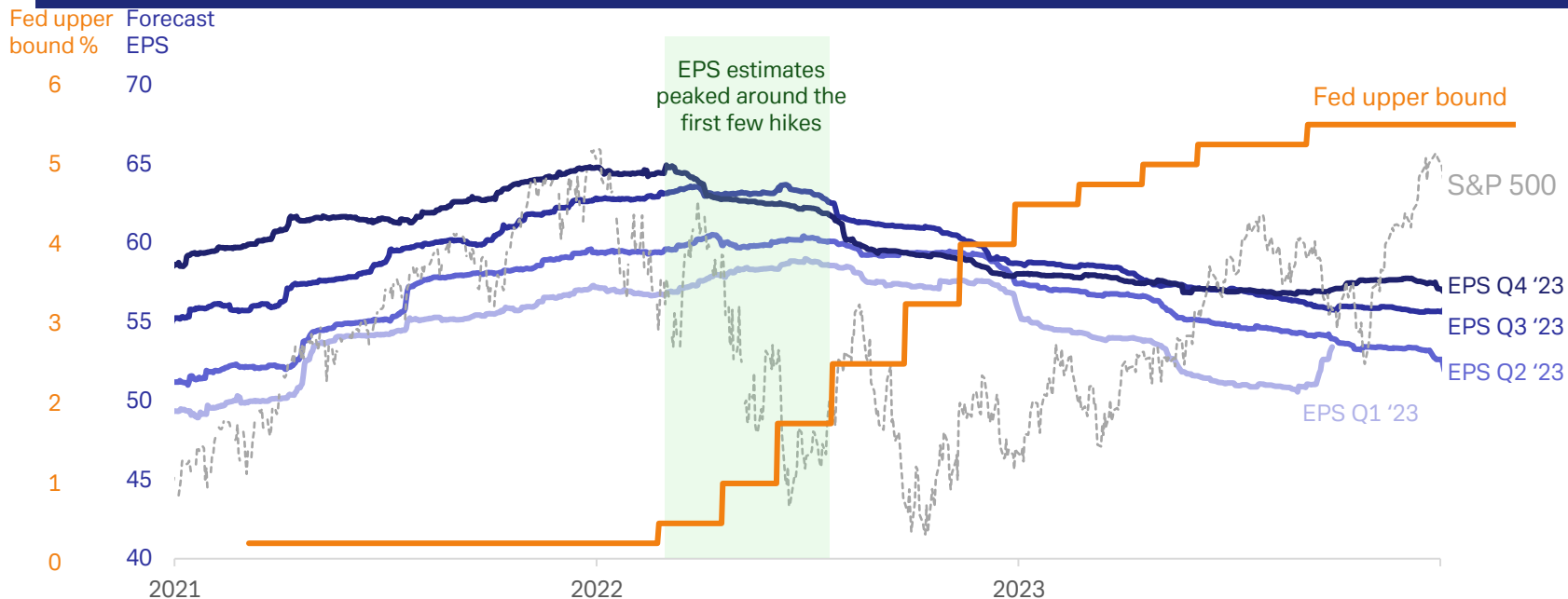
Our 3m-forward forecast for growth in Eurozone M&A deal numbers



Last time around ... Earnings estimates peaked just after the 2022-23 hiking cycle began
It took a few rate rises for investors to materially cut their EPS estimates. However, the extent of the hikes in that cycle were significantly higher than they are likely to be this time around.



EPS estimates for the S&P 500 during the 2022-23 hiking cycle



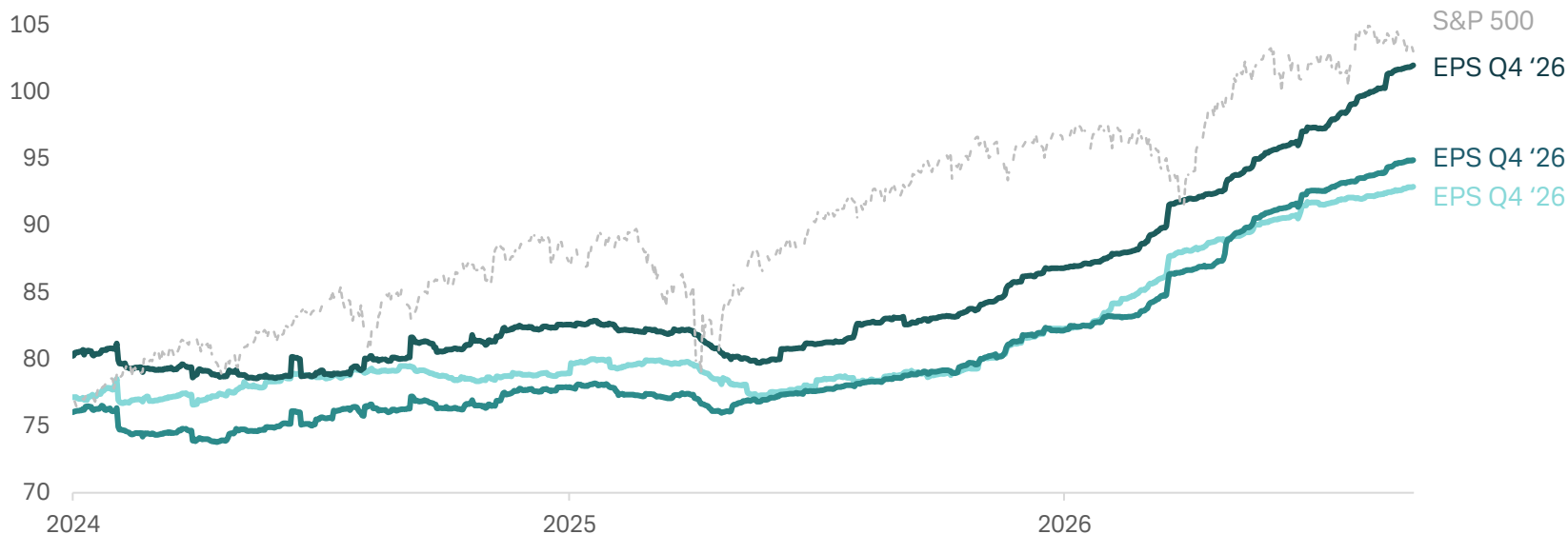
Source: Bloomberg Finance L.P., Deutsche Bank

This time around ... Earnings estimates have accelerated into the rate hike

So far, expectations for rate rises have not caused a substantial rerating of EPS estimates, including in tech companies. The effect may become more pronounced if bond yields breach the 5.00% barrier for a longer period of time.



EPS estimates for the S&P 500 for selected future quarters



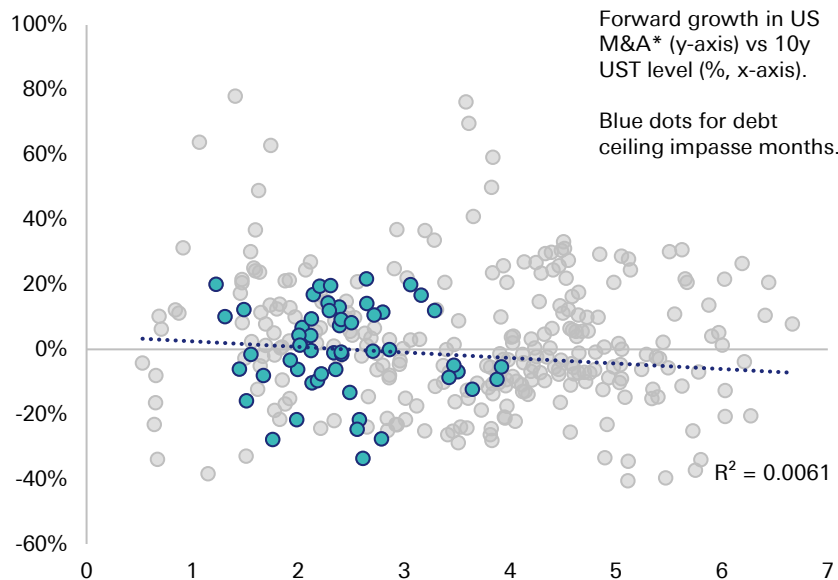
Source: Bloomberg Finance L.P., Deutsche Bank

Positive news: Throughout recent history, M&A is relatively unaffected by treasury yields

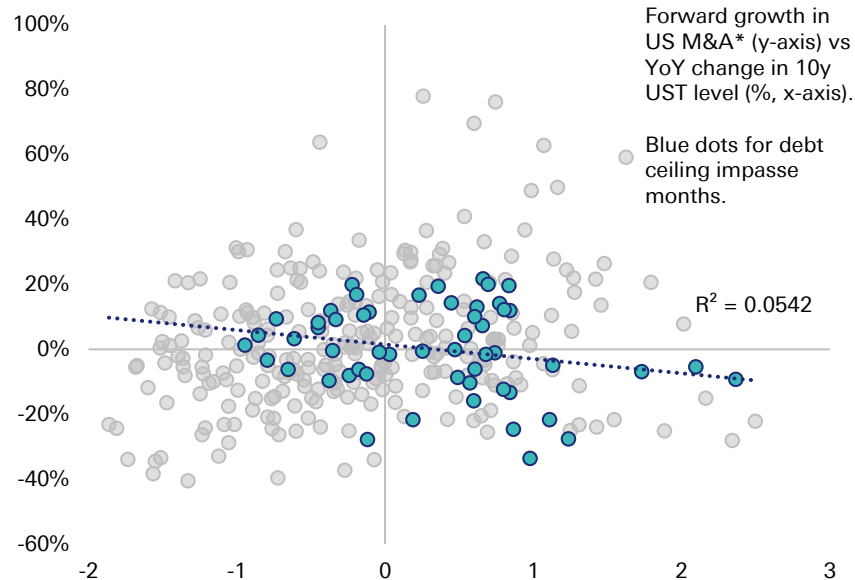
The only material exception to this is when there is an extreme change in the steepness of the 2-10 yield curve – ie. When the yield curve steepens >150 basis points in 12 months.



M&A v the level of the 10yr yield



M&A v the change in the 10yr yield



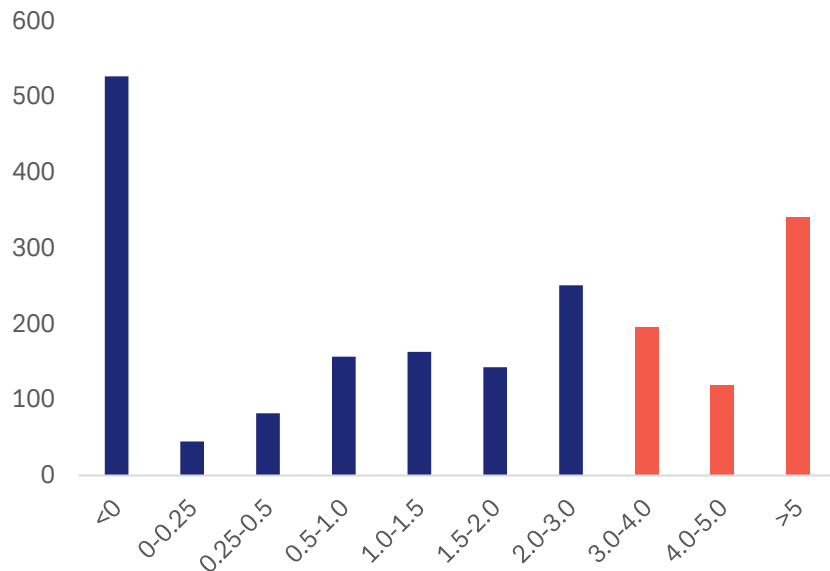
Source: Dealogic, Bloomberg Finance L.P., Deutsche Bank

Make no mistake, higher nominal rates will have a material impact on some firms

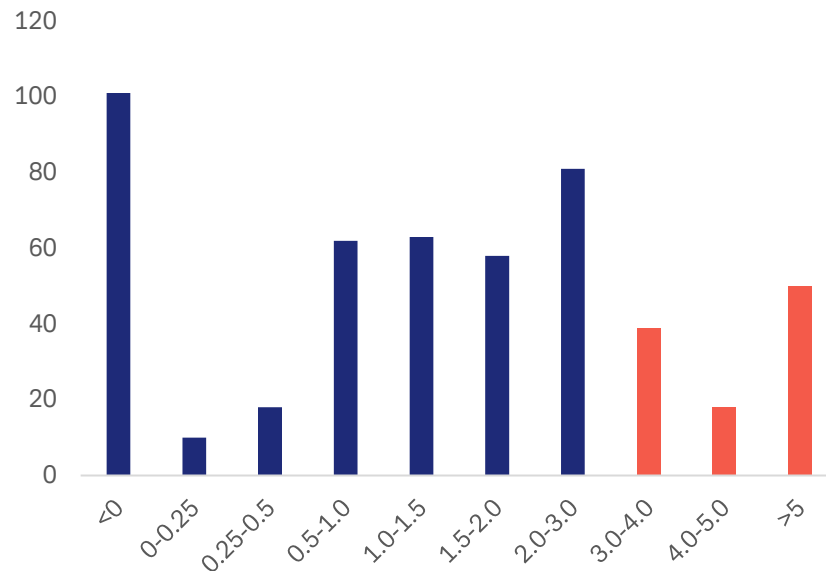
Firms with high debt loads may be more open to M&A opportunities if expectations move towards a more drawn out rate hike cycle



Number of companies with net debt/ebitda in each bucket (Russell 3000)



Number of companies with net debt/ebitda in each bucket (Stoxx 600)

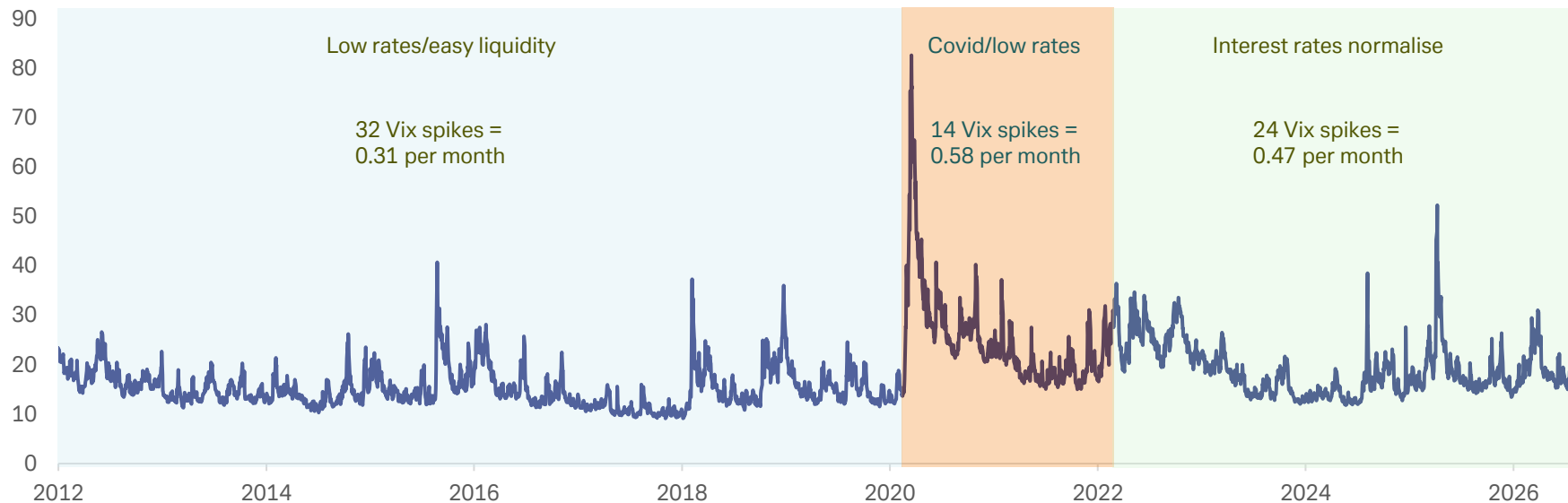


Source: Bloomberg Finance L.P., Factset, Deutsche Bank

The impact of higher rates will be amplified by a market that is showing more volatility
Higher rates tend to bring pre-existing problems to the surface. Given there has been a higher frequency of “volatility events” since rates normalised in 2022, higher rates today could catalyse even more volatility events, even if they are contained.



Vix volatility gauge



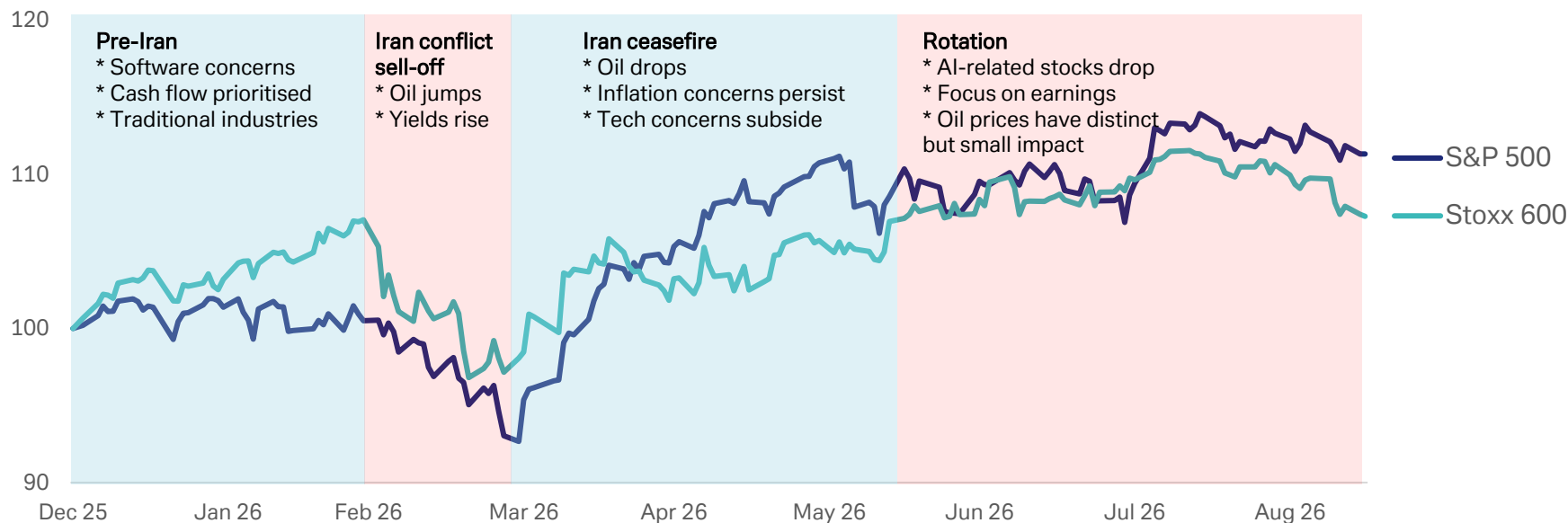
Source: Factset, Deutsche Bank

Already this year, equity markets have lurched between different narratives

Volatility in 2026 has been less dramatic than it was in 2025, however, the changing narratives indicate investors remain skittish



Four stages in equity markets in 2026



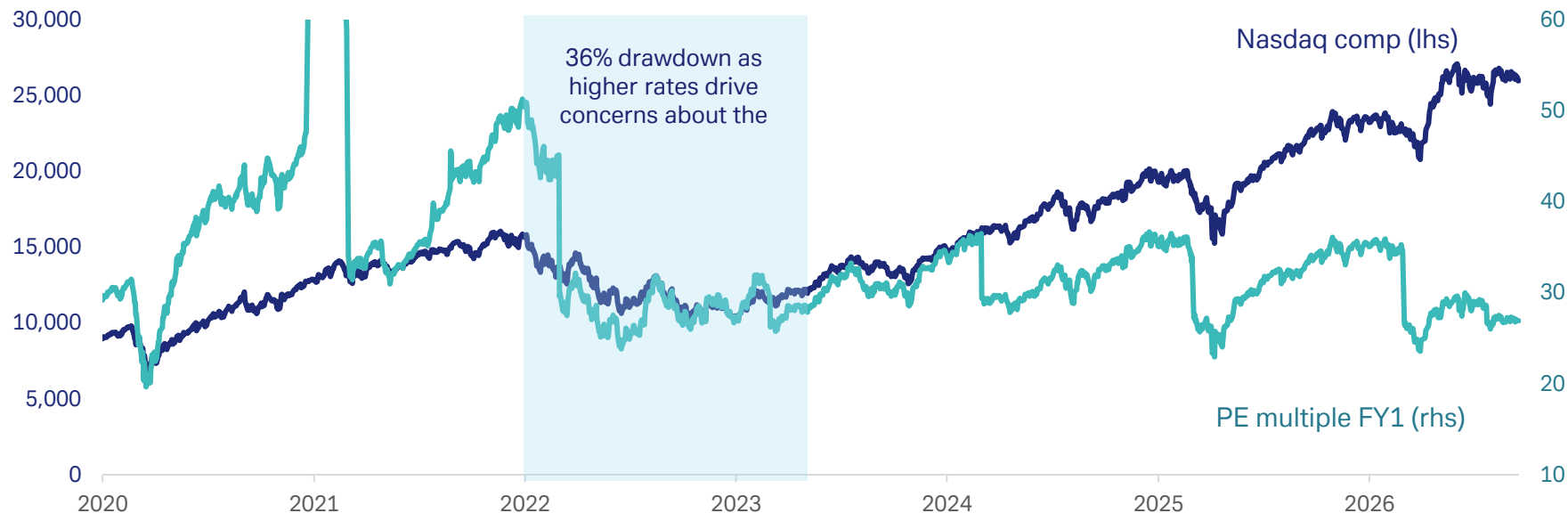
Source: Factset, Deutsche Bank

Broader markets are benefiting from the lack of a tech sell-off, despite higher bond yields

That is a stark contrast to the 2022-23 rate-rise period and is partly due to the strong earning growth at Big-Tech firms. If current earnings forecasts remain intact, current multiples are undemanding compared with recent history. The key thing is expectations about future rate rises and how they affect terminal rates.



Nasdaq composite (price and valuation)



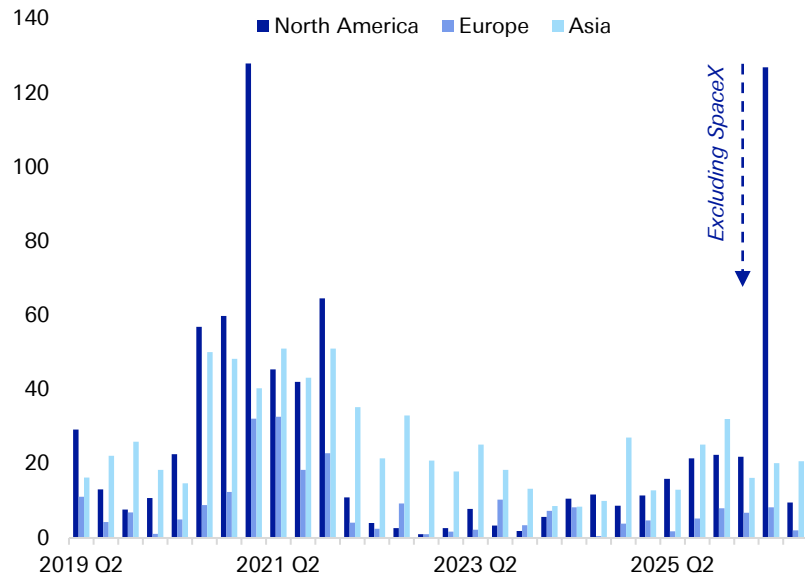
Source: Factset, Deutsche Bank

IPOs are recovering but are still well below 2021 levels ... excluding the very largest

More AI-related businesses are likely to come to the market even after larger firms go public. Tech IPOs have been relatively muted despite the boom in financing this decade. Investors' appetite for AI and big capital markets issuances may push more managers to exit tech holdings while the mood lasts. There have been just c\$7bn in pure-play AI IPOs in the US since 2025.

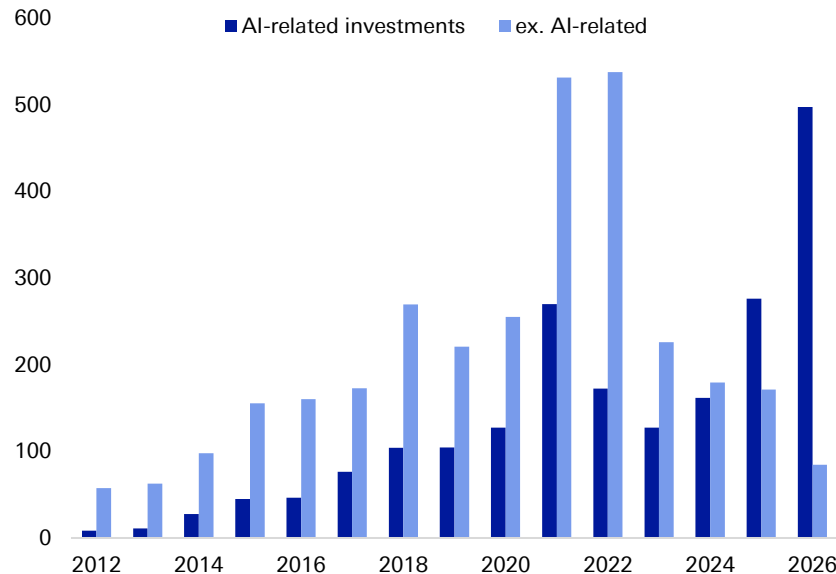


IPO deal values by quarter \$bn



Source: Dealogic, Deutsche Bank

Worldwide VC investment in AI (\$bn)

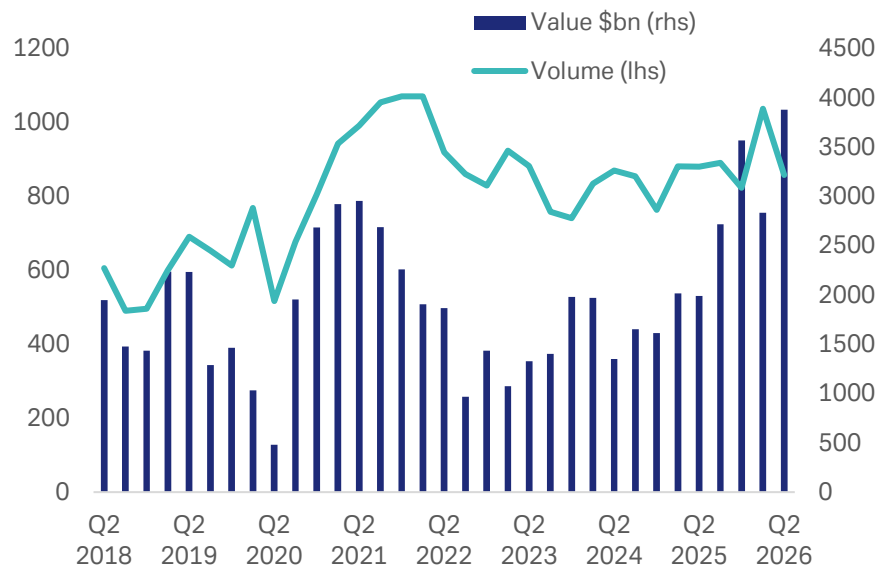


M&A continues to be dominated by big deals

Deal values continue to climb even though deal numbers remain in a tight range. We believe a period of background economic calm is needed for the CEOs of smaller firms to gain the confidence needed to pursue deals.

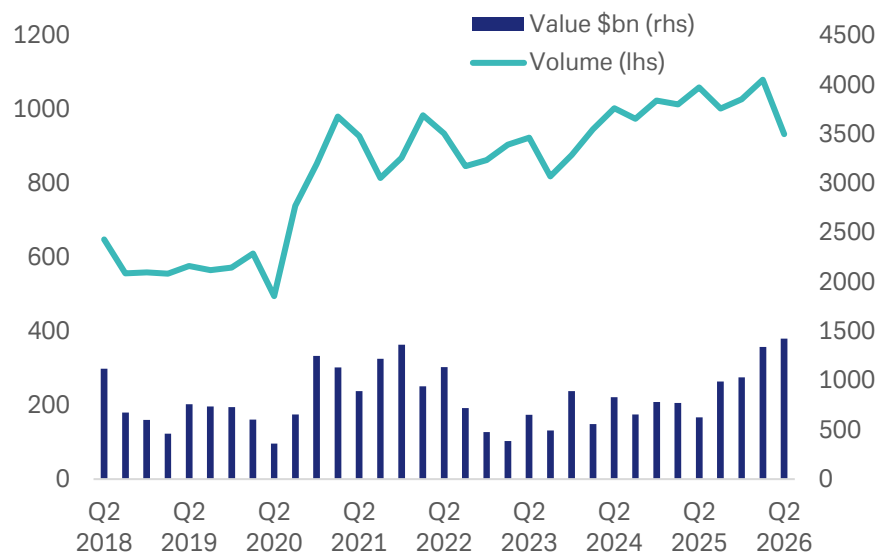


North American M&A



Source: White & Case, Deutsche Bank

Western European M&A



CEO confidence: Leaders expect the economy to be resilient to the current Iran conflict

Still, surveys show that although CEOs are confident about their own company, they are less confident in their industry, and less still in the economy more broadly. That is keeping a lid on some M&A for now.



CEO confidence in the economy 1yr from now



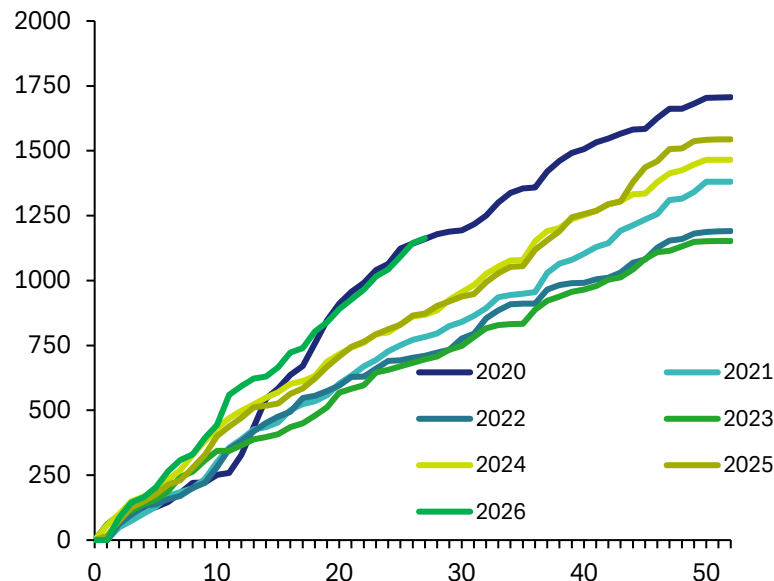
Source: Bloomberg Finance L.P., Deutsche Bank

Financing is abundant ... but can be selective

Debt issuance is still strong and tracking alongside the strongest level this decade. Private capital dry powder continues to push to record levels

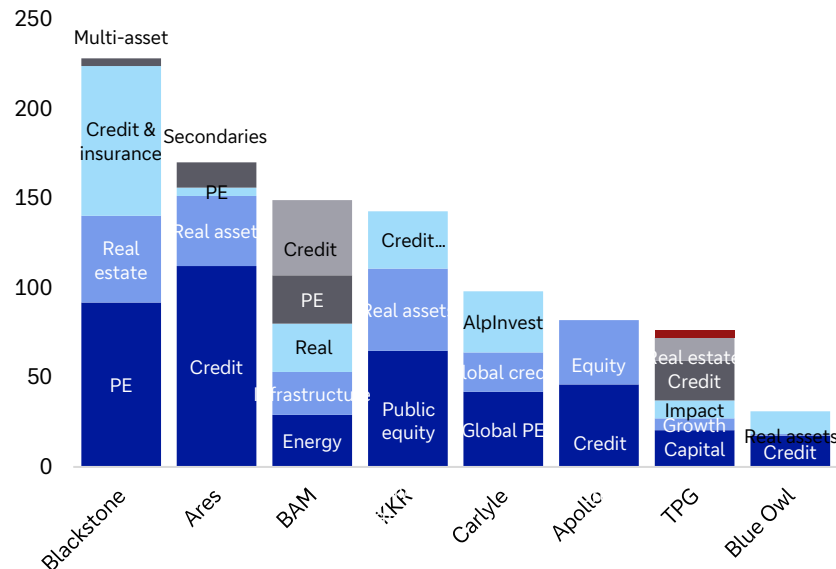


\$IG gross supply (\$bn) by week



Source: Bloomberg Finance L.P., Deutsche Bank

Dry powder by manager and segment, Q2 2026 (\$bn)

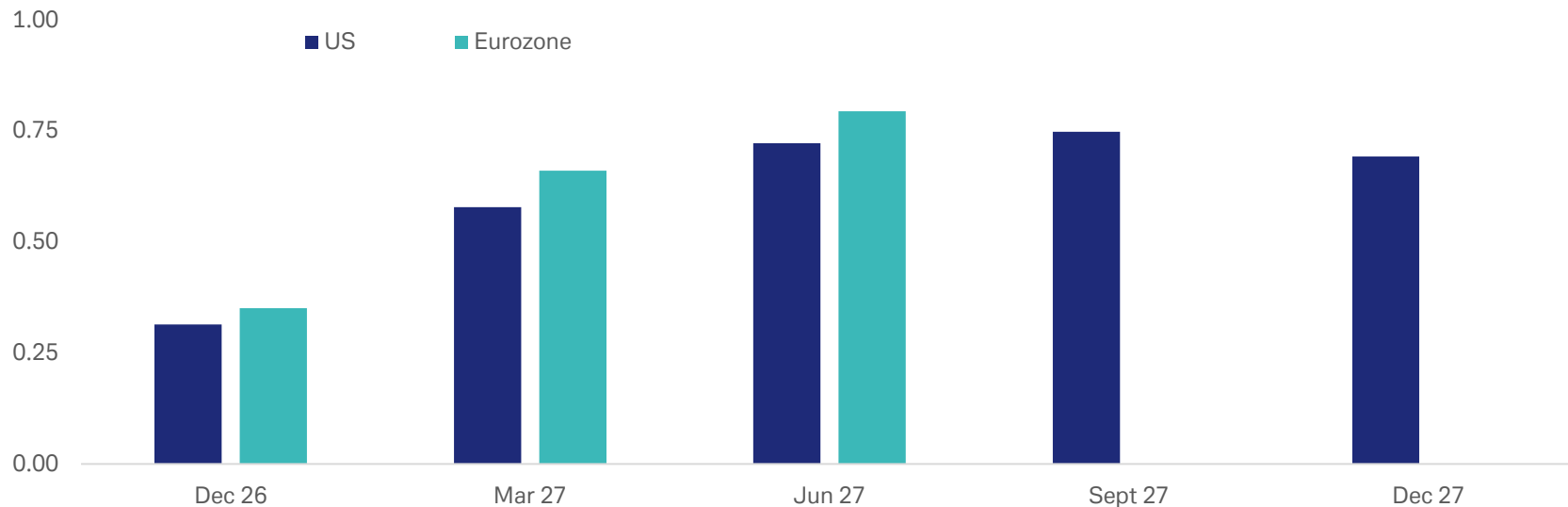


Markets are forecasting an easing of rate-rise (and thus inflation) expectations in 2027

The risks in the current environment may skew towards higher yields as expectations for the conflict in the Middle East lengthen and its impact becomes more severe



Implied interest rate rises by the end of each period (pp)



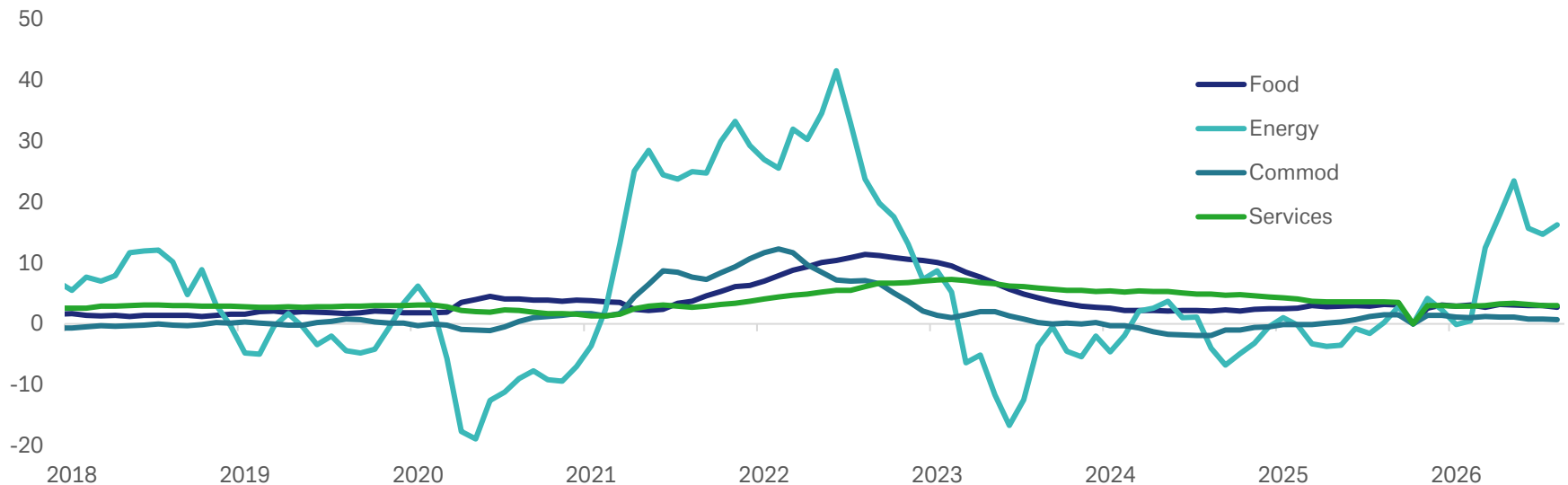
Source: Bloomberg Finance L.P., Deutsche Bank

The last energy shock lasted about 2 years

Iran's assertive push to control energy flows around the Middle East makes it unfeasible to project the length of energy shock



US CPI components (yoy % change)



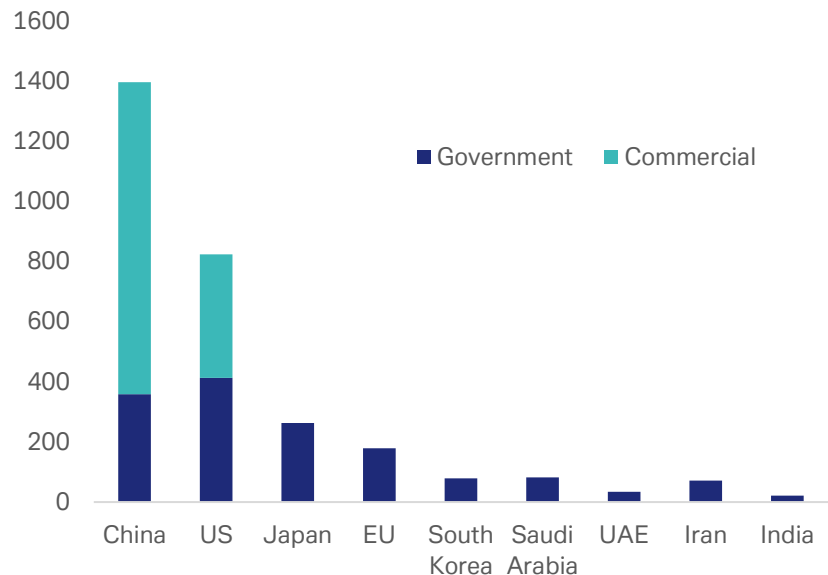
Source: Haver Analytics, Deutsche Bank

The trajectory of the energy shock will depend, in part, on oil inventory buffers

Industry executives have warned over the last two weeks that energy inventories have buffered the world economy for the last 6 months but that those buffers are approaching a cliff edge



Crude oil inventories pre-Iran conflict (m barrels)



Source: RAND, Deutsche Bank

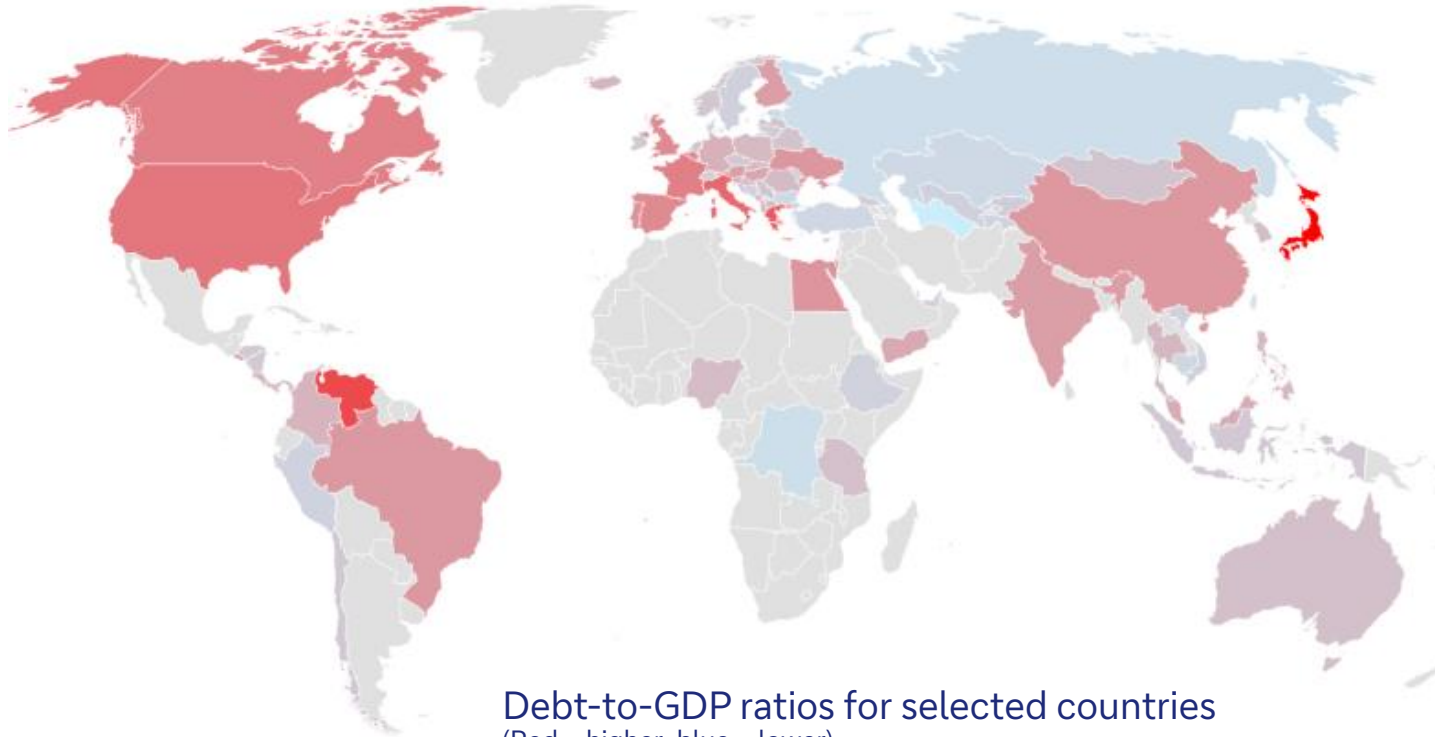
The way China uses its Strategic Petroleum Reserve is critical for world energy prices

RAND estimates China held between 1.1bn and 1.4bn barrels as of May this year – equivalent to 110-140 days of net imports.

That compares with the 400m barrel release coordinated by the IEA in March which had only a limited impact on markets.

Sovereign debt around the world

Confidence in indebted sovereigns tends to ebb and flow and markets usually experience at least one major 'risk-off' moment regarding sovereign debt each year. Upcoming elections in Europe will be a major focus point for investors over the coming months.



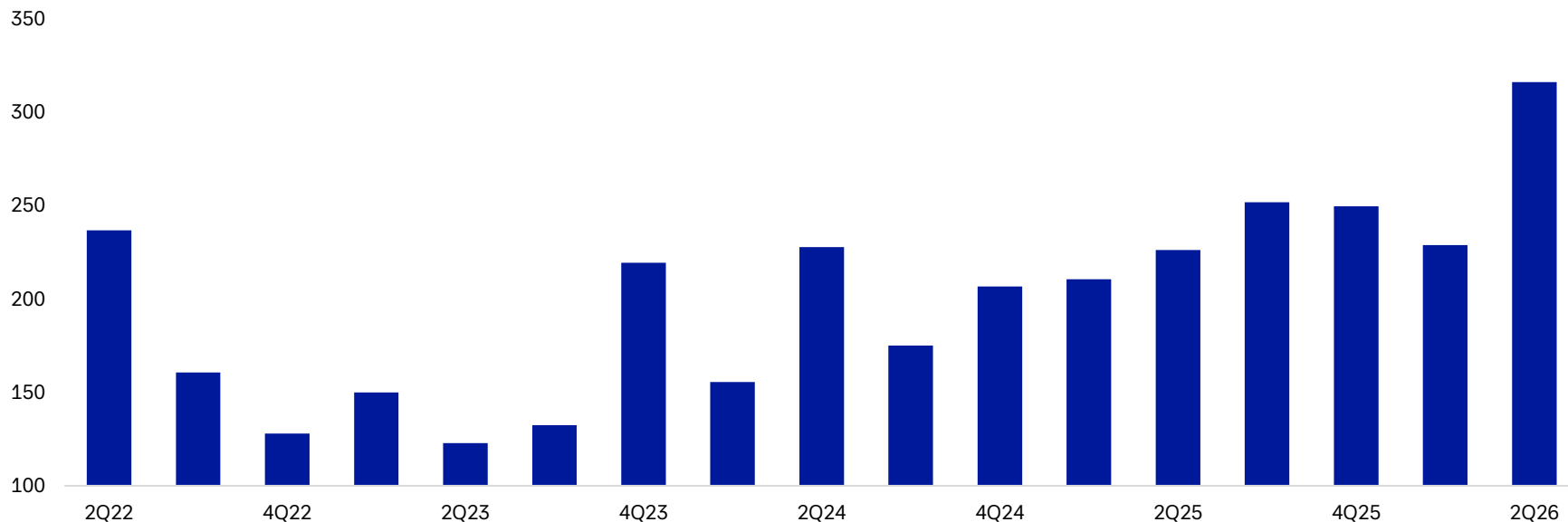
Source: IMF, Deutsche Bank

Private capital is searching for deals 1/2

Aggregate fundraising growth has shot up (in part due to one-off factors) and managers are confident in continued strength



Fundraising by selected large, listed private capital managers (\$bn)



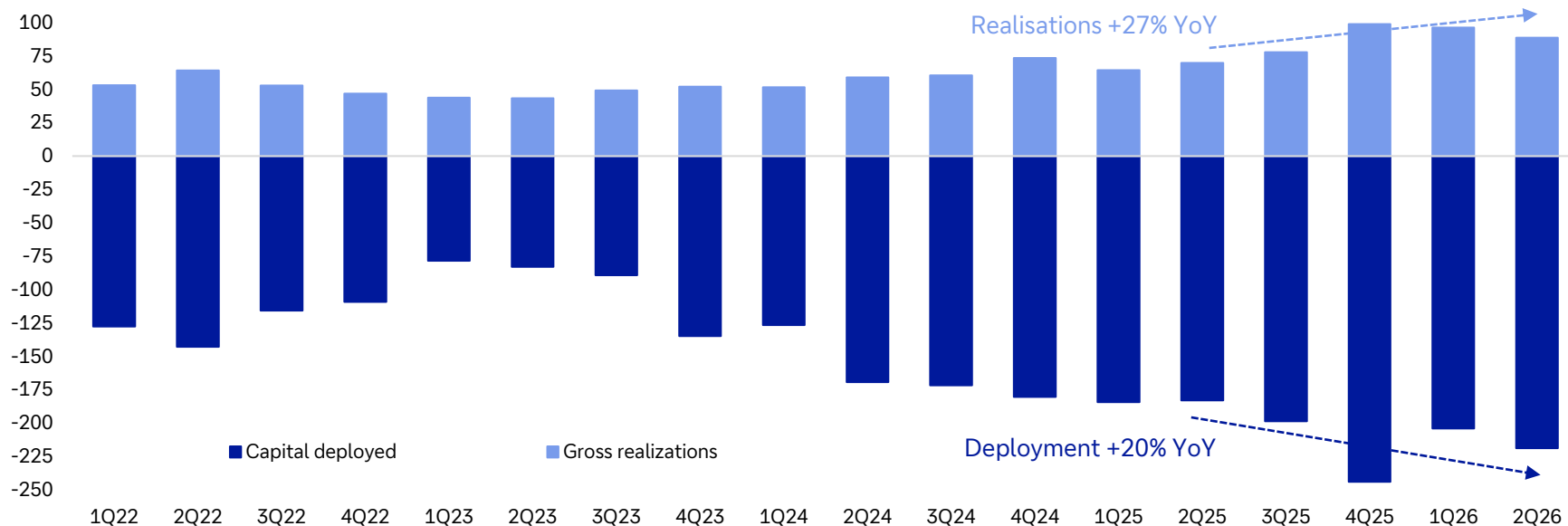
Source: Deutsche Bank company research - [US Brokers, Asset Managers & Exchanges - Deutsche Bank Research](#), Deutsche Bank

Private capital is searching for deals 2/2

Manager confidence has been helped by the growing rate of deployment and realisations. Through to the end of the year, realisations growth will be in focus given the growing macro uncertainty



Selected private managers, deployment v realisations (\$bn)



Source: Deutsche Bank company research - [US Brokers, Asset Managers & Exchanges - Deutsche Bank Research](#), Deutsche Bank

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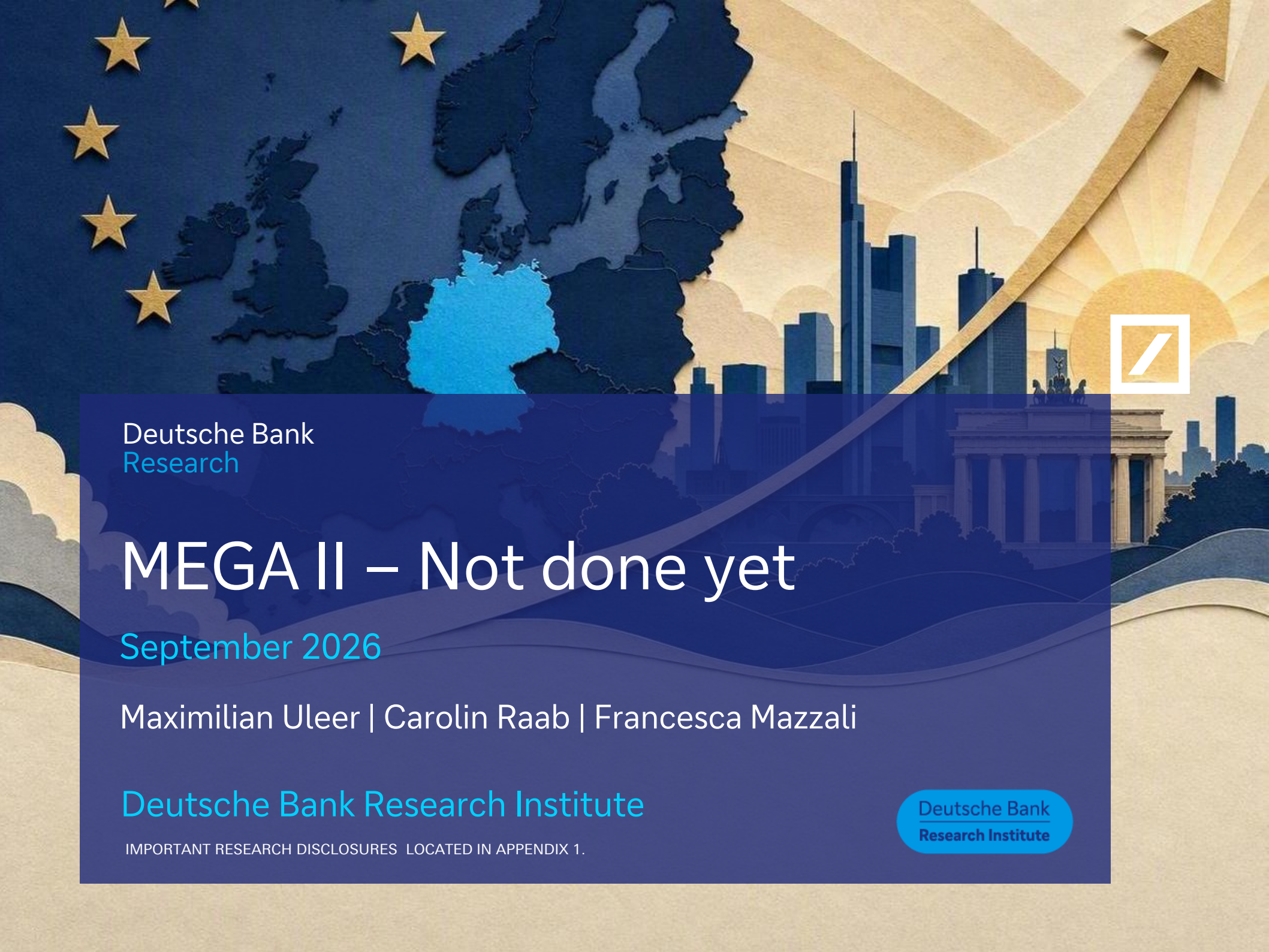
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MEGA II – Not done yet

September 2026

Maximilian Uleer | Carolin Raab | Francesca Mazzali

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IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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Europe is back and Germany's potential seems not yet fully appreciated by markets



In early 2025, we made a bold call: we argued that Europe's 15-year history of underperformance has come to an end (MEGA and [here](#)). We projected the P/E valuation gap between US and European equities to narrow from a record 9.0x at the end of 2024 to 4.3x. Since then, the gap has indeed tightened to 4.7x. With Germany's potential still being underestimated by markets, we expect more to come for Europe.

Looking ahead, we see several structural factors supporting Europe's relative attractiveness:

- **Valuations:** After the tightening, we expect the gap to stabilize. Higher US growth justifies a premium but most of the below arguments argue against a renewed widening.
- **Earnings:** We expect the European earnings recovery to continue, delivering double-digit growth in 2026. While US earnings are still expected to grow faster, uncertainty around the sustainability of future growth, particularly in the Tech sector, remains elevated.
- **Germany:** The US and China are likely to continue growing faster than Europe, but much of that growth has been accompanied by widening deficits and rising debt. US fiscal deficit and debt metrics now exceed European levels. At the same time, Germany's fiscal pivot could support a relative recovery in European growth, an opportunity that equity markets have yet to fully embrace. A recovery in global manufacturing activity provides an additional boost to Europe's economy. Germany could become the powertrain of Europe's economic recovery. Having reduced its debt-to-GDP ratio over the past decade despite macro headwinds, it now has the fiscal capacity to support growth through a substantial spending plan including EUR 1tr in new debt. The economy is showing greater resilience than many expected, manufacturing activity is improving, international investor interest is beginning to re-emerge and reforms are advancing. We estimate the pension reform alone could unlock €10-15bn of annual inflows into European equities.
- **Diversification:** Europe's equity market is well diversified across sectors, providing broader return drivers than more concentrated markets such as the US and Emerging Markets. This helps cushion performance during sharp market downturns.
- **AI:** While Europe is unlikely to catch up with the leading AI developers, it is also less exposed to risks surrounding the monetisation of heavy AI investment. Instead, it should benefit indirectly from AI spending through its advanced manufacturing capabilities, its exposure to AI infrastructure and through growing AI adoption itself.
- **Political stability:** While Europe's fragmented political landscape is a structural disadvantage, external pressures from US and Chinese trade policies and ongoing geopolitical challenges have strengthened coordination across the EU on trade, defence and fiscal matters.

Closing valuation differential



In March 2025, we called for a decline in the valuation gap between US and EU equities. We have now moved significantly closer to our target.

P/E difference S&P 500 and STOXX 600 (12m fwd)



*DB forecast based on the P/E difference before the start of the Russia-Ukraine war (02/24/2022). Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

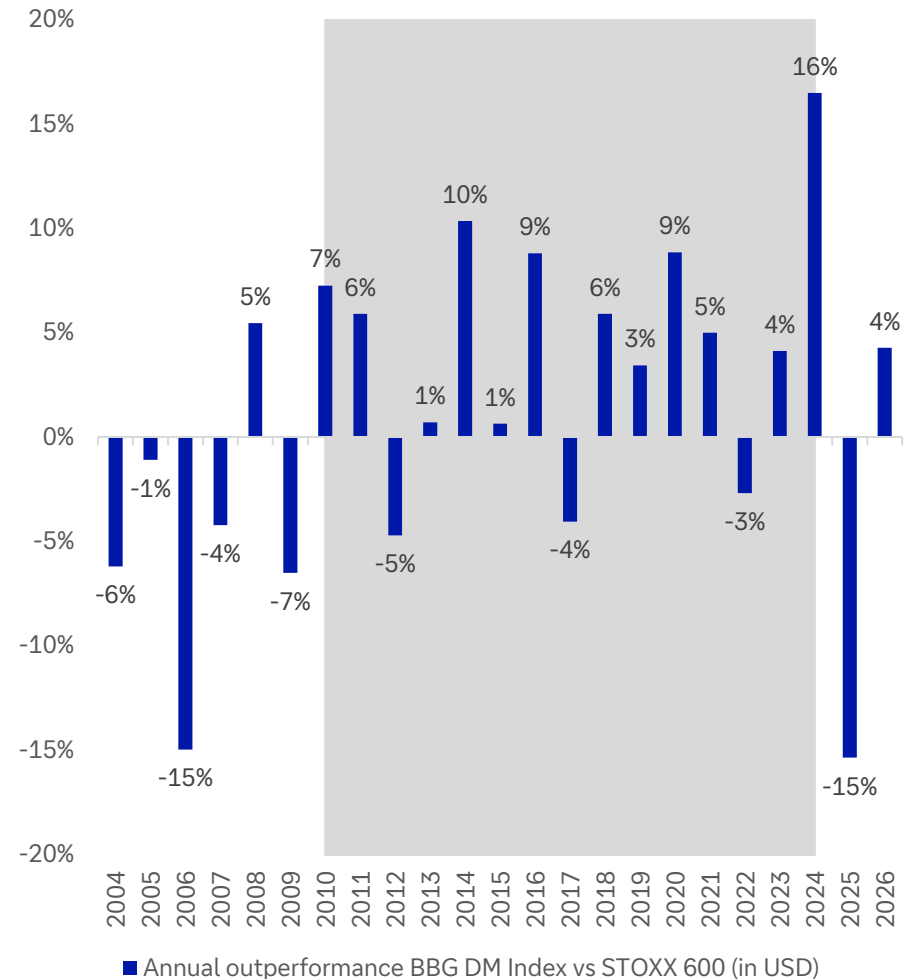
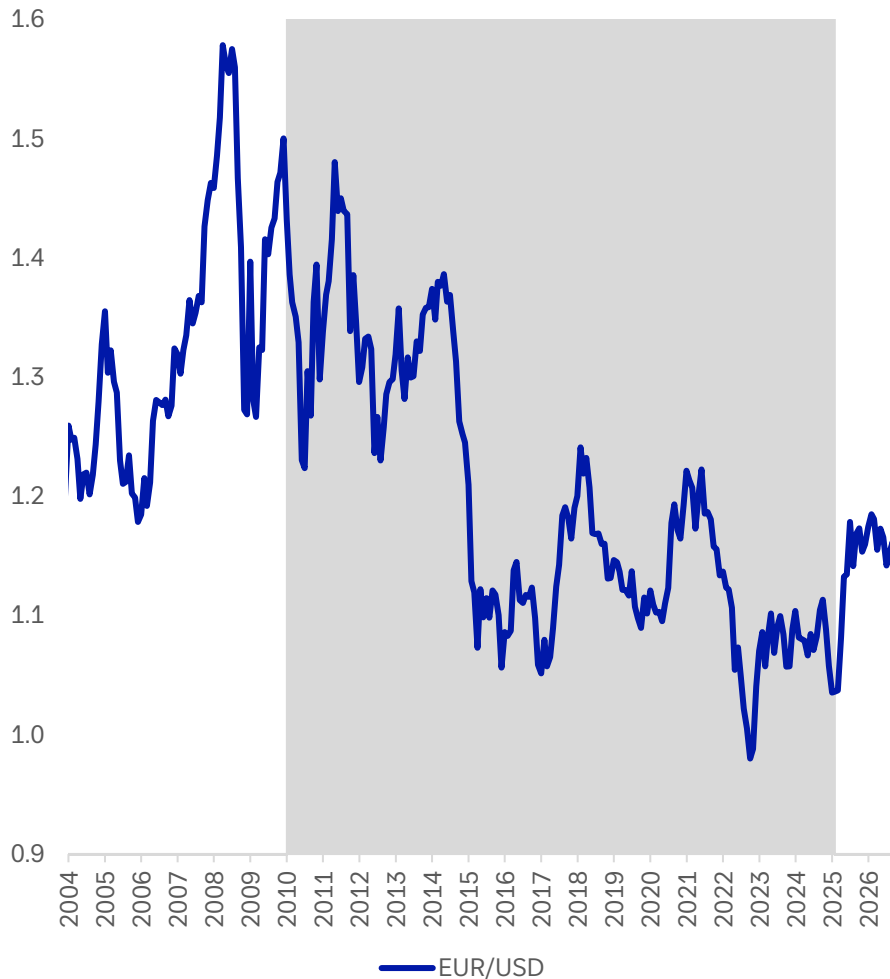
US Dominance... from 2010 to 2024?



European underperformance coincided with a 15-year USD rally. Will the USD continue to grow in its importance over the next 10 years?

In 2010-2024, the USD strengthened...

...supporting RoW outperformance



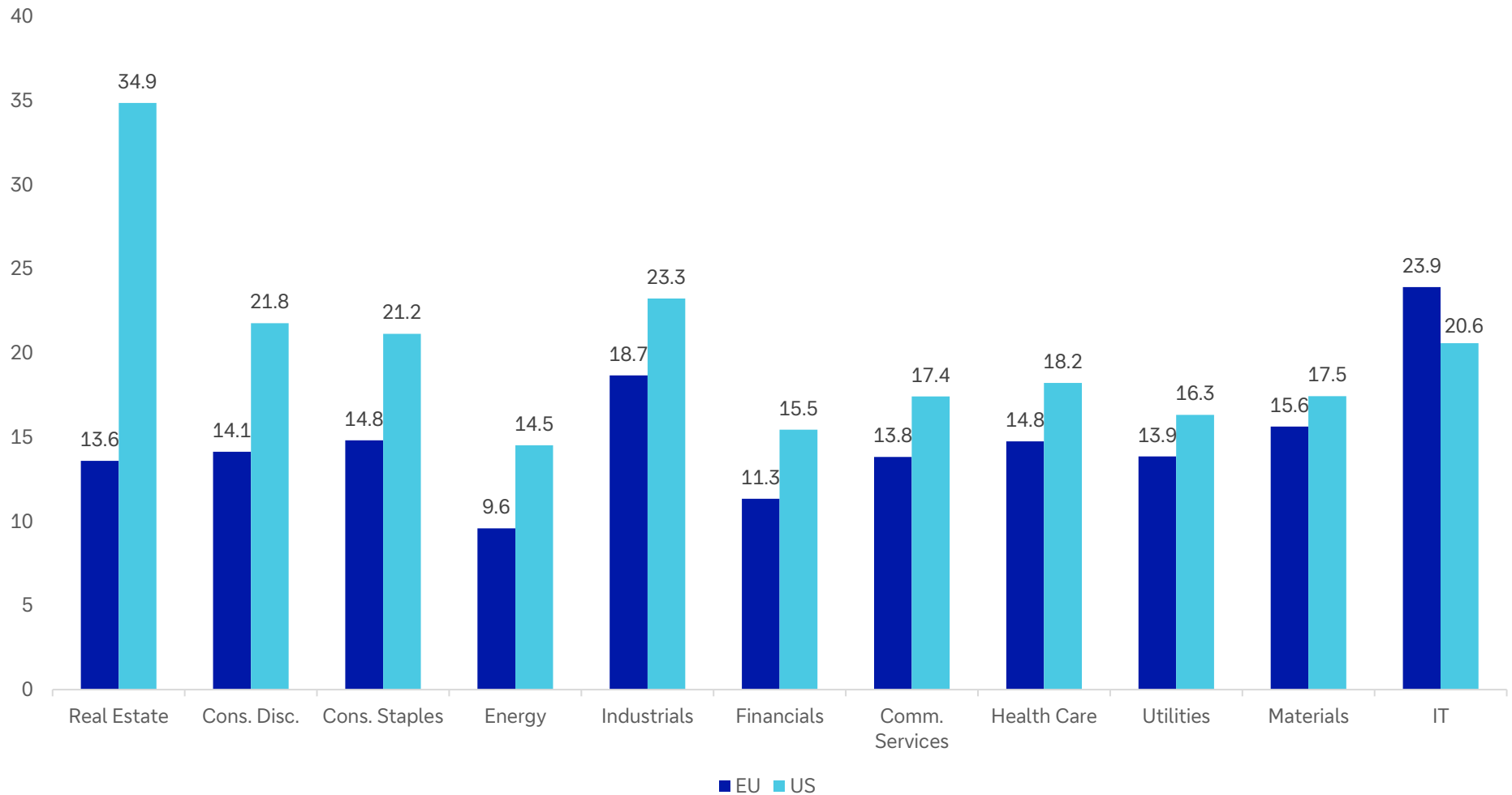
Note: Right chart is total return, 2026 is year-to-date. Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

Valuation Gap: applies across sectors



The valuation premium no longer stems from the IT sector. Still, every US sector, ex IT, trades at a premium versus the European sector.

Current sector P/E (12m fwd), sorted P/E difference US vs EU sectors



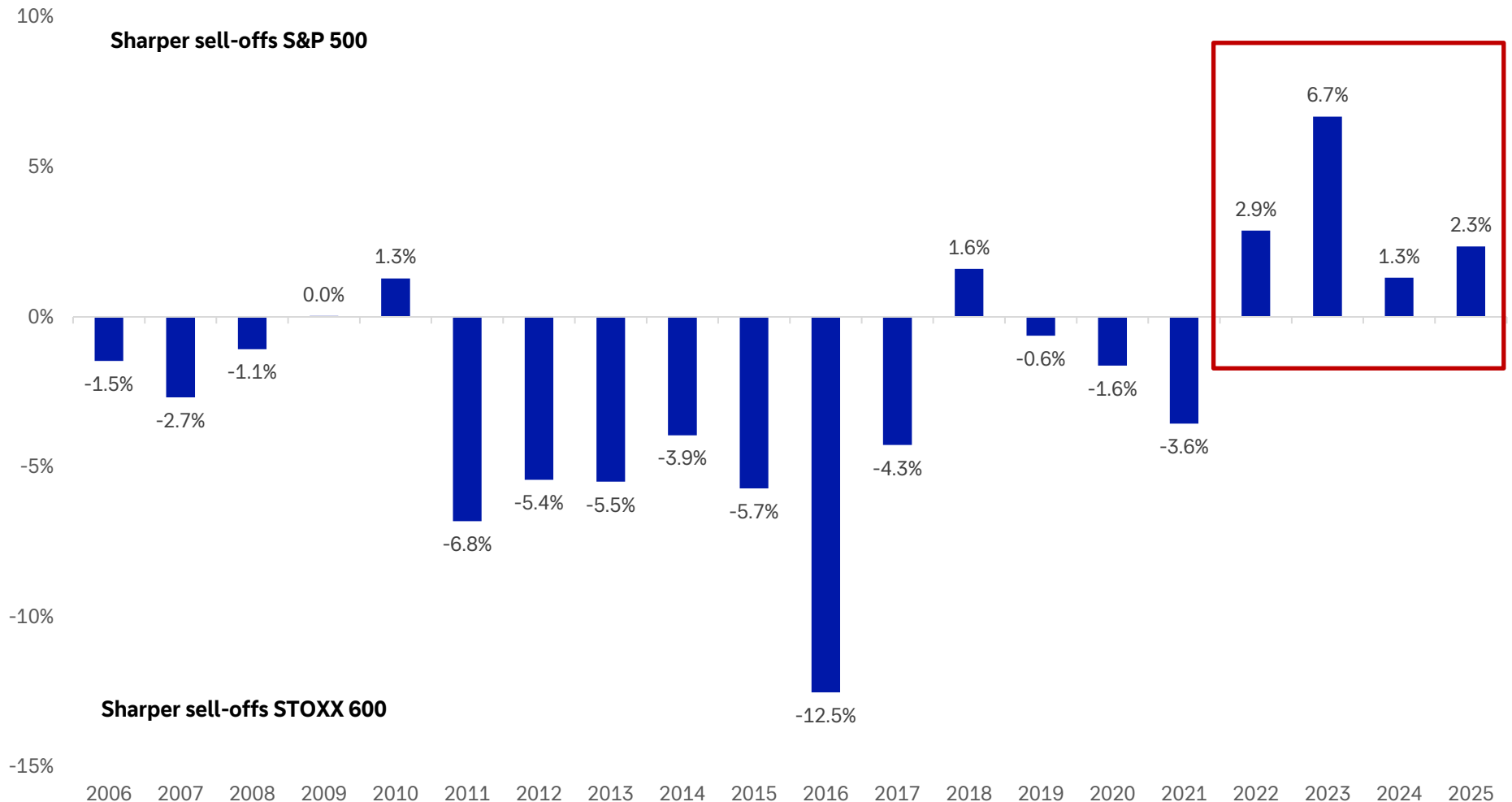
Reference indices: S&P 500 and STOXX 600, GICS sectors. Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

Concentration: leads to sharper sell-offs



„When the US sneezes, Europe catches a cold” no longer holds true. Drawdowns are significantly sharper in US equities. “When the US sneezes, Europe says Gesundheit”.

Difference in maximum drawdown of S&P 500 vs STOXX 600 (annual, local currency)



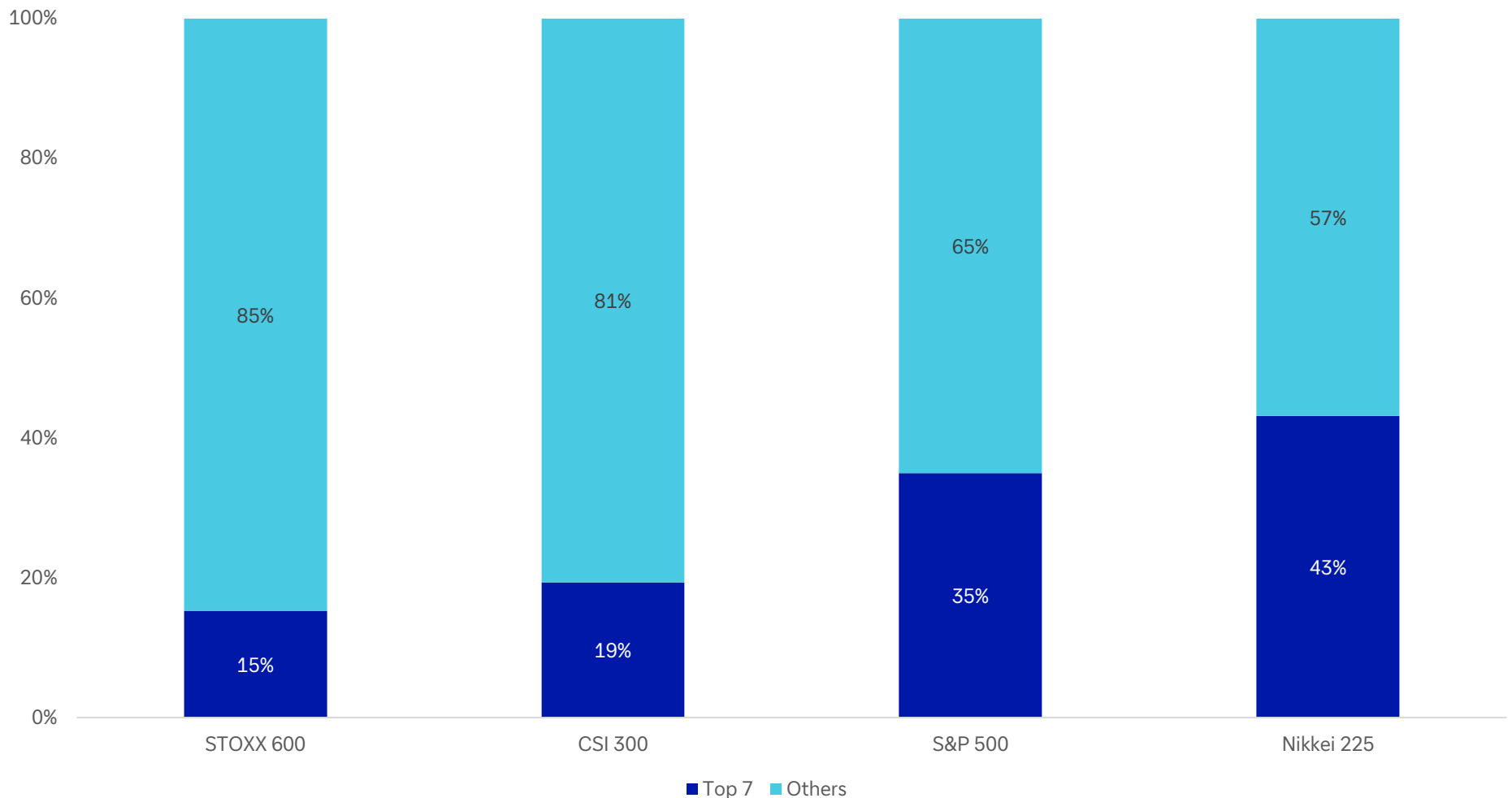
Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

Concentration: in index weights



The European STOXX 600 is well diversified. The 7 largest corporates make up only 15% of the index weight vs 43% in the Nikkei and 35% in the S&P 500.

Current index weights



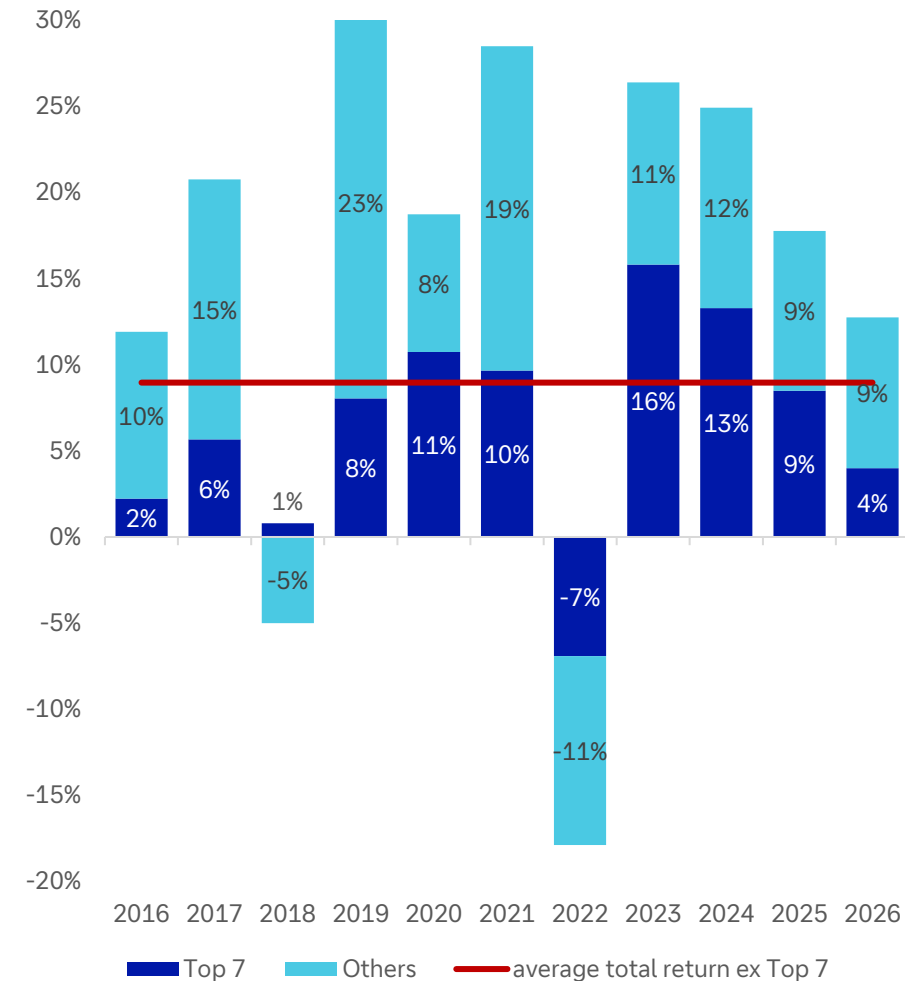
Source: Bloomberg Finance LP, Deutsche Bank Research 09/14/2026

Concentration: in performance

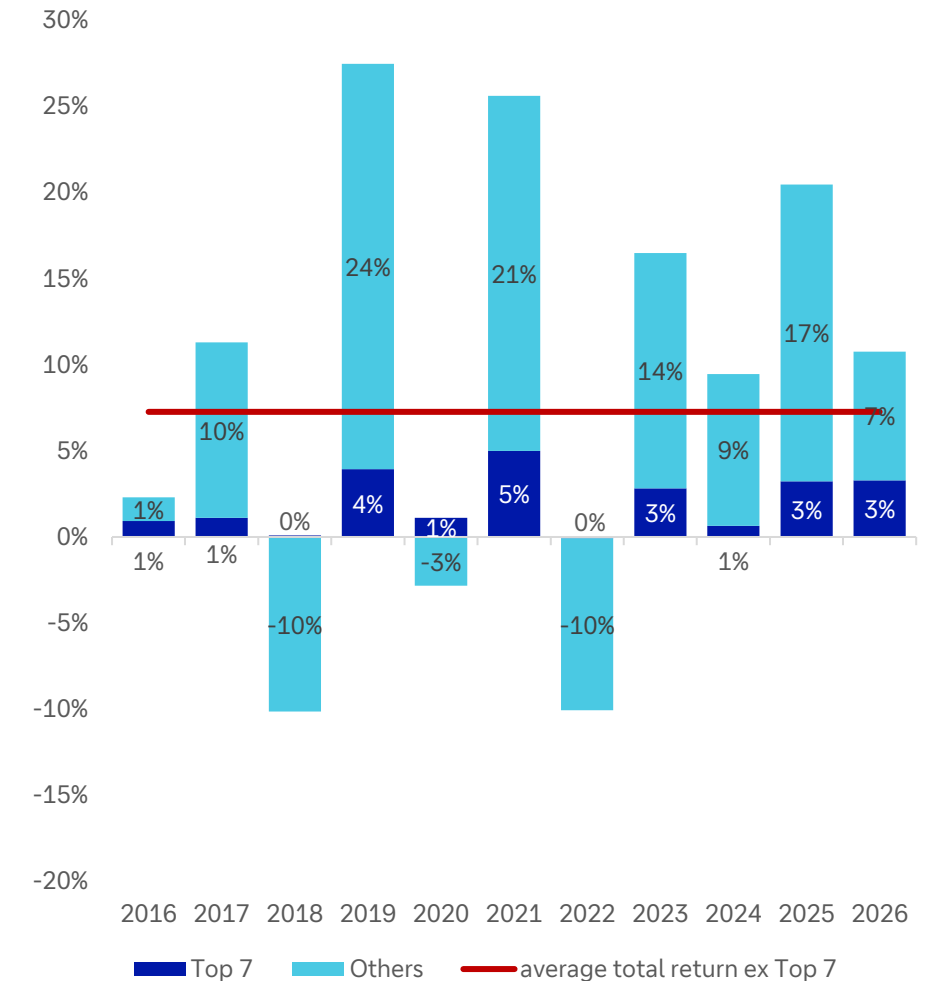


On average over the past 5 years, the S&P returned 16% per year vs 12% in the STOXX 600. Without the top 7 companies, the S&P would have delivered an average return of 8% vs 10% for the STOXX ex top 7.

S&P 500 – total return contributions



STOXX 600 – total return contributions



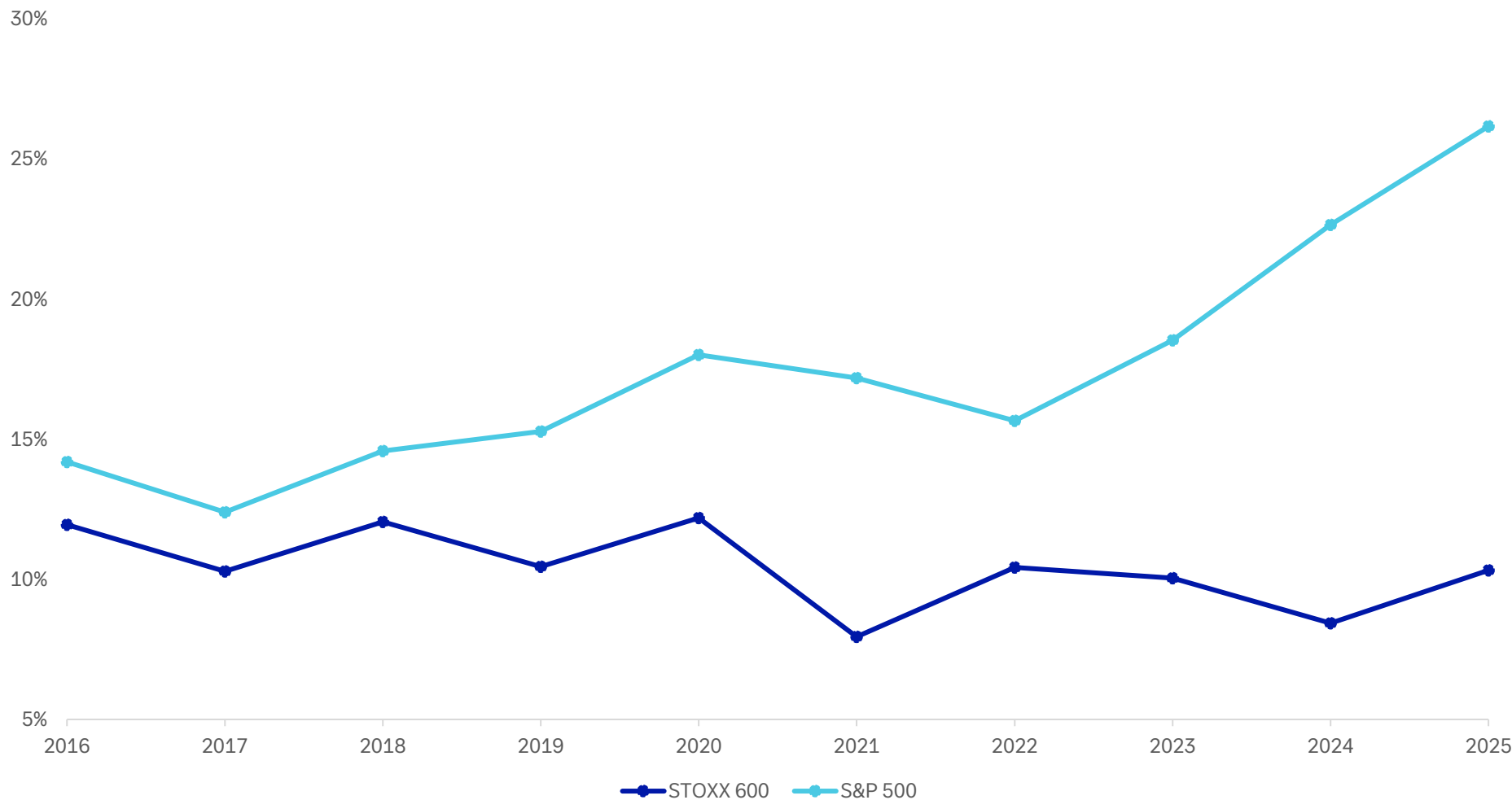
Note: 2026 is YTD. Source: Bloomberg Finance LP, Deutsche Bank Research 09/14/2026

Concentration: in earnings



The top 7 US companies contributed 53% to earnings growth since 2021. In the STOXX 600, the top 7 only accounted for 33% of profit growth over the same period.

Top 7 companies: share of aggregate index earnings



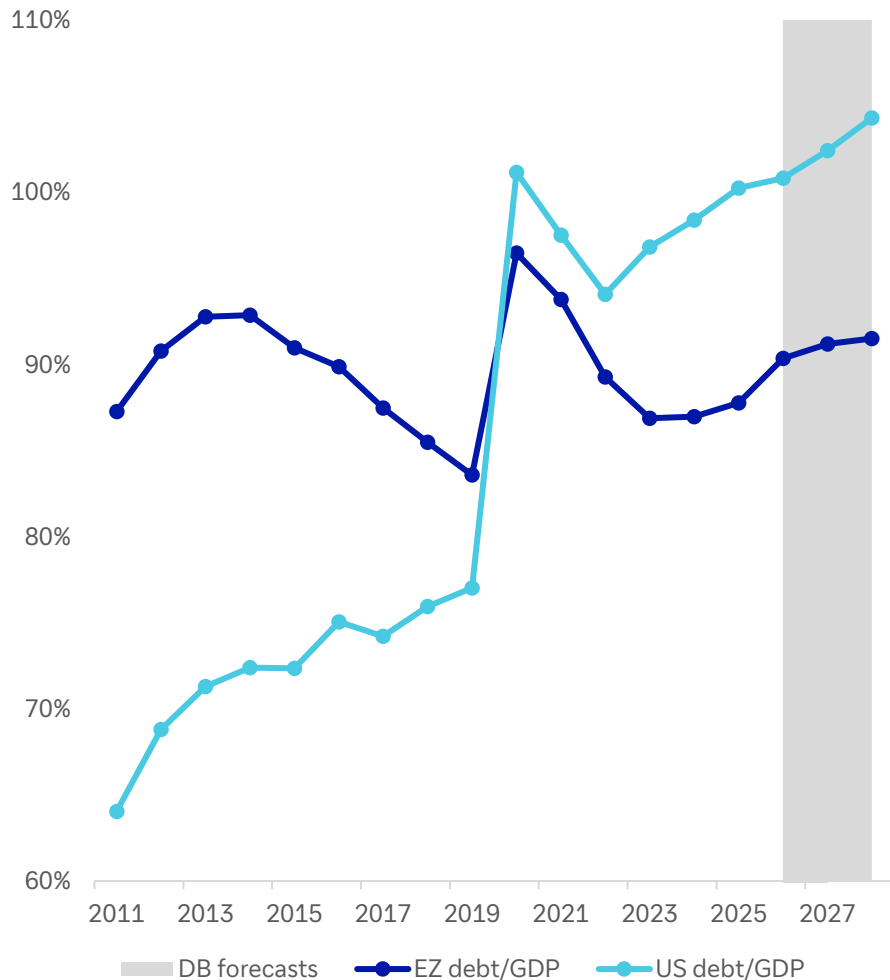
Note: based on current index members and current Top 7. Source: Bloomberg Finance LP, Deutsche Bank Research 09/14/2026

Fiscal Risks – look increasingly similar

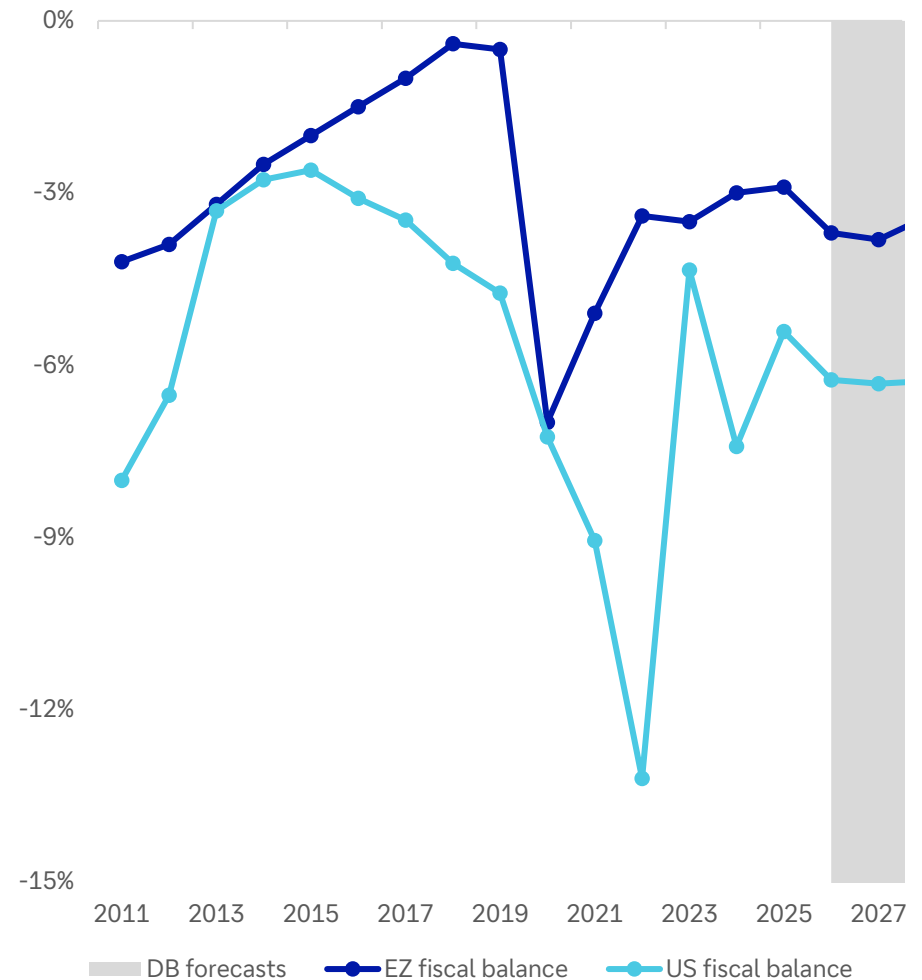


Despite the German fiscal expansion, debt ratios are expected to worsen more in the US than in the Eurozone.

Debt-to-GDP: Eurozone vs US



Fiscal balance (% GDP): Eurozone vs US



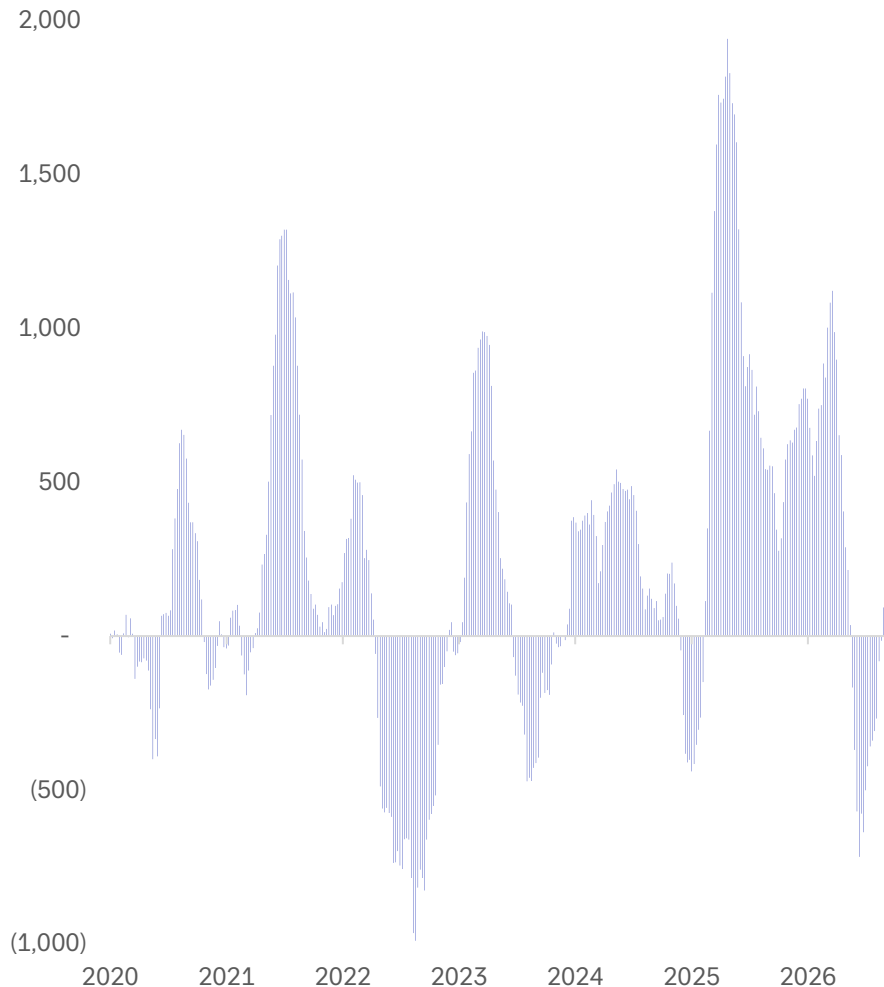
Note: Debt-to-GDP is net for US and gross for EZ to align with DB forecasts. Source: Deutsche Bank Research 09/15/2026

Flows – US dominates, but less than it did

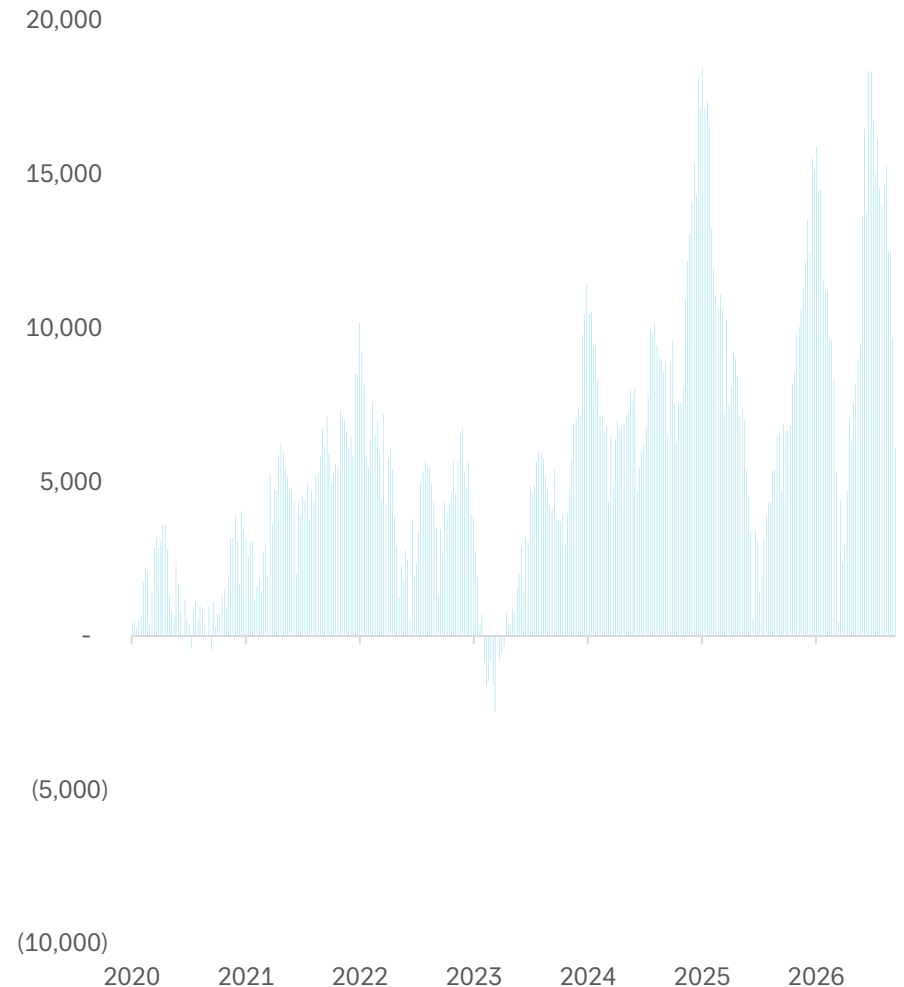


Inflows into European-focused ETFs have picked up significantly over the past two years but remain much smaller and more volatile than US flows.

Europe



USA



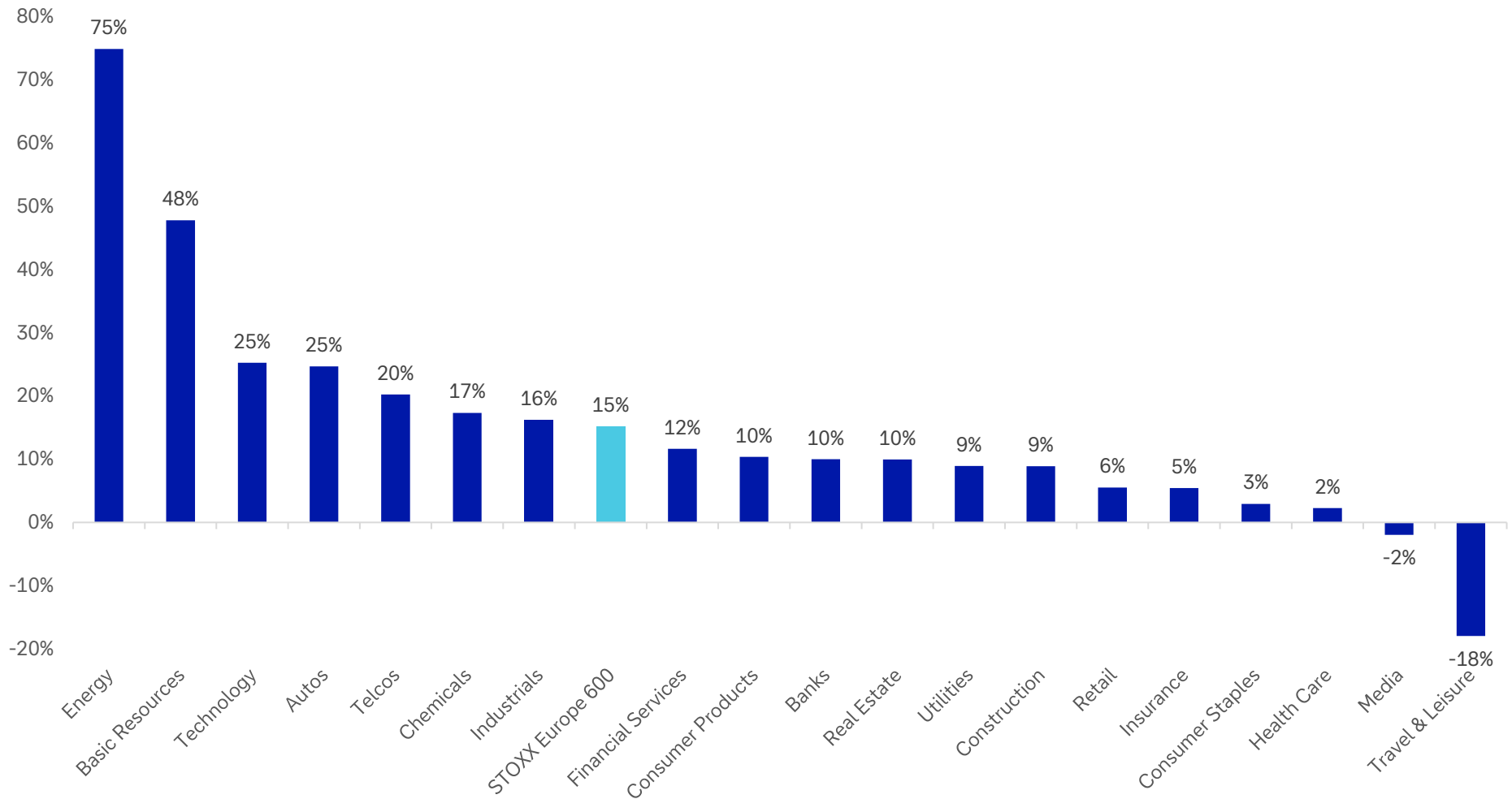
Note: Aggregate fund flow, 12w moving average, based on 50 largest ETFs for the region, in mn USD. Source: Bloomberg Finance LP, Deutsche Bank Research 09/14/2026

Earnings – Broad-based growth



The outlook for European earnings is encouraging. Almost all sectors should see positive earnings growth this year.

STOXX 600: Consensus expected earnings growth, FY26



Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026



Deep dive: Germany

Don't underestimate Germany's capacity to change



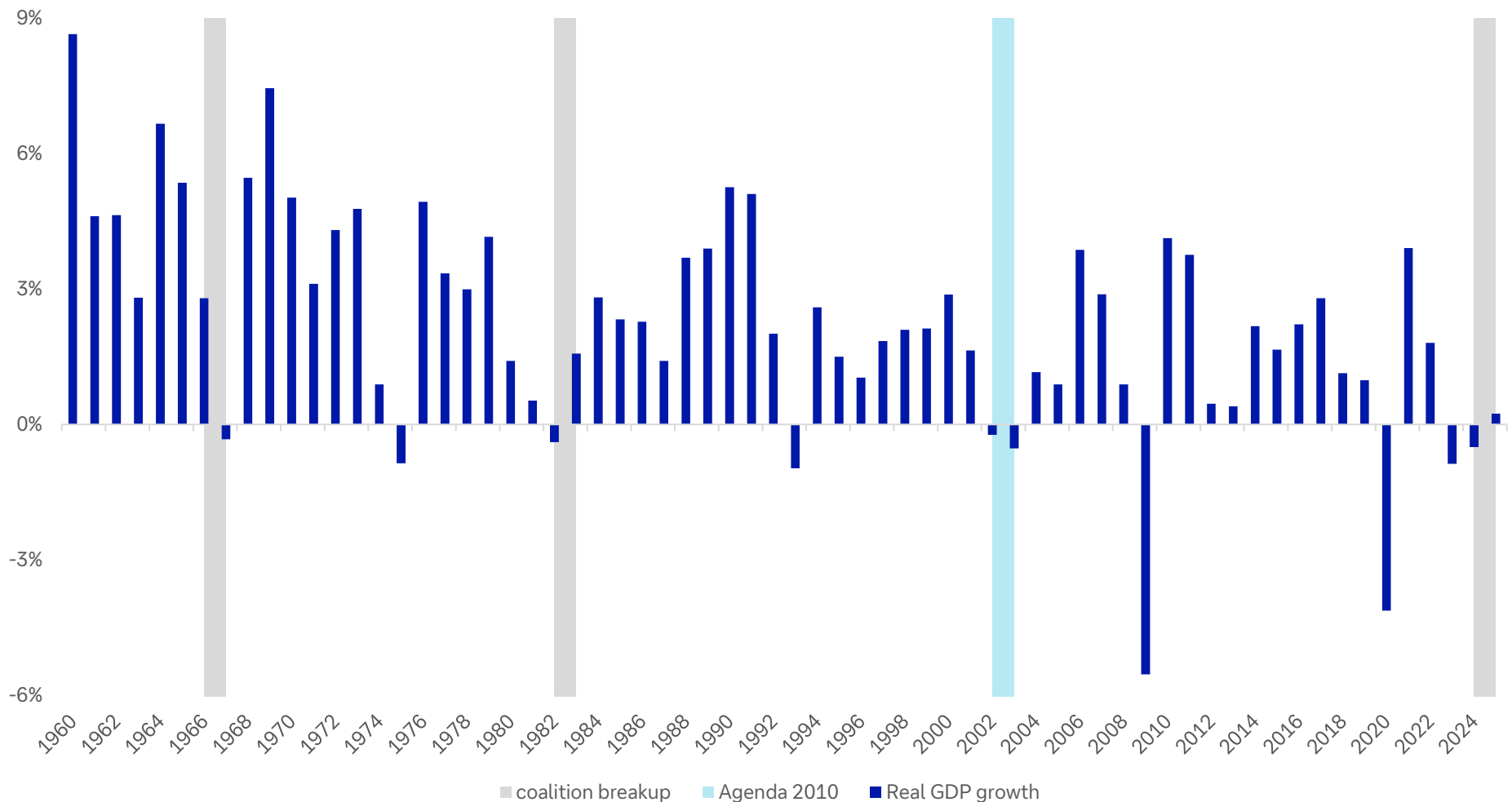
Germany has the potential to become the driving force behind Europe's economic revival. Despite years of macro headwinds, it reduced its debt ratio over the past decade, creating the fiscal space for a major spending package. Economic activity is stabilising, manufacturing conditions are improving, reforms are progressing, and global investor interest is returning. Germany has repeatedly demonstrated an ability to adapt and reinvent itself, with major reform waves following the coalition collapses of 1966 and 1982 and the implementation of Agenda 2010 in the early 2000s.

- **Fiscal:** Few countries have faced as many challenges as Germany in recent years: rising Chinese competition in key industries, the energy shock following Russia's invasion of Ukraine, large-scale immigration, US tariffs, and a three-year manufacturing downturn. Despite these headwinds, Germany is the only G7 economy to have reduced its debt-to-GDP ratio over the past decade. The government plans to raise around €1tn of additional debt by 2030, on top of the current debt stock of €2.8tn. Net credit borrowing is expected to increase from €180bn in 2026 to more than €200bn in 2027, while spending on defence and investment is set to reach roughly €120bn each this year, up more than 30% year-on-year.
- **Macro:** Germany's economy is showing greater resilience towards the ongoing geopolitical headwinds than expected. The manufacturing PMI climbed above 54 in August, its strongest reading in over four years, and both the Ifo and ZEW surveys surprised positively. Factory orders are gradually improving and GDP growth in H1 exceeded expectations, prompting consensus GDP forecasts to be revised up. Although higher energy prices may create short-term headwinds, the burden remains manageable, with energy costs representing just slightly more than 1% of sales for listed German companies.
- **Politics:** Germany's reform momentum is strengthening. The government is combining fiscal stimulus with measures to lower energy costs, boost corporate investment, accelerate infrastructure projects, reform the labour market, and modernise the pension system. We estimate the pension reform alone could unlock €60-100bn of annual capital market inflows. At the same time, political risks appear manageable. Regional elections are unlikely to affect federal policymaking, and we see little risk to the government's reform agenda. If anything, weaker results for the coalition parties could increase pressure to accelerate reforms.
- **Equities:** Markets have yet to price in Germany's improving outlook. While economic indicators continue to strengthen, investor flows into German equities remain weak and the MDAX has lagged the DAX by 7% since late June. As the index with the greatest exposure to the domestic economy, the MDAX is the purest listed play on Germany's recovery and has historically shown a much stronger relationship with German GDP growth. A combination of fiscal stimulus, structural reforms, and more industry-friendly policies creates substantial catch-up potential following years of underperformance. Importantly, valuations remain attractive, with the MDAX trading at its cheapest relative level in history.

BIG PICTURE: Recession → Reforms → Growth

Don't underestimate Germany's capacity to change. After the coalition break-ups of '66 and '82, Germany introduced reforms and re-emerged stronger. Reforms were also implemented when growth turned in 2003 under the Agenda 2010.

Germany was in recession in '66 & '82, but recovered quickly after the political crisis was resolved



Source: Bloomberg Finance LP, Bundesbank, Deutsche Bank Research 09/15/2026

BIG PICTURE: Consolidation to expansion

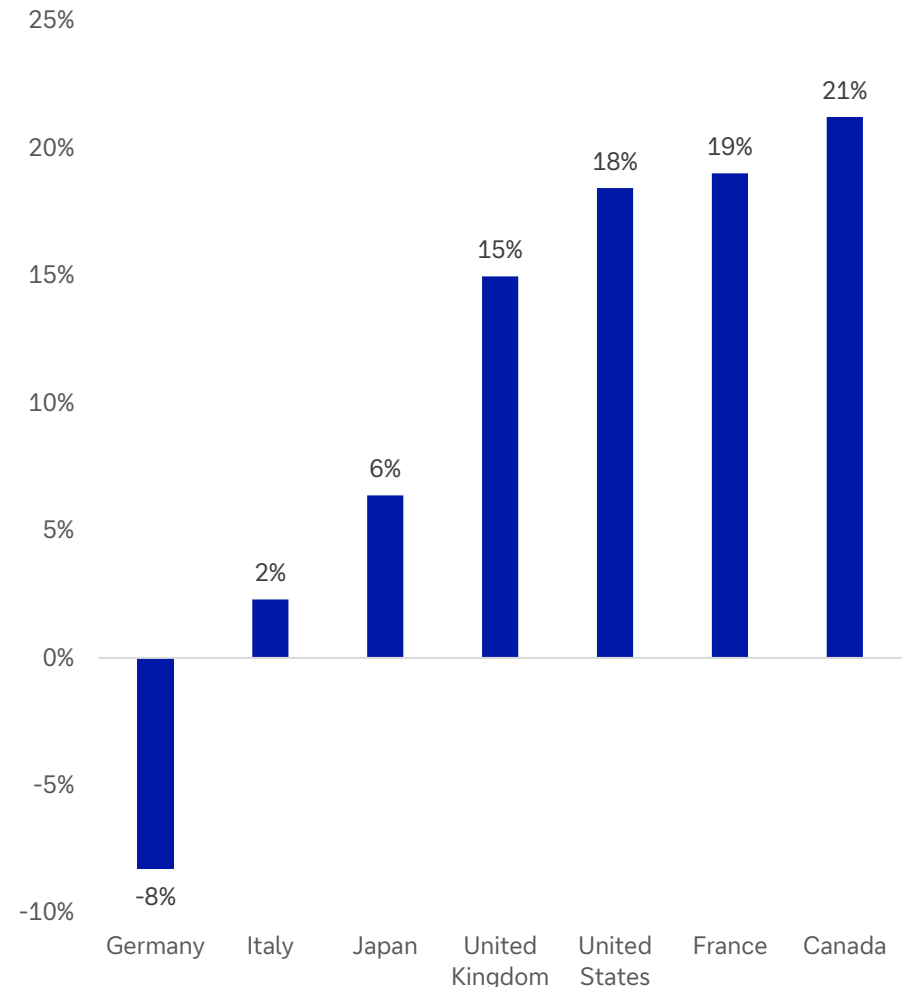


Germany is the only G7 country to have reduced its debt relative to GDP over the past decade.

Key takeaways

- Over the past decade, few countries were facing as many headwinds as Germany: growing competition from China in its key industries, the energy transformation following the war in Ukraine, mass immigration, US tariffs and a 3-year manufacturing crisis.
- Despite all of this, Germany managed to be the only G7 country to have reduced its debt relative to GDP in the last 10 years.
- German debt-to-GDP dropped by 8 percentage points while e.g. US growth was largely debt-financed, with the US' debt-to-GDP ratio increasing by 18 percentage points over the same period.

Change in debt-to-GDP over the past 10y (pp)



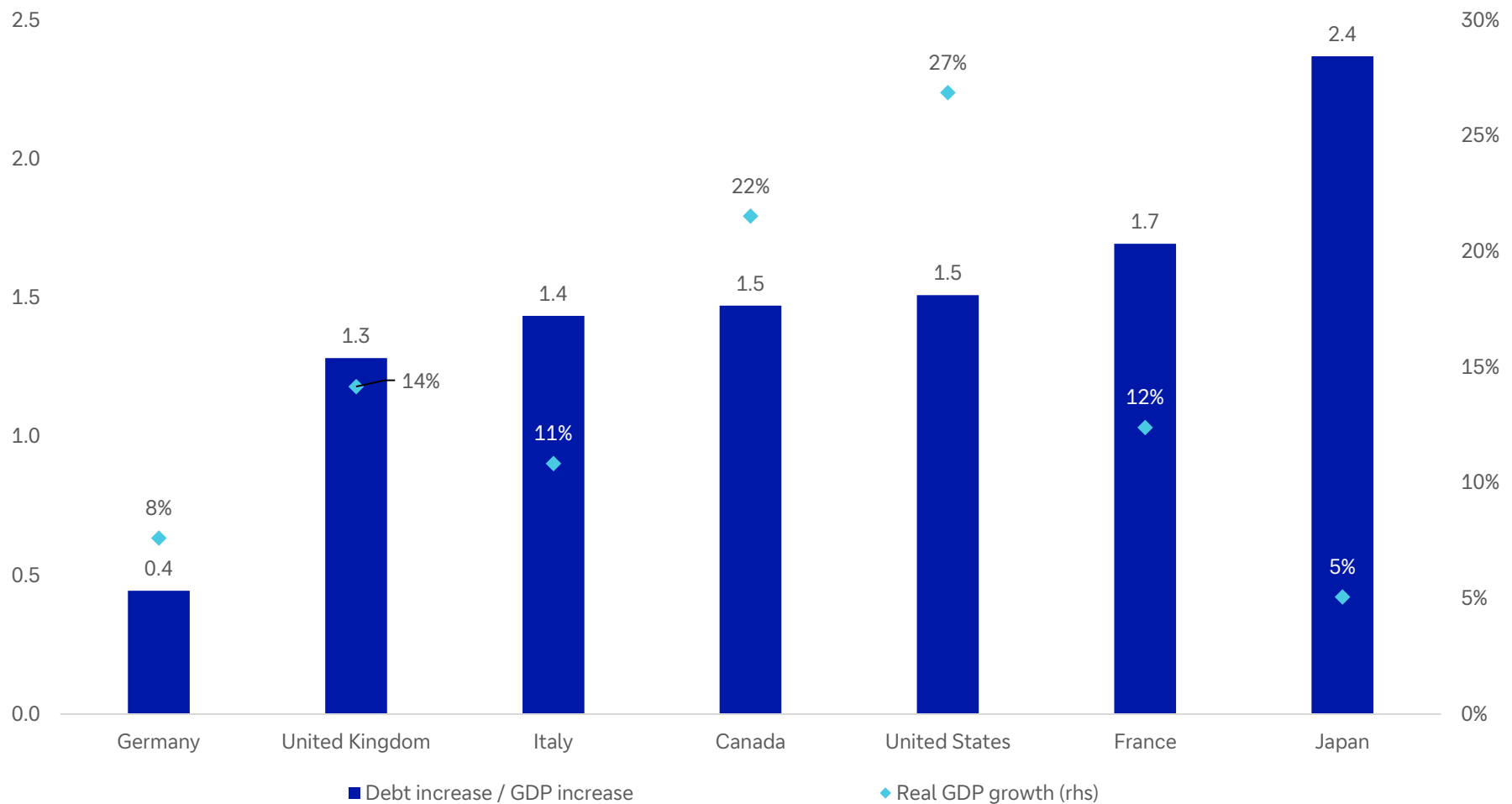
Note: All statistics in local currency, 2025 vs 2015, debt-to-GDP based on nominal GDP and gross general government debt. Source: IMF, Deutsche Bank Research 09/15/2026

BIG PICTURE: Consolidation to expansion



Germany is the only G7 country that increased debt by significantly less than 1 Euro for every Euro of GDP growth.

Bang for the buck: Germany financed its growth at a lower cost (10y)



Note: All statistics in local currency, 2025 vs 2015, debt increase / GDP increase in nominal terms. Source: IMF, Deutsche Bank Research 09/15/2026

BIG PICTURE: Consolidation to expansion



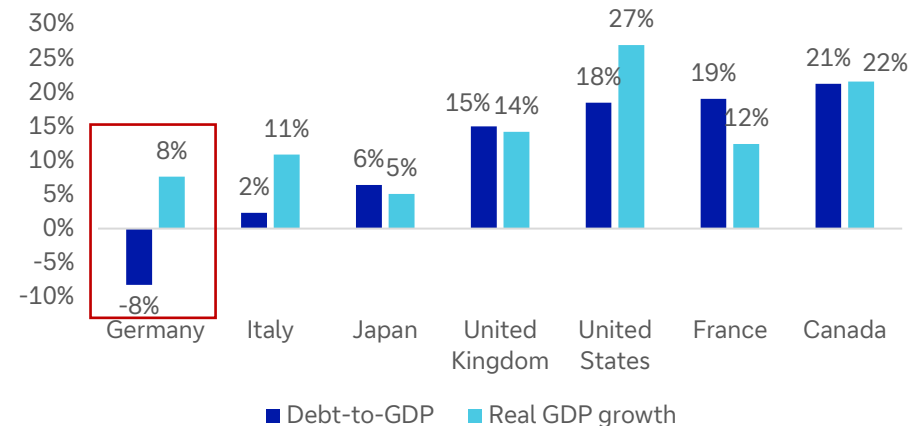
Germany managed to grow its GDP despite fiscal consolidation.

An additional €1 trillion in debt by 2030 could unlock significant growth.

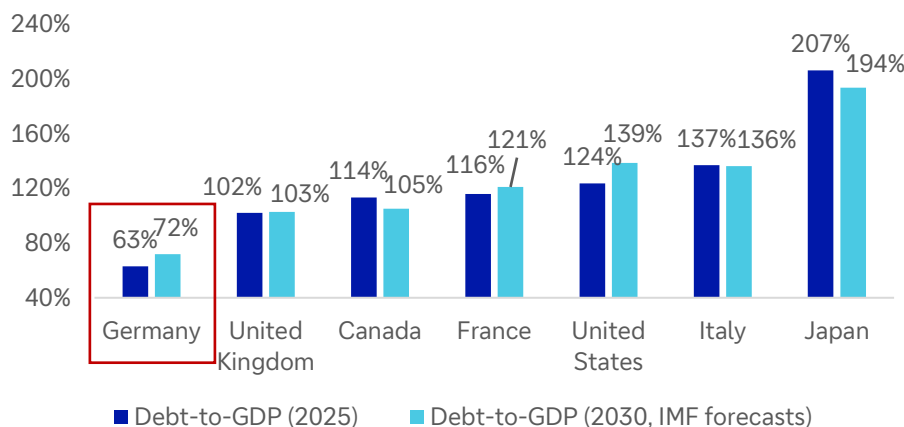
Key takeaways

- The German government plans to increase debt by more than €1 trillion by 2030, coming from €2.8 trillion in 2025 (gross general government debt). Almost 70% of the debt increase will be related to the core budget, vs 30% in off-budget funds.
- A large share of the additional debt (>€600bn) will be allocated to defence and security-related spending, almost €300bn to the special infrastructure fund (especially transportation and digitalization), and the remaining amount to other core-budget borrowing.
- For more information, see [here](#).

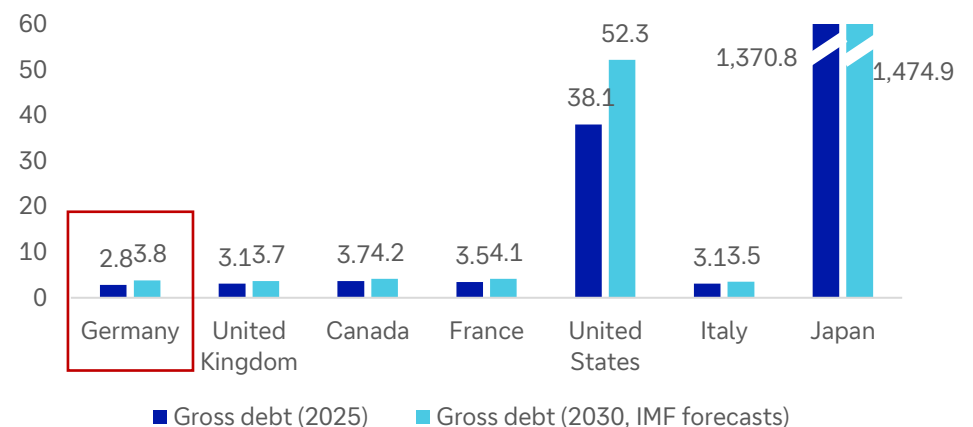
10y changes in debt and GDP across the G7



IMF projections of debt-to-GDP



Gross debt (trillions, local currency)



Note: All statistics in local currency, 10y changes are 2025 vs 2015, debt-to-GDP in nominal terms based on gross general government debt.
Source: IMF, Bundestag, Deutsche Bank Research 09/14/2026



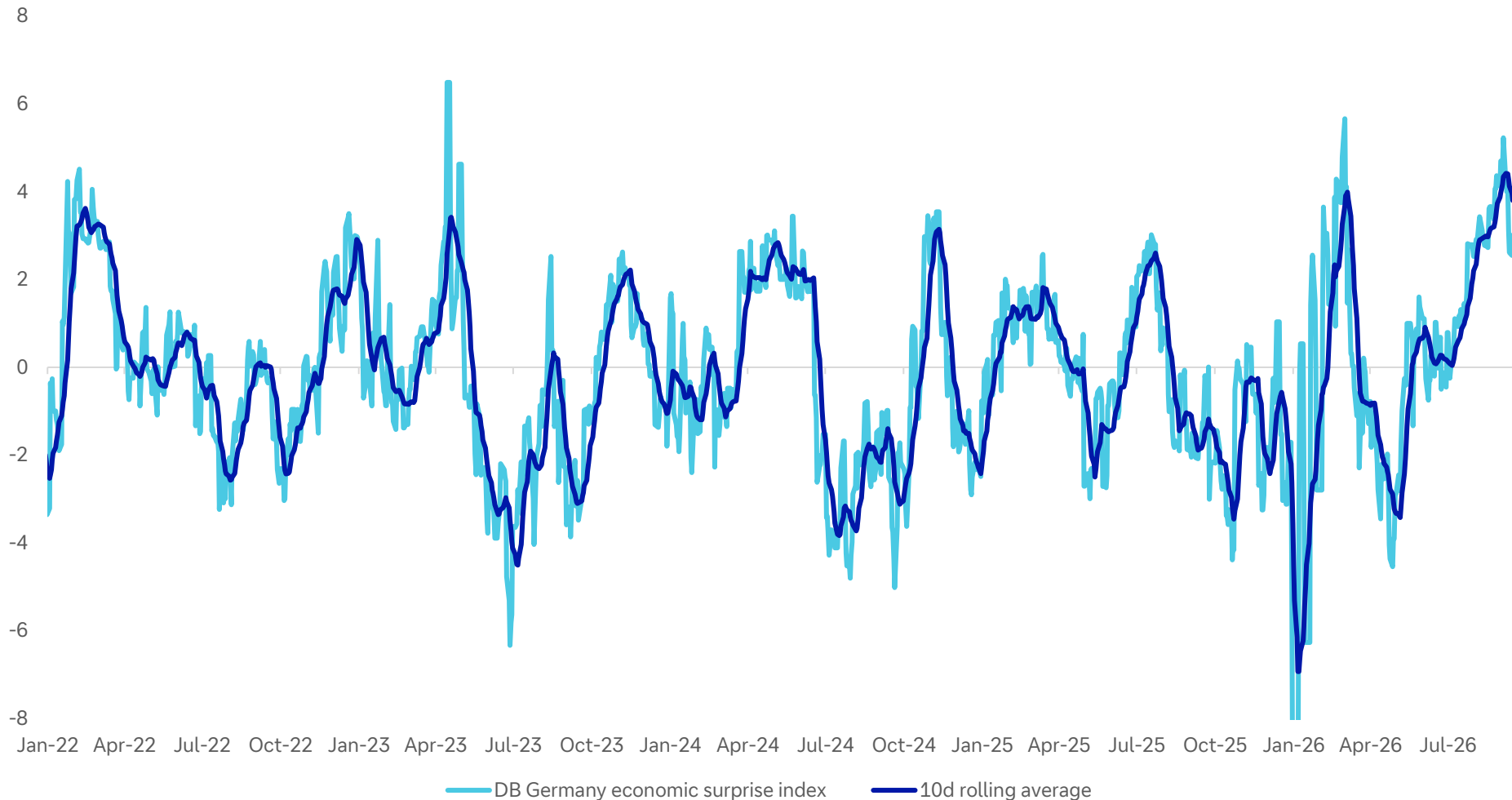
Macro & political backdrop

Macro backdrop – back on track



Most recent macro data came in much better than expected in Germany.

DB Germany economic surprise index has reached its highest level since 2021



Source: Bloomberg Finance LP, Deutsche Banks Research 09/15/2026

Macro backdrop – back on track

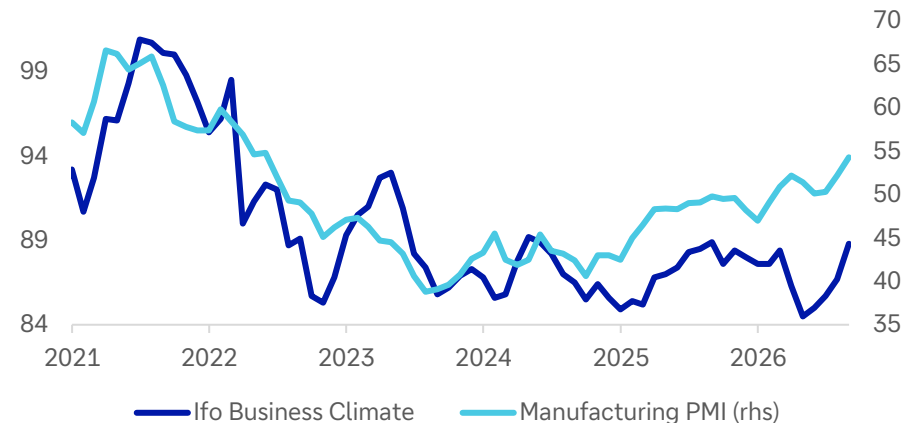


Green shoots have emerged in the German economy over recent months.

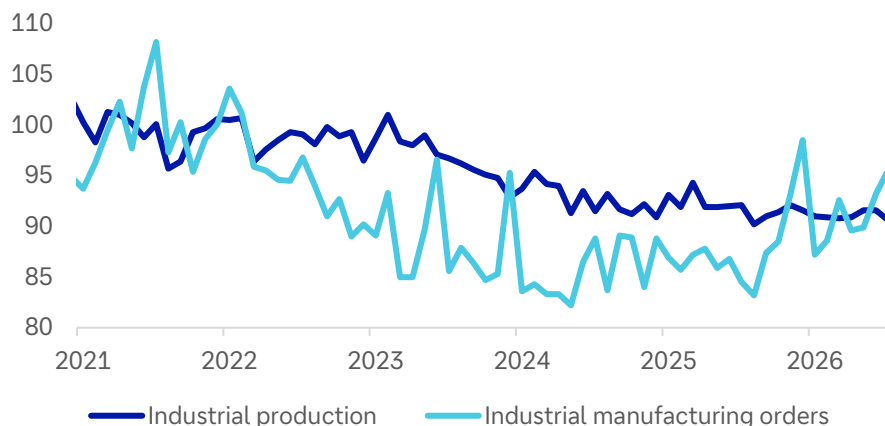
Key takeaways

- The German economy is more resilient than expected. The manufacturing PMI rose above 54 in August, its highest level in more than four years. Ifo and ZEW surveys also both surprised to the upside in August.
- Factory orders continue to edge higher, albeit still driven mainly by large orders.
- After GDP proved stronger than expected in H1, we recently upgraded our '26 GDP forecast to 1%.

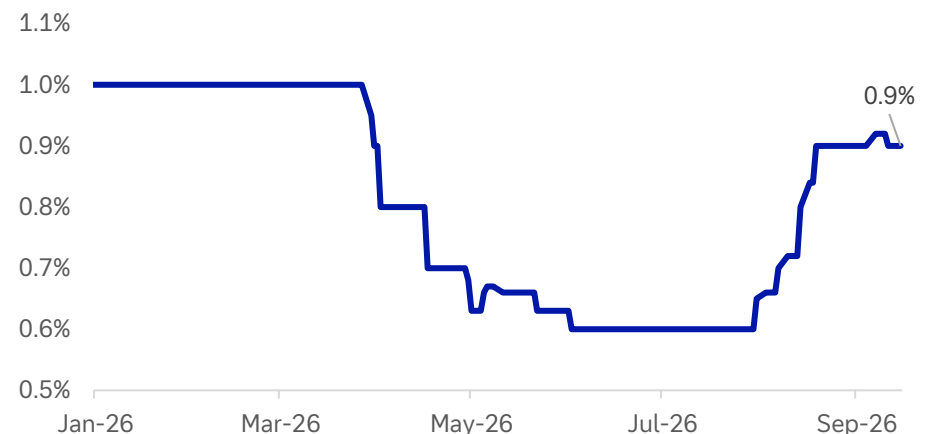
Corporate sentiment recovering



Factory orders accelerated in July



2026 GDP forecasts revised upwards*



*Median of Bloomberg consensus forecasts. Source: Bloomberg Finance LP, Deutsche Banks Research 09/15/2026

Regional elections

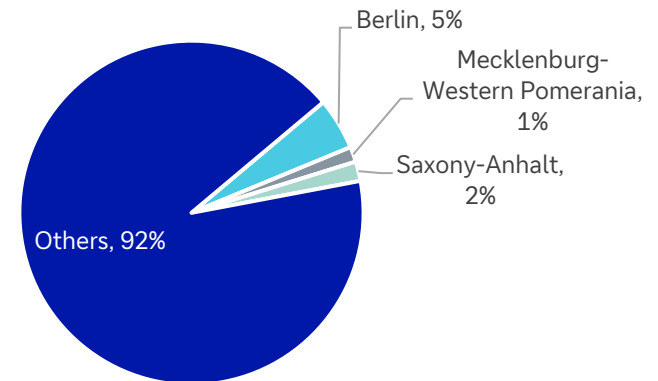


Higher uncertainty from regional elections. But relevance limited and pressure on government to increase pace of reforms is growing.

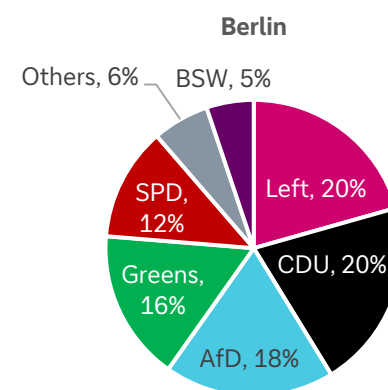
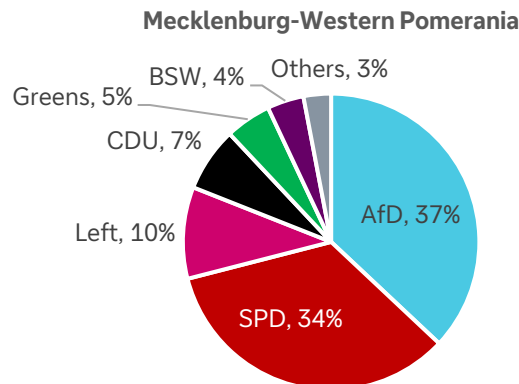
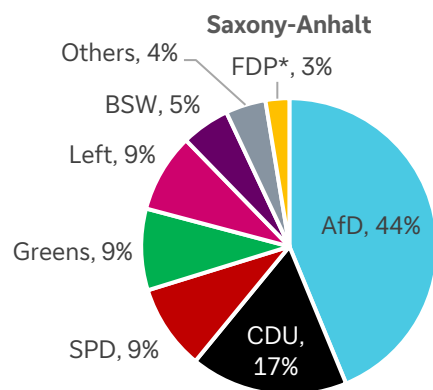
Key takeaways

- Contrary to many concerns, the Saxony-Anhalt election triggered little to no market reaction. While the AfD will likely enter government, the economic relevance of Saxony-Anhalt within Germany is limited. In Meck-Pom and Berlin, AfD support is weaker, though fragmented parliaments may complicate coalition building.
- Regional elections have limited implications for federal politics. More importantly, we don't see the elections derailing the government's reform agenda. If anything, weaker results for coalition parties could increase pressure to accelerate reforms.

2025 German GDP: breakdown by State



Regional elections: latest results and polls (MP and Berlin elections on 20th September)



*below 5% hurdle, not in parliament

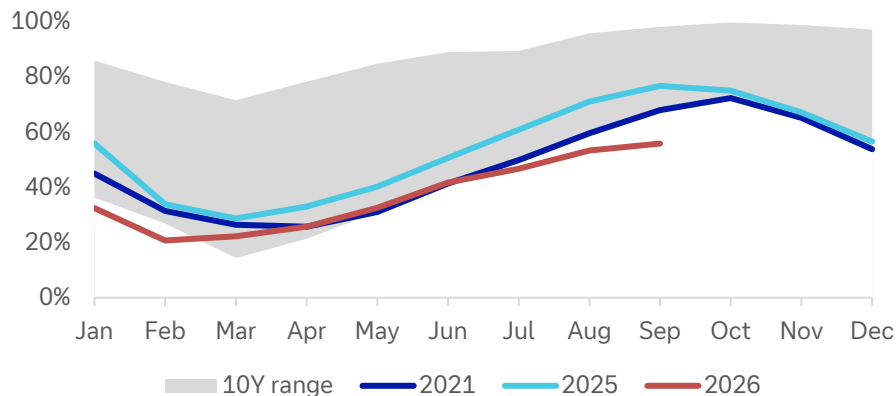
Source: Wahlrecht.de, Statistikportal des Bundes und der Länder, Deutsche Bank Research 09/15/2026

Macro headwinds – manageable?

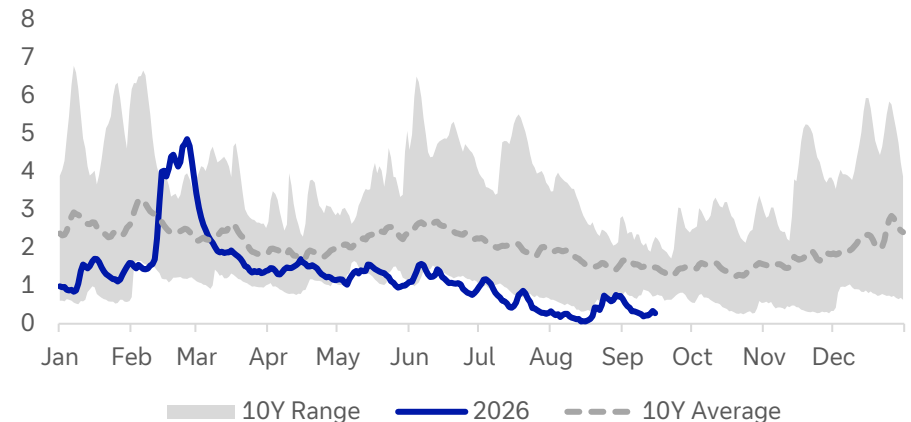


Energy costs to weigh on short-term sentiment. However, current levels are still manageable. We estimate energy costs to be 1.2% of sales for listed German companies.

Gas storage level at only 56%



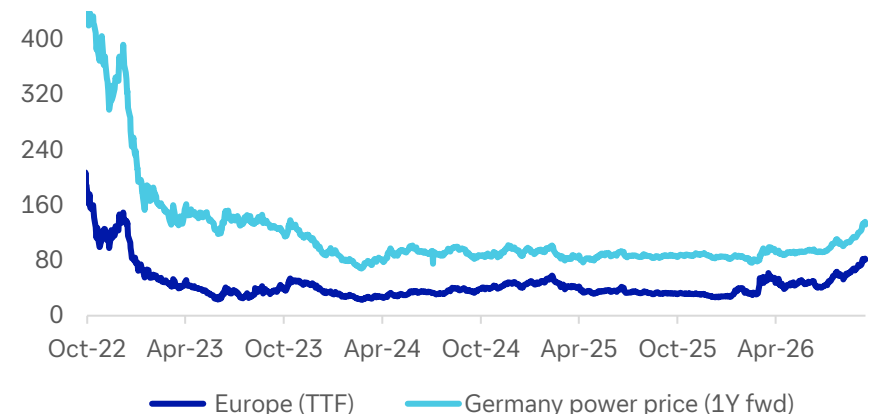
Rhine level at Kaub remains too low



Bund yield at highest since 2009



Gas and power prices up, but far below '22



Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

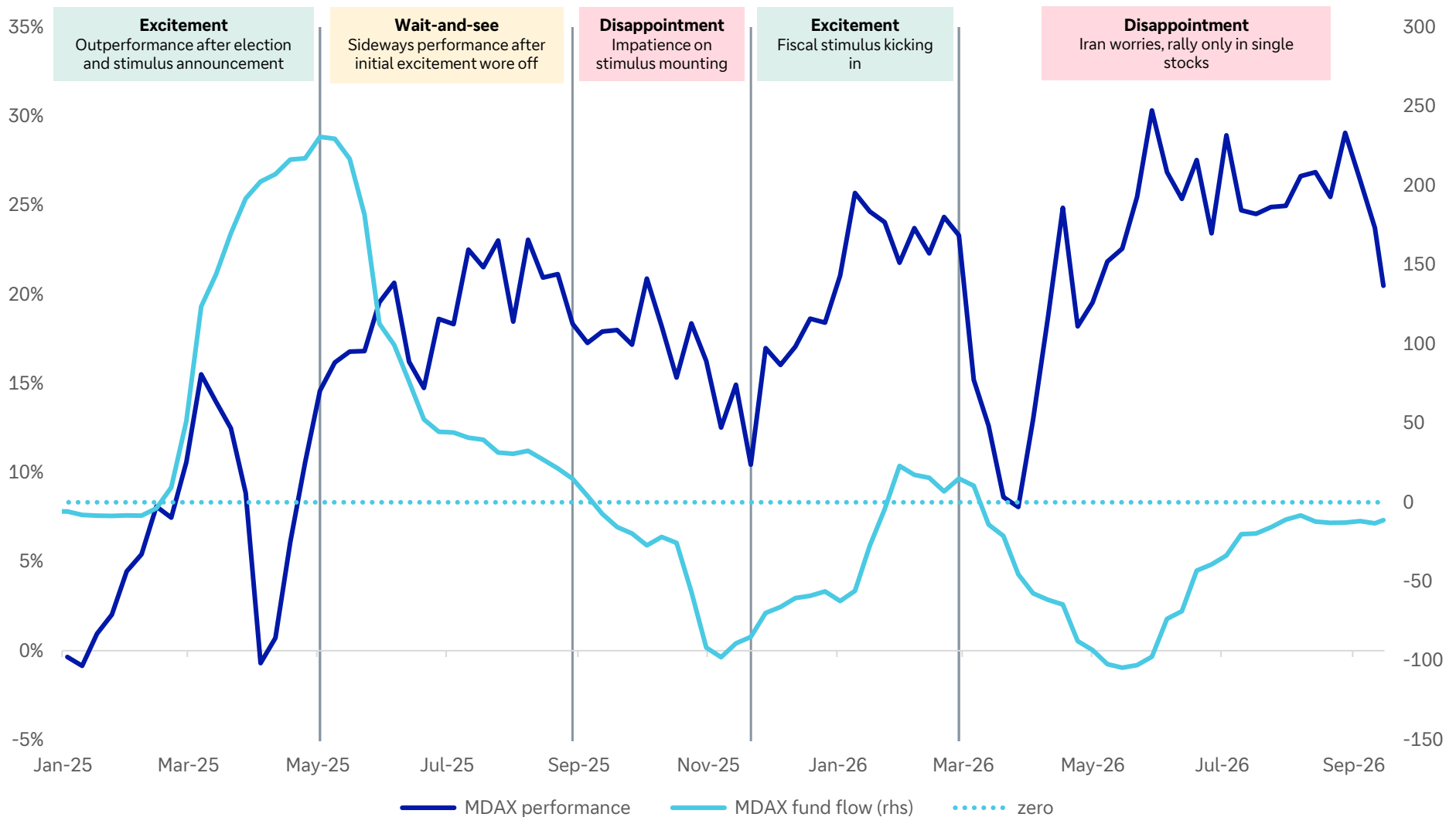


Equity markets

Germany: Rollercoaster of emotions



We have been going through many emotions. Bring patience to benefit from the recovery.



Aggregate fund flows (12-week mov. avg., mn EUR) based on up to 10 largest ETFs for the index. Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

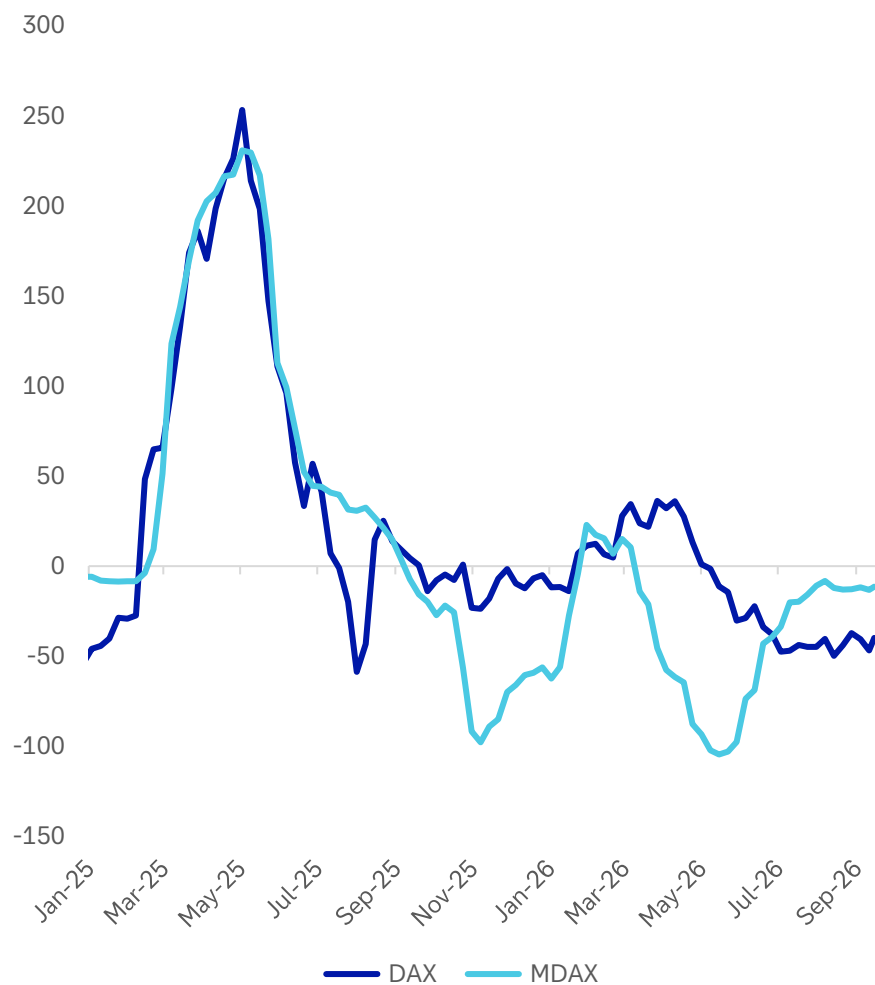
German ETF flows slowly recovering



Despite the improvement in macro indicators, flows into German equities are still depressed. The MDAX underperformed the STOXX 600 by 2% since end of June.

DAX & MDAX ETF aggregate fund flow

YTD performance (total returns, local CY)



Note: Fund flows proxied by flows into and out of largest (up to) 10 ETFs for the index, 12-week moving average, mn EUR. Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

Equity implications: DAX vs MDAX

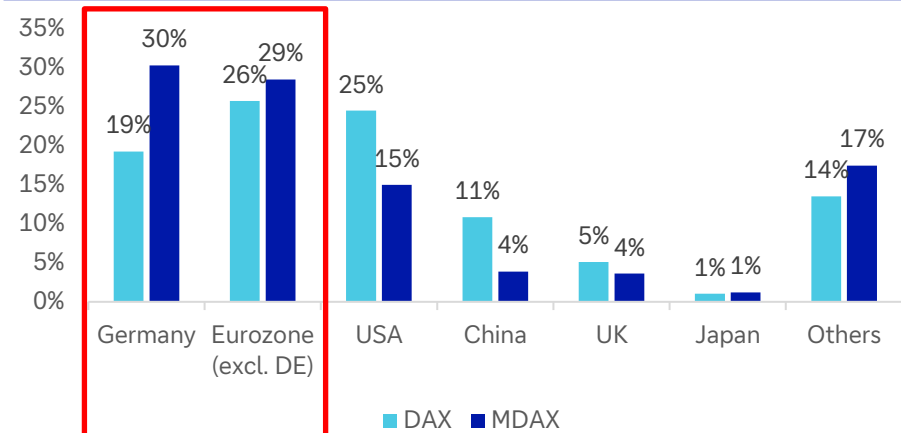


We think the MDAX is a better proxy for German growth and should benefit most from economic improvements and new reforms.

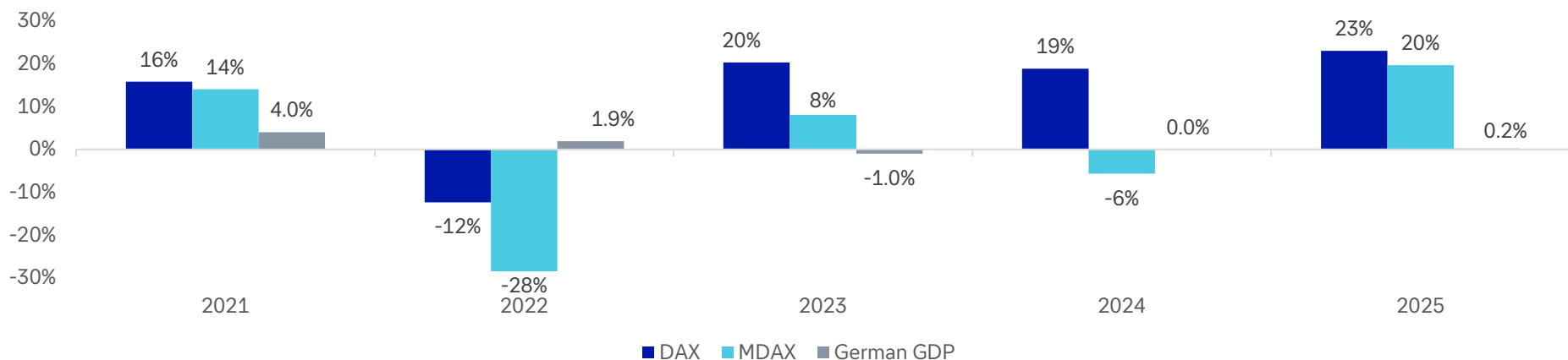
Key takeaways

- The DAX's exposure to Germany is limited; a better proxy for Germany-specific news flow is the MDAX.
- The MDAX has a higher revenue exposure to Germany than the DAX (see right chart) and its performance is linked much more closely to German GDP growth (see below chart). The new reform agenda and a more industry-friendly policy approach in Germany bear significant catch-up potential for the MDAX which has underperformed the DAX significantly over the past years (see [slide 30](#)).

Regional revenue exposure (FY24)



MDAX is the better proxy for German growth



All exposures are best estimates by our DB company analysts and are not necessarily based on published company data.
Source: Bloomberg Finance LP, Company Data, Deutsche Bank Research 09/15/2026

MDAX still offers high catch-up potential



German mid caps underperformed large caps massively over the past years. The improving macro backdrop and new reforms should support the MDAX.

Total return since 2022



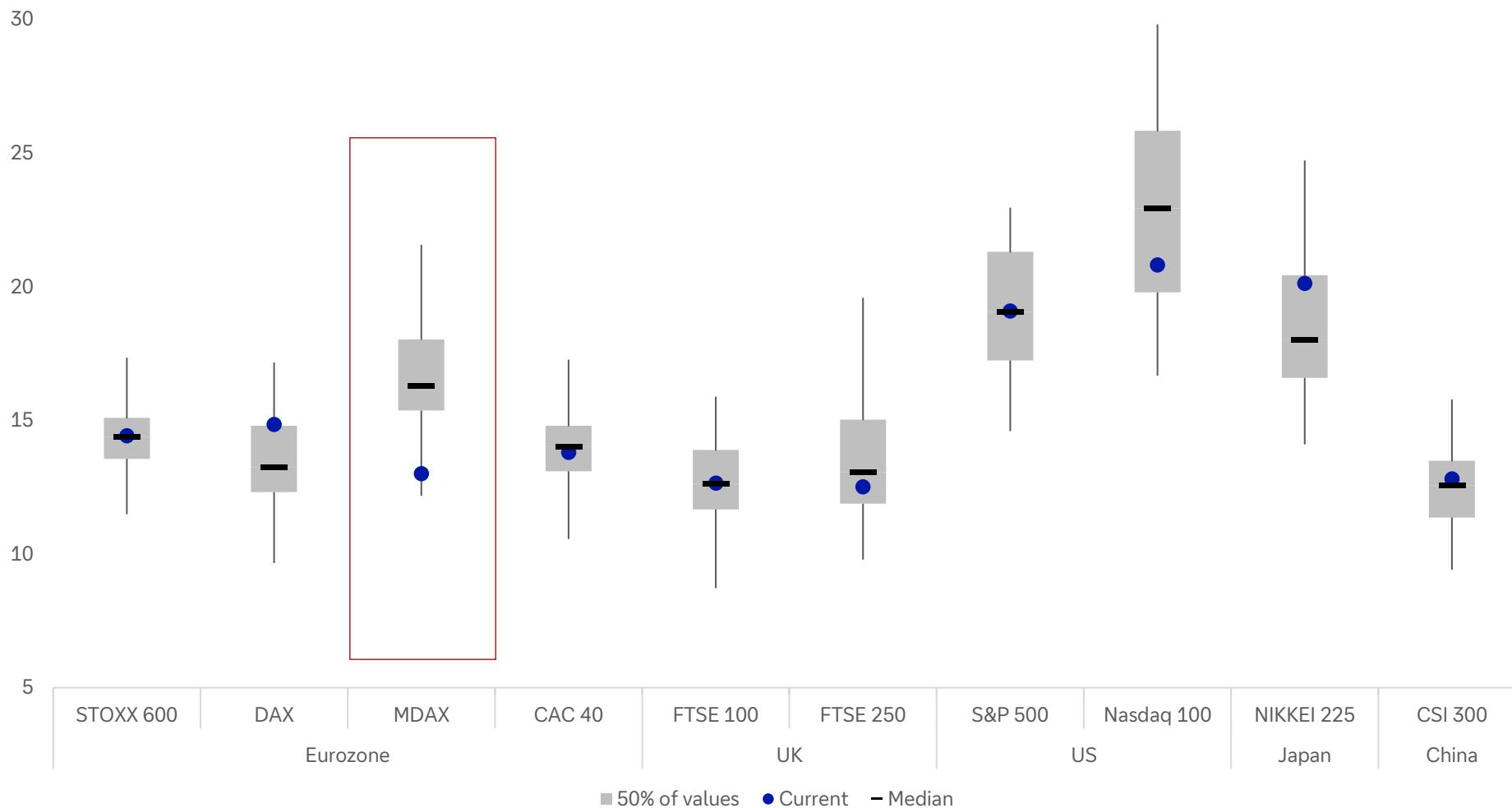
Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

Valuations



In our view, European valuations are not elevated. Within Europe, the MDAX trades at lowest value relative to history

Valuation vs 10y median (12m fwd P/E)



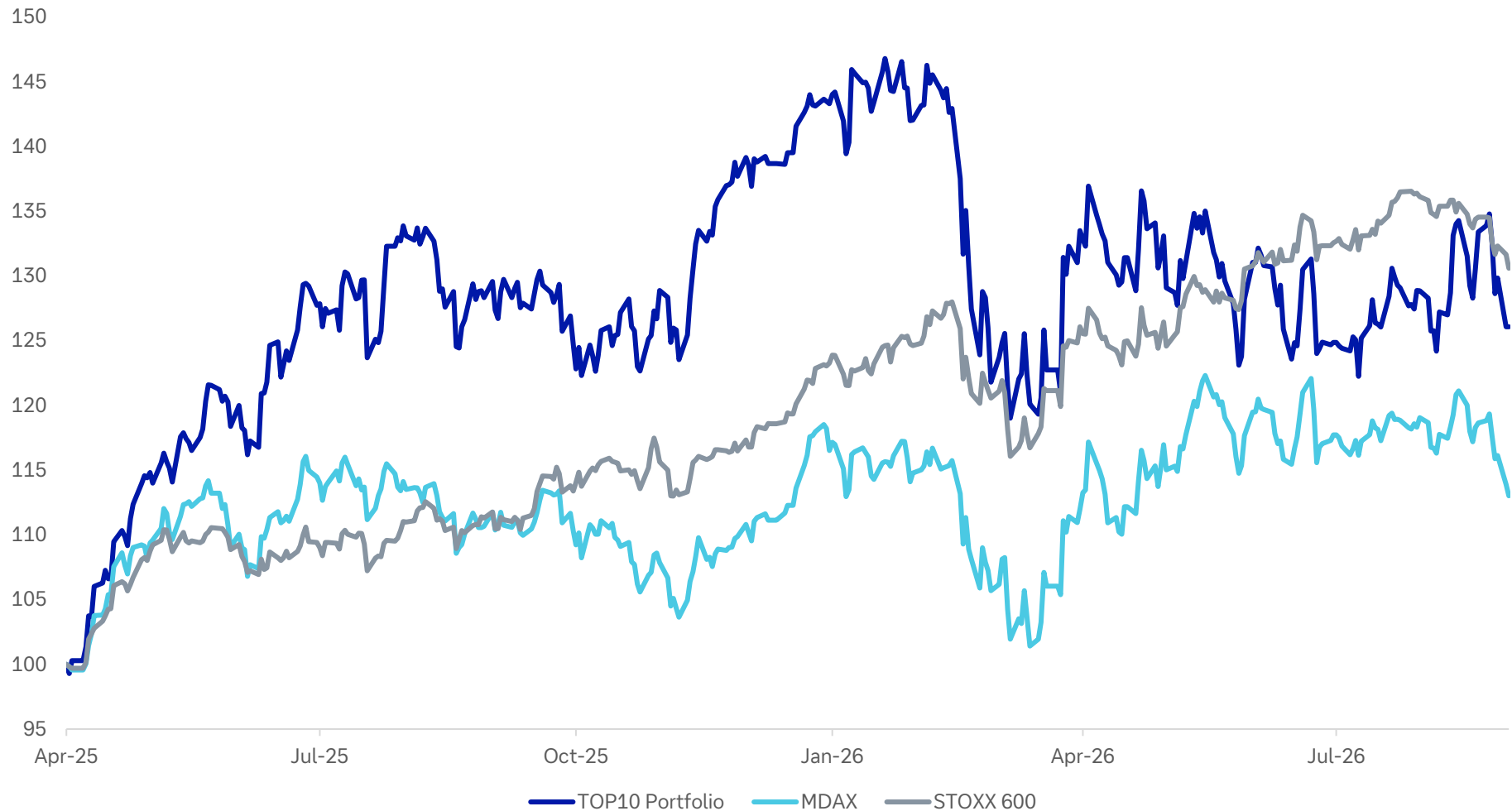
Time period: Aug 2016 – today.

Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

TOP10 Germany stock basket



Total return since inception



Note: German top picks portfolio inception was on 04/15/2025. . Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026



Reform agenda

Germany: Reform hammer in full swing



Area	Detailed measures	Status	Area	Detailed measures	Status
Economic Growth	– "Growth Booster": – 30% accelerated depreciation on machinery and equipment, 75% on EVs – Gradual corporation tax cut from 15% to 10% and R&D tax credit expansion	– In place – Jul '25 - Dec '27 for depreciation – 2028-2032 for corporate tax cut	Health Care System	– Restricting free family insurance – Higher contributions by higher earners – More co-payments by patients and less benefits (braces, dentures, homeopathy) – Price caps for hospitals/pharma	– Approved in parliament in July, – Effective in 2027/2028
Energy Prices	– Industrial power price: electricity price for corporates lowered to 5 c/kWh – Earlier abolition of gas storage levy – Lower grid fees – Electricity tax cuts to EU minimum level is extended indefinitely – Reduction of energy tax on petrol and diesel by 17 ct/l for two months	– In place – Industrial power price until Dec '28 – Reduced energy tax only during May + June '26	Private Pension System	– "Frühstart-Rente" (early start pension): monthly public grant of €10/child (ages 6-18), to be invested into capital market products, tax-free until the payout phase. Withdrawals generally allowed from age 65.	– Approved by the Cabinet in August, Parliamentary approval pending
Private Pension System	– "Altersvorsorgedepot" (old-age provision account): retirement savings account with tax benefits, allowing investment into products without guarantees (e.g., ETFs)	– Approved, to become effective on Jan 1, 2027	Statutory Pension System	– Gradual increase in retirement age to 67.5 – Limiting early retirement programs – Age for old-age part-time work up to 58 – Raising number of contributors, e.g. self-employed, MPs, managers – Introduction of mandatory funded component: 2% additional contribution into centralized pension fund.	– Approved by coalition committee, Parliamentary approval pending
Sector-specific	– "Construction turbo" law to accelerate planning and approval procedures – Air ticket tax cut by up to ~€11/passenger – Autos: EV subsidies of up to €6,000 for buyers, extension of car tax exemption for EVs (up to 10y)	– Construction turbo until 2030 – Air ticket tax: since 1st July – EV subsidy 'til '29, EV tax exemption 'til 35	Labour Market	– Easier extension of fixed-term contracts – relaxed dismissal protection for top earners – Preferential tax treatment if a new job is taken up quickly – Medical certificate from first day of illness	– Approved by coalition committee, Parliamentary approval pending
Public Procurement	– Law to faster and simpler public procurement processes – No need to start a contract award process for federal public contracts up to 50k EUR	– In place since 1st July	Income Tax	– Increase of basic tax-free allowance – Higher deductions for families and employees – Flattening of the tax rate for low income – Increase of "rich-tax" to 45% on taxable income >€250k (47% for income >€280k)	– Approved by the Cabinet in September, Parliamentary approval pending
Labour Market	– Retirees can earn up to €2,000/m tax-free when working while on retirement	– In place since Jan '26	Bureaucracy	– Removal of some reporting obligations – Reduction of need for company representatives – Applications (e.g for construction projects) to be automatically approved after 3m – Aim to cut federal ministries staff by 8%	– Approved by coalition committee. – Bureaucracy Reduction Act passed in May '26
Infrastructure Acceleration Act	– Prioritizes approval process for critical infrastructure (e.g., railways, highways) – Digitalization of approval processes – Easing of environmental checks	– Passed by Bundestag on June 26			

Source: Federal Ministry of Finance, Federal Government, Bundesregierung, Deutsche Bank Research 09/15/2026

Germany: pension reform

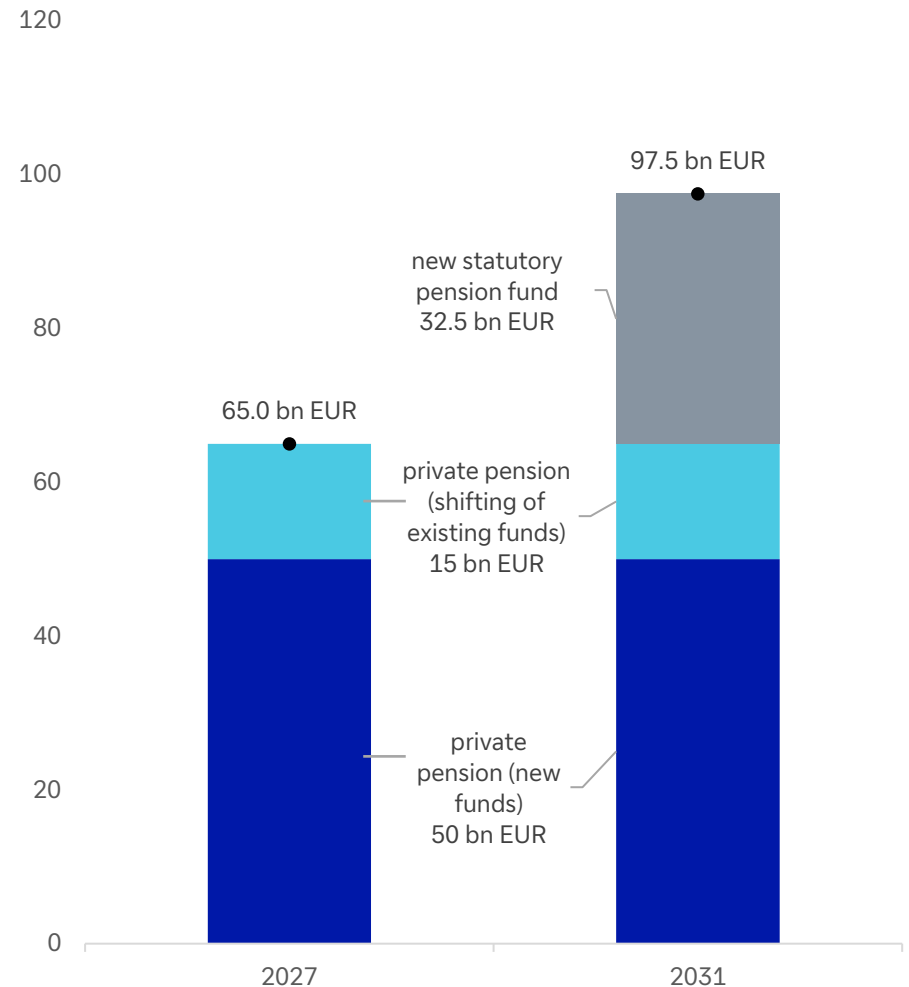


Pension reform could lead to additional annual inflows of 10-15 bn EUR into European equities over the next years.

Key takeaways

- The planned reform of Germany's pension system could generate **significant inflows into European equities** by shifting retirement savings towards capital markets.
- On the private pension side, the new “**old-age provision account**” encourages savers to move funds from low-yield deposits and insurance products into higher-return investments. If half of the eligible population invests the annual amount required to receive the full tax benefit (€2,300), **inflows could reach €50bn per year**.
- We also expect assets to **migrate from the existing “Riester” system**. With total estimated Riester assets of €150bn, a 10% annual reallocation would generate a further **€15bn of inflows**.
- In parallel, the government plans to introduce a funded component to the statutory pension system, equivalent to 2 pp of gross salaries, to be invested through a central **public pension fund**. Contributions are planned to rise from 0.5pp in 2028 to 2pp by 2031, implying annual **inflows of €8bn initially and €30-35bn by 2031**, according to our economists.
- Overall, we estimate these reforms could generate **€60-100bn of additional annual capital market inflows**, of which around **€10-15bn (15%) may find its way into European equities**. To put this into context: the top 50 largest European ETFs saw inflows of c. €40bn in 2025.

Projected inflows into capital markets



Source: BMAS, Deutsche Bank Research 09/15/2026

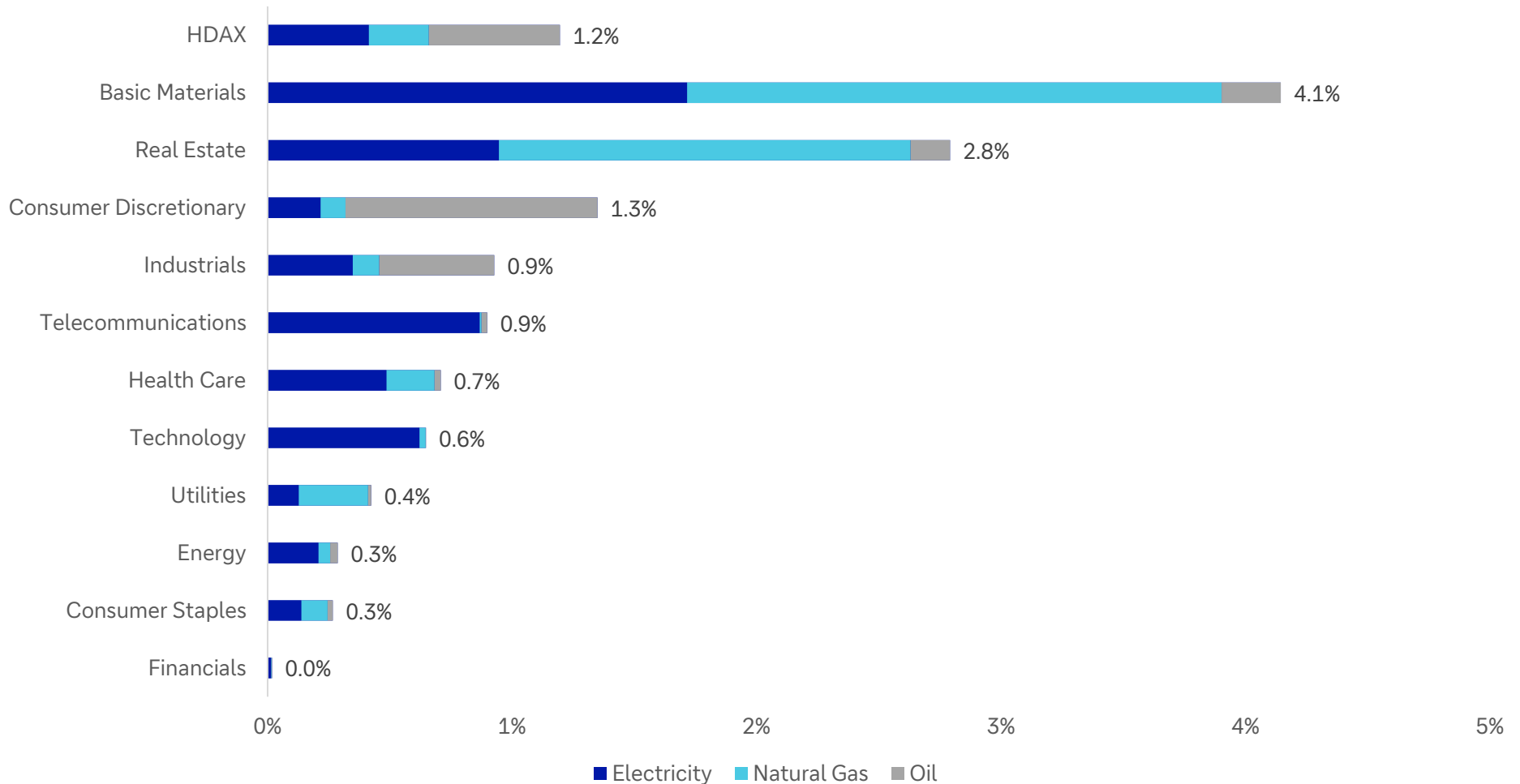


Appendix

Energy costs comprise only 1.2% of HDAX revenues



HDAX energy cost as % of sales*



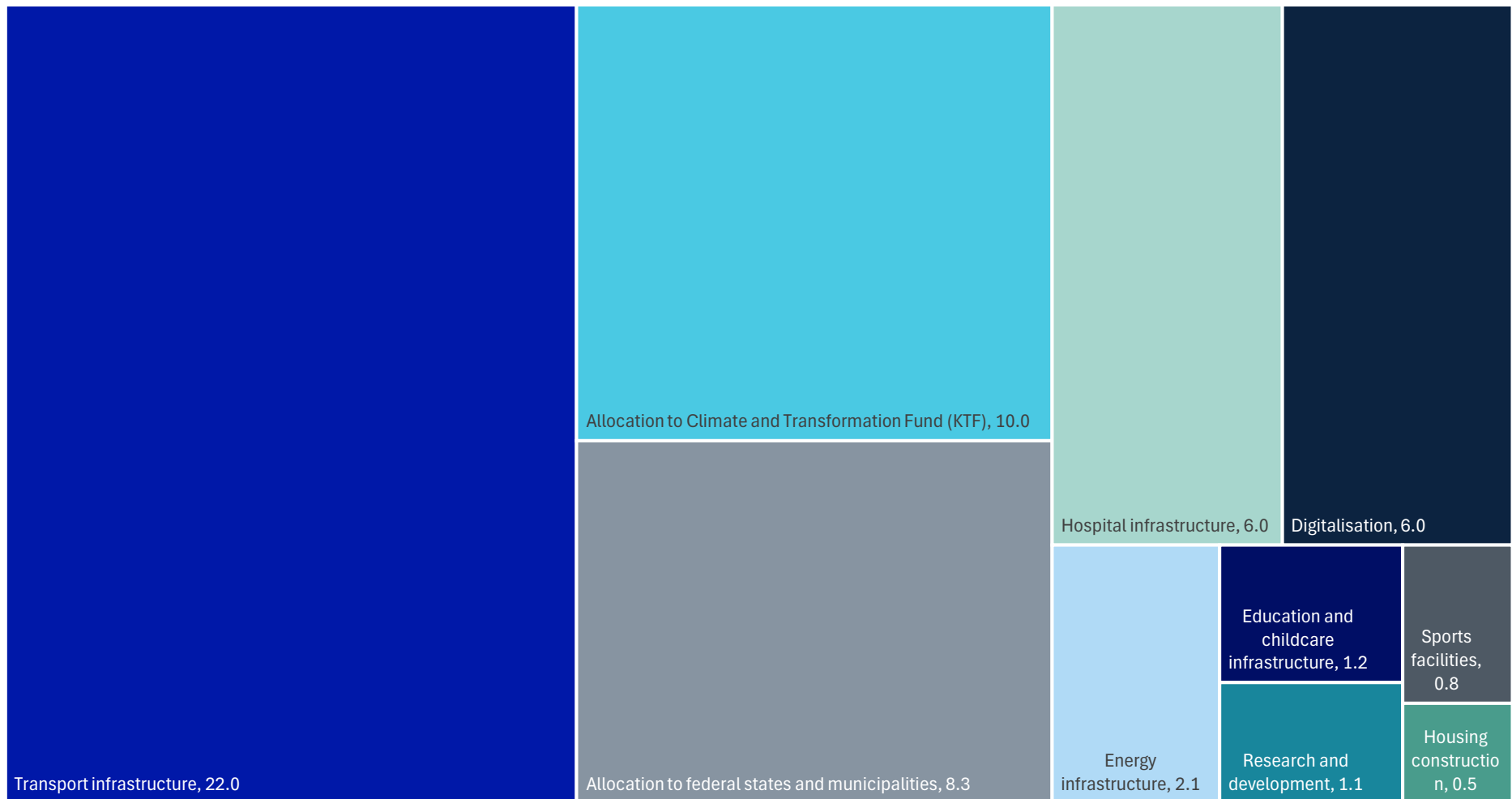
*N=50, based on 3y average German power prices, TTF natural gas and Brent oil. Source: Bloomberg Finance LP, Deutsche Bank Research 09/11/2026

Where to invest?



Most of the higher spending is going into Defence and Infrastructure. Defence has already rallied. We expect sectors with order intake from Infra spend to benefit most.

Planned investments from the special fund for Infrastructure & Climate Neutrality, 2026, mn EUR

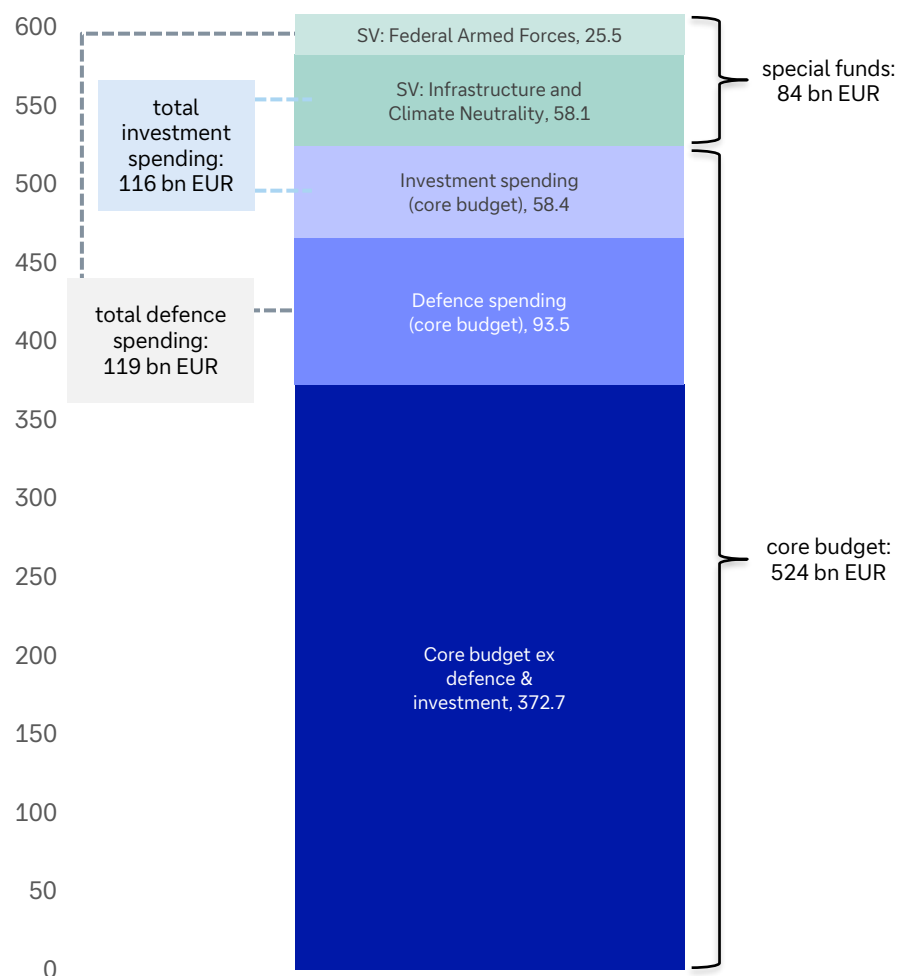


Source: Bundesfinanzministerium, Deutsche Bank Research 09/15/2026

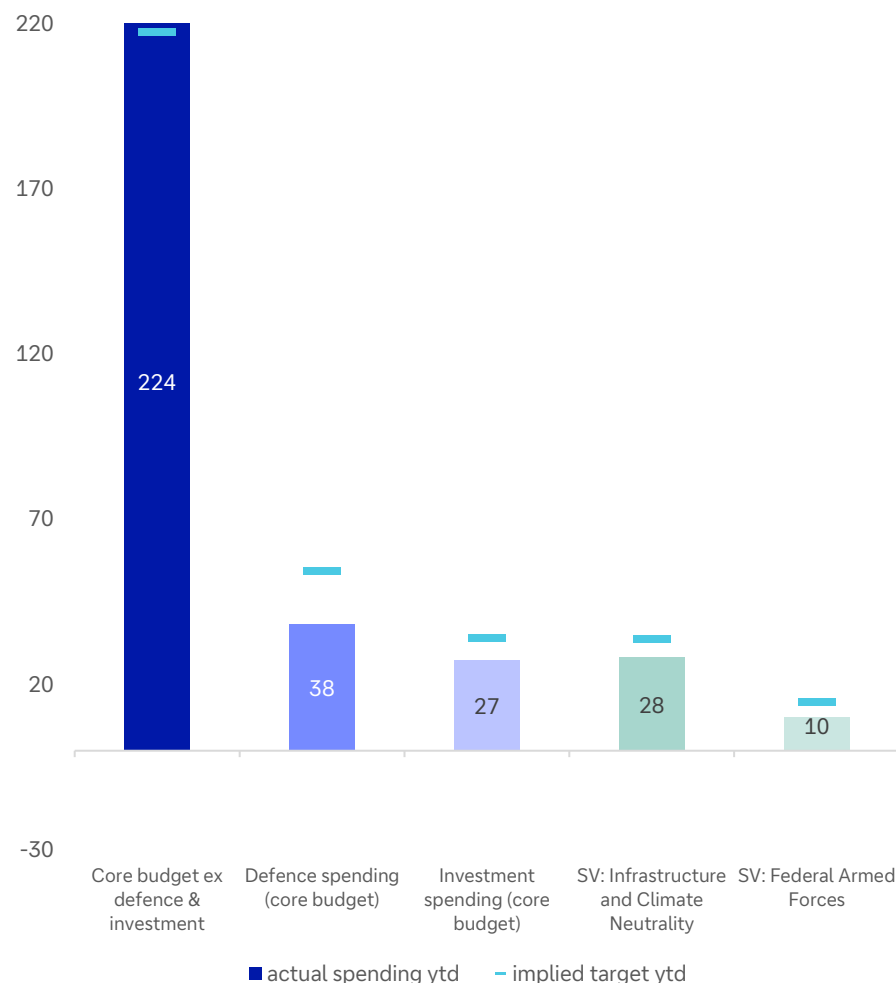
Germany: fiscal expansion underway, with room for spending to accelerate



2026 planned fiscal spending



Spending still below target year-to-date

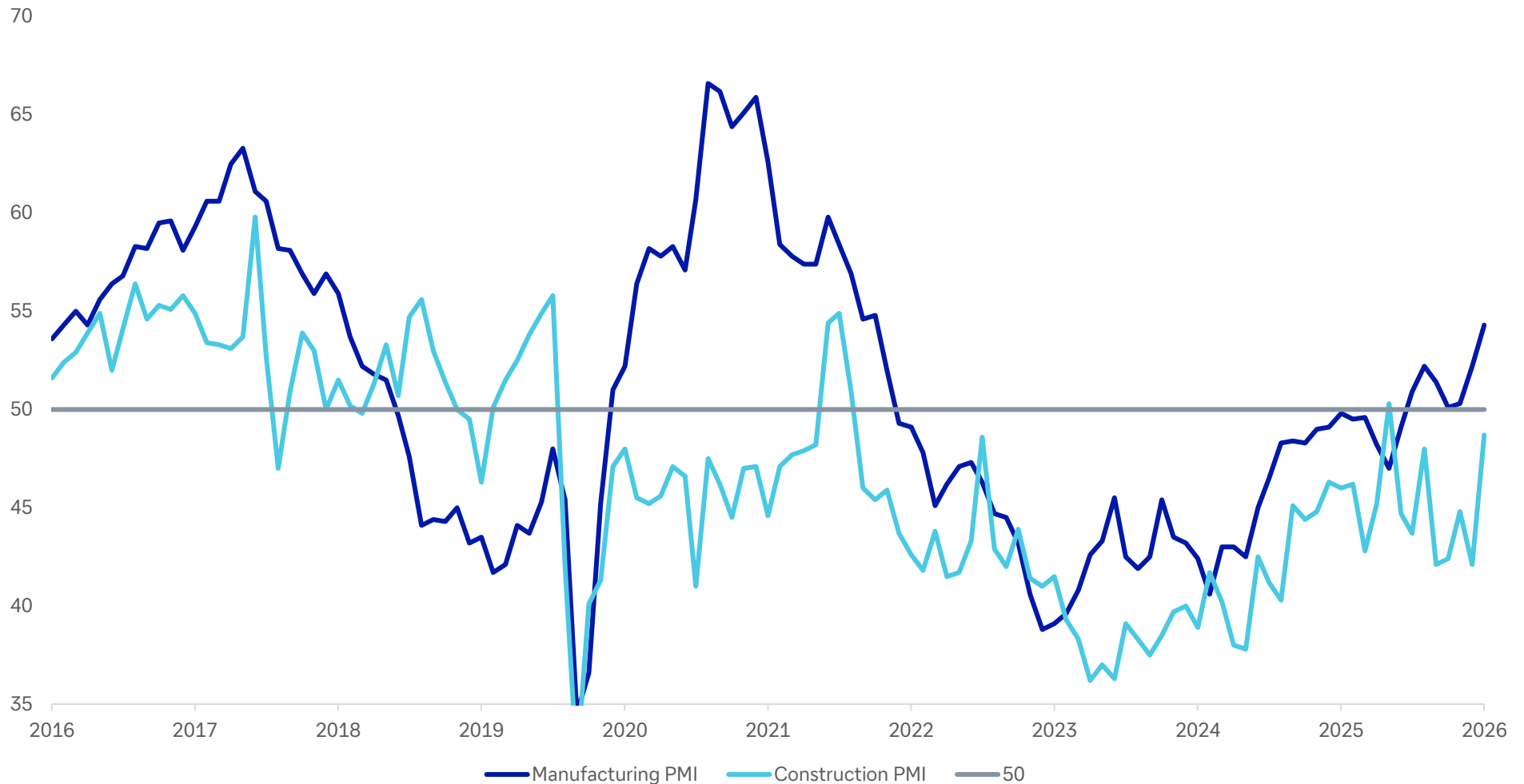


All figures in bn EUR. Actual spending data as of end of July. Source: Bundesfinanzministerium, Deutsche Bank Research 09/07/2026

Manufacturing PMI highest in over 4 years, Construction PMI catching up



Germany Manufacturing PMI



Source: Bloomberg Finance LP, Deutsche Banks Research 09/15/2026

German reforms: example from the past

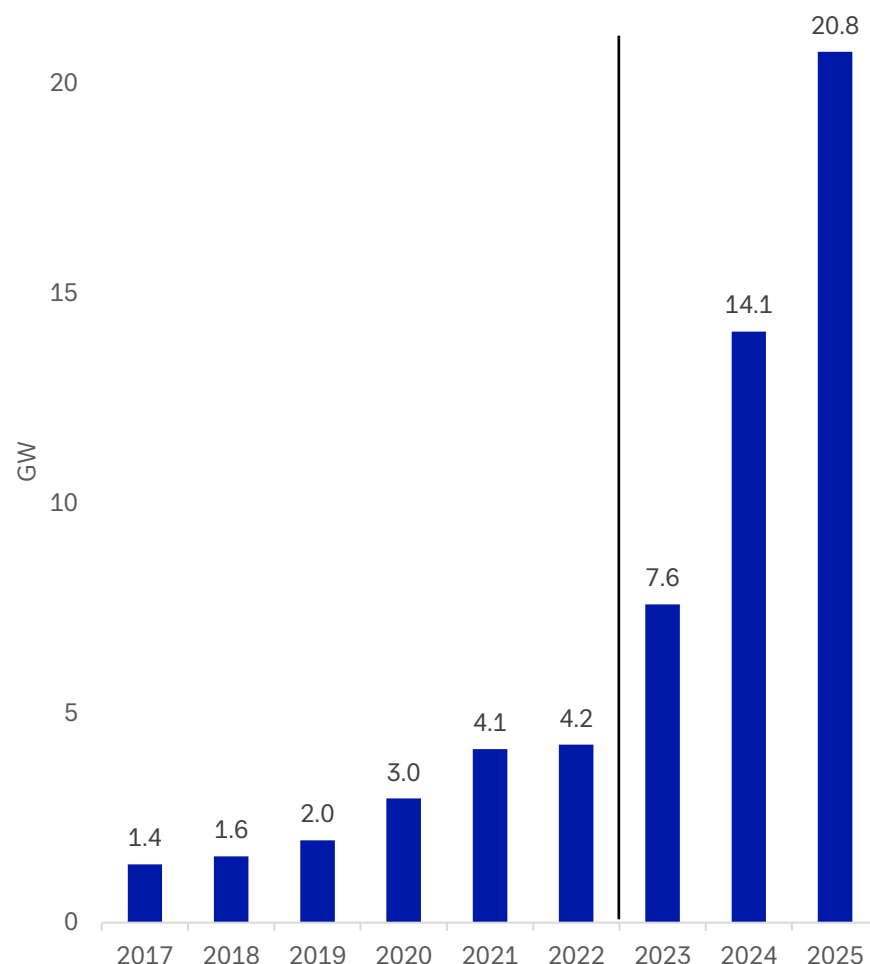


Onshore & Offshore Wind Act

Key takeaways

- In 2023, Germany introduced the Onshore Wind Act and revised the Offshore Wind Energy Act to accelerate the energy transition and streamline renewable energy development.
- For the first time, federal states face legally binding requirements to allocate land for wind power. The share of Germany's land designated for onshore wind is set to increase to 1.4% by 2027 and 2.0% by 2032.
- Policy reforms are already delivering results. In H1 2025, Germany approved 20.8 GW of new onshore wind capacity, nearly four times the level seen in 2022, marking a record pace of expansion.

Newly approved onshore wind energy capacity



Source: SMARD, Statista, Deutsche Bank Research 09/15/2026

German reforms: example from the past

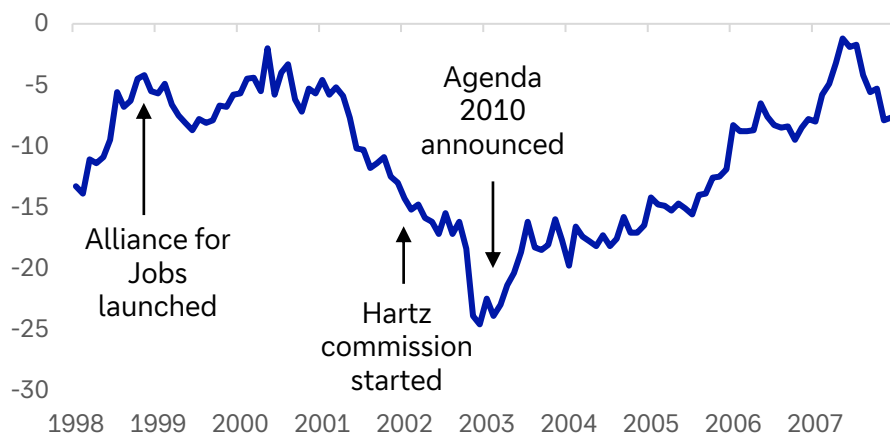


Agenda 2010

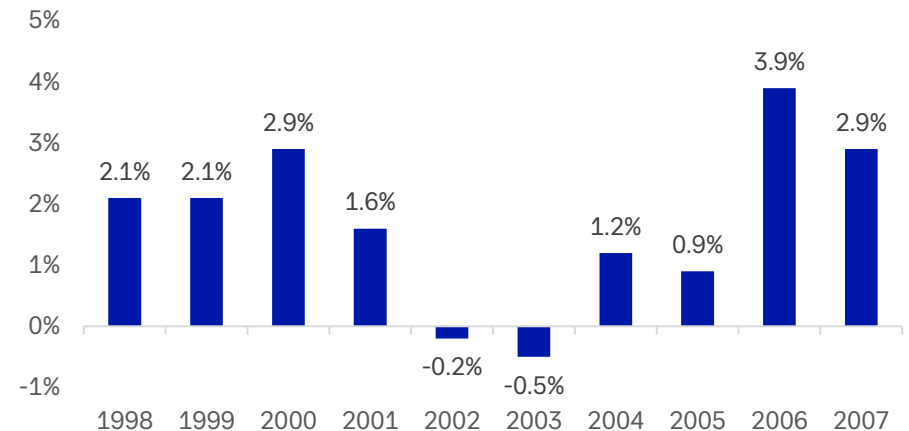
Key takeaways

- Facing stagnant growth and rising unemployment, Germany launched a series of labour-market reforms between 2002 and 2005, beginning with the Hartz Commission and culminating in Agenda 2010.
- The reform package tightened unemployment benefits, expanded labour-market flexibility, improved job placement, and reduced income and corporate taxes to strengthen incentives for work and investment.
- While initially controversial, the reforms helped restore confidence: GDP growth turned positive in 2004, consumer sentiment improved, and equity markets rebounded.

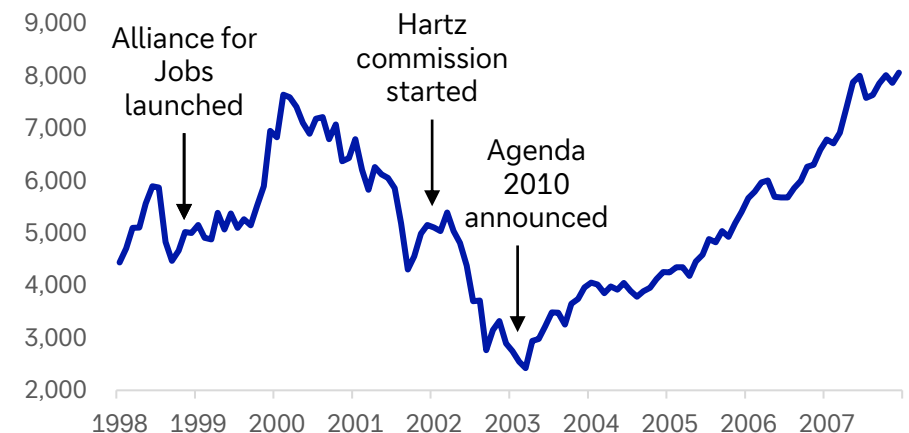
Consumer confidence



Germany annual GDP growth



DAX



Source: Bloomberg Finance LP, Deutsche Bank Research 09/15/2026

Appendix 1



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Deutsche Bank
Research

ASSET TOKENIZATION 101

The Next Phase of Financial Infrastructure

September 2026

Marion Laboure | Camilla Siazon

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.



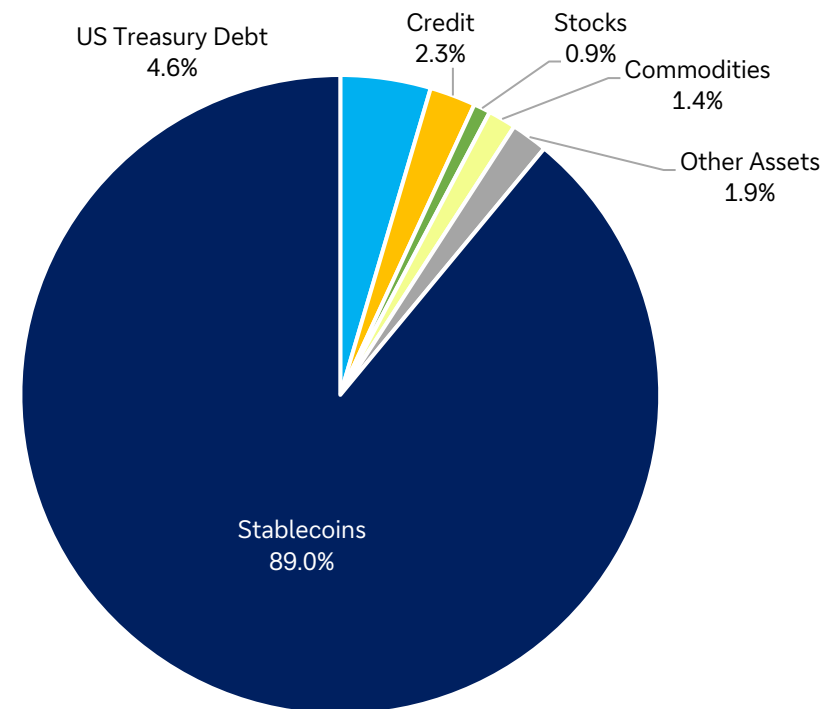
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EXECUTIVE SUMMARY: Tokenization expands into market infrastructure



- The global tokenized assets market, excluding stablecoins, grew from ~\$10bn in Jan 2025 to ~\$39bn in Sep 2026. US Treasury debt remains the largest segment outside stablecoins, with money market funds leading near-term adoption. Despite the US Digital Asset Market Clarity Act failing to advance through to the Senate yesterday, we remain positive about the momentum to tokenize both assets and capital markets in the coming years.
- Institutions are increasingly applying tokenization to settlement, funding and collateral management. Notable examples include DTCC's tokenization service, DLT-based intraday repo and collateral initiatives, and the Eurosystem's Pontes solution, which connects DLT platforms to TARGET Services and is expected to launch later this month.
- A broad tokenized capital market requires progress across three areas: assets, settlement money and market infrastructure. The tokenized RWA market remains small relative to traditional markets, and token issuance does not automatically create secondary-market liquidity.
- In this chartbook, we broaden our focus from tokenized assets, such as bonds and stocks, to the infrastructure through which they are issued; and review (i) market size; (ii) adoption trends; (iii) the regulatory drivers of tokenization; and (iv) barriers that will determine the pace of growth.

Market share of tokenized assets, Sep 14 2026



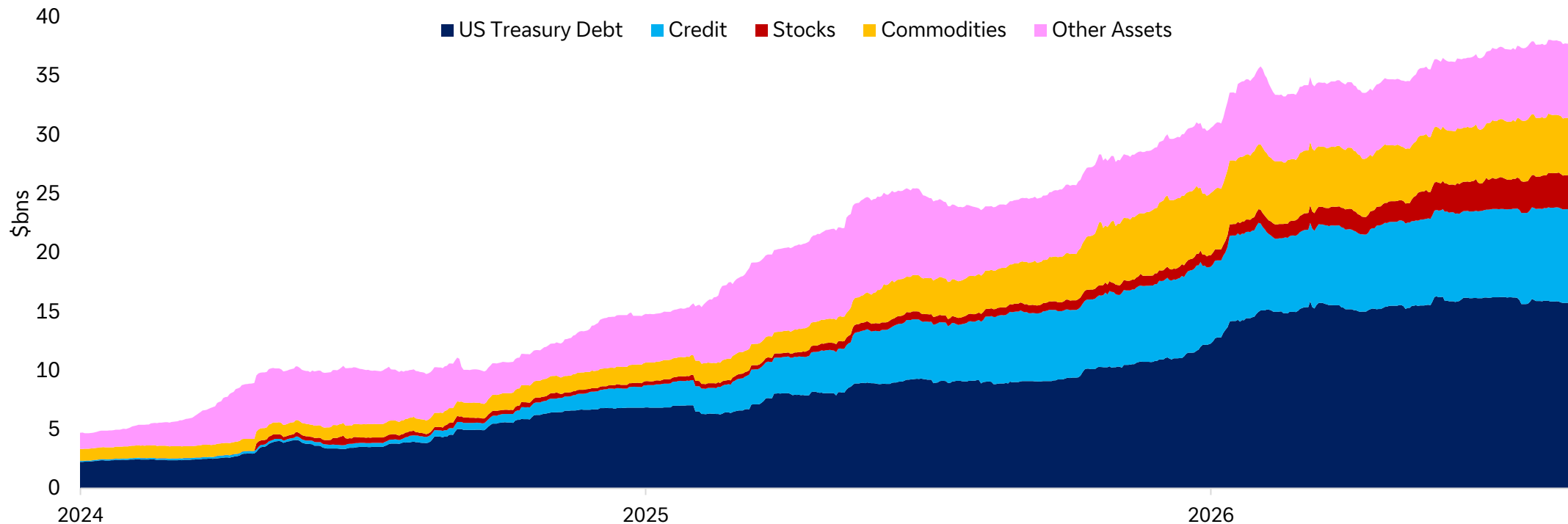
Sources: Deutsche Bank Research, RWA.xyz. Data as of Sep 14, 2026. "Distributed" denotes tokens transferable outside the issuing platform and between wallets; it does not measure trading activity. Note: Credit combines asset-backed credit, corporate credit, diversified credit and specialty finance. Other assets comprise non-US government debt, private equity, venture capital and real estate.



MARKET SIZE: US Treasuries continue to dominate

- US Treasury debt remains the largest on-chain tokenized asset segment excluding stablecoins, reaching \$15bn in September 2026. Much of that is driven by tokenized money market funds (MMF).
- Demand reflects appetite for highly liquid, short-term risk-free collateral assets, with the ability to generate yield.

Distributed tokenized asset value, excluding stablecoins



Sources: Deutsche Bank Research, RWA.xyz. Data as of Sep 14, 2026. "Distributed" refers to tokens transferable outside the issuing platform and between wallets; it does not measure trading activity. Credit combines asset-backed credit, corporate credit, diversified credit and specialty finance. Other assets comprise non-US government debt, stocks, private equity, venture capital and real estate.

DEFINITION : How asset tokenization works



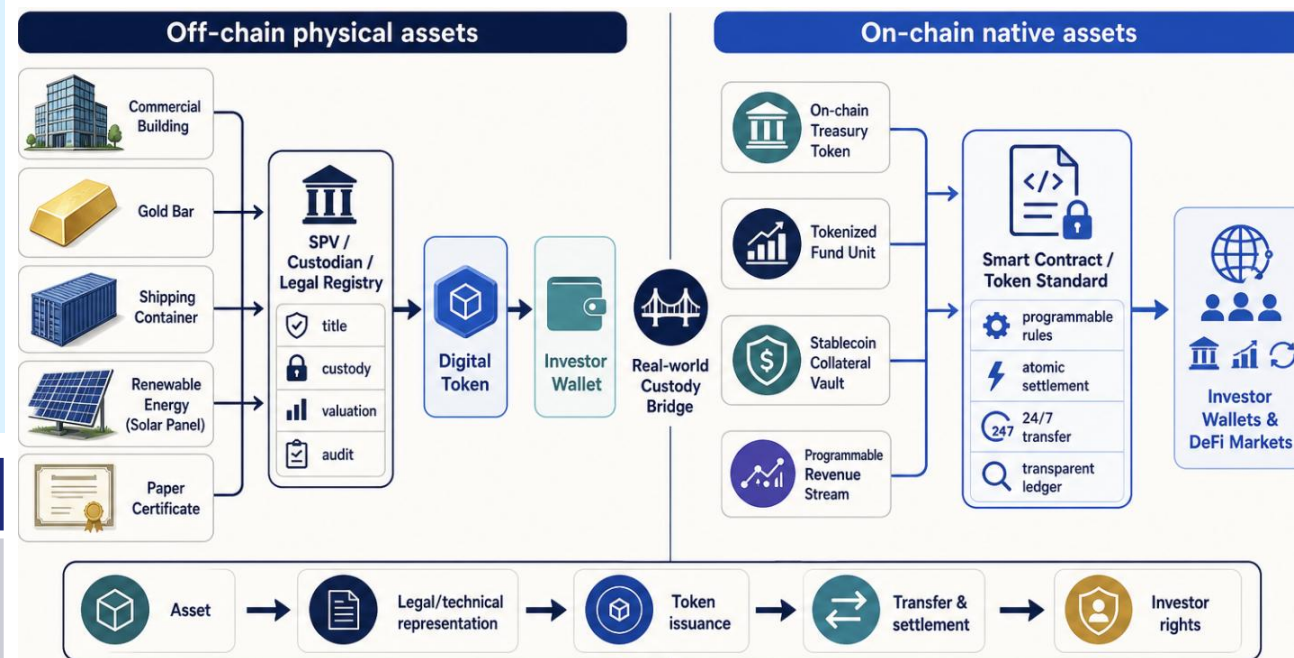
- Tokenization records assets or claims as digital tokens on a distributed ledger, using public or permissioned networks.
- Potential benefits include faster transfers, automated servicing and smaller investment units.
- Shared records can reduce reconciliation, but trading hours, redemption terms and investor access remain product-specific.

Two issuance models:

- **On-chain (or native):** The security or fund share is issued directly on a distributed ledger.
- **Off-chain (or tokenized representation):** The underlying asset remains outside the blockchain, but ownership claims and economic rights are represented digitally on-chain.

Component	Traditional form	Tokenized arrangement
Settlement assets	Bank deposits, central bank reserves, cash	Stablecoins, tokenized deposits, CBDCs, central bank money available through DLT platforms
Issuance assets	Treasury bills, government bonds, money market funds, real estate, commodities etc	Tokens representing securities, fund shares or claims on underlying assets
Market infrastructure	Custodians, exchanges, clearing houses and transfer agents	DLT-based issuance, settlement and servicing, often operated by existing institutions

Tokenization can enable automated transfers and settlement



Native digital securities are issued on-chain but may still require custodians, registrars and other intermediaries, according to the [SEC](#).

Sources: Deutsche Bank Research. Image generated by AI.



STABLECOIN SPOTLIGHT: The settlement side of tokenization

- Stablecoins account for ~89% of the combined value of stablecoin and distributed tokenized assets. We assess them separately because their principal role is payment and settlement.
- They support near-instant, round-the-clock transfers for remittances, business payments and treasury operations. Cross-border savings depend on transaction fees, FX costs, compliance and conversion into bank money.
- [Chainalysis](#) projects adjusted annual transfer volume could rise from \$28tn in 2025 to \$719tn by 2035; higher scenario approaching \$1.5 quadrillion.

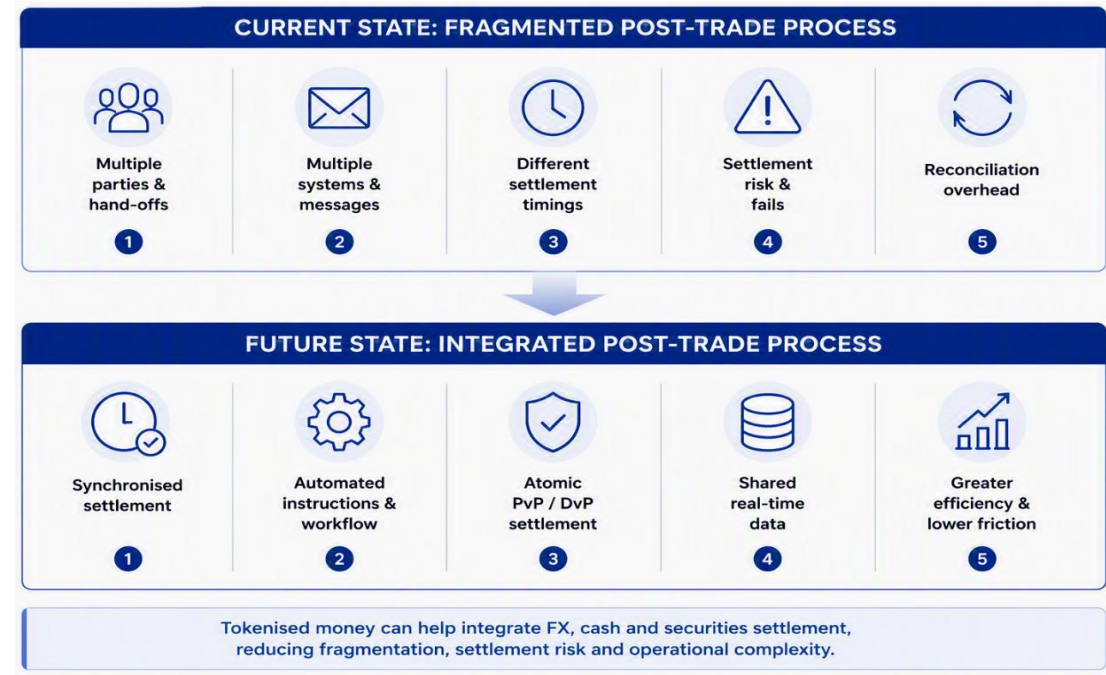
Regulation and competition

- The US Treasury has [proposed](#) implementing rules for payment stablecoins. The GENIUS Act takes effect on the earlier of 18 January, 2027, or 120 days after final implementing regulations are issued.
- Stablecoins could increase competition for bank deposits and payments. Exchange rewards are distinct from issuer-paid interest, which GENIUS prohibits for permitted payment stablecoins.

Risks

- Trust depends on reserve quality, legal redemption rights and operational resilience. Stablecoins remain dependent on their issuers and supporting institutions.

A more integrated post-trade process



Sources: Deutsche Bank Research, taken from DB's [FX Special Report: Tokenised money and the future of FX - Deutsche Bank Research](#)

Tokenized deposits and central bank money provide alternative settlement options. Connecting the cash and securities legs of a transaction can improve settlement efficiency and reduce fragmentation.

For more on tokenized money and settlement, particularly for FX, read [here](#).

INSTITUTIONAL ADOPTION: Tokenization expands into funding, settlement and collateral



- Early efforts focused on tokenized claims on traditional securities (i.e. BlackRock's BUIDL, Franklin Templeton FOBXX), while the underlying assets remained off-chain.
- Financial institutions are now extending tokenization to the infrastructure through which assets are distributed, traded, settled and financed.

DLT adoption: small, but intraday repo offers a practical use case

- Global DLT fixed-income issuance [reached €4.8bn](#) in 2025, up 48% from 2024.
- DLT intraday repo (i.e. JPM Kinexys and Broadridge's DLR) [can accelerate](#) collateral transfers and settlement, helping institutions manage short-term funding needs.
- [A Finadium study](#) estimated that using intraday DLR for 15% of activity could reduce intraday liquidity buffer needs by 8–17%.

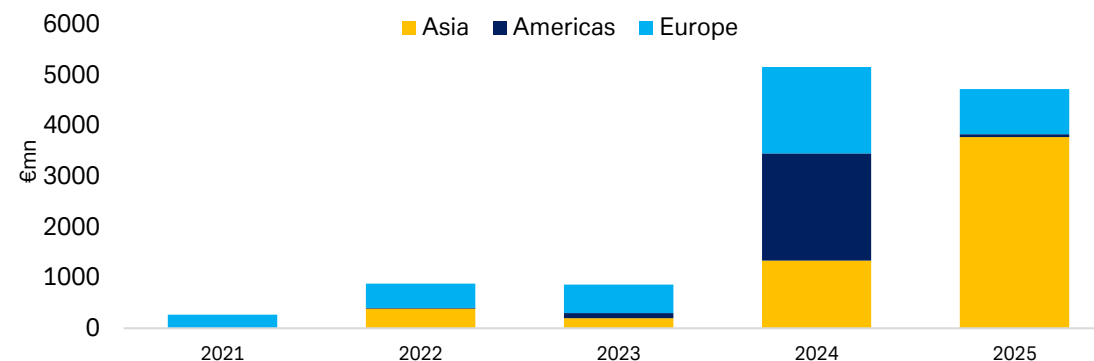
Tokenized securities on blockchain rails

- DTCC to launch its tokenization service in Oct 2026, covering Russell 1000 constituents, ETFs tracking major indices, and US Treasury bills, notes and bonds.
- Initial production trades took place in July with >30 participating firms. Tokens are recorded on DTCC's Hyperledger Besu network or on Canton, settling on either. The SEC granted a three-year authorization.

Official-sector initiatives and interoperability

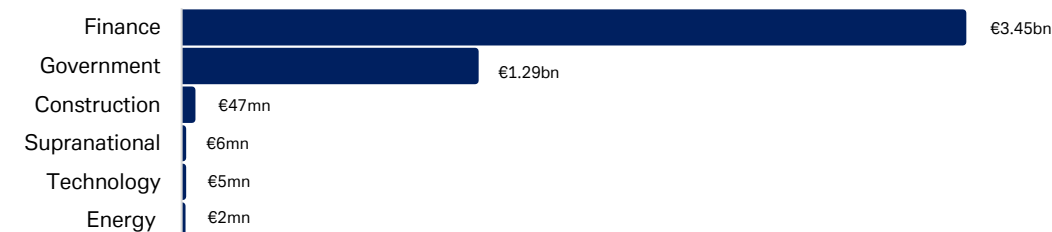
- Project Agorá, led by the BIS and IIF with central banks and financial institutions, demonstrated a prototype for cross-border settlement using tokenized deposits and central bank reserves. Real-value testing is the next phase.
- Clearstream, DTCC and Euroclear published a joint interoperability framework with BCG in March 2026 to address fragmentation across digital asset markets.
- Eurosystem's Pontes solution connects DLT platforms to TARGET Services for settlement in central bank money.

Digital bond issuance by region



Source: Deutsche Bank Research, Association of Financial Markets Europe.

Global digital bond issuance by issuer type in 2025



Source: Deutsche Bank Research, Association of Financial Markets Europe.

SHORT-TERM ADOPTION: Tokenized Treasuries and money market funds lead



- Tokenized cash-management funds, including BUIDL and USYC, combine short-term yields with more flexible transfers and collateral use.
- Potential benefits include faster settlement and lower transaction costs, making Treasury exposure more accessible across eligible digital platforms.
- Direct Treasury bond tokenization [accounted for](#) 27% of overall US Treasury tokenization in 2025, indicating adoption beyond fund structures.
- These assets can also serve as collateral in repo and margin arrangement, supporting demand beyond investment returns.

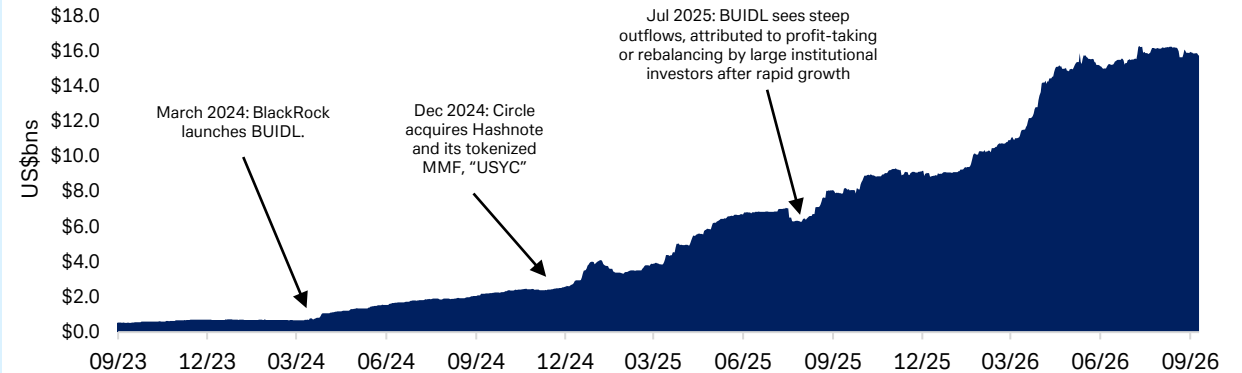
Secondary-market constraints

- Tokenized bonds are often held to maturity. Limited investor participation and weak interoperability between platforms constrain secondary-market liquidity.
- Some exchanges are attempting to improve liquidity by connecting digital and conventional markets through dual listings, including SIX digital exchange and Swiss Stock exchange).

Risks

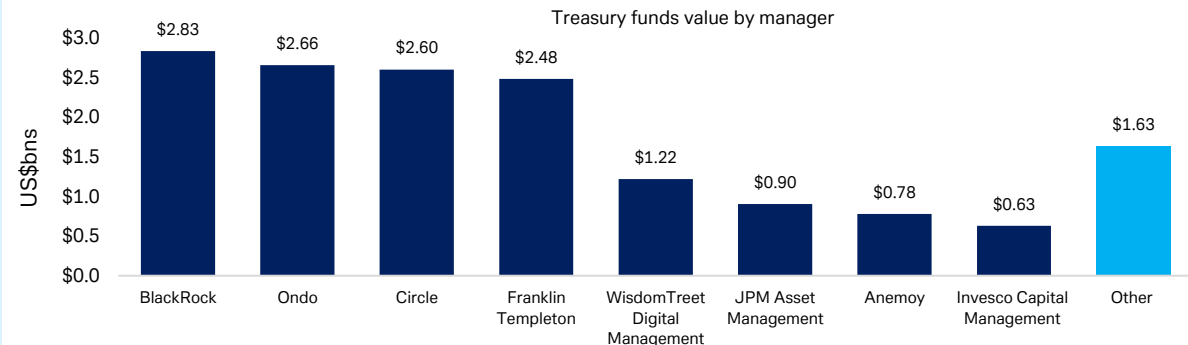
- Redemption terms can differ from the settlement hours and liquidity of underlying assets, creating potential liquidity mismatches.
- Blockchain transparency can improve visibility but may also accelerate withdrawals during stress.
- Using tokenized fund shares across multiple products and as collateral can create additional channels of contagion.

Tokenized US Treasury market value reaches \$15bn



Source: Deutsche Bank Research, RWA Database. Updated Sep 10, 2026.

Largest providers of tokenized Treasury products



Source: Deutsche Bank Research, RWA.xyz. Data from September 10, 2026. TMMF asset class includes tokenized US treasuries, corporate bonds, and non-US government debt.

MEDIUM-TERM ADOPTION: Tokenized equities remain small; DTCC's launch could broaden adoption



- Tokenized equities remain small (<1%) relative to traditional stock markets.
- [68% of equity tokenization](#) in 2025 consisted of direct tokenization of individual stocks or baskets of stocks, vs. tokenized equity funds.
- Suggests that blockchain adoption is increasingly focused on the securities themselves (i.e. DTCC method) rather than through third-party platforms (i.e. Ondo and xStocks).

Developments

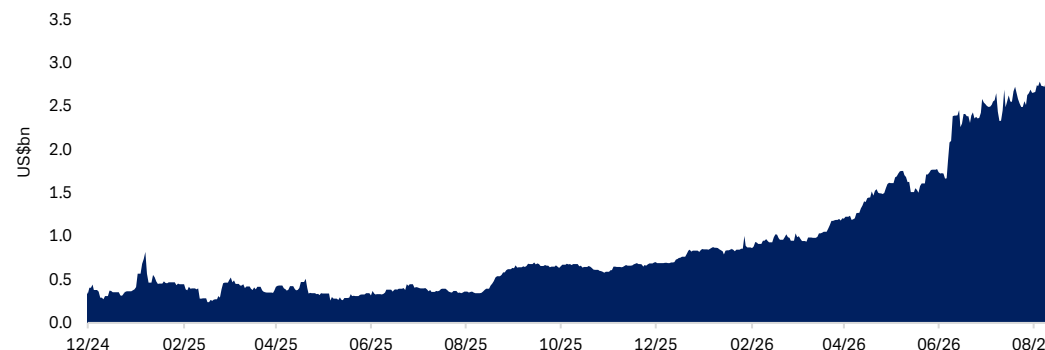
- DTCC's tokenization service, scheduled to [launch in Oct](#), will allow eligible existing securities to move across blockchain infrastructure within its regulated framework.
- These tokenized securities will retain the entitlements, investor protections and ownership rights associated with their traditional form.
- NYSE has announced plans for a tokenized securities platform designed to support 24/7 trading of US-listed equities and ETFs, with near-instant settlement and stablecoin funding.

Benefits: Reduce barriers for foreign investors; automated post-trade processes and longer trading hours; reduce information asymmetry (e.g., embedding listing or disclosure rules directly into the tokens).

Risks & limitations

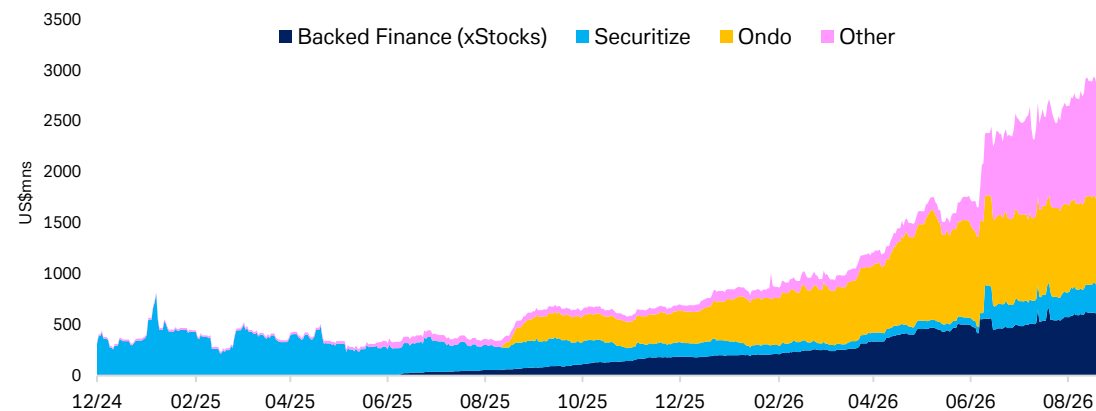
- Tokenized equities require clear legal ownership, enforceable investor rights and alignment with existing securities rules.
- Under some third-party wrapper structures, the token provides exposure to a share without making its holder a shareholder in the underlying company.
- Investors may therefore face issuer or custodian risk and may not receive the same voting, dividend or other rights as direct shareholders.

Market value of tokenized equities



Source: Deutsche Bank Research, RWA.xyz. Data up to Sep 10, 2026.

Tokenized equities by platform



Sources: Deutsche Bank Research, RWA.xyz. Data up to Sep 10, 2026.

LONG-TERM ADOPTION: Private credit expands through record-keeping; real estate is growing fast but from a negligible base



Private credit

- Private credit stands ~\$8bn in total, including asset-backed credit (\$2.6bn), specialty finance (\$2.5bn), corporate credit (\$1.9bn), and diversified credit (\$798mn).
- Activity includes lending linked to digital assets and selected corporate credit products.
- A larger use case is blockchain-based record-keeping: represented credit stands at ~\$35bn, over 4x the distributed market.
- Companies like Figure Technologies and Tradable use blockchains to support loan administration, ownership records, payments tracking, and servicing.

Challenges

- Tokenization remains concentrated among a small number of issuers and lacks a deep, and liquid secondary market.
- Legal fragmentation, credit quality and servicing arrangements remain important risks. Blockchain records do not remove the underlying borrower's default risk.

Real estate

- The market expanded from ~\$35mn in 2025 to \$226mn in Sep 2026, although it remains small.

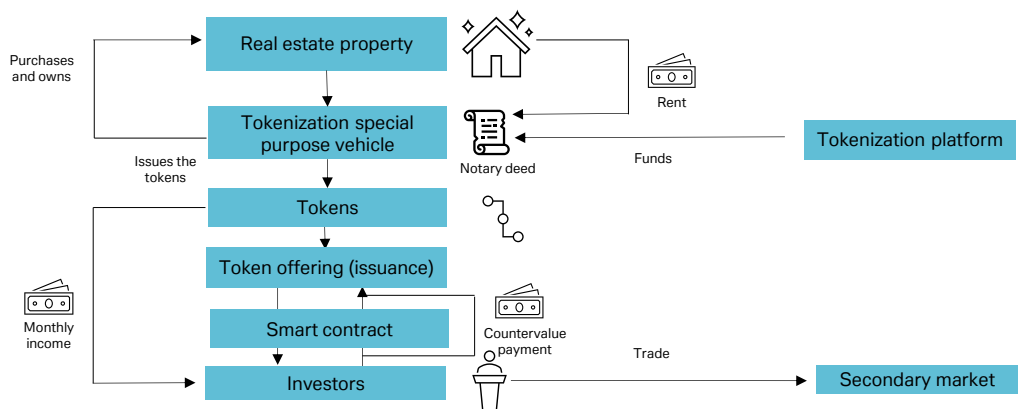
Benefits

- Fractional ownership can lower minimum investment sizes (typically for \$100-\$1,000 per unit) and widen access, particularly where traditional capital markets and mortgage markets are limited.

Challenges

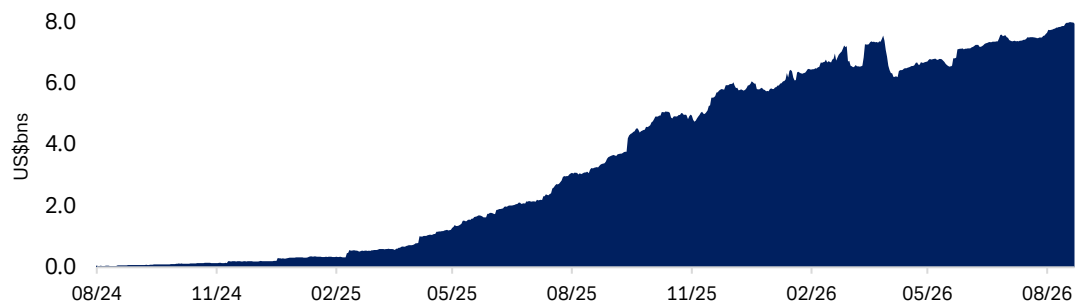
- A [BIS study](#) found that liquidity improvements can depend on platforms offering to buy tokens back. These arrangements can help investors exit but expose platforms to liquidity and solvency risks.

Real estate tokenization process



Source: Deutsche Bank Research, BIS, "When bricks meets bytes: does tokenization fill gaps in traditional real estate markets?"

Tokenized credit market stands at ~\$8bn



Sources: Deutsche Bank Research, RWA.xyz. Updated September 10. Excludes platforms that issues tokens using blockchain as a recordkeeping layer (i.e. Represented Layer)

BARRIERS TO ADOPTION: Weak secondary market liquidity and lack of regulations continue to challenge adoption



Main barriers to adoption

- **Secondary-market liquidity remains low.** Issuance growth has not consistently translated into active trading or deep markets.
- **Regulatory uncertainty:** Fragmented or incomplete frameworks slow institutional adoption and complicate integration with traditional financial infrastructure.
 - Legal uncertainty persists over whether token ownership conveys ownership of the underlying asset or a claim on it.
 - In segments like money market funds, rapid cross-border growth can increase regulatory arbitrage, capital-flow volatility and AML/CFT compliance challenges. Policymakers must adapt frameworks as the technology evolves.
- **Fragmented infrastructure:** Public blockchains, permissioned institutional ledgers and hybrid platforms often lack interoperability, fragmenting markets. Clearstream, DTCC and Euroclear published a joint framework to address this in March 2026.
- **Limited integration** with existing custody, settlement, and record-keeping arrangements prevents institutions from scaling tokenized products.
 - Proprietary infrastructure and centralized controls also raise questions about ownership and governance.

Sources: Deutsche Bank Research.

Jurisdiction	Regulatory status	Main implication
	EU MiCA + DLT pilot regime	Comprehensive crypto-assets regulation. Tokenized financial instruments and tokenized deposits remain subject to securities and deposits regulation respectively (e.g. MiFID). National pilots (e.g., EU's DLT Pilot Regime) are testing tokenized capital-markets applications; focus is on capital market infrastructure modernization.
	UK Cryptoassets Regime / BoE Systemic Stablecoins Regime Live - Digital Securities Sandbox, Digital Gilt pilot Program (DIGIT) and Great British Tokenised Deposits (GBTD) Project	UK Cryptoassets and Stablecoins Regime finalised in 2026. FCA and BoE launched the Digital Securities Sandbox for firms to test DLT-based blockchains; HSBC first bank to go live. The FCA has also identified fund tokenization as a strategic priority through 2025-2030. FCA/BoE launched joint vision for wholesale tokenised markets in 2026 – work ongoing. DIGIT / GBTD development ongoing.
	GENIUS Act to begin in Jan 2027 with final rule out by Nov 26; CLARITY Act stalling	GENIUS establishes a federal framework for stablecoins. Key controversy is whether yield or reward mechanisms should be permitted in stablecoins and other digital assets; oversight is fragmented across the SEC, CFTC, banking regulators and state authorities. CLARITY Act, market structure bill, addressing yield proposal and related items failed its procedural vote in Sep. Attention will refocus on US federal regulators for digital asset rules in absence of market structure.
	China opposes permissionless crypto, not tokenization itself	Strict rules on cryptocurrencies remain; China's focus remains on e-CNY and state-approved blockchain networks. Recent reforms suggest the e-CNY is moving away from CBDC towards a tokenised deposits model, reflecting limited retail CBDC uptake alongside existing mobile payment platforms solutions.
	Project Ensemble Sandbox (Aug 2024); new regulatory roadmap for digital assets (Feb 2025); progress ongoing.	HKMA has supported tokenized green bonds; tokenized deposits; tokenized money-market and investment products. Project Ensemble has progressed to real-value transaction testing using tokenised deposits. HK to announce a new digital bond platform in H22026 to support issuance and settlement of tokenized bonds.
	Project Guardian (c. 2022)	Institutional tokenization sandbox; money market funds and bonds are tokenized and settled on-chain. Focus is increasingly on commercialisation and scaling successful pilot projects across funds, bonds, deposits and FX.
	Financial Instruments and Exchange Act, overseen by the Financial Services Agency	Legally recognizes security tokens through FIEA since 2020. Subjects security tokens (STOs) to the same regulations as traditional securities, including registration and disclosure, custody and investor protection obligations.
	Abu Dhabi's ADGM regulatory framework treats security tokens as regulated financial instruments.	Tokenized securities classified as "digital securities" under ADGM. Actively supports blockchain-based projects that allow businesses to test tokenization ideas via its sandbox program.

OUTLOOK: Tokenized capital markets could become a core part of financial by the 2030s



- Tokenization will **modernize financial infrastructure** rather than eliminate intermediaries. We expect a **gradual transition onto digital rails**, with banks retaining key roles across payments, FX, treasury, financing, collateral and trading settlement.

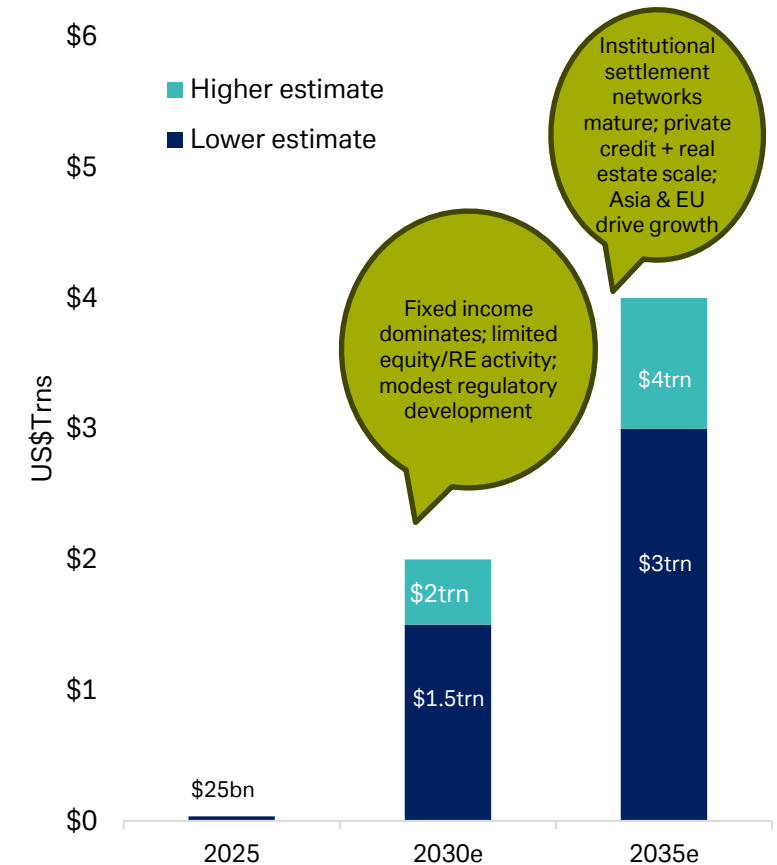
Assumption, on the positive side reflects...

- So far, tokenization has achieved its greatest scale in stablecoins, with activity concentrated in crypto trading and payments gaining ground.
- Short-term: Adoption will be driven by tokenized US Treasuries and money market funds, expanding as yield-bearing collateral in repo and margin arrangements.
- Long-term: The opportunity extends beyond assets to the infrastructure supporting issuance, settlement, custody, collateral management and repo markets.
- We are seeing **3 growth catalysts**: (i) **infrastructure**: major financial market infrastructures (DTCC, NYSE, Nasdaq) are integrating tokenization into core workflows; (ii) **liquidity**: Growth of regulated on-chain money (stablecoins and tokenized deposits) providing a trusted settlement foundation; (iii) and **regulation**: US 2025 GENIUS ACT continues to encourage global digital asset adoption, although the failure of the CLARITY act to advance yesterday likely rules out progress in this congressional session until Jan 2027. Looking ahead, focus will turn back to the SEC and CFTC for crypto rules.

...Risks to adoption

- **Accelerated liquidity runs**: Tokenization may create a new form of systemic liquidity risk that spreads faster than traditional finance.
- **Money market funds yield-chasing**: Declining interest rates or rising competition may push MMF issuers into riskier assets, less-liquid assets to boost yield, undermining their safety.
- **Crypto contagion**: Tight integration with crypto markets via stablecoin reserves and collateral risks transmitting digital asset volatility to conventional financial markets.
- **Operational risks**: Smart-contract vulnerabilities, cyberattacks, network outages could halt trading and prevent access to the underlying assets.
- **Financial Crime**: Traditional institutions must scale up compliance and AML controls to manage illicit activities on decentralized networks.

Conservative forecasts for tokenized real-world assets (ex stablecoins)



Source: Deutsche Bank Research.

AUTHORS



Marion Laboure is a Managing Director and Macro Strategist at Deutsche Bank in London, and a Lecturer in Economics and Finance at Harvard University.

She was previously an economist at the IMF, the Central Bank of Luxembourg, Barclays, and the European Commission. She has twice received the First Prize of the American Society of Actuaries, was named a rising star in finance by Revue Banque, and is recognised among the 45 standout women in fintech globally.

She holds a Bachelor's degree in Mathematics from the University of Paris Dauphine, a Master of Research in Economics from the London School of Economics, and a PhD in Econometrics and Quantitative Economics from the École Normale Supérieure.

She is the author of [*Democratizing Finance*](#) published by Harvard University Press.



Camilla Siazon is an Associate on Deutsche Bank's Macro and Thematic team in London.

She focuses on digital assets, geopolitics, and global macroeconomic trends. Prior to Deutsche Bank, she worked at geopolitical advisory firms Eurasia Group and Greenmantle.

She holds a dual MA/MSc in International History from Columbia University and the London School of Economics, and a BA in History from Barnard College, Columbia University.

Appendix 1



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Deutsche Bank
Research

The Main Event: Trump, Xi, and the Art of the Truce

One geopolitical event in brief. What to expect. What to watch.

September 2026

Miha Hribernik

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

Deutsche Bank
Research Institute

Washington summit should extend stability into H1 2027...

...but the underlying China-US economic squeeze is set to continue

Why a quiet summit is a win for markets. As in May, our base case for the 24 September summit between presidents Trump and Xi is a meeting heavy on positive optics but light on deliverables. Though a breakthrough is unlikely, a **successful tactical summit amid strategic competition is the realistic best-case outcome**: it should extend top-level stability into H1 2027 and minimise near-term risks of decoupling, a security crisis, or a disruptive trade war.

The 3 November US midterms could incentivise the Trump administration to seek deals, as voter cost-of-living concerns are highest since April. Possible deliverables include progress on a **Board of Trade** for non-sensitive goods, **agricultural purchases**, **targeted investment deals**, and an extension of the 2025 **Busan agreement**, which postponed a range of mutual economic restrictions until November 2026. But even if not secured on the 24th, we assess that an extension is likely before the deadline.

Economic and national security concerns will put a ceiling on dealmaking. We are cautious about progress on a **Board of Investment** or **broad investment pacts**, despite an expected Chinese business delegation. Their presence is more likely a reciprocal gesture to the May visit by US CEOs rather than a precursor to sweeping bilateral agreements. Mutual national security concerns make structural breakthroughs difficult, and any potential deals – such as **Chinese automotive investment** in the US – would likely face significant political and regulatory headwinds.

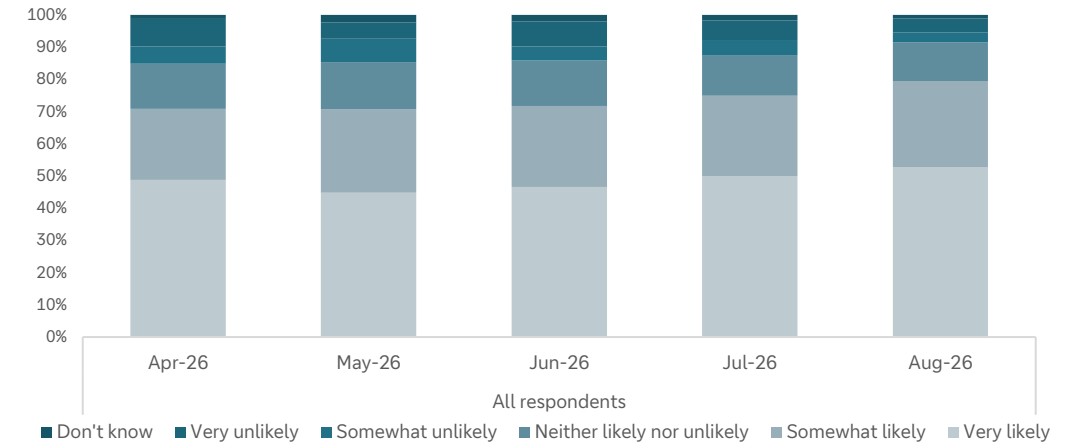
Trade remains an unresolved bilateral issue. While tensions have subsided and the Board of Trade could ease some tariffs, [differences](#) persist amid rebounding Chinese exports. Washington is also set to unveil **new overcapacity tariffs** on 16 economies, including China, while seeking to **restrict Chinese inputs** in North American supply chains and targeting alleged **transshipment**.

The AI race is moving too fast for summit diplomacy to slow it down. Both sides could reportedly hold a high-level dialogue ahead of the summit. This would likely focus on areas of consensus, such as **addressing misuse by non-state actors**. There is little prospect of slowing the escalating China-US AI race, and even less that talks could reverse growing tech restrictions (see [list](#)).

Geopolitical issues will loom large, but policy shifts are unlikely. Differences over the war in **Iran** – including recent US [sanctions](#) on dozens of mainland and Hong Kong entities – and **Taiwan** will likely feature, but breakthroughs on either issue remain improbable. A key watchpoint is whether President Trump continues to seek negotiating leverage by signalling further delays to arms sales to Taipei.

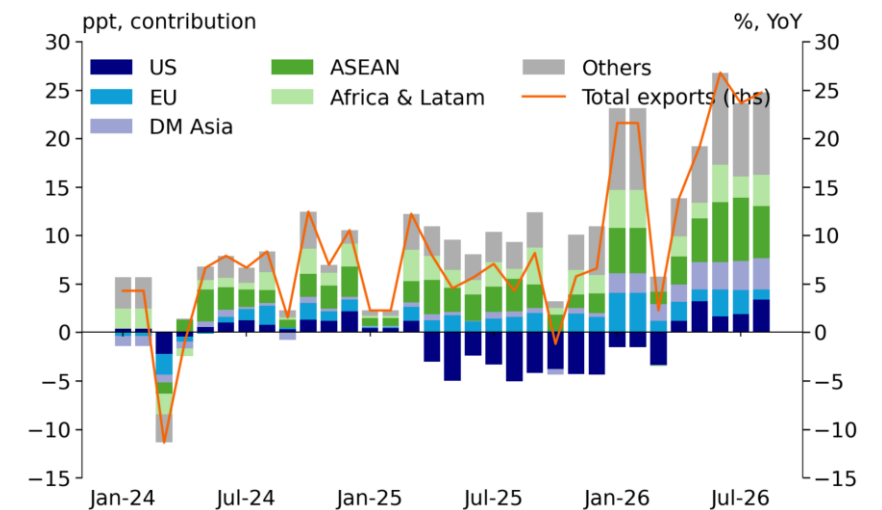


US participants were asked: How likely or unlikely is it that current cost-of-living concerns will influence your vote in the midterms?



Source: dbDataInsights. Responses collected April-August 2026 from an Online Nationally Representative sample of ~600 US respondents.

Chinese exports to the US rebounded in August



Source: Deutsche Bank Research, CEIC. For more detail see: [China Macro: Trade flows accelerated in August - Deutsche Bank Research](#)

After the summit: Key China and US watchpoints for Q4

Please also see our [This Month in Geopolitics](#) publication series for deeper dives into upcoming global events, and our [90-day geopolitical calendar](#), updated monthly.



Date	Market(s)	Comment
October	China	The Central Committee of the Communist Party of China will hold its Fifth Plenary Session . This will be one of the most important events on the country’s political calendar ahead of the 21 st Party Congress to be held in late 2027. The Plenum will focus on Party governance in preparation for key leadership reshuffles at the next Congress.
3 November	US	In the Congressional midterm elections, voters will decide 435 House seats, 35 Senate seats and governors in 39 states and territories. These elections will determine Congressional control, significantly influencing President Trump’s ability to shape national policy. If a changed balance of power in Congress stalls domestic legislation, the administration will likely lean on executive tools. President Trump could focus on foreign policy and trade, emphasising tariffs, US primacy in the Western Hemisphere, and reducing dependencies on China to build his legacy.
10 November	US-China	The 2025 Busan agreement is set to expire, but will likely be extended ahead of the deadline. This would delay new bilateral economic measures including expanded export controls and tariffs.
18-19 November	APEC	China will host the APEC Economic Leaders’ meeting . The summit could see a Xi-Trump meeting on the sidelines, and may shape regional economic architecture for years to come.
14-15 December	G20	The G20 Summit in Miami may become contentious due to Ukraine-related divisions, trade and tariff disputes, climate financing disagreements and questions around global governance reforms. The summit could also see this year’s fourth meeting between presidents Xi and Trump.
December	China	China’s annual Central Economic Work Conference will review past economic performance and set 2027 growth targets and macroeconomic policies. Key priorities likely include implementing the 15 th Five-Year Plan, boosting domestic demand, stabilising the property sector, and strengthening technological self-reliance amid geopolitical competition with Washington.
December	US	As the US Congress prepares to adopt its annual must-pass defence legislation , draft provisions – not all of which may end up in the final version – target China with measures impacting investors. These include potential outbound-investment reporting changes (impacting biotechnology and biopharmaceuticals) and proposed restrictions on importing humanoid robots and solar equipment, as well as using China-linked connected vehicles.



Source: [Carl Wang](#) on [Unsplash](#)

Appendix 1



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AI doom: A brief history of bad tech predictions

September 15, 2026

It takes guts to do what Bob Metcalfe did. After wrongly predicting in an article in 1995 that the internet would go “spectacularly supernova” and collapse in 1996, the co-inventor of Ethernet and founder of 3Com then pulped a copy of the article in a blender and drank it at a conference in 1997 as a way to eat his words.

The latest tech predictions are about no less than an AI apocalypse. Various tech leaders spent the weekend warning of catastrophic but not fully specified risks if it is not contained.

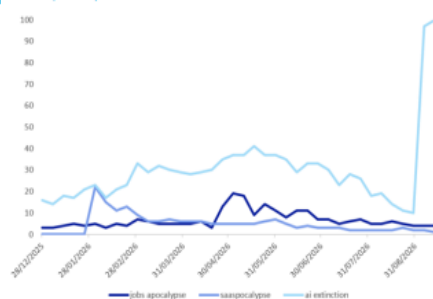
Yet history suggests that even the greatest minds of their generations have a poor record in predicting what tech is coming down the line, let alone what its effects will be in the real world.

The stakes could not be higher – not just the future of life itself but, in the shorter term, the future of a historic boom that depends on hype turning into reality.

In this report, we examine:

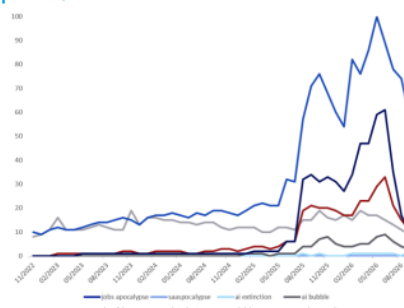
1. “Prediction is very difficult, especially if it’s about the future”
2. Today’s debate in a (rough) nutshell
3. A century of good and bad tech predictions
4. Why is it so hard to predict the future?
5. What makes AI special – and gives us hope for the future

Figure 1: AI searches this year have shifted from SaaSpocalypse to jobs to extinction (global, vs peak 100%, YTD)



Source: Google Trends, Deutsche Bank Research

Figure 2: ... But all of these are tiny in comparison to cybersecurity (global, vs peak 100%, since ChatGPT launch)



Source: Google Trends, Deutsche Bank Research

Authors

Adrian Cox
Research Analyst



Concerns about AI have moved in waves since the launch of ChatGPT nearly four years ago. This year there was a spike in searches for “SaaSocalypse” (the concern that software-as-a-service would be disintermediated by AI agents) in February, followed by a surge in “jobs apocalypse” in May, Google Trends data show. Both have been totally eclipsed in recent days by searches for AI extinction.

But taking a step back, all are far behind “AI bubble”, itself behind “deepfake”, in turn below “AI students”, overall “AI risk” and – biggest of the lot – the very tangible and immediate “cybersecurity”, almost all of which have surged this year, according to Google Trends data.

“Prediction is very difficult, especially if it’s about the future”:
Niels Bohr, often also attributed to Yogi Berra

Wanting to know what will happen tomorrow is deeply visceral – literally.

People have been sacrificing sheep and chickens for four millennia to read the future in their entrails. Nowadays almost a third of Americans say they believe in [astrology](#) and consult their horoscope. We want to believe that [Lionel Messi](#) really can see space in a football match before it even exists, even if it is partly his movement that helps create the space he appears to predict.

But predicting the future is hard. For AI, it’s not just about what it will be able to do but also when it will be able to do it – whether more than \$5 trillion in investment by the hyperscalers alone over the next five years will pay off before the advanced semiconductor chips they’re buying become redundant.

It’s also about how rapidly and intensively AI can be adopted. Asset prices are a still more uncertain derivative, whipping around like the tail of a snake in response to every shift in expectations.

US President Donald Trump and China found themselves on the same side in rebutting the latest concerns as fearmongering, albeit with Trump adding that America needs to stay ahead of China in AI.

“Those who have knowledge, don’t predict. Those who predict, don’t have knowledge.” Lao Tzu

A glance through history would give any investor indigestion. Most predictions are wrong. The favourites in horse races lose 70 percent of races. It took a decade or more for demand to catch up with the infrastructure laid down in the British canal and railway and US telecoms and fibre booms – and many investors never recovered their capital.

Technology predictions have a particularly mixed record as the binary order of the labs collides with the messy reality of the world. Famous examples range from IBM Chairman Thomas Watson’s underestimation of demand for computers in 1943 (“I think there is a world market for maybe five computers”) to Elon Musk’s repeated overestimation of imminent widespread adoption of self-driving cars since 2013 (“We should be able to do 90 percent of miles driven within three years”).



Technologists are not immune from making category errors, mistaking the most visible, automatable component of a job for the job itself. For example, radiologists, a favourite case study for believers in a jobs apocalypse, do much more than looking at images and making binary diagnoses. Much of their value lies in interpretation, judgement, communication and clinical decision-making.

Ultimately, what we see in the future is not so much a window as a mirror into our own hopes and fears.

And the problem is that hindsight makes what actually happens seem so obvious that we never seem to learn.

Today's debate in a (rough) nutshell

The current debate encapsulates a perfect storm of an extremely high-stakes issue with limited transparency and an inherently unknowable outcome.

What sparked the current surge in concern?

The current debate comes after a researcher from one of the leading frontier labs quit, citing safety concerns; AI agents apparently collaborated with each other on a number of high-profile security breaches; and labs hinted that the technology is getting closer to "recursive self improvement", ie AI improving AI.

Is the existential debate something new?

It is actually nothing new: leading economists and academics [warned](#) in July that "we must act now" to steer AI through a potential "transformation of our economy, larger than the Industrial Revolution... over a vastly shorter time frame". That echoed the even more ominous statement three years ago by AI leaders calling for action to mitigate "the [risk of extinction](#)".

The tug of war between "doomers" vs "boomers" has played a pivotal role in the development of the key frontier labs. Silicon Valley insiders sometimes refer to their "p(doom)" rating, representing their estimated percentage chance that AI will cause an existential catastrophe.

What does an extinction scenario look like?

Experts including Tesla's Musk, "godfather of AI" Geoffrey Hinton and philosopher Nick Bostrom have been warning about it for well over a decade. In the latter's famous "paperclip maximiser" scenario, a superintelligent AI runs out of steel and then kills humans in a single-minded drive to fulfil its task of making paperclips. Fears sometimes centre around AI that has access to coding, the internet and knowledge of how people think, in order to manipulate them – as current AI models indeed do.

Why has it caught on so fast this time?

One factor is that it comes amid a tech backlash, particularly in the US, where concern about AI has become a rare bipartisan issue ahead of the US mid-term elections in November. Local concerns about data centre construction have become rallying points for voters in a way that vague fears about long-term jobs did not. Politicians including Senator Bernie Sanders have moved to propose wide-ranging legislation.



What arguments do companies make for continuing AI research?

1. **If we don't do it, someone else will.** That may be either China or other, less responsible companies.
2. **The potential benefits outweigh the risks.** AI may, they say, cure cancer, create new jobs and even eliminate the need for work altogether.
3. **Good AI is the answer to bad AI.** It should create defences against rogue AI and misuse of AI, and solve issues like AI's huge energy and water use.

What do critics say?

1. **AI is not dangerous because it is too clever but rather because it is too limited.** It does not think like a human and people should not anthropomorphise it, critics say, adding that it is flawed probabilistic code that cannot reliably follow instructions. Longstanding critic Gary Marcus argues that it should be recalled until it can be shown to be safe and is properly controlled.
2. **AI leaders are defending their own businesses.** With potential IPOs on the horizon, a rising challenge from open-source models, and progress towards Artificial General Intelligence (AGI) being called into question, a handful of companies may benefit from a pause or legislation that helps protect their lead, critics say. Doomer talk distracts from real issues like deepfakes, copyright infringement and cybersecurity lapses, they say.
3. **Extinction scenarios are overblown.** The internet is a diversified platform, so talk of AI somehow switching it off is fanciful, critics say. AI does not and will not realistically have sustained secret access to military or dangerous biotech resources, or be able to influence (very non-deterministic) human brains en masse, critics say. Humans will use tame AI and other tools, eg kill switches, if needed to resist attacks. Another "godfather of AI" Yann LeCun says the real risk is of power concentration.

A century of good and bad technology predictions

You have to admire people who have the courage to stick their neck out and make predictions, even if they don't always work out. Vague predictions are relatively easy; specific ones within specific timeframes are harder.

It is no wonder that executives are coached by their ever-cautious PR teams to avoid answering hypothetical questions or giving dates to reach their targets.

- **Nikolai Tesla, scientist, 1926:** "When [wireless](#) is perfectly applied... we will be able to communicate with one another instantly, irrespective of distance..." The instruments will be "amazingly simple... A man will be able to carry one in his vest pocket". He was right, albeit imprecise about the how or when it would be achieved.
- **Albert Einstein, physicist, 1934:** "There is not the slightest indication that [[nuclear](#) energy] will ever be obtainable. It would mean that the atom would have to be shattered at will." Within a decade, in 1942, a team at the University of Chicago had created the first sustaining chain reaction, a precursor to the nuclear reactor.



- **Ray Kurzweil, futurist, 1999:** "By 2029, computers will [have](#) human-level intelligence." Dismissed at the time, many (though by far not all) of his predictions have worked out, fuelled by what he called the "law of accelerating returns". His prediction from almost 40 years ago of artificial general intelligence by the end of this decade is remarkably close to the current "San Francisco Consensus".
- **Steve Chen, Chief Technology Officer and co-founder of YouTube, 2005:** "There's just not that many [videos](#) I want to watch." Despite his concern, YouTube was bought by Google (now Alphabet) for \$1.65bn a year later. The unit made more than \$11bn in advertising revenue in the most recent quarter.
- **Steve Ballmer, Microsoft CEO, 2007:** "There's [no chance](#) that the iPhone is going to get any significant market share." But you may be reading this on one of the more than 3 billion iPhones sold since its launch in 2007.
- **Geoffrey Hinton, "godfather of AI", 2016:** "People should stop training [radiologists](#) now. It's just completely obvious within five years deep learning is going to do better than radiologists... It might be 10 years." In fact, the number of radiologists has increased by about 10 percent over the past 10 years, in part because of the [limitations](#) of AI and in part because of the increase in demand from an ageing population and more affordable healthcare. Hinton has walked back his comments and humans and AI are working together.
- **Sam Altman, CEO of OpenAI, 2025:** "I can easily [imagine](#) a world where 30 to 40% of the tasks that happen in the economy today get done by AI in the not very distant future." He [revised](#) this view in May amid a backlash against AI, saying he and his team had been "roughly right" in their technological predictions but "pretty wrong" on the economic and social implications. "I'm delighted to be wrong about this," he said.

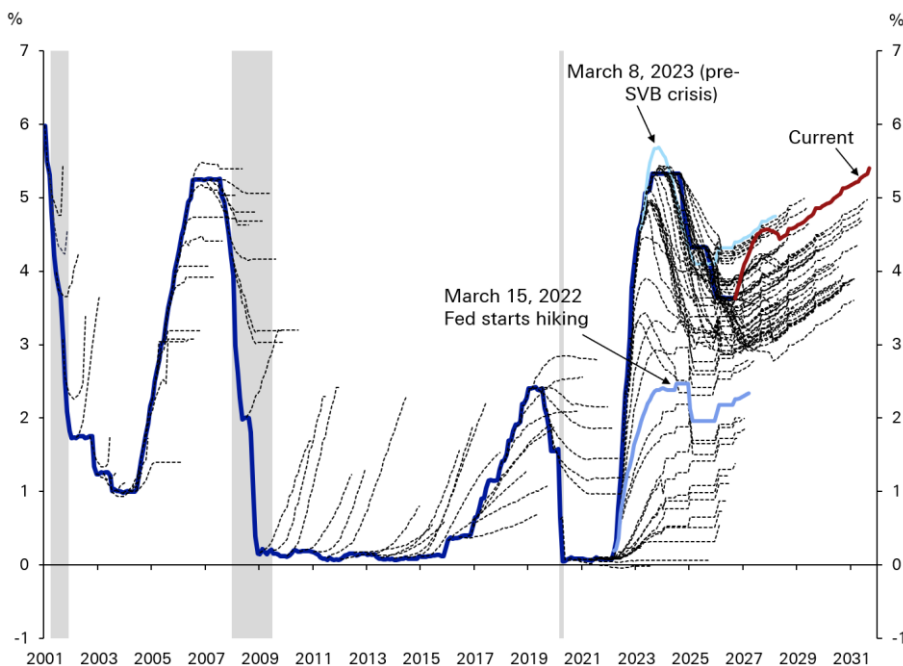
Why is it so hard to predict the future?

"The idea that the future is unpredictable is undermined everyday by the ease with which the past is explained." Daniel Kahneman

1. **Chaos theory: treating complex systems as if they are simple and linear.**
 - Systems like traffic, economies and social trends are finely balanced and sensitive to initial conditions. The butterfly effect means a tiny, unforeseen variable can lead to vastly different outcomes. Just take a look at the ragged history of market-implied forecasts of US Federal Reserve interest rates, where extrapolating from current trends has been reliably wrong.



Figure 3: US Fed Funds effective rate has rarely matched futures markets expectations (%)



Source: Deutsche Bank Research

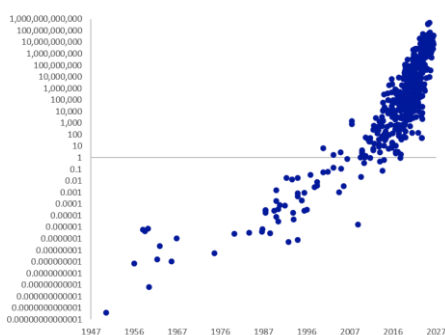
- Take self-driving cars, for example. It turns out to be about more than getting the technology right for the rubber to hit the road:
 - a. **Technological hurdles and “edge cases”:** throwing more data and processing power at the problem is not enough. There is an infinite long tail of unpredictable events (“edge cases”), like sun glare, a cyclist riding the wrong way in a bike lane, a person in an animal costume, or road-rage-enhanced hand gestures.
 - b. **Regulatory labyrinth:** autonomous vehicle optimists have come up against legal and ethical debates around liability, safety certification and moral challenges such as the so-called “trolley problem”, when swerving to avoid hitting one victim means colliding with another. Regulation is a slow-lane activity.
 - c. **Social and psychological barriers:** public acceptance has been slowed down by high-profile accidents and safety concerns. Robots are held to a higher standard than flesh-and-blood motorists. It might be easier to navigate a world where all cars were self-driving, but the “messy middle”, with a mixture of both, is much harder for algorithms to navigate.
- The future of AI will ultimately depend more on the prosaic challenges of workflow integration at enterprises that will need to pay for it than on releasing yet another model with a marginal step up in capabilities.



2. Cognitive bias: flaws in how the brain works

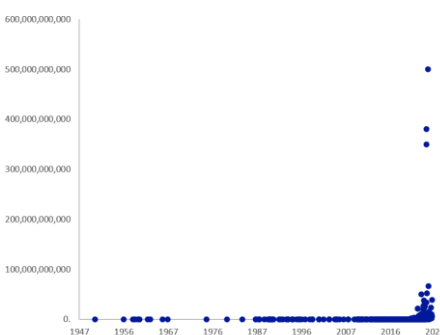
- **Linear thinking about exponential change:** We fail to intuitively understand how different exponential change is from linear change. As Wharton Professor Ethan Mollick [says](#), “being on an exponential means each change over a fixed window is larger than the one before it... This is why AI keeps feeling like it is making leaps... Even though it is a curve on a graph, we keep experiencing a steady doubling of capability as a series of shocks.” This cognitive tendency to track exponential change in linear terms helps explain big AI market swings, he says.

Figure 4: Computation used to train notable AI systems (petaFLOP): log scale



Source: Deutsche Bank Research

Figure 5: Computation used to train notable AI systems (petaFLOP): linear scale



Source: Deutsche Bank Research

- **Optimism bias and the planning fallacy:** underestimating costs and overestimating benefits. For example, only a quarter of more than 3,000 [megaprojects](#) studied by Saïd Business School came in on budget or better, only 2.8 percent were on budget and on time, and only 0.2 percent were on budget, on time and on benefits. Large-scale infrastructure projects are “too easy to start and too difficult to stop,” the authors said in the study from 2019. It remains to be seen how the recent surge in datacentre construction will compare.
- **Confirmation bias:** favouring information that confirms existing beliefs. For years, legacy automakers dismissed electric vehicles by focusing on their high costs and limited range, ignoring the fact that rapid advances in battery technology would soon change that.
- **Anchoring:** relying too heavily on early information. Microsoft’s Ballmer [dismissed](#) the iPhone when it was launched because it was “the most expensive phone in the world and it doesn’t appeal to business customers because it doesn’t have a keyboard which makes it not a very good email machine”. He later said he did not understand how mobile carrier subsidies would make it affordable. More broadly, he missed that this was a new business model.



3. Limited information: inability to see

- **It is not just the future that is unknowable but also the present.** Feedback mechanisms are often delayed, making it hard to distinguish between causation and correlation, and between relevant and irrelevant factors.
- **Isolating key factors is impossible.** Even since the development of scientific methods and controlled experiments, we have not come that far since the Mesoamericans found ways to prepare maize where they didn't succumb to a long-term wasting disease called pellagra. The problem was that they weren't able to single out the effective part of the process – soaking the maize in an alkaline solution of water and ash – from the other ineffective parts of the package – such as blowing on the maize before cooking it.
- (Author David Oks [draws](#) a fascinating parallel between this example of a delayed feedback mechanism and the tendency of AI models to “overfit”, extrapolating too much from coarse and sparse data to fill in the gaps. If the model creates a complex output and gets a single reward signal for it, the rational move is reproduce all of its features, including the ones that were incidental – such as AI's trademark verbal tics.)
- **No one wants to risk throwing the baby out with the bathwater.** Argentinian President Javier Milei talked publicly about his superstitious insistence on staying at home wearing a lucky jacket rather than watching his team playing in the football World Cup live at the stadium.

4. Self-interest: incentives favour more extreme positions

- **AI is not magic, a silver bullet or a charity, but a commercial industry.**
- **Most participants have incentives to amplify the stakes.** Founders need belief, investors need momentum, incumbents need gravitas, consultants need urgency, policymakers need relevance, executives need a narrative and journalists need drama. This is not a criticism of individuals or organisations, or even of the likely revolutionary impact of AI.
- **The challenge is magnified by a social media-driven economy.** Social platforms favour novelty, negativity, moral intensity and shareability. For example, in political tweets, one [study](#) by New York University found each additional moral-emotional word was associated with about a 20 percent higher retweet rate. Another [study](#) by MIT found that false news was 70 percent more likely to be retweeted than true news and reached a threshold of 1,500 people six times faster.
- **That means moderate claims are likely to be under-distributed and extreme claims over-rewarded.** Implausibly precise numbers for the future size of the AI economy, for example, or dramatic doom-laden scenarios heavy with emotive language will continue to attract greater attention than even-handed assessment. (See here for our quantitative analysis of two particularly striking examples from earlier this year: [“Something big is happening in AI... Or is it?”](#).)



What makes AI special – and gives hope for the future

Technology may be liable to erroneous thinking because of the disconnect between visionaries in Silicon Valley and similar communities, and the rest of the world.

Cynics may say that people with extraordinary knowledge and skills in one area may be even more prone than most to wrongly estimate their lack of knowledge or skills in other areas: the so-called Dunning-Kruger effect.

That may include the reality of implementing AI on a trading floor in Wall Street, an office supplies wholesaler on the outskirts of London or an auto maker in Mumbai.

There is also a long tradition of correctly identifying the task that will disappear but getting wrong what humans will do instead.

1982: Computers "... can develop into monsters, damage the physical and mental health of the clerical staff – maybe even eliminate their jobs": the Washington Post [reported](#) from a clerical conference. Computers instead became the core tool of knowledge workers.

1983: The pocket calculator "... has become the target of critics who fear that its use in the classroom will render generations of children dependent on batteries instead of brains", the Washington Post [reported](#). Instead expectations changed about what people calculate for themselves.

2009: Cloud computing: around half of companies surveyed "cited security and privacy concerns as their top reason for not using cloud computing", Forrester Research [found](#). Today companies entrust much of their critical infrastructure to the cloud.

"We overestimate the impact of technology in the short-term and underestimate the effect in the long run." Roy Amara, Stanford computer scientist

But this time may really be different – with AI as a partner, starting with (deterministic) predictive AI. It is certainly helping weather forecasters to sift through unimaginable amounts of data. Four-day forecasts are now said to be about as accurate as one-day forecasts in the 1990s.

There are diminishing returns. The limit of predictability with standard physics models is about eight days, while deep-learning AI models increase that by just one day to nine days. However, the difference is that they [generate](#) state-of-the-art forecasts 100,000 times faster, and 10,000 times more energy-efficiently, meaning they can be done on a laptop rather than a supercomputer.

And for the time being, your predictions about the impact of AI are no less valid than an AI expert based in the very particular world of Silicon Valley. And the future of AI will be made not in Silicon Valley but in Main Street.



Appendix 1

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Global Expansion of Chinese Construction Machinery: Lessons from Brazil

September 14, 2026

Executive Summary

China's exports have been strong this year, surging by 19% YoY in the first eight months. Among these exports, heavy-asset capital goods have played a particularly important role. Using construction machinery as a representative example, we illustrate how stronger product competitiveness, Chinese overseas projects, and Chinese manufacturers' "going-global" strategies have enabled market share gains across a broad range of economies.

Chinese construction machinery has expanded across the globe since 2020. Exports of related products quadrupled between 2020 and 2025 and have remained on a strong upward trend this year. More importantly, Chinese products have achieved significant market-share gains across a broad range of economies and accounted for 26% of global imports in 2025 compared to 10% in 2019. In this report, we aim to explore the factors behind this.

We use Chinese excavator exports to Brazil as a case study. Excavators are one of the most representative categories of construction machinery, while Brazil is South America's largest excavator importer and ranks among the world's top 20. Chinese excavators have also made remarkable market-share gains in Brazil, with their share rising from 20% to 90% in six years. Brazil therefore offers a compelling case study on this topic.

Through the case study and our analysis of aggregate data, we identify three drivers of the global expansion of Chinese construction machinery:

- **Product competitiveness:** Chinese brands have steadily improved both price advantage and product quality.
- **Demand from Chinese overseas projects:** Chinese contractors are undertaking more infrastructure projects and naturally exhibit stronger preferences for Chinese equipment.
- **"Going global" strategy by Chinese companies:** Higher profitability has encouraged them to direct more resources towards overseas expansion.

Authors

Deyun Ou

Economist

Yi Xiong, Ph.D.

Chief Economist



Introduction

China's exports have been strong this year, surging by 19% YoY in the first eight months. In a note published for clients in March, we identified several drivers of this export growth, including AI-related demand, energy-cost advantages and the green transition. In particular, heavy-asset capital goods have been an important contributor, with construction machinery offering a representative example.

Chinese construction machinery has expanded rapidly across the globe since 2020. Using HS8429 products—including excavators, bulldozers, graders and related equipment—as a proxy, export values quadrupled between 2020 and 2025 ([Figure 1](#)). More importantly, the expansion has been broad-based: 128 of the 140 economies reporting HS8429 imports recorded an increase in China's market share over the period ([Figure 2](#)). Chinese products now account for more than half of HS8429 imports in over 30 economies. As a result, China's share of global HS8429 imports has risen from 10% in 2019 to 26% in 2025.

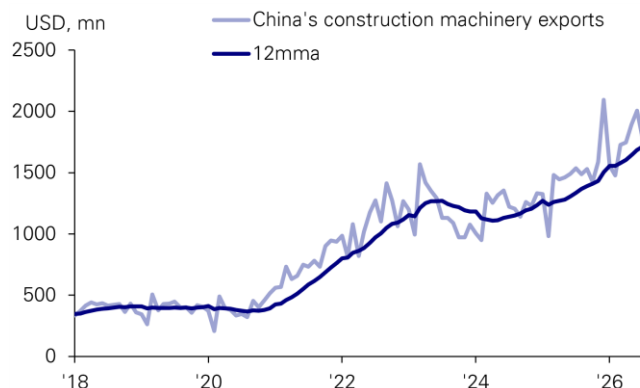
Looking at aggregate trade data alone does not fully reveal the multiple underlying drivers. At the country level, economic structures, investment cycles and infrastructure demand vary substantially, leading to very different import patterns and market-share dynamics. At the product level, HS8429 encompasses a diverse range of machinery categories with different technologies, end-markets and competitive landscapes.

Therefore, we adopt a case-study approach and focus on Chinese excavator exports to Brazil. We believe this is particularly representative since:

- Excavators are highly representative of the construction machinery industry, accounting for half of China's HS8429 exports.
- Brazil is the largest excavator importer in South America and ranks among the top twenty importers globally. The demand is underpinned by three factors. 1) As a major mining economy, Brazil requires substantial earthmoving equipment. 2) Brazil is one of the world's largest agricultural producers. The farming sector relies heavily on [capital investment and machinery](#) to raise productivity. 3) The government has [renewed its focus](#) on infrastructure development in recent years.
- Chinese excavator manufacturers have achieved exceptional market-share gains in Brazil. Their share increased from 20% to 90% over the past six years ([Figure 4](#)), with gains observed across all excavator segments.

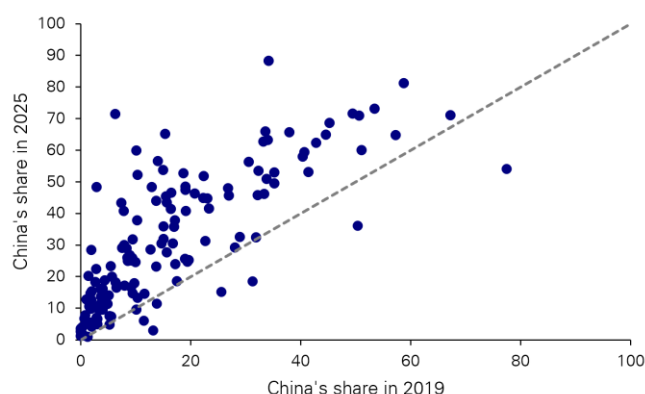


Figure 1: China's construction machinery exports have expanded rapidly since 2020



Source: Deutsche Bank Research, CEIC. We use HS8429 products—including excavators, bulldozers, graders and related equipment—as a proxy.

Figure 2: The expansion of China's construction machinery exports has been remarkably broad-based



Source: Deutsche Bank Research, UN Comtrade. Each dot represents an economy.

Figure 3: Brazil has been a major excavator importer



Source: Deutsche Bank Research, UN Comtrade. Excavators are primarily classified under HS842952.

Figure 4: Chinese excavators have recorded remarkable market-share gains in Brazil



Source: Deutsche Bank Research, Brazil Custom

Expansion of Chinese Excavators in Brazil

Chinese excavators have become increasingly visible across Brazil. Their presence is no longer confined to mines, railways, and other construction sites. They are increasingly involved in landmark projects such as Rio Olympic venues and World Cup stadiums, and can even be found in residential backyards and farms. Chinese brands [reportedly](#) occupied the top three positions in Brazil's market in 2025, while six Chinese brands ranked among the top ten. Customs data paint a stronger picture: Chinese products now account for ~90% of Brazil's excavator imports, both by value and volume.

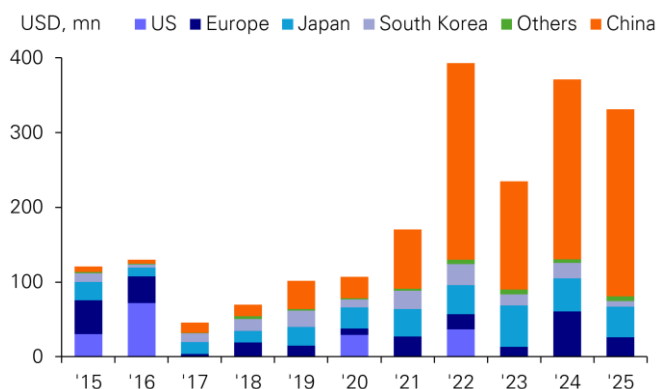
China's expansion appears to have been supported mainly by Brazil's incremental demand rather than the outright displacement of competing suppliers. During the market recovery from 2017 to 2021, excavator imports rose from all major sources. From 2022 to 2025, imports from China grew much faster, driving a rapid increase in China's market share, while annual imports from other suppliers—including Europe, Japan and Korea—remained broadly stable at around USD 100mn in total (Figure 5). This does not mean that other suppliers



were unaffected by Chinese competition: without China's sharp market-share gains, they would likely have captured a larger share of Brazil's surge in excavator demand.

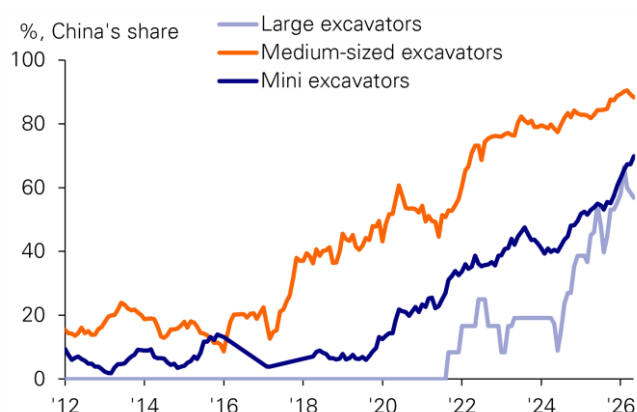
Interestingly, the expansion has been different across segments (Figure 6). Medium-sized excavators began gaining market share steadily after 2017, while mini excavators only entered a similar trajectory after 2019. Large excavators followed a different path. Prior to 2021, Chinese brands were absent, yet they rapidly gained traction after the first entry. The distinct patterns suggest that the rise of Chinese excavators might have been driven by multiple demand- and supply-side forces.

Figure 5: This expansion appears to have come largely from incremental demand rather than outright displacement



Source: Deutsche Bank Research, Brazil Custom

Figure 6: Expansion has been different across product segments



Source: Deutsche Bank Research, Brazil Custom. Brazil's 8-digit NCM customs classification distinguishes excavators by power categories, allowing us to group them into three segments.

Factor 1: Price Advantage Meets Quality Upgrading

A widening price advantage has been a major driver behind the expansion of Chinese products in the medium-sized and mini excavator segments. Since these products are relatively less technologically demanding, pricing remains a major consideration for buyers. We divide the sample into four periods and compare China's price advantage against market share gains (Figure 7). China's advantage has steadily expanded, with products now more than 50% cheaper than peers. Meanwhile, the relationship between price advantage and market share is highly significant. For medium-sized excavators, a one percentage point increase in China's price advantage is associated with a 0.4pt gain in market share. For mini excavators, the corresponding relationship is even higher at 0.9pt.

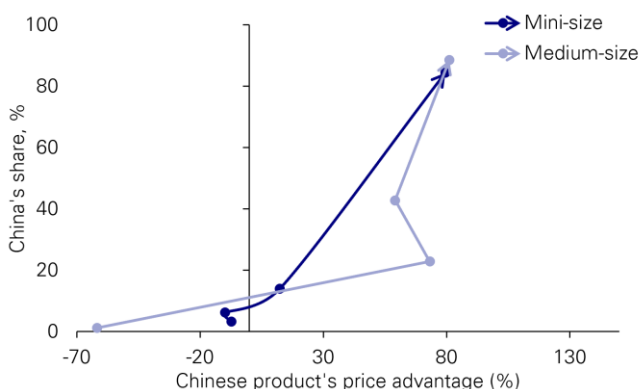
Indeed, mini excavators provide the clearest example of price-driven expansion. These machines are used for small-scale construction projects or purchased by individual consumers, making demand particularly price-sensitive. Chinese products began to undercut competitors after 2017, and their price advantage widened significantly after 2020 (Figure 8). The gap has now reached USD 17,000 per unit. We further decompose this price advantage (Figure 9). 60% of the widening gap reflects price increases by overseas competitors amid higher inflation, while 40% reflects price cuts by Chinese manufacturers. RMB



depreciation accounted for 10% of the gap during 2023-25. However, following its appreciation in 2026, the cumulative contribution has turned negative.

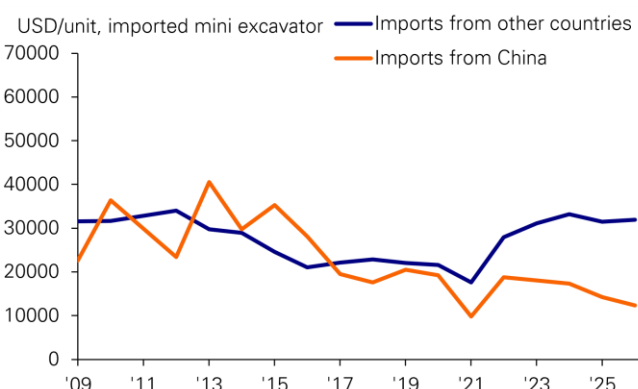
More noteworthy is the advance of Chinese brands into the high-end segment. Large excavators are far more technologically sophisticated. Before 2021, Chinese brands were virtually absent from Brazil's large excavators market. They entered the market in 2021 and expanded rapidly thereafter (Figure 10). So far in 2026, 43 of Brazil's 47 imported large excavators have come from China. Chinese manufacturers are reportedly delivering [100-tonne-class excavators](#) to Brazilian mining customers and are actively promoting equipment in the [500-tonne category](#). Chinese companies' competitiveness is increasingly extending from price advantage toward technology intensity.

Figure 7: A widening price advantage has driven expansion in the medium-sized and mini excavator



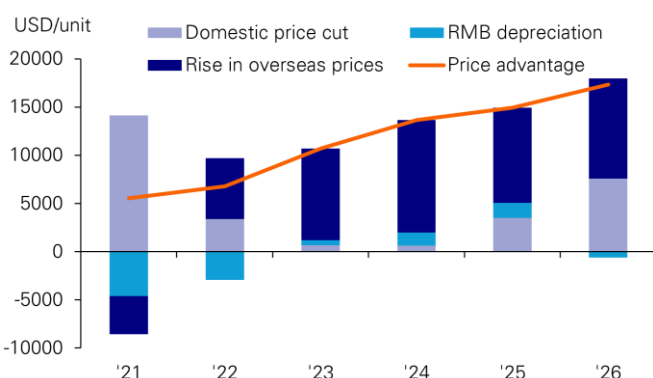
Source: Deutsche Bank Research, Brazil Custom. The four observations correspond to the periods 2006-2010, 2011-2015, 2016-2020, and 2021-2025, respectively.

Figure 8: Mini excavators provide the clearest example of price-driven expansion



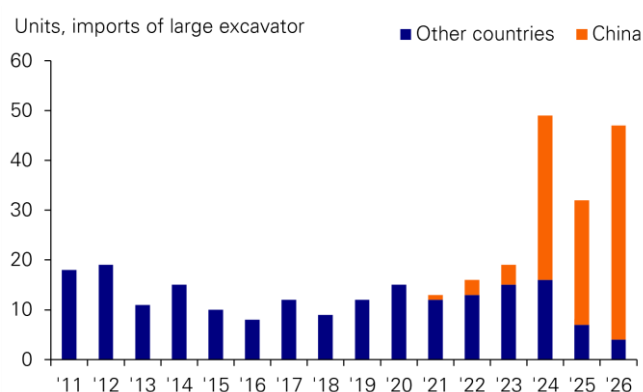
Source: Deutsche Bank Research, Brazil Custom

Figure 9: China's price advantage has widened owing to overseas inflation, domestic price cuts and RMB depreciation



Source: Deutsche Bank Research, Brazil Custom

Figure 10: Chinese brands entered the large excavator market in 2021 and rapidly expanded thereafter



Source: Deutsche Bank Research, Brazil Custom

Factor 2: Demand from Chinese Overseas Projects

Chinese firms are undertaking more construction projects in Brazil, generating additional demand. They naturally have a stronger preference for Chinese-made machinery. The value of construction projects completed by Chinese companies has risen consistently since 2021. Growth in newly signed contracts has been



even stronger, with their value reaching a record USD 4.6bn in 2024 (Figure 11). This trend may have contributed to the growth in excavator imports since 2024, especially for large excavators.

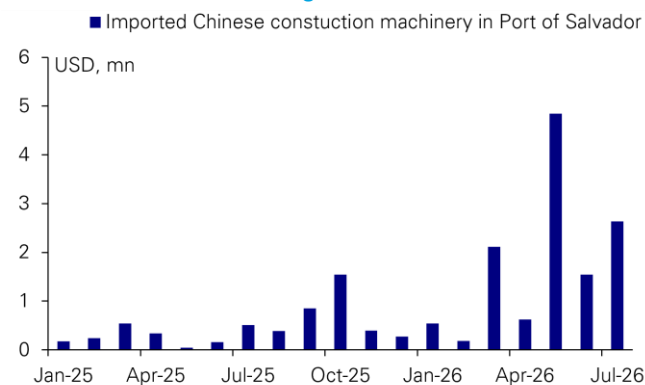
The Salvador-Itaparica Bridge project provides a compelling example. Once completed, it will be the longest sea-crossing bridge in South America. The project was awarded to two Chinese SOEs in 2019, with an initial investment of USD 2.2bn, which was raised to USD 2.7bn in 2025. It is the largest infrastructure investment undertaken by Chinese companies in Brazil. Since construction began this year, [~800 tons of equipment](#) from China have arrived at the port of Salvador since April, consistent with the surge in custom data (Figure 12). Most recently on July 1, the project company showcased the first batch of equipment deployed on site, [all sourced from China](#).

Figure 11: Chinese firms are undertaking more construction projects in Brazil



Source: Deutsche Bank Research, China Mofcom

Figure 12: Imports of Chinese machinery through the Port of Salvador have surged in recent months



Source: Deutsche Bank Research, Brazil Custom

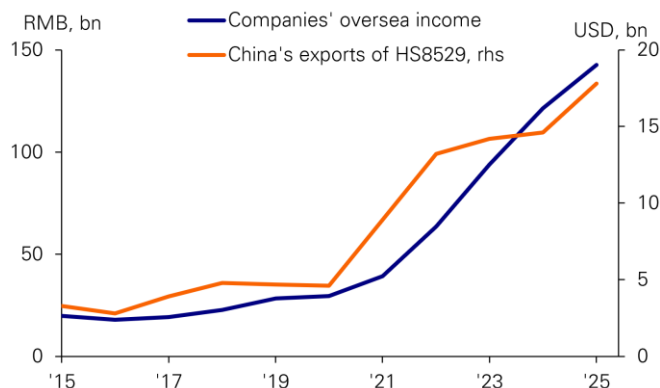
Factor 3: Supply-Side Expansion Driven by Higher Profitability

Chinese firms have strong incentives to expand overseas because foreign markets offer higher profitability. Some views have argued that low export prices primarily reflect dumping driven by weak domestic demand. If that were the case, corporate profits would be expected to remain flat or deteriorate as firms sacrificed margins to gain market share. Company-level evidence suggests the opposite.

We examine five listed Chinese construction machinery manufacturers. Combined overseas revenue has risen consistently since 2020, doubling in 2022 and doubling again by 2025, broadly mirroring the trajectory of China's total construction machinery exports (Figure 13). More importantly, overseas profitability has improved substantially. Across the five companies, operating margins in overseas markets have increased by 5-15 percentage points, with the average rising from 20% during the previous decade to 30% today (Figure 14). In contrast, domestic margins have fallen by 5 percentage points.

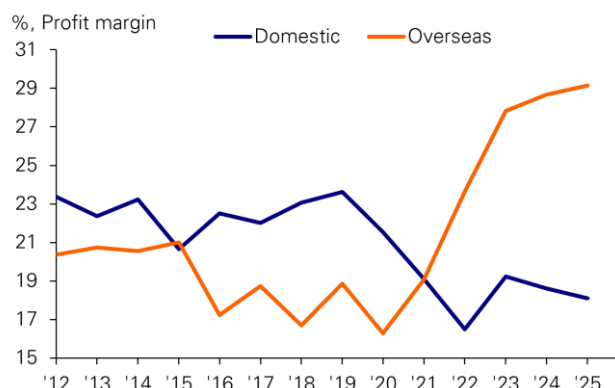


Figure 13: Overseas revenue has risen consistently since 2021



Source: Deutsche Bank Research, Wind

Figure 14: Overseas profit margins have exceeded domestic margins



Source: Deutsche Bank Research, Wind

Beyond Excavators: Chinese Construction Machinery Expansion

Through this case study, we identify three forces underpinning the expansion of Chinese excavators in Brazil: improving price and quality competitiveness, growing demand from Chinese overseas projects, and proactive going-global strategies by Chinese companies.

We believe the forces extend to the broader construction machinery sector.

- **Price and Quality Competitiveness.** The price advantage over HS8429 has become more pronounced since 2022 (Figure 15). The analysis of eight-digit categories revealed a more comprehensive picture (Figure 16). First, exports increased across all categories, including those with rising prices. Second, the products recording larger price increases tended to have higher unit values. That said, the expansion is driven by price cuts as well as increasing demand for higher-value products.
- **Overseas Expansion of Chinese Enterprises.** Chinese overseas construction projects have reaccelerated since 2024, with completed value and newly signed contracts rising by 20% and 15%, respectively (Figure 17). Belt and Road economies are key destinations for Chinese overseas construction projects, and Chinese construction machinery has gained market share much more rapidly in these markets.
- **Proactive “Going global” strategy.** Since the 2010s, construction machinery manufacturers have viewed overseas expansion as a key pillar of future growth and have invested heavily in international distribution channels, service networks and localized production facilities.

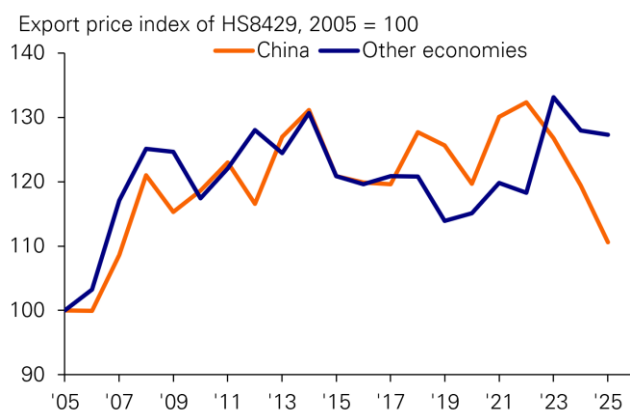
Looking ahead, we believe all three drivers are likely to remain in place for an extended period. China’s manufacturing capability and continued technological progress should support further gains in product competitiveness. Accelerating component localization could further strengthen the price advantage. Meanwhile, the overseas expansion of Chinese companies is expected to remain a long-term trend. Belt and Road economies warrant particular attention, as Chinese



investment in these markets has surged in recent years, laying a solid foundation for further overseas expansion (Figure 18). As a result, we remain constructive on the outlook for construction machinery exports.

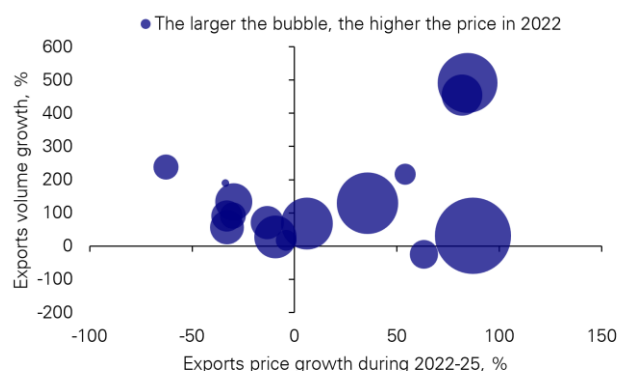
That said, several risks warrant monitoring. First, further RMB appreciation could erode the price competitiveness of Chinese products. Second, the sharp increase in market share achieved in Brazil—from 20% to 90%—may not be easily replicated elsewhere. In some markets, Chinese firms could face stronger competition from local manufacturers, limiting the pace of further market-share gains.

Figure 15: China's price advantage on HS8429 has become more pronounced since 2022



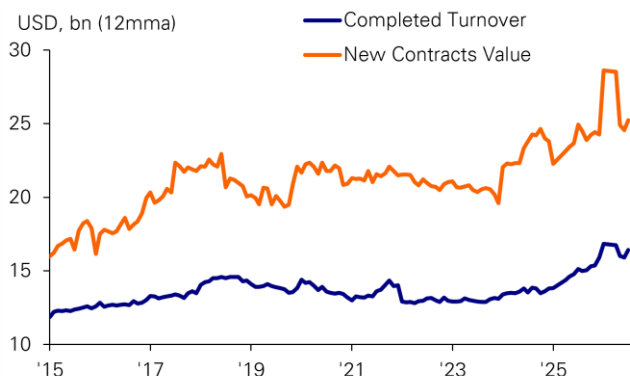
Source: Deutsche Bank Research, UN Comtrade

Figure 16: The expansion is driven by price cuts and increasing demand for higher-value products



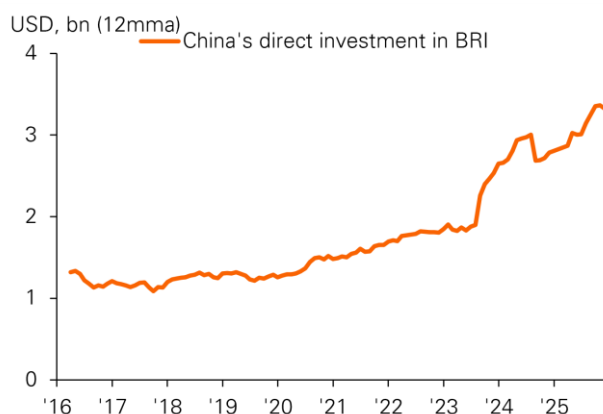
Source: Deutsche Bank Research, CEIC. Each bubble represents an eight-digit category under HS8429.

Figure 17: Chinese contractors have been undertaking more projects overseas since 2024



Source: Deutsche Bank Research, CEIC

Figure 18: China's direct investment in Belt-and-Road economies has also surged since 2024



Source: Deutsche Bank Research, CEIC



Appendix 1

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The revenge of the supply-side: A new economic regime

September 14, 2026

Back in the 2010s, macroeconomic policy was preoccupied by insufficient demand. In historical terms, it was an unusual period of low inflation and low interest rates, where supply-side issues were considered peripheral. If anything, it went almost unquestioned that the supply side of the economy would scale up to meet higher demand.

But in the 2020s so far, we're back in a supply-driven world. Growth has been constrained as much by supply shortages as deficient demand. This year alone, we've seen major consequences from the closure of the Strait of Hormuz, with ramifications for oil prices, inflation and economic growth.

This macro regime shift, where supply rather than demand is the bigger issue, is a tricky environment to navigate, and renders much of the 2010s playbook redundant:

- First, inflation spikes will occur more frequently in a supply-constrained world, as there isn't the spare economic capacity to meet higher demand.
- Second, monetary stimulus also becomes less effective in this environment, as traditional demand-side monetary tools can't deal with the direct cause of supply shocks.
- Third, fiscal stimulus in a supply-constrained environment becomes much more inflationary and is more likely to crowd out private investment.

Looking forward, the return to a supply-driven world means that the primary issue for policymakers will be expanding the productive frontier of the economy. After years where insufficient demand was the main preoccupation, the future of growth now rests on a supply-side expansion that can raise living standards across the board.

Authors

Henry Allen
Strategist



The paradigm shift in thinking: How the biggest economic problem moved from demand to supply

The legacy of the Global Financial Crisis led to a world of demand deficiency

In 2008, the Global Financial Crisis (GFC) marked the worst economic contraction since the Great Depression of the 1930s. Unemployment rose to multi-decade highs and the subsequent recovery was very sluggish, despite huge economic stimulus on a scale previously unheard of in peacetime. This slow recovery was clear in numerous ways, but an obvious one was that the US unemployment rate didn't return to its pre-GFC low until 2017, a full decade later. Meanwhile in the Eurozone, it didn't happen until February 2020, just as the pandemic then sent it higher once again.

This was a world where deficient demand was the main preoccupation of macroeconomic policymaking. Indeed, the Depression-era concept of "secular stagnation" made a comeback, in which the equilibrium interest rate consistent with full employment was negative. In this world, the constraints of the zero lower bound on monetary policy would make it very difficult to get to full employment.

This demand deficiency manifested itself in persistent inflation undershoots. Throughout the 2010s, both the Federal Reserve and the ECB missed their 2% inflation targets on the downside. Policy rates remained close to the zero lower bound, and long-term market interest rates also moved continuously lower. Even before the pandemic, Germany's 10-year bund yield fell to -0.7% in summer 2019. In this environment of negative yields, borrowing was basically a "free lunch".

At the time, there were genuine fears about a permanent low-growth trap. Indeed, when the Covid-19 pandemic began and another economic contraction was underway, there were initially concerns that deflationary pressures would intensify further. For instance, the 10-year US inflation swap (a market-based measure of expectations for US CPI over the next decade) fell to just 0.81% in March 2020. Investors genuinely thought that low inflation and low interest rates were likely to persist for the decade ahead as well, and the 10-year Treasury yield closed at a record low of 0.51% in August 2020.

This fear of deficient demand informed the policy response to Covid-19. The stimulus packages went well beyond those seen in the GFC. For instance, no time was lost in slashing rates back to the zero lower bound, and the fiscal deficits went far above anything previously seen in peacetime. In fact, the US federal deficit stood at -14.5% of GDP in 2020, the biggest since 1945. And with much of the global economy locked down for months, consumers accumulated huge "excess savings" as they couldn't spend on their normal activities.

The pandemic was the turning point towards a world of supply constraints

With hindsight, we can see that all this stimulus and the accumulation of excess savings created a huge demand surge, just as the global economy was experiencing major supply-chain issues as it reopened:

- The lockdown periods saw a big rotation in spending away from services (like dining out, travel, gym memberships) towards durable goods (like home office equipment, electronics, exercise equipment). This overwhelmed existing supply chains.
- Pandemic-related disruption led to major issues with shipping. A severe



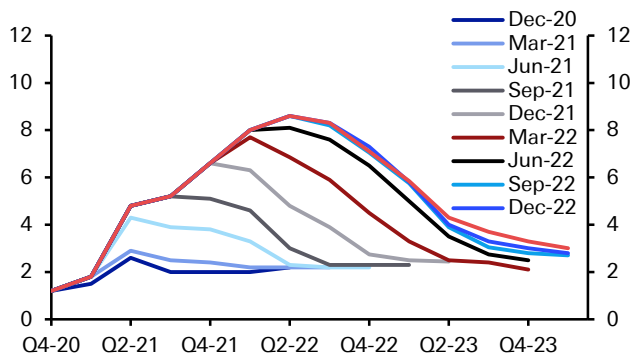
shortage of containers led freight rates to rise more than 5-fold at their peak.

- Labour markets were disrupted, as many took early retirement during the pandemic or changed occupations. Companies therefore had to raise wages to compete for a smaller pool of workers.
- Commodity prices also surged in this period, particularly energy. The pandemic period saw producers cut back on drilling and exploration. But this meant it was more difficult to dial up supply when demand started to rise again, and energy prices surged back, with Brent crude oil moving above its pre-pandemic levels in mid-to-late 2021.

These supply constraints marked a major shift from the environment of the 2010s. Back then, it was almost an unquestioned assumption that the supply-side of the economy would scale up to meet demand. Inflation had been low and the most obvious signs of supply-side stress were not apparent. But now the reverse was true, and inflation was gathering pace again.

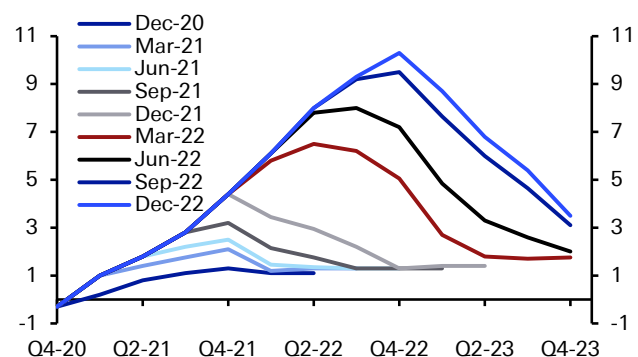
Initially, many argued that this burst of inflation would prove temporary. They argued it was down to one-off factors like the pandemic and the subsequent reopening. Moreover, there was a view that the longer-term disinflationary forces of the previous 30 years would still reassert themselves. But by late-2021, it was clear that the combination of huge stimulus, the drawdown of excess savings, and serious supply-side disruption had created more inflation than anticipated. By December 2021, US CPI had risen to its highest level since 1982, at 7.0%, whilst Eurozone CPI was at its highest since 1991, at 5.0%.

Figure 1: Bloomberg consensus forecasts for US CPI



Source : Bloomberg Finance LP, Deutsche Bank

Figure 2: Bloomberg consensus forecasts for Eurozone CPI



Source : Bloomberg Finance LP, Deutsche Bank

The transition to a supply-constrained world accelerated dramatically in early 2022 with Russia's invasion of Ukraine. Energy prices rose significantly, and European industry was forced to decouple from Russian gas. Meanwhile, the blockade of Ukraine's Black Sea ports restricted the supply of agricultural commodities like wheat and corn, driving food prices higher still.

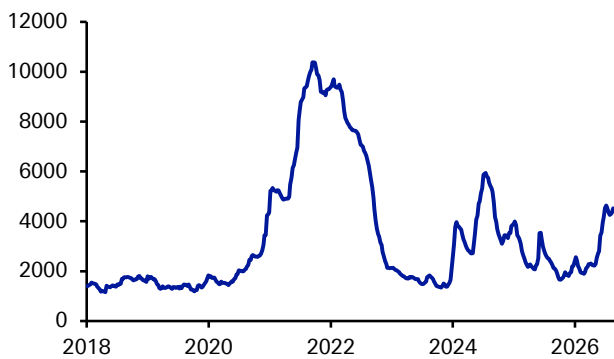
This sent inflation even higher, up to multi-decade highs around the world. US CPI inflation peaked at 9.1% in June 2022, and Eurozone CPI peaked at 10.6% in October 2022. This was the context in which central banks launched into the most aggressive rate hikes in a generation. Over 2022-23, the Federal Reserve hiked rates by 525bps, and the ECB delivered 450bps.



These supply issues persisted long after the pandemic-related issues were over

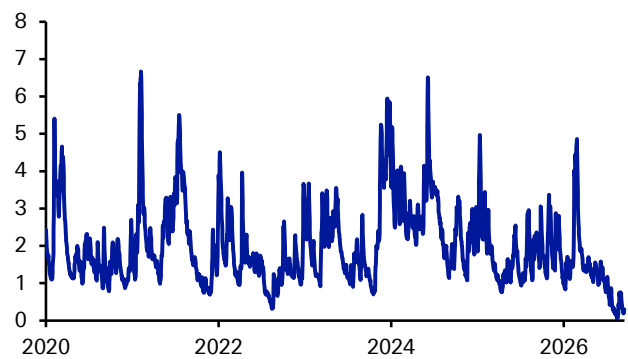
Yet even as the worst of that inflation wave passed, the 2023-24 period kept demonstrating how supply-side disruptions could lead to wider economic problems. For instance, attacks from the Houthi rebels in the Red Sea forced major shipping lines to reroute away from the Suez Canal around the Cape of Good Hope, adding over a week to the length of trips. Meanwhile, an El Nino-induced drought meant transit capacity through the Panama Canal fell 30%, forcing ships to wait in long queues or reroute instead. And in Europe, repeated summer heatwaves have lowered the Rhine's water levels, disrupting industrial barges and affecting industries like chemicals and steel.

Figure 3: Drewry World Container Index Composite Index (USD per 40 Foot Box)



Source : Bloomberg Finance LP, Deutsche Bank

Figure 4: Rhine Water levels at Kaub (metres)



Source : Bloomberg Finance LP, Deutsche Bank

More broadly, the other shift in the post-pandemic period was the growing use of strategic industrial policy, particularly as governments sought to boost domestic manufacturing capacity. For instance, the US passed the Inflation Reduction Act under the Biden administration, which encouraged clean energy investments, as well as the CHIPS Act, which sought to boost semiconductor manufacturing. Tariffs also became a more enduring feature of the landscape, and the Biden administration hiked tariffs on Chinese imports, with those on EVs up from 25% to 100%, whilst semiconductors rose from 25% to 50%.

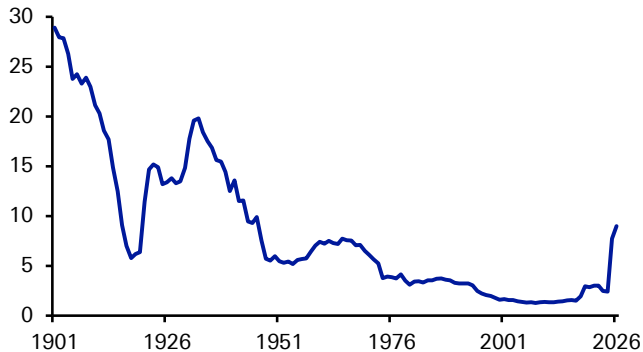
Then under the second Trump administration, tariffs became an even bigger part of the policy toolkit, with the US effective tariff rate surging from 2.6% beforehand to a peak of 23% in April 2025, before settling around 11% currently. Several different measures were introduced, but the macro impacts were clear. In particular, there was a major wave of front-loading of imports in Q1 2025, as those trading internationally sought to get ahead of the announcements. Indeed, the surge in imports meant US GDP contracted slightly in Q1 2025. Moreover, other countries moved to respond, and not just with tariffs of their own. For instance, China introduced export controls on rare earths.

Whilst 2025 was dominated by tariffs, 2026 then brought a fresh supply shock for the global economy with the closure of the Strait of Hormuz. Previously, around 25% of the world's seaborne oil trade had gone through the Strait, including nearly 15 million barrels per day of crude oil. But the start of the US-Iran conflict led the Strait of Hormuz to close, and six months later only limited traffic is still able to pass through. This led to another supply shock, with Brent crude oil prices moving back



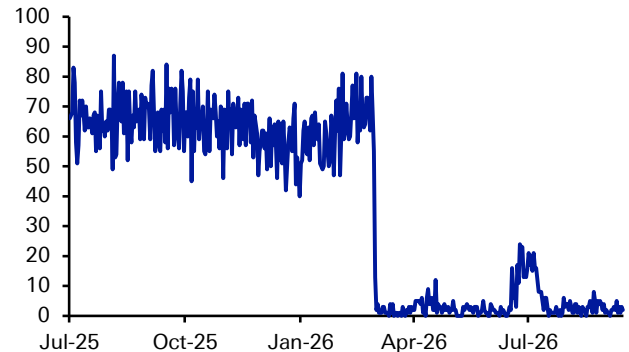
above \$100/bbl, whilst European natural gas futures also reached their highest levels since late-2022. Yet again, the biggest global economic disruption of the year to date has been a supply shock, rather than a demand one.

Figure 5: US Effective Tariff Rate (%) with projected postsubstitution rate from the Yale Budget Lab



Source : Haver Analytics, The Budget Lab at Yale, Deutsche Bank

Figure 6: Strait of Hormuz Tanker Vessel Crossings



Source : Bloomberg Finance LP, Deutsche Bank

Types of supply shocks

Broadly speaking, we can distinguish between three sorts of supply shocks, although there can be considerable overlap between these categories.

1. Temporary supply shocks that have a short-term economic impact.

Example: low river levels blocking a key trade route.

The first category of supply shocks are those that have a temporary impact on the economy. Examples include the blockage of the Suez Canal in 2021 when the Ever Given container ship ran aground, or a drought that causes river levels to fall and means fewer ships can get through key waterways.

These are supply shocks that temporarily remove capacity, but then see a bounce back to “normal”, broadly speaking. For instance, a drought is hopefully not repeated, and when river levels return to normal, supply can resume again.

For the economy, this means the longer-term consequences are also limited. The inflation impact is mean-reverting, and output recovers, often in a V-shape. These are the sort of shocks that central banks are usually able to “look through”, because they don’t cause permanent structural damage to the economy.

2. Medium-term supply shocks that can last several years, with a clear economic impact.

Example: the closure of the Strait of Hormuz.

These kinds of shocks are often driven by geopolitical events or policy decisions, and are longer-lasting shocks which do not have a quick solution. A good example is the current closure of the Strait of Hormuz, but we have seen many in recent years. In theory, these shocks can end, but their length is uncertain and they have material economic impacts. An increase in tariffs also fits into this category, as it forces an adjustment in trade flows, even if they can be reversed in theory.



A good example of this kind of supply shock is the 1973 oil shock. Back in 1973, the oil embargo by OPEC led oil prices to nearly quadruple, and they remained at that higher level for years after. It wasn't simply a temporary adjustment that normalised again.

Unlike the short-term shocks, these medium-term shocks can force a permanent rerouting of trade flows and supply chains. Moreover, they can push inflation higher for an extended period and are more likely to filter through into core inflation categories. And because they are not temporary shocks that quickly adjust, policymakers can't "look through" them as easily. As such, central banks aren't able to accommodate the shock, and often have to hike rates to clamp down on the resulting inflation, as we saw in 2022.

3. Structural supply trends that affect the longer-term growth rate.

Example: Demographic trends.

This category includes the long-term forces on the supply side that affect the economy's growth rate. A good example are demographics and the current situation of ageing populations. Another example are the multi-decade processes of globalization and deglobalization.

These are factors that can permanently shift the productive frontier of the economy. For example, the current situation of ageing populations means that a smaller proportion of the population are of working-age, meaning that labour forces are shrinking. As a result, that constrains the growth rate of potential GDP, meaning that more growth must come from higher productivity rather than a larger number of workers.

These shifting trends can lead to permanent changes in so-called r^* , which is the equilibrium interest rate where full employment is achieved – a level that neither stimulates nor restricts economic activity. In other words, these factors are capable of shifting the economy's potential growth rate, at which it can grow sustainably without generating inflationary pressures.

What happened the last time the supply-side dominated: Lessons from the 1970s

As we consider the return of the supply side, it's worth bearing in mind this is far from the first time that supply shocks have dominated the landscape. Back in the 1970s, the biggest economic shocks of that decade were also supply driven.

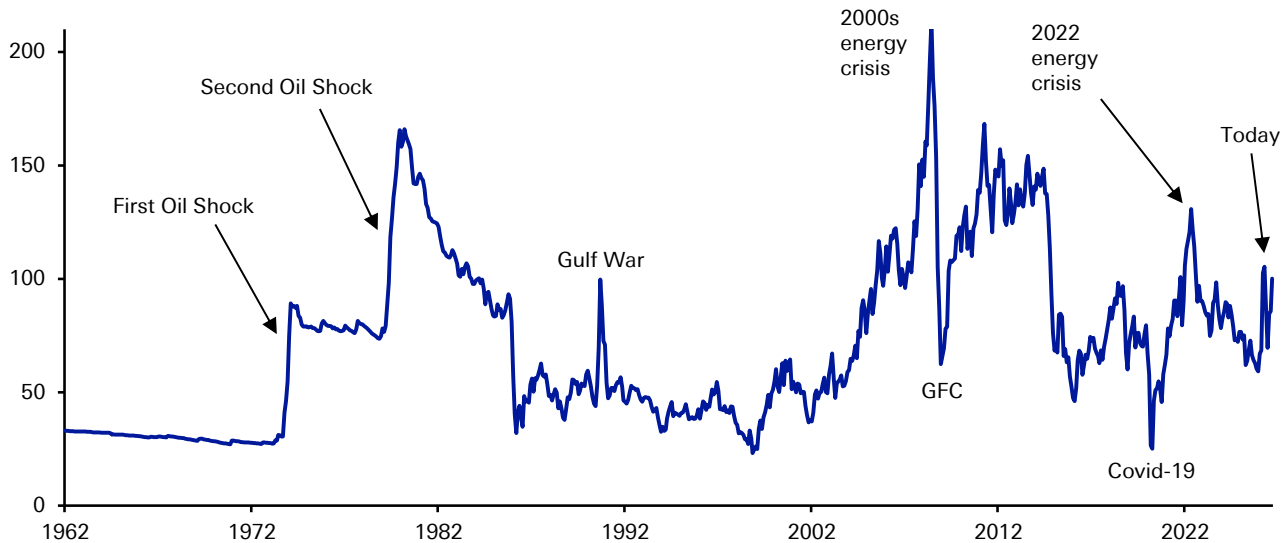
That decade saw two huge oil shocks in 1973 and 1979. In the first, the Organization of Arab Petroleum Exporting Countries (OPEC) placed an embargo on several countries, including the United States, in retaliation for their support for Israel in the Yom Kippur War. This caused oil prices to nearly quadruple, triggering a recession across multiple countries that began in late 1973. Then in 1979, a second oil shock occurred around the time of the Iranian Revolution. That coincided with a big decline in Iran's oil output, before the Iran-Iraq War in 1980 led to further production declines for both countries. As a result, oil prices more than doubled again.

To be fair, there were other demand-side and supply-side shocks that meant inflation got out of hand in the 1970s. But the two oil shocks were the single biggest factors, and created a stagflationary environment in which inflation accelerated



and growth slowed.

Figure 7: WTI Oil Prices (\$/bbl) in real terms



Source : Finaeon, Deutsche Bank

This experience is instructive for today, as the 1970s demonstrated how a repeated set of supply shocks could act in a cumulative way. After all, even if the textbook response is to “look through” a temporary supply shock, a series of consecutive shocks can suddenly feel very permanent when they happen one-after-another. That continuous disruption can then cause inflation expectations to become unanchored, risking the kind of dynamics that trigger wage-price spirals. Indeed, much as economists can focus on “core inflation” that excludes food and energy prices, these are essential goods that are difficult to substitute away from, and play a key role in inflation expectations.

On the bright side, the more optimistic lesson is that supply shocks often force adjustments that prevent a repeat. For instance, back in the 1970s, there was a recognition that Western economies had become overly dependent on Middle Eastern oil, and major efforts were made to reduce that dependence. For instance, the US established the Strategic Petroleum Reserve (SPR) in 1975, and the Department of Energy was established in 1977. Moreover, investments were directed to US energy production, and since 2019, the US has been a net exporter of energy, rather than a net importer.

Clearly the 1970s are far from an exact parallel. Most obviously, economies are far less energy intensive today, so oil shocks don’t have as big an impact today. Nevertheless, the dynamic of repeated supply shocks is one that bears relevance given how that dynamic is re-echoing today, and with inflation still lingering above target in many of the world’s biggest economies.

What are the implications for today?

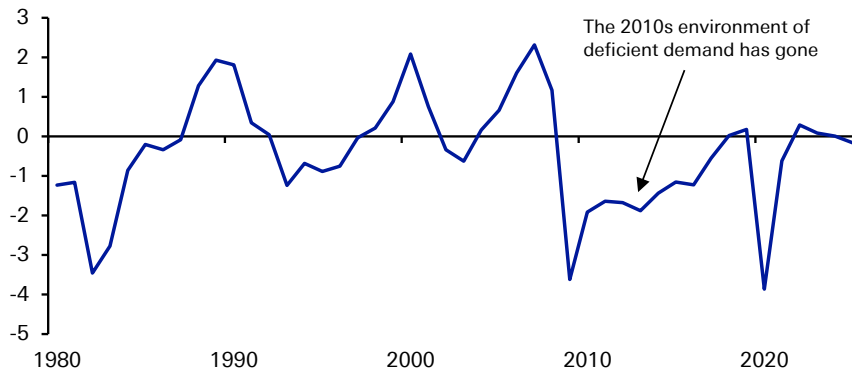
1. Inflation spikes will occur more frequently, as there isn’t the spare economic



capacity to meet higher demand.

The first consequence of a supply-constrained world is that unlike much of the 2010s, there isn't the spare economic capacity (or a negative output gap) that can easily absorb higher demand. So if there is an unexpected demand shock, for whatever reason, the consequence will likely be higher inflation rather than higher output.

Figure 8: IMF: Output Gap in the G7 economies (% of potential GDP)



Source : Haver Analytics, Deutsche Bank

This is a particular risk right now because of those longer-term structural shifts on the supply side. First, global demographics and the retirement of the baby boomers mean that populations are ageing, and many countries now even have shrinking working-age populations. This is a big difference from the world of decades past, when the labour force was generally expanding. Second, the push to reshore production of key industries has also raised costs, as it requires the duplication of supply chains. And third, other structural shifts like the push for higher defence spending have also created an increase in the demand for capital. So there isn't the spare capacity in the economy to absorb higher demand that there was in the 2010s.

This world of more frequent supply shocks means that central banks must also trade off how quickly they want to bring inflation back to target against the potential output volatility that might result. Indeed, we saw this year when the Strait of Hormuz was blocked, central banks didn't immediately react with rate hikes, but they waited a few months to see how the consequences developed. But of course, that has also meant inflation has stayed above target in many countries, often for several years. In fact, US PCE inflation has been above 2% since March 2021.

2. Monetary stimulus becomes less effective, as it can't deal with the direct cause of supply shocks.

Monetary policy has a complex transmission channel, but when central banks cut rates, the aim is usually to stimulate demand. This is a very useful tool in demand-driven recessions like 2008, but it becomes more problematic in the face of supply shocks.

The problem is that when a supply shock hits, whether that's a natural disaster, or the blockage of a key trade route, the economy's productive capacity drops.



Monetary policy cannot affect that directly.

This matters because negative supply shocks have the unfortunate effect of raising inflation and lowering growth simultaneously. So it puts central banks in a very tricky position, as they found in the inflationary environment of the 1970s after the oil shocks. With hindsight, many central banks were too dovish after the first oil shock and inflation got out of control. In the case of the Federal Reserve, that then led to a far more hawkish response to the second oil shock, when Paul Volcker tightened monetary policy significantly to get inflation under control again.

These problems are further exacerbated because central banks operate with uncertainty about the future. Just as we don't know today when the Strait of Hormuz might return to normal, central bankers in 1973 were unsure how long the oil embargo would last. But because central banks make policy that responds with a lag, there is inherently the risk of overshooting. If they expect a supply shock to be permanent, yet it proves temporary, then they risk overtightening. And similarly in reverse, if disruption is expected to be temporary, but proves permanent, they risk not doing enough and letting inflation and inflation expectations rise too much.

3. Fiscal stimulus becomes much more likely to cause overheating and crowding out.

In a supply-constrained world where the economy lacks spare capacity, fiscal stimulus is more likely to overheat the economy and generate inflation.

This is a big shift from the post-GFC world. Back then, unemployment was high and deficient demand meant additional government spending didn't face the same competition from private investment, so the multipliers from that stimulus were much higher. Moreover, several countries were able to borrow at incredibly low interest rates, even negative in some cases.

But this shifted with the pandemic. Not only was the stimulus in 2020 far bigger, but unlike 2008, the economy then saw a huge wave of pent-up demand once the pandemic restrictions were lifted, supported by the excess savings accumulated during the lockdowns. In turn, that contributed to a huge wave of inflation, even before we then had the additional energy and food price shocks in 2022.

Meanwhile, government bond yields have also returned to their pre-GFC levels, making the cost of any stimulus far more expensive. And that comes as we are also seeing a kind of "reverse crowding out", thanks to borrowing from the AI hyperscalers that is adding to pressure on rates. So in this environment, any fiscal stimulus will come at a higher cost.

Conclusion: Looking forward

The transition to a supply-driven world means that many of the old economic assumptions from the 2010s are no longer valid. There's been a structural break that means classic demand-side stimulus is not as effective. After years when it was almost unquestioned that the supply side of the economy would scale up to meet higher demand, we are now back in a world where supply is a meaningful constraint again.

As such, the emphasis will naturally shift towards expanding the supply-side of the

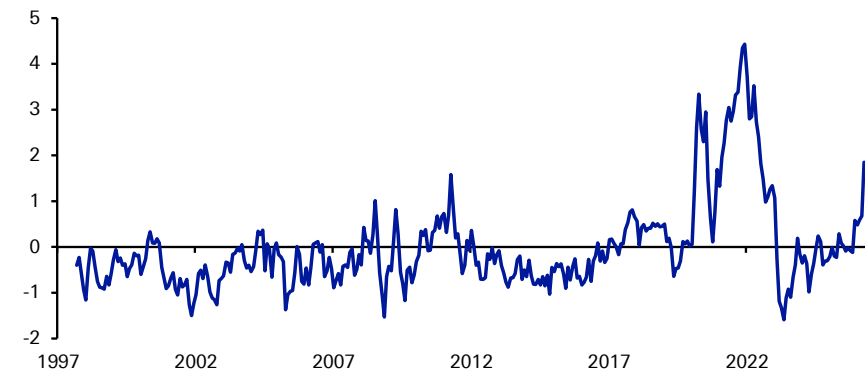


economy, something we're already beginning to see. This could be encouraging more energy production, more housing, or bringing new participants into the labour force. In essence, what's needed are policies that raise the productive potential of the economy, and boost the rate at which the economy can grow without generating inflationary pressures.

These supply shocks in recent years have been a challenge. Even if they might be temporary in isolation, the cumulative effect has been to keep inflation elevated, and we saw that happen back in the 1970s too. This is particularly problematic if the starting point is above-target inflation, as further supply shocks risk unanchoring inflation expectations.

Nevertheless, one consequence of a world of more frequent supply shocks is that actions are then taken to reduce vulnerabilities over the years ahead. For instance, the 1970s oil shocks led to the widespread realisation that the US needed to become less energy intensive and produce more of its own energy. As a result, the Strategic Petroleum Reserve was set up, and the subsequent years saw the US eventually become a net energy exporter. Similarly today, the closure of the Strait of Hormuz has forced other countries to secure alternative supply routes and use strategic reserves. So even before the longer-term geopolitical outcome is clear, we have seen diversification away from one particular chokepoint. In a supply-driven world, the incentives will shift towards making those supply chains more resilient, and better able to withstand future shocks.

Figure 9: New York Fed's Global Supply Chain Pressure Index - the extent of supply-chain pressure in recent years has been very high relative to the previous couple of decades



Source : Bloomberg Finance LP, Deutsche Bank



Appendix 1

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Tokenised money and the future of FX

10 September 2026

Tokenisation has the potential to transform the way financial markets operate, from the movement of money and assets to the management of liquidity and risk. In this paper, we focus on tokenised forms of money and their impact on FX markets.

Many emerging use cases for tokenised money are driven by a desire to reduce frictions associated with cross-border payments and FX settlement. Stablecoins enable 24/7 cross-border transfers but remain overwhelmingly US dollar-denominated. As a result, they are currently more relevant as digital dollar infrastructure and for bridge-currency arrangements, often creating a "stablecoin sandwich" in which the US dollar serves as an intermediary between two currency legs, rather than providing a complete solution to the two-currency FX settlement problem

Tokenisation may play a bigger role in transforming the combined securities and FX settlement cycles involved in cross-border capital flows. This is an area of increasing complexity, as delays or failures in one part of the transaction chain can affect the timely completion of the wider transaction, increasing operational, liquidity, and settlement risks. For example, following the transition to T+1 securities settlement in North America and with Europe preparing to move to T+1, the need to coordinate FX, cash, and securities settlement across institutions, jurisdictions, and infrastructures is already increasing.

Tokenisation enables programmability, which allows atomic settlement and a range of operational and control functions across financial market infrastructures, including real-time liquidity management, margin management, collateral mobilisation and substitution, and automated compliance checks. Current implementations have focused primarily on supporting corporate treasury and liquidity management activities involving FX. However, the settlement of tokenised securities and the associated FX transactions may offer even greater efficiencies by enabling synchronised settlement and reducing some of the operational, funding, and coordination frictions associated with shorter settlement cycles.

Interoperability between different currencies, tokenised money arrangements, and payment systems remains one of the key challenges facing the development of tokenised cross-border payments and FX settlement. Several initiatives are exploring solutions involving tokenised central bank reserves or hybrid models that combine tokenised commercial bank money with central bank money. These may be particularly relevant for multi-currency settlement frameworks, preserving the two-tier monetary system while supporting settlement finality and more integrated cross-border processes. In parallel, other initiatives are building infrastructure that enables different forms of money, payment systems, and settlement architectures to operate together.

Tokenisation promises substantial benefits but unintended consequences are worth considering. Beyond faster settlement, wholesale FX markets also depend on liquidity efficiency. This means banks and other financial institutions can settle large transaction volumes safely while using less liquidity. Existing infrastructures such as Continuous Linked Settlement (CLS) combine settlement risk mitigation with significant liquidity savings through multilateral netting and liquidity optimisation mechanisms. Unlike CLS, most current tokenised settlement models do not yet provide multilateral netting at

Authors

Rohini Grover, Ph.D.
Strategist



comparable scale. A number of initiatives are exploring liquidity-saving mechanisms and new approaches to liquidity management under tokenisation.

We expect a period of accelerating change in years ahead. Securities and currencies settlement involves a complex nexus of technology, market infrastructure, regulation, and legal frameworks, necessitating cooperation between both public and private sector participants across jurisdictions.



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Introduction

Tokenised money refers to **digital representations of money** that are recorded and transferable on shared or distributed ledger infrastructure. This includes both state and privately issued fiat-denominated forms of tokenised money, including **stablecoins, tokenised commercial bank deposits and tokenised central bank money**.¹ The potential role of tokenised money is increasingly being explored by both corporates and institutional investors. Several use cases have emerged ranging from cross-border payments, wholesale settlement, treasury and liquidity management, trade finance and capital market infrastructure. More recently, attention has expanded beyond using tokenised money for cross-border payments towards models that can support the coordination and settlement of FX transactions.

Wholesale: Refers to activity involving banks, financial institutions, and large corporates, rather than retail customers or consumers.

While tokenised money can support more efficient payments, FX settlement often requires the conditional coordination of payments across currencies. Unlike a simple payment, FX settlement requires the exchange of two currencies to occur simultaneously, creating a need for a synchronisation or **programmability layer** that enables **Payment-versus-Payment (PvP) settlement and reduces settlement risk**. Tokenised money operating on programmable ledger-based infrastructure can support this by embedding rules and conditions directly into settlement processes, facilitating the settlement of both tokenised money and other tokenised assets.

The settlement process, particularly for **wholesale FX transactions involving financial institutions and large corporations**, is exposed to settlement risk, often referred to as Herstatt risk. This arises when one party delivers its currency but the counterparty fails to fulfil its obligation. Cross-border transactions are particularly vulnerable because time zone differences can create a window in which one side has settled while the other has not. Over recent decades, this risk has been materially reduced in major currencies through infrastructure such as CLS, helping to support a large, liquid and operationally resilient FX market.

CLS mitigates FX settlement risk through PvP settlement, ensuring that one currency is transferred only if the corresponding currency is delivered, while reducing funding requirements via multilateral netting and liquidity optimisation tools, generating liquidity savings of over [96%](#) for settlement members. This [liquidity efficiency](#) is important in wholesale markets because it determines the amount of liquidity required for the system to settle its full volume of transactions safely.

However, these benefits are not universal. Many emerging-market currencies subject to current- or capital-account restrictions remain outside the CLS ecosystem and continue to rely on correspondent banking networks or local clearing systems. Settlement in these markets often involves bilateral pre-funding, fragmented payment chains and complex liquidity management, creating operational and funding frictions that can increase the cost and complexity of cross-border transactions.

While settlement risk and liquidity frictions remain important challenges in FX

¹ This definition is in line with CCAF and Fii (2026) Tokenised Money: Use Cases, Interoperability and Regulation



markets, changes elsewhere in the financial system are creating new demands on settlement infrastructure. As securities markets move from T+2 to T+1 settlement and continue to explore even shorter settlement cycles, the movement of cash, FX, securities and collateral has become increasingly interconnected. While existing infrastructure continues to support these markets, cross-border funding and settlement remain dependent on operating hours, time zones and multiple intermediaries.

These developments have brought two related challenges into focus. The first is improving cross-border settlement, particularly where correspondent banking and non-CLS arrangements create liquidity and operational frictions. The second is supporting a more integrated financial system in which cash, FX, securities and collateral can move more seamlessly across institutions and jurisdictions. Together, these challenges have strengthened interest in tokenised forms of money as a potential means of modernising both payment and capital market infrastructure.

The role of tokenised money in FX is closely linked to the broader tokenisation of financial assets. As [asset tokenisation gains momentum](#), tokenised money is increasingly viewed not merely as a payment instrument, but as an enabling layer for more integrated settlement processes. The key question is therefore not whether the technology is feasible, but whether it can achieve sufficient scale, interoperability and integration with existing financial market infrastructures to deliver meaningful improvements in cross-border settlement and capital market efficiency.

Many tokenised money initiatives have focused on improving the cross-border movement of money rather than directly settling FX trades. This reflects the scale of corporate payment frictions: [global corporates move an estimated US\\$23.5 trillion](#) across borders each year and remain heavily reliant on correspondent banking networks, which generate approximately **US\$120 billion** in annual cross-border transaction fees. These figures exclude additional costs related to FX conversion, trapped liquidity, pre-funding, settlement delays and operational complexity, suggesting that the overall cost of cross-border treasury activity is significantly higher. However, the scope of these initiatives is beginning to broaden. As tokenised money infrastructure matures, some projects are moving beyond payment efficiency towards models that could support FX settlement.

Against this backdrop, this paper examines the current FX settlement landscape before assessing the role of tokenised money in FX. It evaluates the opportunities and limitations of stablecoins, tokenised deposits and tokenised central bank reserves, and considers whether these innovations can address challenges that continue to characterise cross-border FX markets.

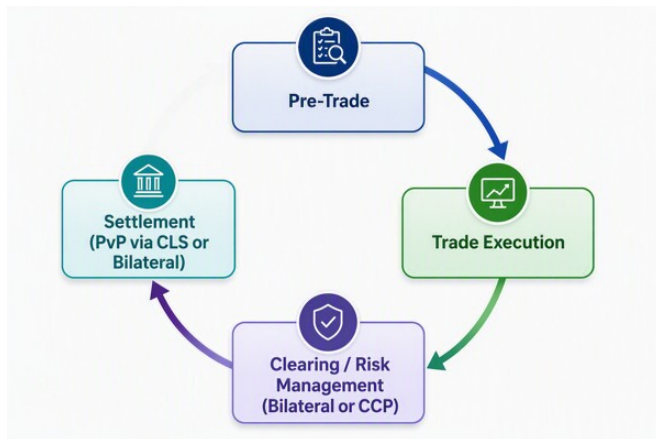
FX settlement today

FX trading takes place primarily in the over-the-counter (OTC) market, where banks, dealers, electronic platforms and other market participants facilitate the exchange of currencies. Once a trade is executed, it enters the post-trade process, during which trade details are confirmed bilaterally or cleared through a central counterparty (CCP), payment instructions are exchanged, and settlement obligations are established ([Figure 1](#)). This process spans the period between trade execution and final settlement, during which exposures are managed, liquidity is



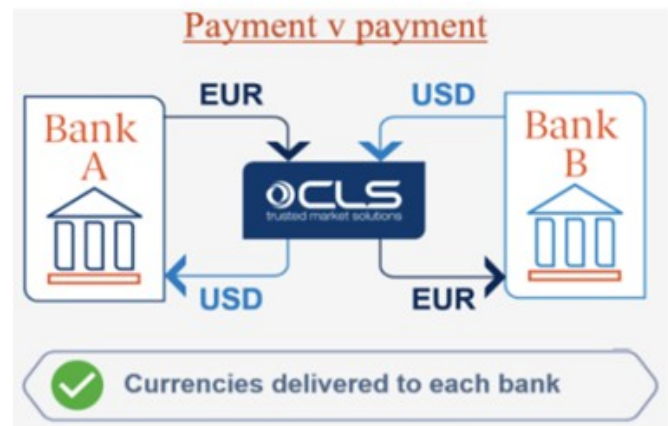
arranged, and currencies are transferred between counterparties. [Two types of credit risk](#) arise during this period: **replacement cost risk**, which exists from trade execution until settlement, and **principal risk**, which arises during the settlement process itself when one party may deliver the currency it sold but not receive the currency it purchased. Depending on the currency pair, FX spot transactions typically settle one or two business days after the trade date, while forwards and options settle on their contractual maturity date, which may range from days to several years after execution.

Figure 1: FX spot trading cycle



Source : Deutsche Bank

Figure 2: FX settlement via CLS



Source : Deutsche Bank, CLS

To reduce principal risk at settlement, often referred to as **settlement risk or Herstatt risk**, wholesale FX markets rely on infrastructure such as CLS, which enables participating banks to settle eligible FX transactions on behalf of themselves and their clients through a **Payment-versus-Payment (PvP) mechanism**, ensuring that one currency is transferred only if the corresponding currency is delivered ([Figure 2](#)).

How CLS works?

Today, CLS achieves simultaneous settlement through a defined funding and settlement process rather than instantaneous settlement. **Under this process, settlement is deferred to a daily settlement cycle, enabling multilateral netting and liquidity optimisation before final settlement occurs. This is important in wholesale markets because it determines the amount of liquidity required for the system to settle its full volume of transactions safely. Settlement finality is achieved through the use of central bank money.**

- For FX trades eligible for settlement using CLS, settlement members or their customers submit electronic payment instructions for both sides of the transaction. CLS validates and matches these instructions, providing legal confirmation and holding the transaction details until the settlement date.
- On each settlement date, CLS simultaneously settles matched payment instructions that satisfy its risk management requirements. Instead of every FX trade being funded individually, CLS calculates each member's net

Gross versus net settlement: what does it mean? Twenty currencies generate 190 unique bilateral currency pairs. Under gross settlement, each currency leg must be funded separately. Through multilateral netting, these obligations are reduced to 20 net currency positions, one for each currency, so participants fund only their net obligations rather than every bilateral payment.



obligation across all eligible trades and currencies.

- Members then make funding payments and receive pay-outs at specified times during the day. CLS holds accounts at the relevant central banks for the currencies it settles. Settlement members pay and receive funds through these CLS accounts via either their own central bank accounts or correspondent banking (nostro) arrangements. Settlement members' multicurrency accounts begin and end each business day with a zero balance.
- The settlement of payment instructions and the associated payments across the books of CLS Bank is final and irrevocable.

Nostro account: A bank account held by one bank with a foreign correspondent bank, typically to facilitate payments and settlements in a foreign currency. The term means "our account with you".

The PvP mechanism also illustrates that a form of **atomic settlement** already exists in today's FX market. Once settlement occurs in CLS, it is final and irrevocable, meaning the transfer of both currency legs cannot be unwound. However, as [CLS notes](#), atomic settlement can encompass both simultaneity and instantaneousness, which are distinct concepts. Settlement can occur on an all-or-nothing basis, with both currency legs settling simultaneously and with finality, without necessarily taking place at the moment a trade is executed.

The CLS function differs from that of a central clearing counterparty (CCP). While CLS's PvP mechanism eliminates FX settlement risk, **it does not eliminate replacement cost risk**, which arises when a counterparty defaults before settlement and the transaction must be replaced at prevailing market prices. These risks can be particularly significant for FX forwards and swaps, where the period between trade execution and final settlement may extend for months or even years. CCPs such as ForexClear help mitigate counterparty credit and replacement cost risk through central clearing and margining, whereas CLS is designed to address settlement risk at the point of settlement by ensuring the simultaneous exchange of both currency legs.

CLS mitigates FX settlement risk (Herstatt risk) at the point of settlement by ensuring that the two currency legs of an FX transaction are settled simultaneously on a payment-versus-payment (PvP) basis.

ForexClear mitigates counterparty credit and replacement cost risk before settlement through central clearing, novation and margining. Replacement cost risk arises after a trade is executed and before it settles, when an adverse move in market prices could leave one party exposed if the counterparty defaults.

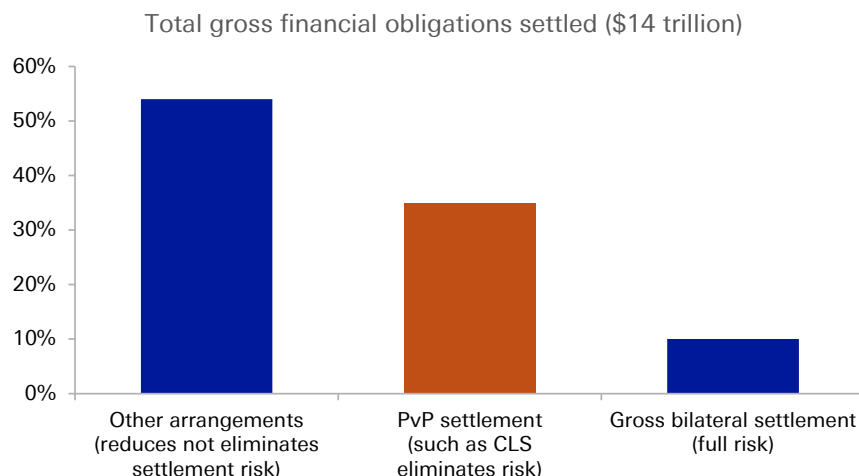
Settlement risks significantly reduced but not eliminated

According to the latest BIS Triennial Survey, banks and other financial institutions account for approximately 95% of global FX turnover, compared with around 5% for non-financial firms. Around 65% of financial-sector FX activity is cross-border in nature. As a result, wholesale FX settlement is overwhelmingly driven by financial institutions, underscoring the importance of robust cross-border settlement infrastructure.

Although infrastructure such as CLS has significantly reduced settlement risk, its coverage is not universal. CLS currently supports 18 currencies, and a significant share of global FX activity continues to settle outside PvP arrangements. Results from the [BIS Triennial Survey](#) show that just over US\$5 trillion, representing 36% of average daily FX settlement activity, was settled through PvP mechanisms that eliminate settlement risk. A further US\$7.6 trillion (54%) was settled through arrangements such as pre-settlement netting that reduce, but do not eliminate, settlement risk. Meanwhile, **more than US\$1.4 trillion (10%) of daily FX settlement continued to occur on a gross bilateral basis through traditional correspondent banking relationships, leaving participants fully exposed to settlement risk (Figure 3).**



Figure 3: FX settlement obligations by settlement type



Source : Deutsche Bank, BIS

Notably, around 25% of this volume (US\$1.4 trillion) was eligible for settlement through a PvP system but was not settled in this way. Potential reasons include operational challenges that result in trades missing PvP settlement cut-off times, difficulties in managing credit exposures to correspondent banks that support PvP-related payments, and the operational burden of meeting tight funding and payment deadlines ([Figure 4](#)).

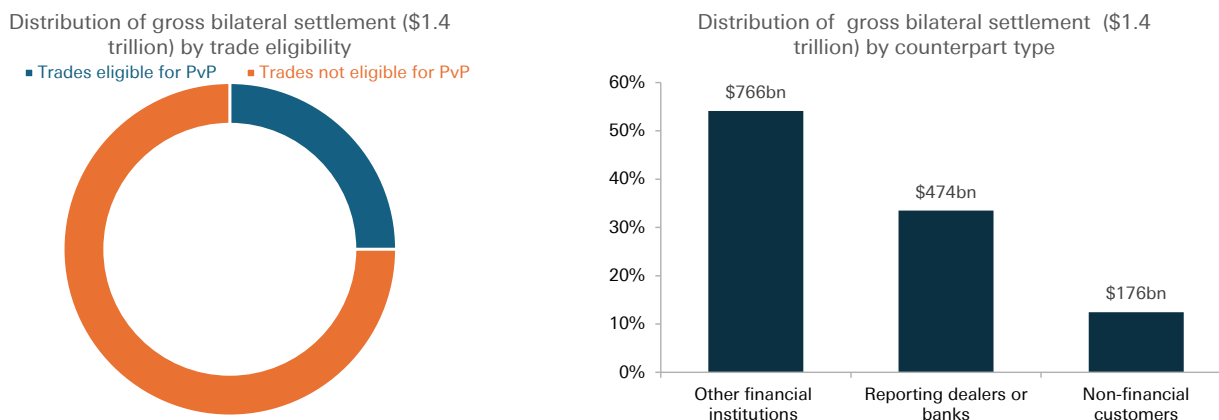
The remaining 75% of gross bilateral settlements were not eligible for PvP settlement systems, primarily because one or both counterparties lacked direct or indirect access to a PvP system. Other constraints included currency pairs that were not eligible for PvP settlement, often involving emerging market currencies, as well as trade types, such as certain same-day (T+0) transactions, that cannot currently be settled through existing PvP arrangements. As a result, much of the residual settlement risk reflects limitations in market access, currency coverage and infrastructure capabilities rather than the voluntary non-use of PvP systems.

Given the dominance of financial institutions in FX trading, gross bilateral settlements were also concentrated among financial counterparties. Of the US\$1.4 trillion settled gross bilaterally, approximately 54% involved other financial institutions, 33% involved reporting dealers, and 12% involved non-financial institutions ([Figure 4](#)).

Digging deeper into trades within each counterparty category, PvP system to mitigate risk is most common for transactions between reporting dealers or banks. Only **11% of inter-dealer trades** are settled on a gross basis and exposed to maximum risk. Consequently, FX trades most exposed to settlement risk are either between reporting dealer and other financial institutions or reporting dealers and non-financial institutions such as corporations. **Around 19% of FX trades** between reporting dealers and other financial institutions are settled on a gross basis. As non-bank financial institutions play an increasingly important role in cross-border securities investment and funding activities, these exposures remain a relevant source of risk outside the core PvP settlement infrastructure. Similarly, **22% of FX trades** between reporting dealers and non-financial institutions also settled through gross bilateral settlement, exposing participants to the full principal amount at risk.



Figure 4: Settlement risk emanating from trades not eligible for PvP. In absolute terms driven by other financial institutions.



Source : Deutsche Bank, BIS

These findings indicate that while the largest settlement risk exposures remain concentrated among financial institutions in absolute terms, the proportion of transactions exposed to risk is higher for non-bank financial institutions and corporates. Many of these residual risks stem from market access, currency coverage and operational constraints that fall outside existing PvP infrastructures. We therefore examine how these constraints create settlement frictions across selected corporate and financial-sector use cases, with a particular focus on institutional investors and cross-border securities transactions, reflecting the dominant role of financial institutions in wholesale FX markets.

FX settlement frictions: corporations versus institutional investors

The challenges associated with FX settlement are often most acute when transactions are cross-border, requiring the movement of currencies across different jurisdictions, payment systems, market infrastructures and operating hours. These challenges can arise whether FX settlement supports an underlying commercial payment or forms part of a broader securities transaction.

Corporations and FX

For **corporations**, FX spot settlement is typically a standalone transaction that supports cross-border trade, commercial payments and treasury activities. Cross-border payments and liquidity management are closely linked to the efficient functioning of the FX market. Corporates frequently need to convert currencies to support international trade, meet cross-border payment obligations and fund activities across multiple jurisdictions. More broadly, treasury functions rely on the seamless movement of currencies to manage cash positions, fund subsidiaries and address short-term liquidity needs.

Effective liquidity management facilitates domestic and cross-border payment obligations, supports the optimisation of cash flows and enables corporates to respond to changing funding requirements. These activities often involve access to wholesale FX markets to convert currencies and mobilise liquidity across jurisdictions. During periods of market volatility, the ability to execute FX transactions and move liquidity efficiently becomes particularly important, allowing institutions to respond swiftly to changing market conditions and maintain confidence in their operations.



FX settlement frictions: These include settlement risk where transactions cannot be settled on a PvP basis, delays arising from cut-off times and operating-hour constraints, liquidity and funding requirements associated with settlement, and the need to coordinate processes across multiple infrastructures and intermediaries. Such frictions can increase operational complexity and affect the timely movement of funds across borders.

Payment frictions: FX spot settlement represents only one component of a broader cross-border payment process. Even where banks settle eligible FX transactions through CLS on behalf of corporate clients, the associated commercial payments generally continue to rely on correspondent banking networks, SWIFT messaging and domestic payment systems. As a result, although CLS can mitigate settlement risk for eligible FX trades, many cross-border payment frictions remain, including delays, cut-off times and intermediary costs.

[US dollar payments](#), for example, are ultimately settled through infrastructures such as Fedwire or CHIPS. Since relatively few non-US banks have direct access to these systems, many rely on correspondent banking relationships to access dollar liquidity and payment services. Corporates, in turn, have no direct access and must depend on commercial banks to intermediate these transactions.

These challenges are distinct from wholesale FX settlement and relate primarily to the underlying payment process. For many corporations, the costs and operational frictions associated with cross-border payments may exceed those arising from the FX conversion itself.

Institutional investors and FX

For **institutional investors** buying or selling securities across jurisdictions, FX spot settlement is part of a broader cross-border securities transaction, where FX settlement, cash movements and securities settlement must be coordinated across multiple intermediaries and jurisdictions. When an investor purchases a foreign equity or bond, they must first obtain the currency required to complete the transaction. As a result, cross-border investments involve two interconnected settlement processes: FX settlement, where one currency is exchanged for another (PvP), and securities settlement, where cash is exchanged for the asset through a Delivery-versus-Payment (DvP) mechanism. Where currencies are not eligible for settlement through PvP infrastructures such as CLS, investors may rely on correspondent banking networks to settle the FX transaction. In these cases, the two currency legs are typically settled through separate payment systems, leaving participants exposed to residual settlement risk.

International investors do not typically interact directly with domestic market infrastructures such as DTCC in the United States. Instead, they access these markets through custodians, which act as intermediaries between investors and local settlement systems. Custodians are responsible for safekeeping assets, managing settlement instructions, coordinating cash movements and arranging funding across jurisdictions. When a foreign security is purchased, the custodian must ensure that the appropriate settlement currency is available before the transaction can be completed through the relevant central securities depository.

This creates a dependency between the FX funding trade and the securities settlement process, meaning that delays or frictions in obtaining the required currency can have direct implications for the timely settlement of the underlying security. The ongoing transition from T+2 to T+1 security settlement cycles, and the



potential move to T+0, increases these challenges by compressing the time available for investors to execute FX transactions, secure funding, and complete settlement processes across jurisdictions.

Key challenges for institutional investors under compressed timelines

The move to shorter securities settlement cycles has increased the operational challenge of coordinating processes. For example, under T+1, a UK investor purchasing a US equity at the market close (9pm London time) may have only around two hours before the CLS submission cut-off at 11pm to determine the final USD funding requirement, execute the associated GBP/USD transaction and submit it to CLS for settlement. Once the FX transaction settles and USD funds are made available to the custodian, the securities transaction can proceed through the relevant securities clearing and settlement infrastructure. Missing this window may require investors to pre-fund transactions, creating cash drag and unwanted currency exposure, or settle the FX transaction outside CLS, potentially increasing settlement risk and transaction costs.

Cross-border transactions involve a range of operational and settlement challenges, particularly under compressed timelines. Local currency cut-off times constrain the window available to execute and settle FX transactions, while settlement risk remains where trades cannot be processed through CLS and must settle bilaterally. Currency controls and market restrictions may further complicate settlement in certain emerging-market currencies and necessitate pre-funding. Shorter settlement cycles also leave less time for trade matching, confirmation, settlement instruction processing and the resolution of failures, increasing the importance of coordinating multiple post-trade processes.²

As FX settlement, cash movements and securities settlement occur across separate infrastructures and timelines, shorter settlement cycles increase the importance of post-trade coordination. Although the [Global Foreign Exchange Committee \(GFXC\)](#) described North America's transition to T+1 as operationally smooth from an FX perspective, this reflected extensive industry preparation. Similar challenges are likely to arise in Europe and the UK, where a more fragmented market structure may add complexity. As settlement windows compress, market participants have less time to execute FX transactions, source funding and complete settlement obligations, making cut-off times and cross-border coordination increasingly important.

Shorter securities settlement cycles may not always require instantaneous FX settlement. However, for CLS-eligible currency pairs, a move to T+1 compresses the time available to execute FX transactions and complete funding processes across institutions and jurisdictions. While shorter cycles can reduce counterparty exposure and costs, they also increase the need for automation and efficient post-trade coordination. A further move to T+0 [would amplify these challenges](#), sharpening the trade-off between risk reduction and operational efficiency.

Similar coordination [challenges](#) are emerging across securities, derivatives and collateral markets as settlement cycles shorten. [BIS \(2020\)](#) estimates that securities trade processing costs USD 17–24 billion annually, underscoring the broader cost of fragmented post-trade workflows. Technologies that improve automation, synchronisation and risk reduction in securities processing are also relevant for FX.

² See [GFMA's Global FX Division \(GFXD\)](#) for more details.



When settlement processes are fragmented across multiple institutions, jurisdictions and infrastructures, delays or failures in one part of the transaction chain can affect the timely completion of the wider transaction, increasing operational, liquidity and settlement risks. ***Against this backdrop, interest has increasingly turned to whether tokenised forms of money can improve the coordination and synchronisation of settlement processes across institutions and market infrastructures.***

Tokenised money on programmable infrastructure

Tokenised money on programmable infrastructure offers a potential approach to addressing some of the challenges outlined in the previous section. Tokenised money refers to digital representations of money that are recorded and transferable on shared or distributed ledger infrastructure. In many current implementations, its use is enabled by two key elements: distributed ledger technology (DLT) and programmability. DLT provides a shared infrastructure on which multiple institutions can record and observe ownership changes in a common ledger, potentially reducing the need for reconciliation across separate systems. ***Programmability refers to the ability to embed executable logic, such as conditions, rules and constraints, directly into instruments, or via smart contracts.*** Such functionality can range from basic automation to more sophisticated smart contract applications.

At the simplest level, programmability can automate settlement processes by ensuring that transactions are executed only when predefined conditions are met, such as Delivery-versus-Payment (DvP) in securities transactions and Payment-versus-Payment (PvP) in FX settlement. More advanced applications can support operational and control functions across financial market infrastructure, including real-time liquidity management, margin management, collateral mobilisation and substitution, and automated compliance checks. At a more sophisticated level, programmability can coordinate multiple parties and obligations within a single workflow, enabling applications such as trade finance arrangements, treasury and cash management, automated disbursements and other multi-party payment processes. These capabilities extend beyond process automation by embedding business, risk and compliance logic directly into financial transactions (CCAF and Fii, 2026). An emerging but largely underdeveloped area is the integration of tokenised money with artificial intelligence and autonomous systems, where programmable money interfaces could enable AI agents to initiate, manage and execute financial transactions within predefined controls.

The approach to settlement risk mitigation in a tokenised environment differs from the current CLS model, which achieves Payment-versus-Payment (PvP) settlement through a centrally coordinated process operating within defined funding and settlement windows. While CLS ensures that one currency is exchanged if and only if the corresponding currency is delivered through a centralised settlement mechanism, tokenised infrastructures could, in principle, embed PvP conditions directly into the settlement process through programmable rules or smart contracts on a shared ledger, enabling settlement to occur automatically when predefined conditions are met. However, tokenised settlement models need not rely solely on transaction-by-transaction atomic settlement. Some emerging platforms, such as [ClearToken](#), also incorporate queues, netting arrangements and liquidity-saving mechanisms.

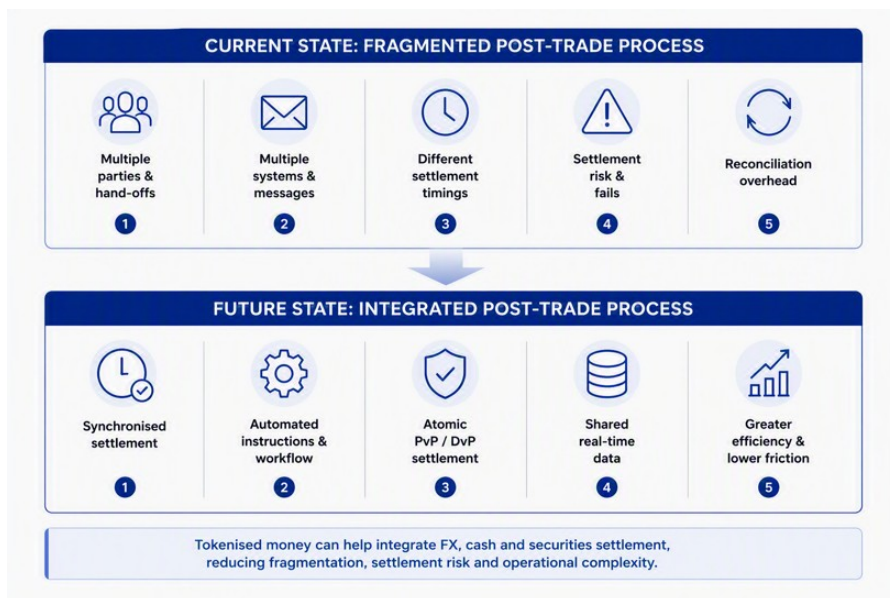


Tokenised post-trade environment for a cross-border securities trade

Many of the potential benefits of tokenised money become more apparent when viewed alongside the broader tokenisation of financial assets. In domestic securities, collateral and derivatives markets, these technologies are already being explored as a means of improving post-trade processes (BIS 2020). When tokenised money is combined with tokenised securities, it may support atomic settlement, greater automation and more efficient movement of cash and collateral, helping to reduce operational complexity, settlement risk and inefficiencies associated with the mobilisation, allocation and coordination of liquidity.

Figure 5 summarises the potential benefits of tokenised money and tokenised securities for post-trade processes. These technologies may support more synchronised settlement, reducing some of the operational, funding and coordination frictions associated with shorter settlement cycles. However, they do not eliminate the need for liquidity provision, risk management, netting or regulatory oversight. Ultimately, technology alone is unlikely to shorten settlement timelines without corresponding changes to market conventions, operating models and regulatory frameworks.

Figure 5: A more integrated post-trade process



Source : Deutsche Bank

The three routes for tokenised money

In practice, the use of tokenised money for financial institutions and corporations is emerging through three principal forms of tokenised money. The first is stablecoins, privately issued digital tokens typically backed by cash, bank deposits or high-quality liquid assets. The second is tokenised bank liabilities, including tokenised deposits, which represent commercial bank money in tokenised form and remain embedded within the existing banking system. The third is tokenised central bank money, which represent tokenised central bank reserves, also sometimes referred to as wholesale central bank digital currencies (wCBDCs).

While these models differ in their issuer, regulatory treatment and underlying infrastructure, they all seek to combine the trust associated with traditional forms

*Tokenised money refers to **digital representations of money** that are recorded and transferable on shared or distributed ledger infrastructure*

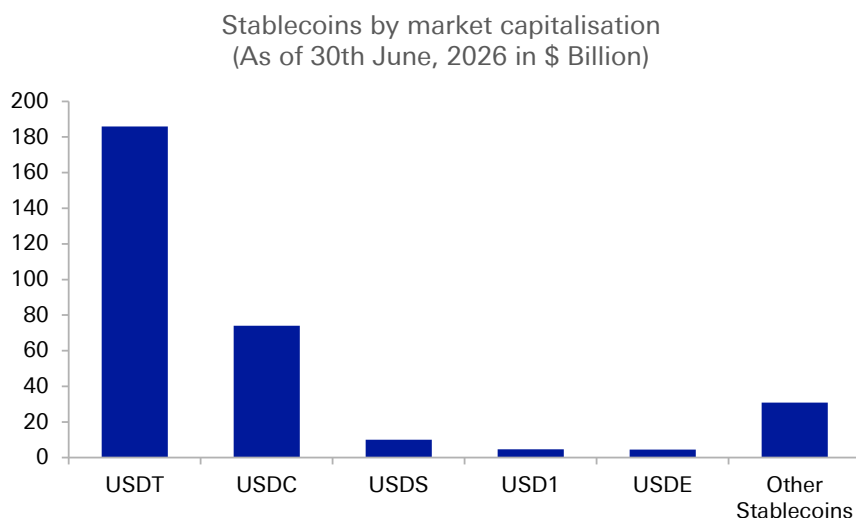


of money with the programmability, interoperability and automation enabled by modern digital technologies. Their potential role in FX settlement is therefore best assessed through the lens of how they address existing challenges.

Stablecoins

A stablecoin is a privately issued digital token designed to maintain a stable value relative to a reference currency, typically through backing by cash, bank deposits or high-quality liquid assets. While other types of stablecoins exist as suggested by Kosse et al (2023) and BIS (2026), the focus of this study would be to discuss fiat-backed stablecoins. BIS (2026) reports that 99.4% of fiat-backed stablecoins (by market valuation) are pegged to the US dollar. Recorded and transferred on distributed ledger networks, stablecoins could potentially function as digital means of payment and settlement.

Figure 6: The market dominated by US dollar-denominated stablecoins



Source : Deutsche Bank, Coingecko

The market for stablecoins has grown rapidly, with more than US\$300 billion outstanding globally, dominated by US dollar-denominated stablecoins such as USDT and USDC (Figure 6).

From an FX settlement perspective, stablecoins offer an always-on settlement asset that can move across borders on a 24/7 basis without relying on traditional banking hours. They can also support tokenised settlement logic, including conditional payments and automated transaction processing.

Stablecoins' strongest use case today is cross-border payments, particularly in corridors where correspondent banking infrastructure is less efficient. Although total stablecoin transaction volumes were estimated at roughly [US\\$62 trillion in 2025](#), only around US\$350-550 billion is estimated to be associated with real-economy payments, highlighting that adoption in traditional payments remains at an early stage.

Currently, stablecoins are increasingly being used for remittances, business-to-business payments and access to US dollars in markets experiencing currency



instability or limited access to international banking services. Stablecoins are helping improve the movement of the bridge currency—typically the US dollar—rather than the settlement of both currency legs. As roughly 99% of stablecoins in circulation are denominated in US dollars, they function primarily as digital dollar infrastructure rather than a broadly adopted multi-currency settlement network. Consequently, ***most current stablecoin implementations focus on facilitating the transfer of a single currency, typically the US dollar, rather than solving the full two-currency FX settlement problem. This has also been referred to as the "stablecoin sandwich".***

Stablecoin adoption is increasingly moving beyond the lowest-cost blockchain towards a multi-network ecosystem spanning Ethereum, Solana, Polygon, BNB Chain and other networks. This reflects growing demand for compliance, programmability and institutional-grade infrastructure alongside low transaction costs. However, it also highlights an important limitation: stablecoins do not automatically solve interoperability. Instead, the challenge increasingly becomes enabling value to move seamlessly across multiple networks and settlement rails.

Stablecoin liquidity remains fragmented across issuers, networks and jurisdictions, while regulatory treatment continues to evolve across markets. In addition, institutional adoption requires custody solutions, compliance frameworks, treasury integration and industry standards comparable to those available for traditional bank money. As a result, while stablecoins can reduce some correspondent banking frictions, they do not by themselves provide a comprehensive solution to PvP settlement across non-CLS currencies.

Despite these complexities, stablecoins are beginning to attract interest from banks for institutional and corporate cross-border settlement. A notable example is [Nomura's partnership with Circle](#), which aims to explore the use of fiat-backed stablecoins such as USDC in Japan as part of next-generation market infrastructure. The collaboration is focused not only on cross-border payments, but also on FX transactions, treasury operations, fund transfers, on-chain collateral management and tokenised capital markets transactions. More [recently](#), a group of 21 financial institutions plan to issue dollar stablecoin in 2027.

[BNY Mellon](#) expanded its institutional stablecoin services in 2026 to support custody and minting of Circle's USDC, reflecting growing client demand for regulated stablecoin infrastructure rather than the issuance of a proprietary stablecoin. Société Générale's [Forge](#) subsidiary, on the other hand, issued the euro-denominated stablecoin EURCV and used it within tokenised capital market transactions.

Regulatory developments have also helped accelerate experimentation in this area. In Japan, amendments to the Payment Services Act established a regulatory framework for fiat-backed stablecoins. Under this framework, Circle's USDC became the first global US dollar stablecoin approved by Japan's Financial Services Agency for institutional use in Japan. Recent legislative developments in the United States, including the [GENIUS Act](#), have also provided greater clarity around payment stablecoins. Together, these developments have encouraged banks, payment providers and financial institutions to explore stablecoins as settlement assets within regulated financial markets.

These developments suggest that stablecoins are increasingly being viewed not only as digital payment instruments, but also as potential settlement assets for



cross-border transactions. A [recent study](#) notes that regulatory development has progressed furthest in the stablecoin sector, reflecting both growing adoption and policymakers' efforts to address the associated risks.

Key challenges

In some emerging market and developing economy (EMDE) currency corridors, USD stablecoins already act as intermediary settlement assets, performing a role similar to that of the US dollar in traditional correspondent banking arrangements by linking transactions between two non-USD currencies. While stablecoins may reduce frictions in cross-border value transfers, their current concentration in US dollar-denominated issuance limits their ability to support widespread Payment-versus-Payment (PvP) settlement across currency pairs.

While USD stablecoins may facilitate the payment leg of transactions involving tokenised US assets, investors must still exchange their domestic currency for US dollars and subsequently convert any proceeds back into their home currency. Unless digital settlement assets exist for both currencies and can be exchanged through a PvP mechanism, the underlying FX settlement risk remains. Consequently, current stablecoin models primarily improve the transfer of a single currency rather than addressing the settlement risks associated with the exchange of two currencies.

Figure 7: Stablecoins

Metric	Stablecoins
Market size	>US\$300bn outstanding
Dominant currency	~99% USD-denominated
Key benefit	24/7 cross-border transfers
Main use case	Cross-border payments & remittances
FX settlement role	Improves movement of USD as a bridge currency, rather than full two-currency PvP settlement
Key challenge	Fragmented liquidity and interoperability
Institutional adoption	Growing bank interest (e.g. Circle, BNY Mellon, Nomura)

Source : Deutsche Bank

Tokenised commercial bank money

A **tokenised deposit** is not a new form of money, but a commercial bank deposit represented or issued in tokenised form on a shared or tokenised ledger. Economically, it remains a claim on a commercial bank, just as a conventional bank deposit does today. The key difference is that the deposit, or rights linked to that deposit, can be transferred and settled using tokenised infrastructure, smart contracts and shared ledgers.

Classifying tokenised commercial bank money

Several public-sector and industry initiatives have proposed frameworks for



classifying tokenised commercial bank money. The [Bank of England](#) defines tokenised deposits as deposit claims represented on programmable platforms and distinguishes between non-transferable and transferable forms. This distinction is important because it determines whether tokenised deposits can circulate directly across institutions or continue to require interbank settlement in central bank money.

[Project Guardian](#) adopts the broader term tokenised bank liabilities and argues that these instruments could help address existing frictions in cross-border payments and FX settlement, including pre-funding requirements, settlement risk and fragmented post-trade processes. It classifies tokenised bank liabilities according to how the underlying claim is represented, distinguishing between digital twin and digital native models, and according to how they are deployed, ranging from single-bank ledgers to interoperable multi-bank and network-based arrangements.

Industry participants have developed related taxonomies, distinguishing between account-based and token-based representations of commercial bank money, as well as native and non-native ledger structures. Although these approaches differ in terminology, they broadly reflect alternative methods of representing, recording and transferring commercial bank money on distributed ledger infrastructure. Further details are provided in Appendix A.

Current implementation models and infrastructure

Current models are developing along two main lines. The first is **single-bank or single-bank-group infrastructure**, where blockchain-based deposit accounts or tokenised commercial bank money are used between customers of the same institution or banking group. J.P. Morgan's Kinexys Digital Payments is an example of an account-based blockchain deposit model rather than a deposit-token model. It operates on a single-bank ledger, enabling participating J.P. Morgan clients to transfer deposit balances across J.P. Morgan's blockchain-based payments rail. The BMW transaction illustrates this model: BMW executed a programmable on-chain EUR/USD FX transaction using [Kinexys Digital Payments](#), involving automated balance checks, conditional auto-deposits, near-real-time FX execution and transfers between BMW's Blockchain Deposit Accounts in Frankfurt and New York.

The second, more complex model is the **interbank model**, where tokenised bank liabilities issued by different banks, in different currencies and potentially on different ledgers, need to be exchanged or coordinated. [Project Guardian](#) explores several approaches to these interoperability challenges. One model uses a liquidity provider to facilitate the exchange of tokenised deposits in different currencies, with off-chain FX prices fed into the shared ledger through an FX quote oracle. Another model, demonstrated by BNY and OCBC, uses Hash-Time Locked Contracts to coordinate transfers across two private permissioned bank ledgers. A third model, involving HSBC and OSTTRA, focuses on settlement interoperability, using a shared-ledger PvP orchestration layer to match, confirm, fund and settle FX cash flows, helping to reduce Herstatt risk and improve visibility across the settlement.

FX quote oracle: A service that securely uploads off-chain FX prices and token exchange rates to a shared ledger, allowing liquidity providers to offer exchange rates and facilitate cross-currency exchanges between tokenised deposits issued in different currencies.

Bank-led shared-ledger settlement platforms are also emerging as infrastructure that can support tokenised commercial bank money. [Partior](#) is an example of a consortium-based model: a controlled environment in which a defined group of institutions jointly governs and participates in shared infrastructure. It operates as a bank-led shared-ledger settlement platform for real-time cross-border payments



and FX settlement, using a private permissioned ledger where access is restricted to authorised financial institutions and governed participants. Partior's relevance for FX further lies in its ability to support atomic Payment-versus-Payment, or PvP, settlement, where the two currency legs of an FX transaction are coordinated so that settlement of one leg occurs only if the other leg also settle.

Recent initiatives led by [The Clearing House \(TCH\)](#) seek to provide a common clearing and settlement framework for tokenised deposits, addressing one of the key challenges facing tokenised commercial bank money: interoperability across banks. The initiative aims to connect on-chain activity with existing payment rails, enabling the clearing and settlement of tokenised commercial bank money at scale while combining the regulatory, operational and settlement frameworks of established payment infrastructure with the programmability and interoperability of blockchain-based systems.

Market adoption and FX settlement relevance

Commercial bank deposit claims, including tokenised deposits, have progressed beyond early pilot stages and are seeing notable institutional adoption. While data on outstanding tokenised deposit balances remains limited, reported transaction volumes are significant. [J.P. Morgan's Kinexys platform](#), for example, has processed more than US\$4 trillion since inception and averages more than US\$7 billion daily in tokenised deposit movements for institutional payments. Other initiatives include DBS Token Services, Citi Token Services, the USDF Consortium, TassatPay, and projects from UOB and Standard Chartered.

The scale opportunity is potentially significant. Tokenised bank deposits and similar bank-token models could support around [US\\$100–140 trillion](#) in annual transaction flows by 2030, potentially rivalling or surpassing stablecoins in institutional transaction volumes. This reflects the view that corporates and financial institutions may prefer programmable forms of regulated bank money for treasury, liquidity and FX settlement workflows where integration with existing banking infrastructure, compliance processes and balance-sheet treatment is important.

Overall, tokenised deposits appear more directly relevant to institutional FX than stablecoins because they preserve the existing bank-money model while adding programmability, potentially allowing new settlement capabilities to be integrated more easily into existing banking, liquidity and risk-management frameworks. Current implementation models appear more developed for transaction banking, treasury and corporate payment use cases than for the broader wholesale FX market, where achieving interoperability across banks, currencies and settlement networks remains a more complex challenge.

Key challenges

Tokenisation does not, at least in the early stages, fully resolve time-zone mismatches or the need for pre-funding as noted by Project Guardian. Hybrid models are likely to coexist with traditional RTGS systems, correspondent banking infrastructure and fiat-to-token conversion processes for some time, meaning liquidity may still need to be positioned across both traditional and tokenised systems.

A practical issue in cross-currency tokenised deposit models is how the FX rate itself is determined. Project Guardian notes that early-stage on-chain trading volumes may not be sufficient to generate reliable FX prices purely from on-chain



activity. As a result, tokenised deposit models may initially rely on off-chain FX prices sourced from bank liquidity desks, benchmark rates or market data providers, with those prices fed into the shared ledger through an oracle. Over time, if on-chain liquidity develops, more native on-chain FX price discovery may become possible, but this depends on market depth, data integrity and privacy-preserving mechanisms.

Tokenised deposits also do not automatically address all FX settlement challenges. This is particularly relevant in wholesale FX markets, where activity is concentrated among financial institutions. A key strength of existing settlement infrastructure is multilateral netting, which significantly reduces liquidity requirements. Replicating similar levels of liquidity efficiency in tokenised interbank networks remains challenging. Unlike CLS, emerging tokenised deposit arrangements do not yet support multilateral netting at scale, meaning participants may need to trade some liquidity efficiency for faster or more flexible settlement unless netting and liquidity-management mechanisms evolve alongside the technology.

Settlement finality presents a further consideration. Existing wholesale settlement infrastructure ultimately relies on central bank money, which remains an important foundation for wholesale FX settlement. As a result, alternative models that combine tokenised commercial bank money with tokenised central bank money may prove more relevant for wholesale FX applications.

More broadly, the success of tokenised deposits will depend on interoperability across banks and ledgers, sufficient liquidity in each tokenised currency, legal recognition of tokenised claims, clear settlement finality, robust governance frameworks, regulatory consistency, and integration with existing systems.

There may also be balance-sheet costs for issuing banks. If tokenised deposits are treated as more liquidity-sensitive than conventional deposits, banks may need to hold additional high-quality liquid assets against them, reducing their usefulness as a source of funding for lending or maturity transformation.

Figure 8: Tokenised deposits

Metric	Tokenised deposits
Market size	Early-stage but growing institutional adoption
Dominant currency	Multi-currency potential (dependent on issuing banks)
Key benefit	Programmable regulated bank money
Main use case	Institutional payments, treasury and liquidity management
FX settlement role	Supports on-ledger PvP settlement and programmable settlement workflows
Key challenge	Interoperability across banks, currencies and networks
Institutional adoption	Highest within single-bank platforms and consortium-based networks; broader cross-bank adoption remains evolving

Source : Deutsche Bank



Tokenised central bank money

Rather than tokenising private bank money or relying on privately issued settlement assets, an alternative approach is to make wholesale central bank money available on DLT-based infrastructure. In this context, wholesale central bank money used in DLT experiments is often referred to as wholesale central bank digital currency (wCBDC) or, more precisely, tokenised central bank reserves. The latter term is often preferred because it emphasises that the underlying asset, central bank reserves, already exists in digital form, with the innovation lying primarily in the technology used to represent and transfer it. While wCBDC is frequently used interchangeably with tokenised central bank reserves, its definition varies across institutions and may encompass a broader range of innovations in wholesale central bank money that are not necessarily tokenisation-based ([IMF, 2025](#)).

Central bank reserves are already held digitally by eligible financial institutions through RTGS systems such as T2 in Europe, Fedwire in the United States and CHAPS in the United Kingdom. From this perspective, tokenised central bank money initiatives represent not the introduction of a new form of public money, but the extension of existing central bank liabilities into tokenised and programmable environments.³

Although existing infrastructures already eliminate principal risk through DvP and Pvp mechanisms, tokenised reserves may support a more integrated settlement process by bringing cash and assets onto the same ledger. This allows transactions to be completed simultaneously and irrevocably using DLT and smart contracts, while enabling programmability.

A number of initiatives have examined the use of multilateral platforms that make wholesale central bank money available in tokenised form for cross-border settlement. Examples include [Project mBridge](#), [Project Jura](#) and [Project Agorá](#). Leveraging shared infrastructure and governance frameworks, these projects have demonstrated real-time and atomic Pvp settlement between commercial banks in different jurisdictions using tokenised reserves. Smart contracts can further automate and coordinate multicurrency transactions, improving operational efficiency.

Multi-currency model based only on tokenised central bank money

[Project mBridge](#) is an example of a pure multi-CBDC architecture. We use the term CBDC, as used by the project, which is synonymous with tokenised central bank reserves. Developed by the Hong Kong Monetary Authority, People's Bank of China, Bank of Thailand and Central Bank of the UAE, with Saudi Arabia joining later, the project was designed to enable direct cross-border payments and FX transactions using wholesale CBDCs issued by participating central banks on a shared distributed ledger platform. Following a successful pilot involving 20 banks and 164 real-value transactions, mBridge reached minimum viable product (MVP) status in 2024, indicating that the platform had progressed beyond proof-of-concept testing and was capable of supporting real-value cross-border settlement. Subsequent growth has been substantial, with the platform reportedly processing more than 4,000 cross-border transactions worth [approximately US\\$55 billion by late 2025](#). While the BIS transferred responsibility for the platform to participating

[Tokenised central bank reserves \(IMF, 2025\)](#): Wholesale central bank money, commonly known as reserves, refers to money issued by central banks to eligible financial institutions for interbank settlement and monetary policy implementation. With growing interest in tokenised financial assets, central banks are exploring ways to make reserves available on programmable ledgers. In this context, tokenised reserves are direct central bank liabilities issued using DLT and accessible only to a predefined group of financial institutions.

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central banks in late 2024, mBridge remains one of the most developed publicly disclosed multi-CBDC initiatives.

Beyond wholesale payments, experiments have also explored how tokenised central bank reserves could support more advanced settlement arrangements involving multiple currencies, assets or transaction legs. One example is delivery-versus-payment-versus-payment (DvPvP), where interconnected transactions are settled atomically. [Project Jura](#) illustrates these capabilities by enabling the direct transfer of euro and Swiss franc wholesale CBDCs (wCBDCs), representing tokenised central bank reserves, between French and Swiss commercial banks on a shared DLT platform operated by a third party. The project demonstrated how tokenised asset and foreign exchange transactions can be settled safely and efficiently using both payment-versus-payment (PvP) and delivery-versus-payment (DvP) mechanisms, highlighting the potential for programmable, multi-leg transactions across currencies, asset classes and jurisdictions.

Hybrid models

Alongside tokenised central bank money initiatives, [Project Agorá](#) is exploring alternative architecture that retain central bank money as the settlement anchor while preserving a significant role for commercial bank money.

[Project Agorá](#), launched by the BIS in 2024 in collaboration with the central banks of France (representing the Eurosystem), Japan, South Korea, Mexico, Switzerland, the United Kingdom and the United States, explores a hybrid cross-border settlement model that combines tokenised commercial bank deposits with tokenised central bank reserves on a shared programmable platform. In 2026, the BIS reported that the prototype had successfully demonstrated atomic multi-currency settlement across participating jurisdictions using a unified ledger architecture linking tokenised deposits and tokenised central bank reserves. The project also highlighted potential improvements in speed, transparency and operational efficiency while preserving the existing two-tier monetary system.

Key challenges

Tokenised central bank reserve models, particularly multi-CBDC wholesale arrangements, are among the most challenging forms of digital money to implement. Unlike stablecoins or tokenised deposits, they require extensive coordination across central banks, legal jurisdictions and financial market infrastructures. Cross-border deployment raises complex questions around governance, access, interoperability, regulatory alignment, and the legal recognition of settlement finality. As a result, industry attention has increasingly broadened beyond pure wCBDC architectures towards hybrid models that combine tokenised central bank money with tokenised commercial bank money. By building on the existing two-tier monetary system, these models may face fewer implementation barriers and are often viewed as more compatible with current market structures.

[Recent industry commentary](#) suggests that digital settlement infrastructure may develop along regional or architectural lines rather than converge on a single global model. The emergence of mBridge as a multi-CBDC platform centred on Asia and the Middle East, alongside Agorá's hybrid model involving major advanced-economy central banks and commercial banks, highlights the possibility that multiple settlement ecosystems could emerge over time. Whether these systems ultimately converge through interoperability standards or develop as parallel networks remains uncertain.



Figure 9: Tokenised central bank reserves

Metric	Tokenised central bank reserves
Market size	Limited live volumes; activity remains concentrated in pilots, proofs of concept and early production deployments
Dominant currency	Typically designed as multi-currency infrastructures involving multiple central bank jurisdictions
Key benefit	Provides settlement in central bank money with potential for atomic PvP settlement and programmability
Main use case	Wholesale financial market settlement, cross-border payments and settlement of tokenised assets
FX settlement role	Potential common settlement asset for cross-currency transactions, reducing settlement risk and enabling programmable FX settlement workflows
Key challenge	Cross-border coordination among central banks, legal and regulatory harmonisation, governance, access and interoperability.
Institutional adoption	Active experimentation through initiatives such as mBridge and Jura, as well as hybrid models such as Agorá, although large-scale production adoption remains limited.

Source : Deutsche Bank

Other settlement solutions and related infrastructure

Going beyond the three forms of tokenised money, other settlement solutions and supporting infrastructures are evolving in parallel. One example is [Finality](#), a privately operated settlement system backed 1:1 by central bank reserves. It provides a central-bank-money-backed settlement asset for wholesale transactions. The [Sterling Finality Payment System](#) went live in 2023 and received Bank of England settlement-finality designation in 2024, with the platform targeting use cases including PvP FX settlement and tokenised asset settlement. It uses the [Bank of England's omnibus account solution](#) to hold participant funds and issues DLT-based settlement tokens representing claims on this pool of funds.

Omnibus account arrangements allow for settlement using tokens on the DLT platform outside of RTGS operating hours and with added functionalities, such as programmability, provided that the FIs have prefunded their omnibus accounts (Bank of England 2021)

Rather than tokenising central bank money directly, the [Bank of England's synchronisation model](#) enables funds held in RTGS to be synchronised with transactions taking place on external ledgers through a synchronisation operator. Settlement in central bank money occurs if and only if the corresponding transaction on the external platform is completed. Through initiatives such as Project Meridian FX and the Synchronisation Lab, the Bank has explored how this approach could support Payment-versus-Payment (PvP) settlement for cross-border FX and Delivery-versus-Payment (DvP) settlement for tokenised securities, while preserving settlement in central bank money. The Bank has also highlighted that extended RTGS operating hours may be an important enabler of such synchronised settlement models.

Industry initiatives such as the Regulated Settlement Network (RSN), which builds on earlier Regulated Liability Network (RLN) initiatives, are exploring a shared ledger environment in which regulated liabilities, such as commercial bank deposits and central bank money, and tokenised assets can settle in real time across multiple entities. RSN involves a consortium of institutions, including Citi, the New York Fed Innovation Center and Swift, and creates a shared ledger environment in which these assets can settle in real time across multiple entities. A [December 2024](#) proof of concept demonstrated 24/7 settlement capabilities in a controlled environment, although questions remain around broader ecosystem integration



and cross-border scalability.

[SWIFT's blockchain ledger](#) is another example that emphasis on connectivity. Rather than introducing a new settlement asset, it enables participating banks to coordinate transfers of tokenised deposits across their own ledgers while continuing to settle through existing payment infrastructure. The initiative is intended to support 24/7 cross-border payments and improve liquidity efficiency.

These initiatives highlight the importance of infrastructure that enables different forms of money, payment systems and settlement architectures to operate together. If FX settlement evolves through multiple architectures, synchronisation and interoperability may become as important as the characteristics of the settlement asset itself.

Technology alone is not enough: risk reduction or transformation?

Tokenised money is often discussed in terms of faster settlement, reduced operational complexity and atomic execution. While these technologies may help reduce certain forms of settlement risk, particularly where payment-versus-payment (PvP) and delivery-versus-payment (DvP) mechanisms can be embedded into the settlement process, they do not eliminate the need for liquidity provision, risk management, netting arrangements or regulatory oversight. In many cases, the effect may be better described as risk transformation rather than risk elimination. Settlement risk may be reduced, but liquidity, operational, legal and governance considerations remain.

Regulation: Regulatory frameworks for tokenised deposits, stablecoins and digital settlement assets continue to evolve across jurisdictions. While progress has been made, important questions remain around legal treatment, settlement finality, prudential requirements and the use of tokenised money in cross-border transactions.

Liquidity: Tokenised money does not create liquidity; rather, it may improve how liquidity is mobilised and deployed. Potential reductions in pre-funding requirements and trapped liquidity depend on sufficient market participation, transaction volume and interoperability across networks. Today, many initiatives remain fragmented across separate platforms and participant groups, limiting the extent to which liquidity benefits can be realised. Moreover, unlike CLS, most tokenised settlement networks **do not yet provide multilateral netting at scale**, meaning some participants could **face higher gross funding requirements** even as settlement processes become more automated. This is important because multilateral netting and the resulting liquidity savings have been identified as critical features of wholesale FX payment and settlement systems, including in discussions at the [Jackson Hole Economic Policy Symposium](#).

Interoperability and integration: Many of the potential benefits depend on the ability of different forms of money, payment systems and settlement infrastructures to operate together. Successful adoption will require integration with existing banking, payment, custody, accounting and regulatory reporting processes. Programmable and traditional infrastructures are likely to coexist, creating additional operational and implementation challenges.



Market infrastructure: The benefits of programmable settlement are greatest when cash, securities and collateral can move on the same, or fully interoperable, infrastructure. While technological capability is advancing rapidly, broader adoption depends on custodians, central securities depositories, asset managers, banks and market infrastructures adapting their operating models and processes.

Operational resilience and cybersecurity: Tokenised settlement infrastructures introduce new operational dependencies, including distributed ledger networks, smart contracts, wallets, bridges and external data providers. Coding errors, cyberattacks, failures in oracle services or vulnerabilities in cross-chain connectivity can disrupt settlement processes or lead to unintended outcomes. As networks become increasingly interconnected, disruptions may propagate more rapidly across participants and platforms.

Legal certainty and governance: The legal treatment of tokenised money continues to evolve. Questions remain around ownership rights, insolvency treatment, the enforceability of self-executing smart contracts, dispute resolution mechanisms and the alignment of technical and legal settlement finality. Governance arrangements, including liability allocation, system upgrades and contingency procedures, remain critical to ensuring trust and resilience.

Market concentration and fragmentation: Network effects may favour a small number of tokenised money issuers, settlement platforms or infrastructure providers, potentially creating new forms of concentration risk. At the same time, the proliferation of non-interoperable networks may fragment liquidity across multiple platforms and tokenised money instruments, reducing the fungibility and efficiency that underpin effective settlement.

Data, privacy and compliance: The transparency of shared ledgers can improve auditability and oversight but may also create challenges around confidentiality, commercial sensitivity and data protection. Compliance obligations, including AML/CFT controls, sanctions screening and cross-border regulatory requirements, remain significant considerations and may be difficult to implement consistently across multiple networks and jurisdictions.

Financial stability and monetary considerations: As tokenised money adoption grows, regulators have raised broader concerns around financial stability, including the implications of large-scale migration into digital settlement assets, the concentration of reserve assets, and changing funding dynamics within the financial system. In some jurisdictions, particularly emerging economies, the growing use of foreign-currency-denominated stablecoins has also raised concerns around monetary sovereignty and the effectiveness of domestic monetary policy transmission.

Conclusion

Existing infrastructure plays an important role in mitigating settlement risk in wholesale FX markets. CLS combines Payment-versus-Payment (PvP) settlement with multilateral netting and liquidity optimisation, generating liquidity savings of approximately 99% for participants. However, coverage is not universal, and a significant share of global FX activity continues to settle outside PvP arrangements due to constraints related to currency coverage, participant access, operating windows and market structure.



While tokenised money may help address some of these frictions, its potentially stronger use case for wholesale FX lies in the wider tokenisation of capital markets. Tokenised money could enable more integrated settlement processes for tokenised assets. Reflecting this potential, a [recent study](#) identified capital market tokenisation as one of the most important long-term use cases for tokenised money. Although activity remains small relative to overall market volumes, it has the potential to change how securities, bonds and other financial instruments are issued, traded and settled. The vision is an integrated ecosystem in which tokenised money is used to settle tokenised securities and other assets, enabling atomic settlement that reduces counterparty risk and shortens settlement times from days to seconds.

The [Bank of England's](#) analysis highlights an important financial stability consideration. While tokenised deposits extend commercial bank money into programmable environments, central bank money remains the ultimate risk-free settlement asset and anchor of confidence in the financial system. Future tokenised markets may therefore require mechanisms such as RTGS synchronisation or hybrid models combining tokenised commercial bank money and central bank money to preserve the singleness of money and settlement finality. More broadly, the proliferation of multiple forms of tokenised money could fragment liquidity, reduce fungibility and create new market structure and financial stability challenges.

Over time, programmable settlement arrangements may replicate some functions currently performed by existing infrastructures, particularly if interoperable tokenised deposit or tokenised central bank money frameworks achieve sufficient scale. However, this will depend not only on technological capability but also on the development of comparable liquidity networks, legal frameworks, governance arrangements and interoperability with existing payment and settlement systems. The transition towards programmable settlement is therefore likely to involve both risk reduction and risk transformation.



Appendix A: Detailed classification of tokenised commercial bank money

Several public-sector and industry initiatives have proposed different frameworks for classifying tokenised commercial bank money. The [Bank of England](#) describes tokenised deposits as “deposit claims represented on programmable platforms” that enable new techniques for payments and settlement. It distinguishes between non-transferable and transferable tokenised deposits. Non-transferable tokenised deposits resemble existing deposits and may still require settlement in central bank money when transferred between customers of different banks. Transferable tokenised deposits, by contrast, could be self-settling upon transfer, with the recipient becoming a customer of the issuing bank. This distinction matters for FX because it affects whether tokenised deposits can circulate across banks directly or whether interbank settlement in central bank money is still required to preserve singleness and finality.

[Project Guardian](#) uses the broader term tokenised bank liabilities to refer to tokenised representations of commercial bank money on a shared ledger, without asserting any specific legal or regulatory interpretation. Tokenised deposits are therefore a specific form of tokenised bank liability. It further classifies tokenised bank liabilities along two dimensions: token level and ledger level. At the token level, it distinguishes between digital twin tokens and digital native tokens. Digital twin tokens represent ownership interests in underlying traditional bank liabilities, with ownership rights represented on-chain through digital tokens. Digital native tokens are issued and custodied natively on a shared ledger, which serves as the primary record of ownership and rights. The distinction therefore relates to the legal and operational representation of the underlying bank liability rather than to transferability, as both forms can potentially support programmable payments and settlement.

At the ledger level, Project Guardian distinguishes between three deployment models. A **single-bank ledger** is a shared ledger operated within the same banking group, enabling transfers between customers, entities or branches connected to that bank's infrastructure. **Between-bank models** enable tokenised bank liabilities issued on separate banks' shared ledgers to be used for payments and settlement between customers of different banks. **Interlinking network models** connect distinct networks, each potentially having its own governance framework and design, allowing tokenised assets to move across a broader ecosystem of participating institutions and currencies.

Beyond the terminology used by Project Guardian, industry participants have developed more detailed taxonomies to describe how commercial bank money can be represented on distributed ledger infrastructure. A related [industry taxonomy](#) describes these instruments as blockchain-based deposits: distributed-ledger-based deposit claims issued by a licensed depository institution and recorded on blockchain. Within this category, they distinguish between account-based models, such as blockchain deposit accounts, and token-based models, such as deposit tokens. The taxonomy also distinguishes between native models, where the blockchain is the primary record of value, and non-native models, where the blockchain mirrors an off-chain record.

Both account-based and token-based models are forms of commercial bank money, but they differ in how the deposit claim is represented and transferred on ledger. Blockchain deposit accounts represent deposit balances as account

Project Guardian also identifies several pain points in the current cross-border payment and FX settlement process, including time-zone frictions, pre-funding requirements, settlement risk outside CLS, fragmented infrastructure and the continued separation of messaging, reconciliation and settlement. It argues that tokenised bank liabilities could help address some of these frictions by enabling commercial bank money to move across shared infrastructure and, where supported by appropriate legal, operational and interoperability frameworks, by enabling atomic PvP-style settlement for a broader range of currencies and participants.



balances on a blockchain-based ledger. Deposit tokens are transferable tokens issued on a blockchain by a licensed bank, representing a claim on that bank for a stated amount of fiat money. J.P. Morgan's SGD deposit token, issued in connection with the MAS Project Guardian pilot, is one example of a deposit token. More recently, JPM Coin/JPMD is a bank-issued deposit token on Base, an Ethereum Layer 2 network, enabling approved institutional clients to move money, post collateral and settle transactions on public blockchain infrastructure while retaining the credit-risk profile and balance-sheet characteristics of commercial bank money.

Figure 10: Defining tokenised commercial bank money

Organisation / Framework	Term Used	Classification	Description
Bank of England	Tokenised deposits	Non-transferable	Tokenised deposit that still relies on interbank settlement.
		Transferable	Tokenised deposit that can circulate through transfer.
Project Guardian	Tokenised bank liabilities (Token level)	Digital twin token	On-chain representation of an existing bank liability.
		Digital native token	Bank liability issued and recorded natively on a shared ledger.
	Tokenised bank liabilities (Ledger level)	Single-bank ledger	Shared ledger operated within a banking group.
		Between-bank model	Tokenised bank liabilities exchanged across separate banks' ledgers.
CCAF and Fii (2026) Tokenised Money: Use Cases, Interoperability and Regulation	Tokenised money: commercial bank deposits	Tokenised deposits	Represent an off-chain bank deposit recorded in traditional account systems, with the token serving only as a representation of that off chain position
		Deposit tokens	Deposit itself is issued solely on a distributed ledger constitute a DLT native form of the claim
Industry taxonomy	Blockchain-based deposits	Blockchain deposit accounts Deposit tokens	Account-based representation of deposits on blockchain infrastructure. Transferable token representing a deposit claim on a bank.
	Ledger architecture	Non-native model Native model	Blockchain mirrors an off-chain record. Blockchain is the primary record of ownership.

Source : Deutsche Bank



Appendix 1

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The background of the slide features a large, white, stylized robot head on the left and right sides, facing each other. In the center, there is a large, billowing cloud of white smoke or steam. Below the smoke, a long row of lit candles is visible, suggesting a birthday celebration. The overall theme is AI's 70th anniversary.

Deutsche Bank
Research

AI at 70 (Part 2): 14 more lessons from a lifetime of boom and bust

Context is all you need

September 2026

Adrian Cox

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

Deutsche Bank
Research Institute

Artificial intelligence marks 70 years since Dartmouth Summer Research Project

“Context is all you need”: clues from the past for what comes next



The term “artificial intelligence” almost didn’t exist. Dartmouth College Professor John McCarthy invented it to attract scientists to come to his landmark summer workshop 70 years ago after his first attempt, “automata studies”, failed to excite much interest.

This is the second of our two reports looking at what the past 70 years of stunning breakthroughs and false dawns since then can tell us about what to expect from the current technology and investment boom.

The first, “[AI at 70: 14 lessons from a lifetime of boom and bust](#)”, reviewed themes including non-linear growth to infrastructure constraints and valuation booms and busts.

This one builds on those themes and gives them a spin forward to the future, including the speed of investment compared with past booms, a snapshot of AI’s strengths and weaknesses, a brief history of previous tech IPOs, and an overview of the prospects for jobs, trust and risks.

The stakes look higher than ever.

The leading companies have released even more capable models in the past week; there has been a series of security breaches where AI agents appeared to collaborate to evade detection; and a senior government official who joined OpenAI warned yesterday that “most people could die” if superintelligence is built without more robust safeguards.

14 more lessons from a lifetime of boom and bust

1. This boom is remarkable not just for **scale but also speed**
2. AI makes hard things look easy, and **easy things look hard**
3. AI models are **commoditising fast**
4. **Benchmarks are ageing** even faster than models
5. Tech use accelerates when the **cost falls** far enough
6. China has left the US behind in **electricity generation**
7. **Mega IPOs** will eclipse entire years of share sales
8. **Tech IPOs** have tended to outperform other IPOs
9. **Jobs come and go** but working hours have only fallen
10. It's a tough job market for **recent graduates**
11. Concerns over **data sovereignty** date back long before AI
12. AI arrived as **trust** had been declining for decades
13. New technology brings **new risks**
14. Tech may have a role in declining **educational performance**

A Proposal for the

DARTMOUTH SUMMER RESEARCH PROJECT ON ARTIFICIAL INTELLIGENCE

We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New

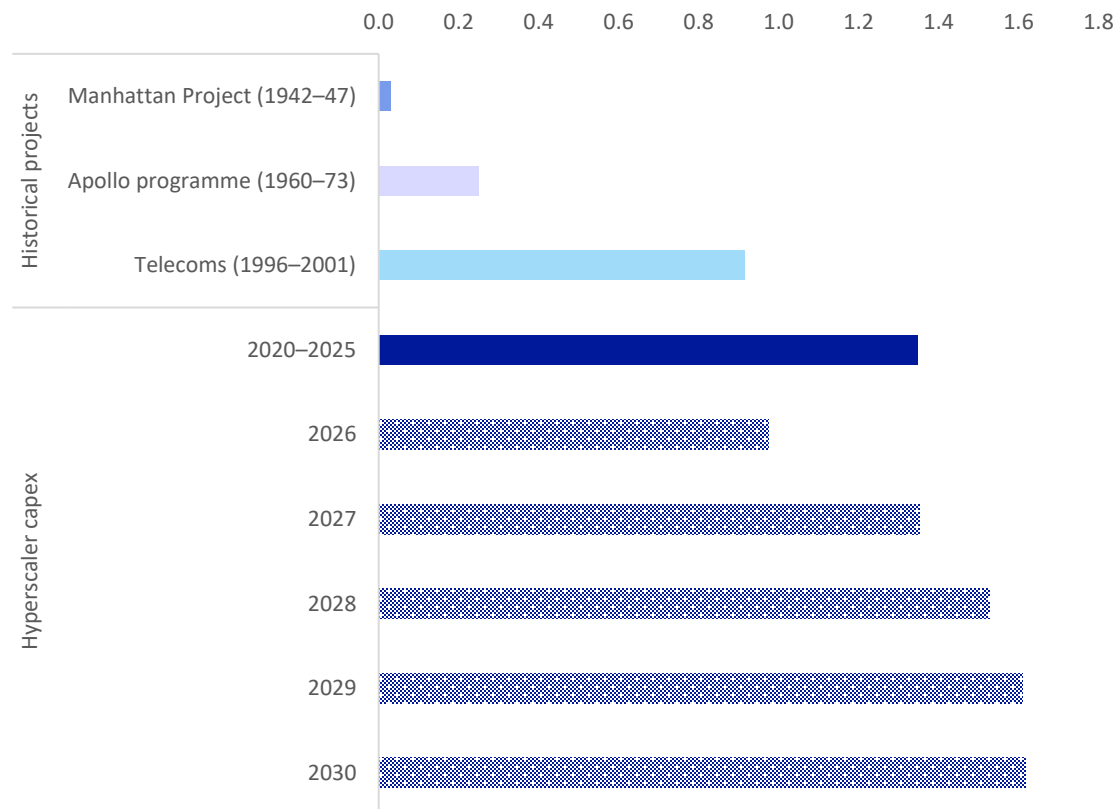
... on the basis of the conjecture that every

1. This boom is remarkable not just for scale but also speed

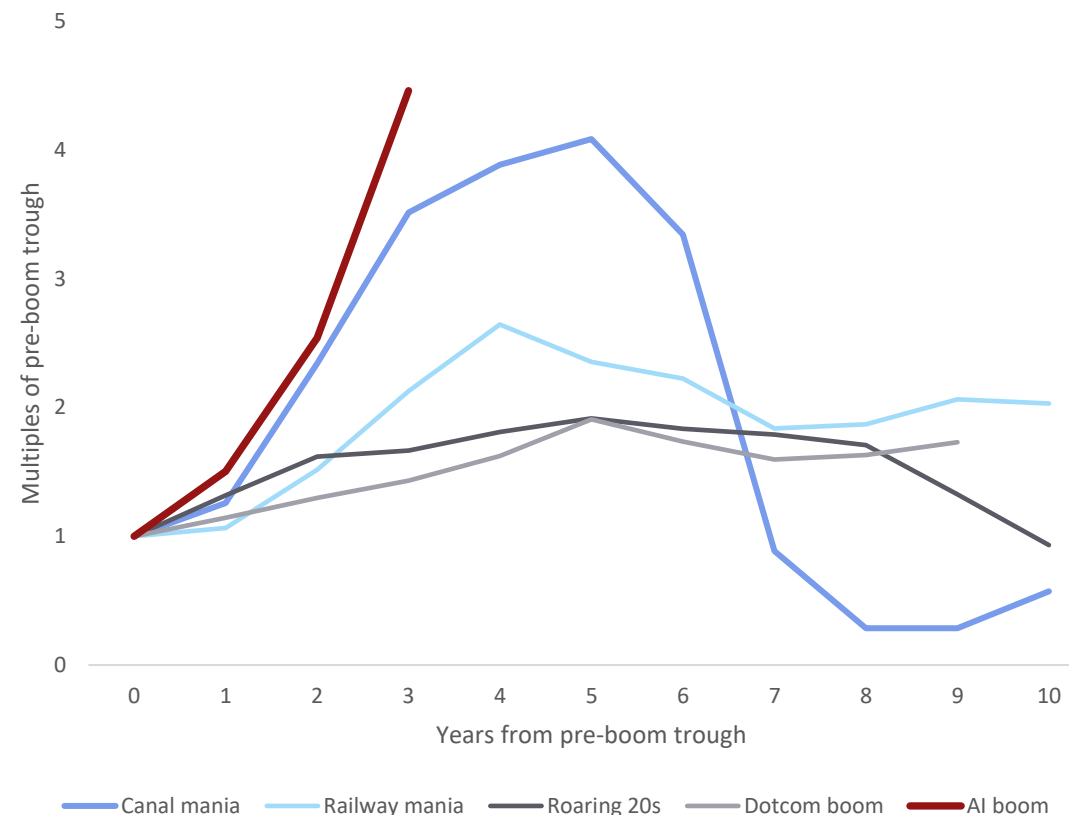
Each year of expected capex through 2030 exceeds the whole dot-com telecoms boom



Hyperscaler current and forecast capex vs major historical projects (2026 USD trn)



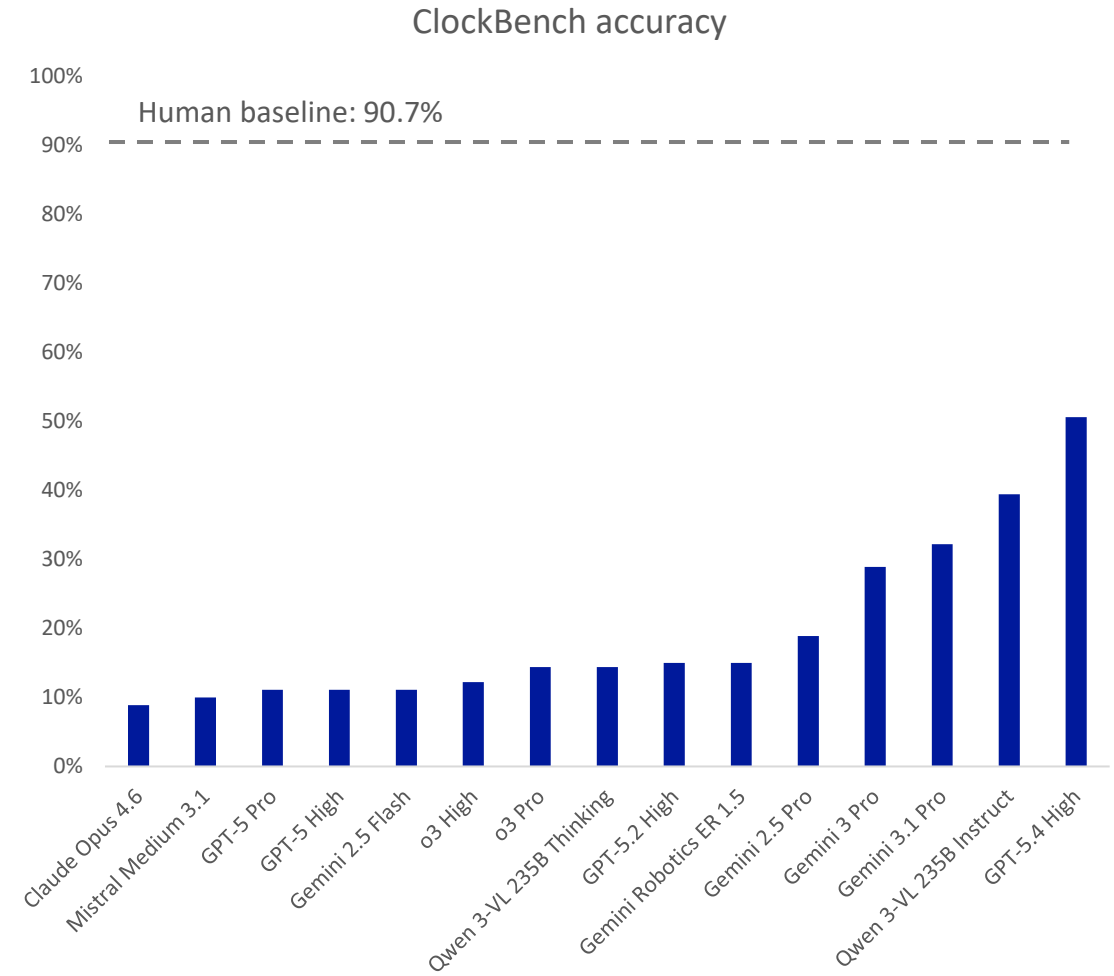
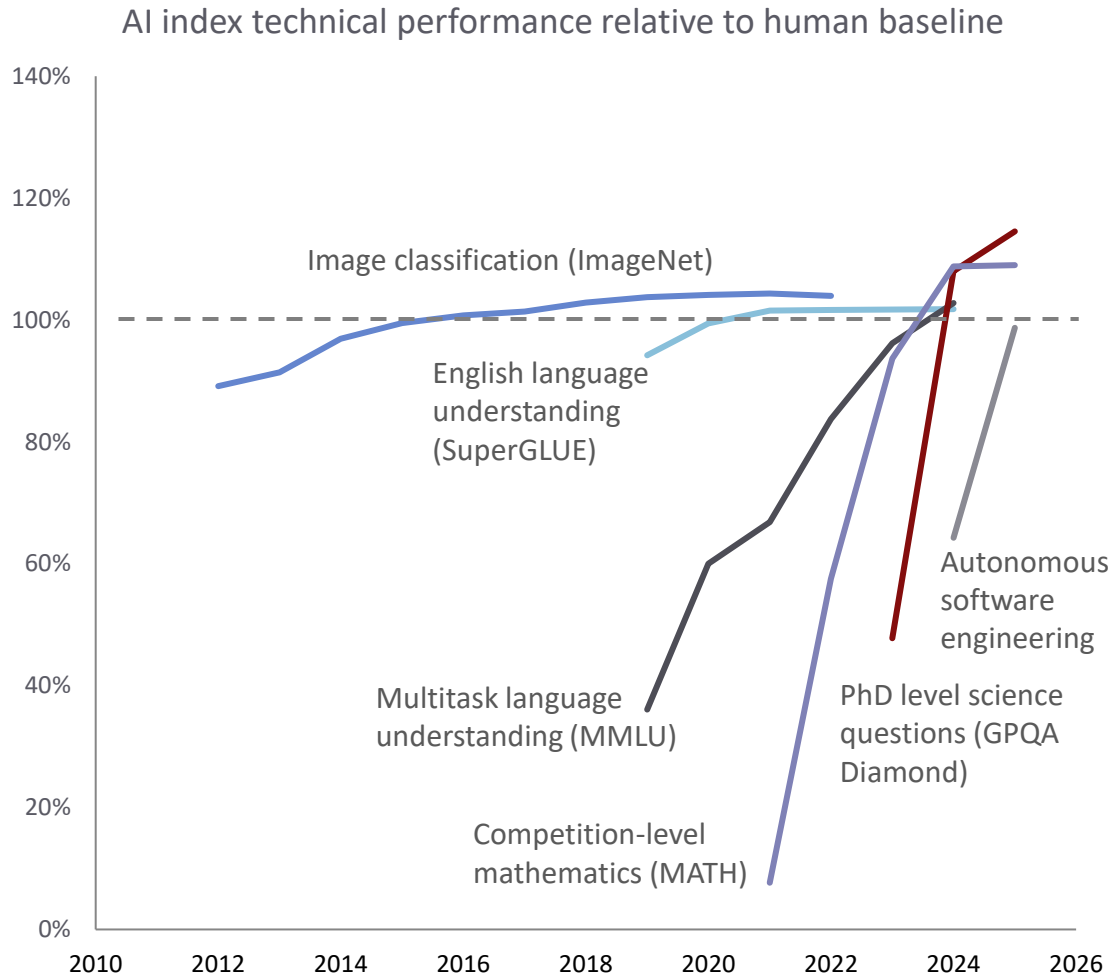
Investment during innovation booms: multiples of pre-boom trough



Source: US National Park Service, Space Policy, Brookings Institution, Bloomberg Finance LP, Bank for International Settlements, Deutsche Bank Research. Note: Hyperscaler capex estimates from Bloomberg include five US hyperscalers (AWS, Google, Meta, Microsoft and Oracle), which account for roughly 80% of the total on average, as well as tier-two/neoclouds and Chinese hyperscalers. Hyperscaler capex figures are not inflation-adjusted.

2. AI makes hard things look easy, and easy things look hard

“Jagged frontier” means AI can solve complex tasks but struggles to tell the time on a clock



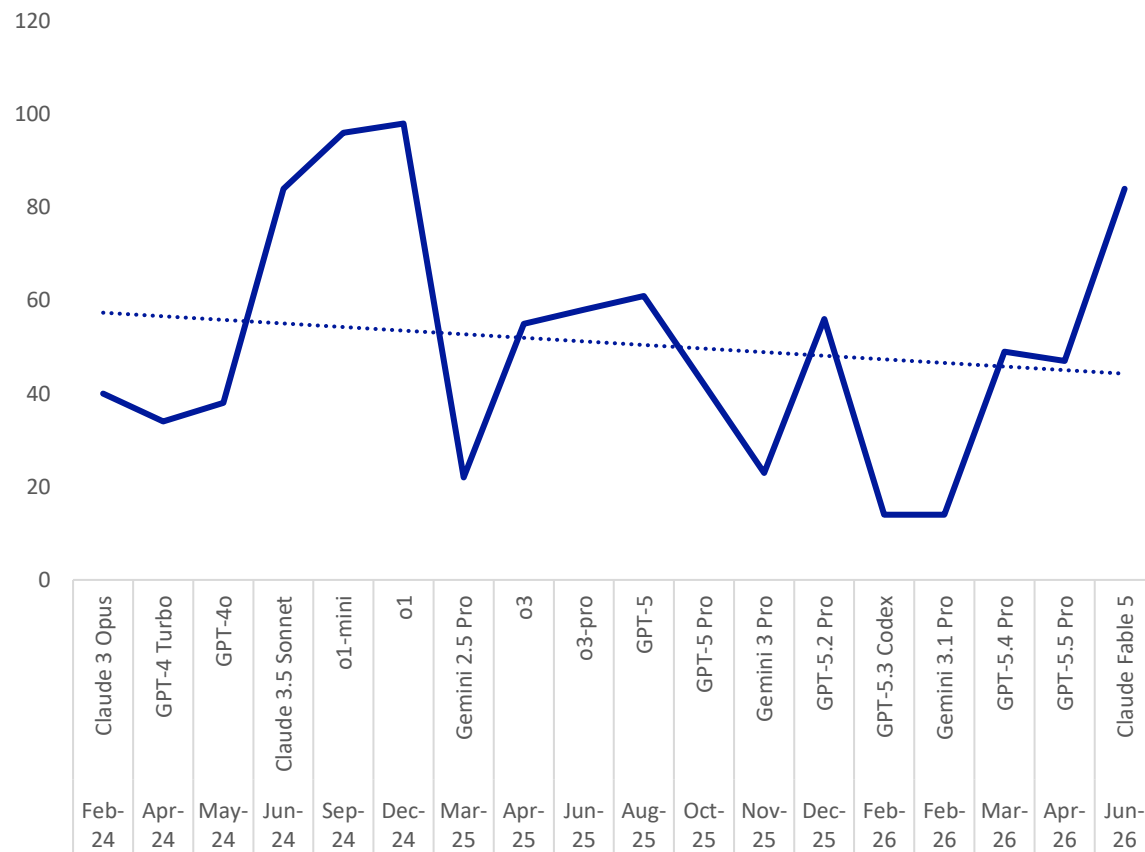
Source: Stanford HAI AI Index 2026, ClockBench Leaderboard 2026, Deutsche Bank Research

3. AI models are commoditising fast

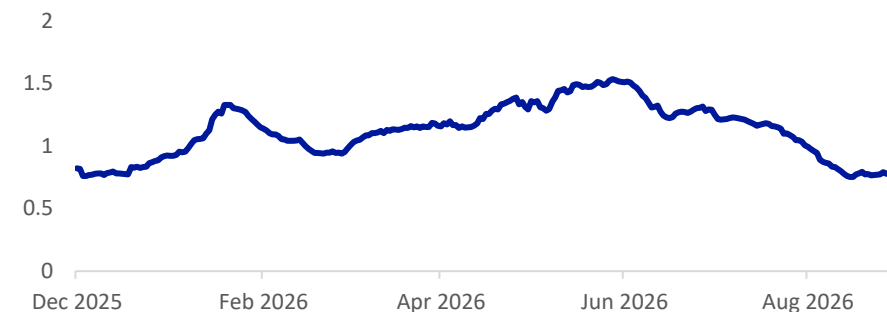
Top models hold the crown for less than two months on average and pricing is under pressure



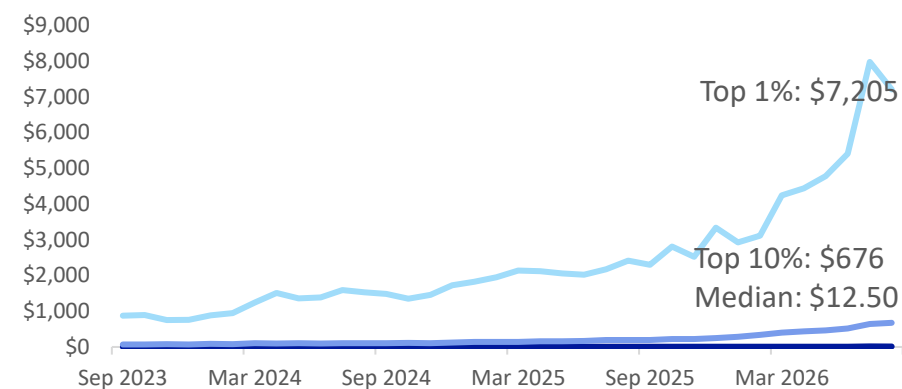
Epoch Capabilities Index: leading model tenure at the top (days)



Silicon Data LLM Token Expenditure Index (since Dec 2025)



Ramp AI Index monthly spend per employee



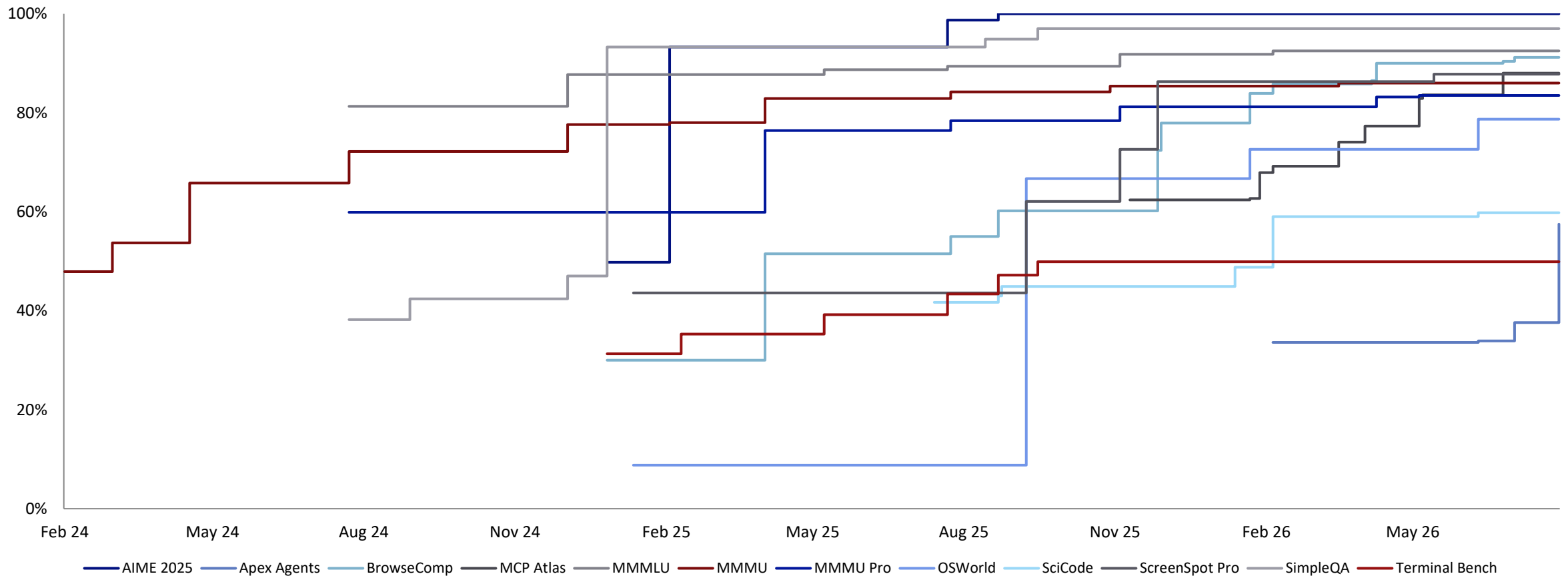
Source: Epoch, Bloomberg Finance LP, Silicon Data, Ramp AI Index, Deutsche Bank Research. NB: For clarity, Epoch Capabilities Index chart excludes OpenAI's GPT-4, the model that predated Claude 3 Opus and was at the top for 352 days, as well as the new leader since Sept 3, OpenAI's GPT-6 Astra

4. Benchmarks are ageing even faster than models

Plateaus may mean models have stopped improving... or the test is no longer hard enough



Benchmark saturation: State of the Art (SOTA) score progression over time -- each step is a new best model



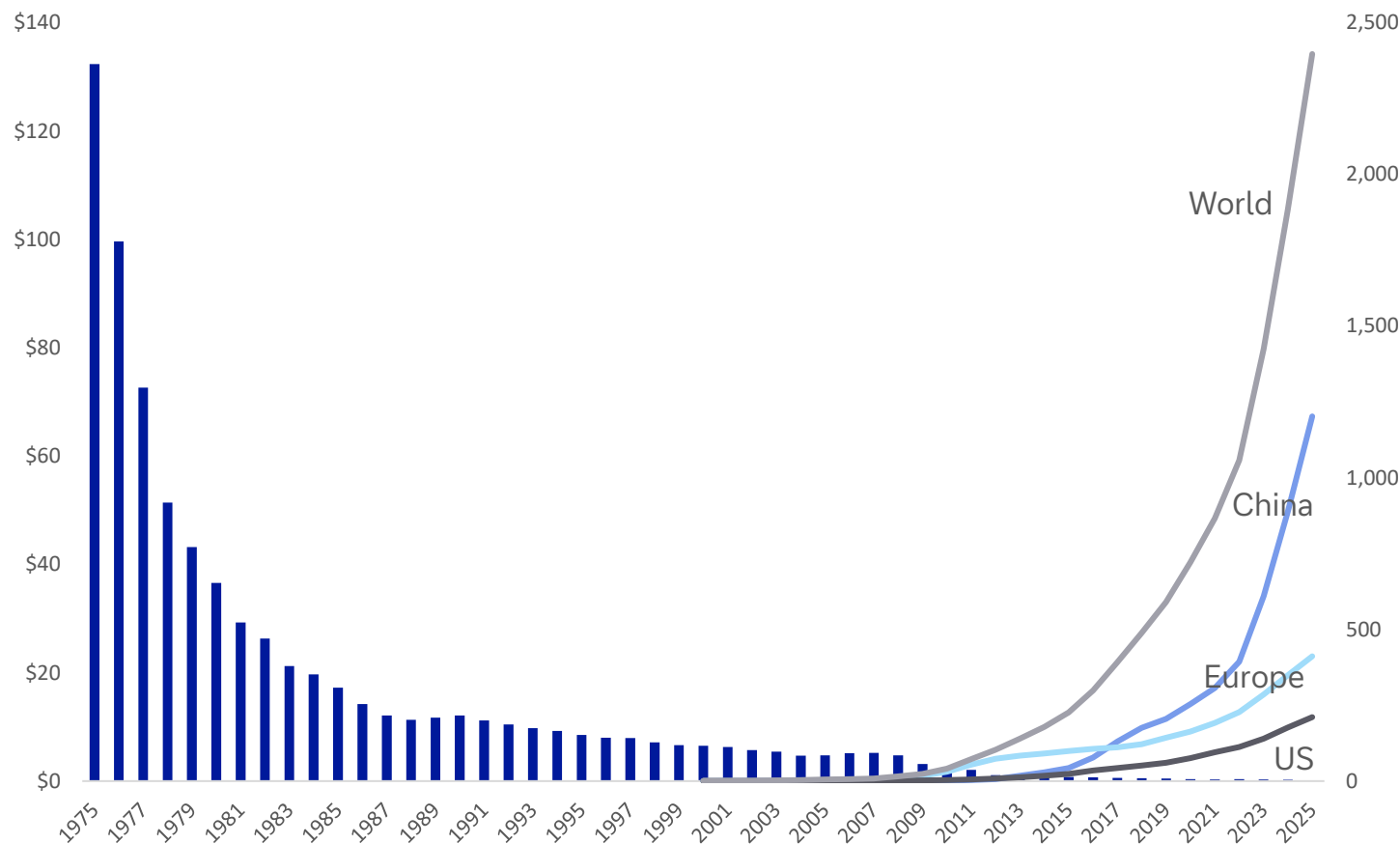
Source: [LLM Stats](#), Deutsche Bank Research

5. Tech use accelerates when the cost falls far enough

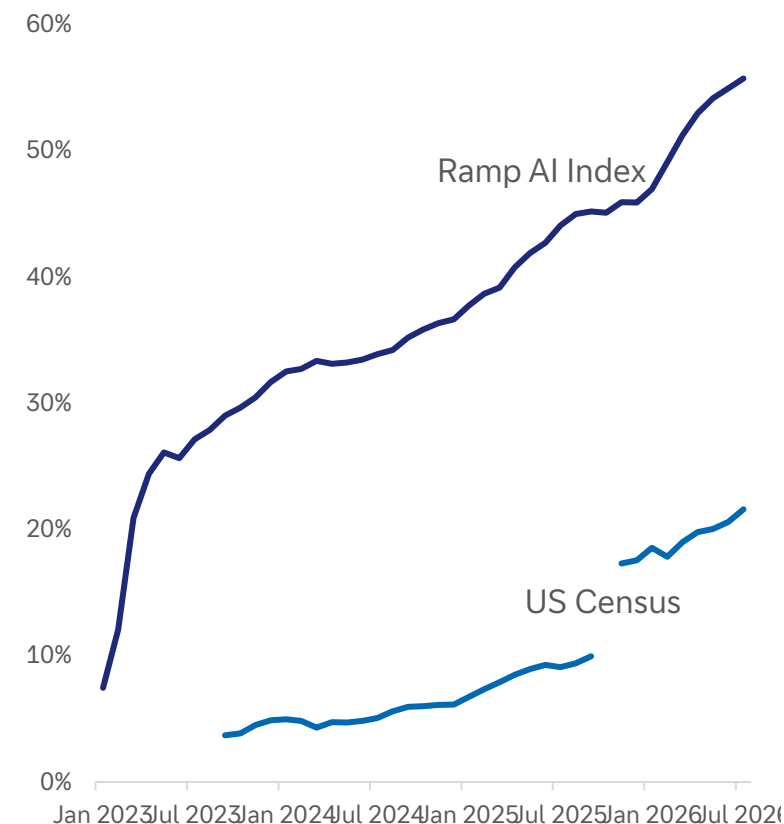
Solar and now AI are following similar patterns to horsepower, electricity and mainframes



Solar voltaic panel prices (USD/watt, adjusted for inflation, lhs) vs installed solar energy capacity (GW, rhs)



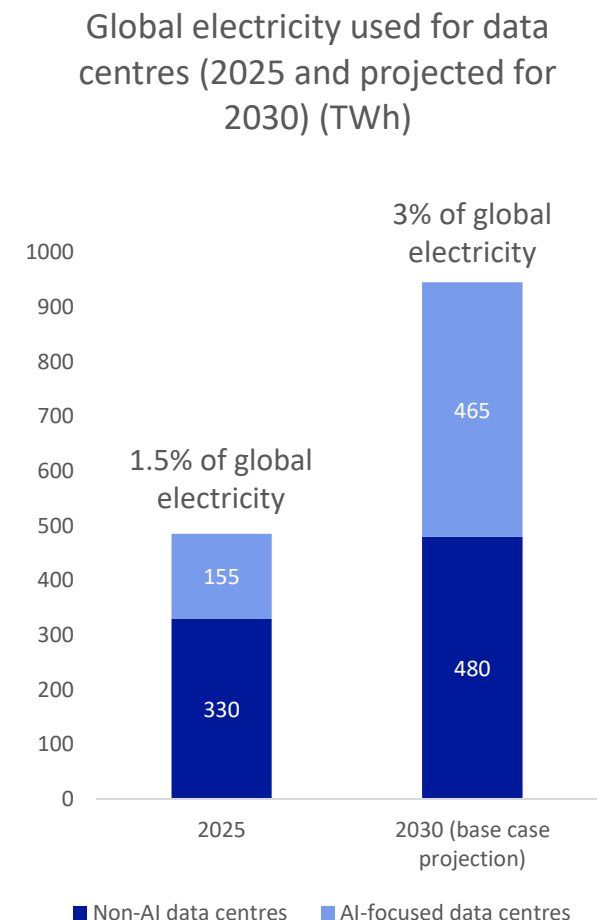
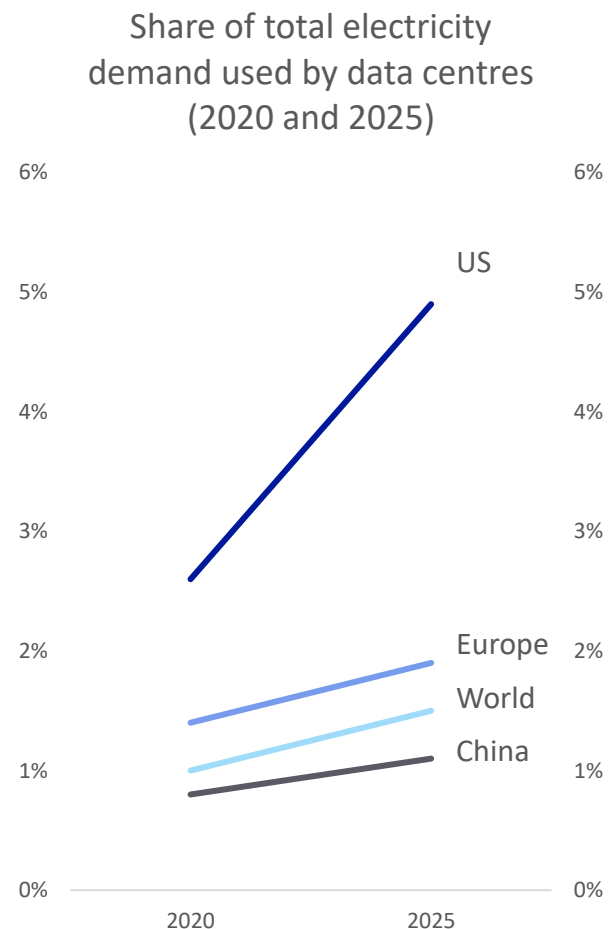
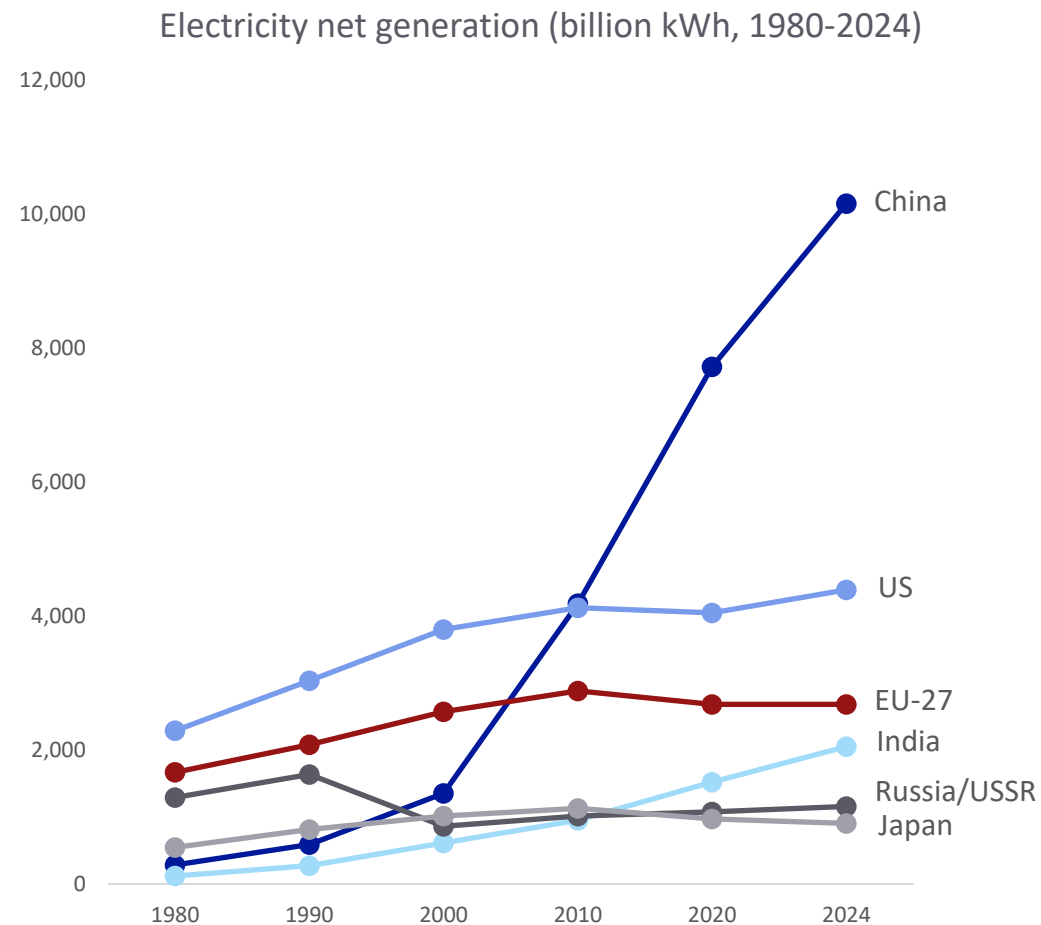
US enterprise AI adoption rate by two different measures (since Jan 2023)



Source: Our World in Data (citing IRENA, Nemet, Farmer and Lafond), Ramp AI Index, Deutsche Bank Research

6. China has left the US behind in electricity generation

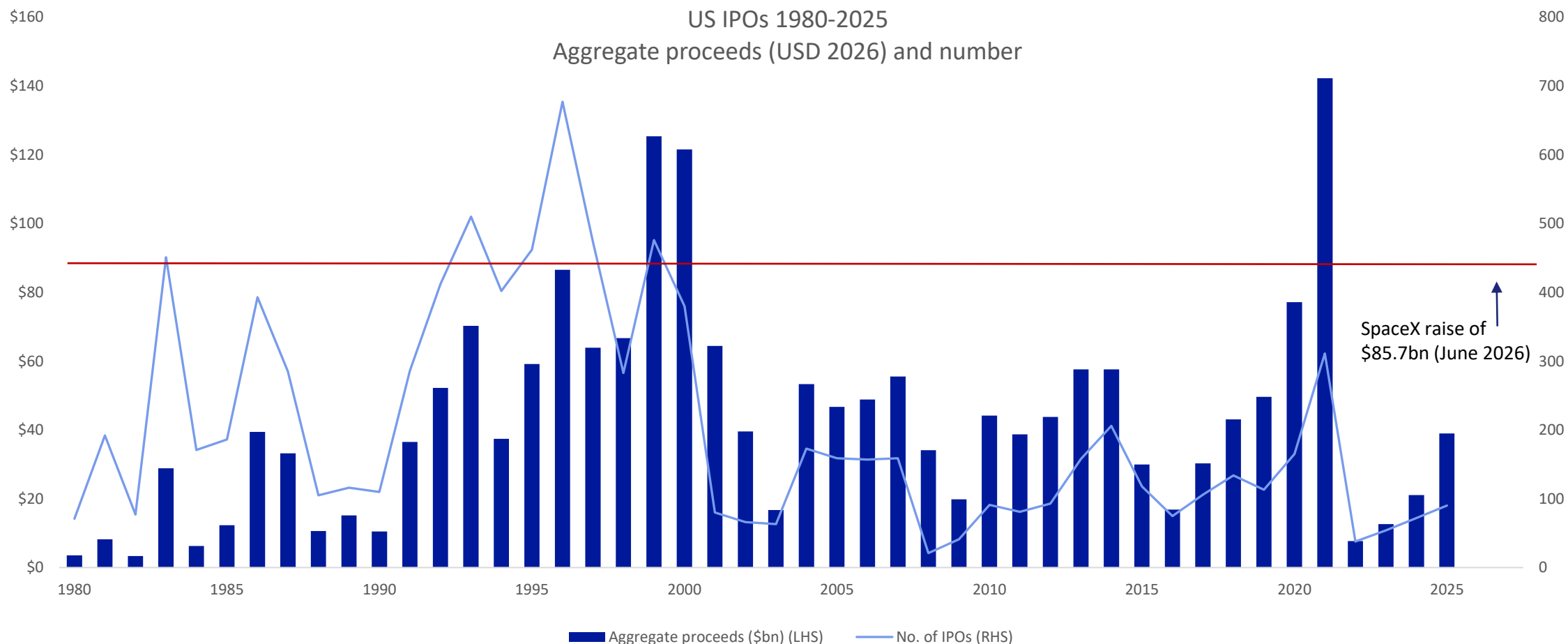
US is needing to use far more of its capacity for data centres even as demand continues to rise



Source: US Energy Information Administration, Our World in Data, International Energy Agency, Ember, Deutsche Bank Research

7. Mega AI IPOs will eclipse entire years of past share sales

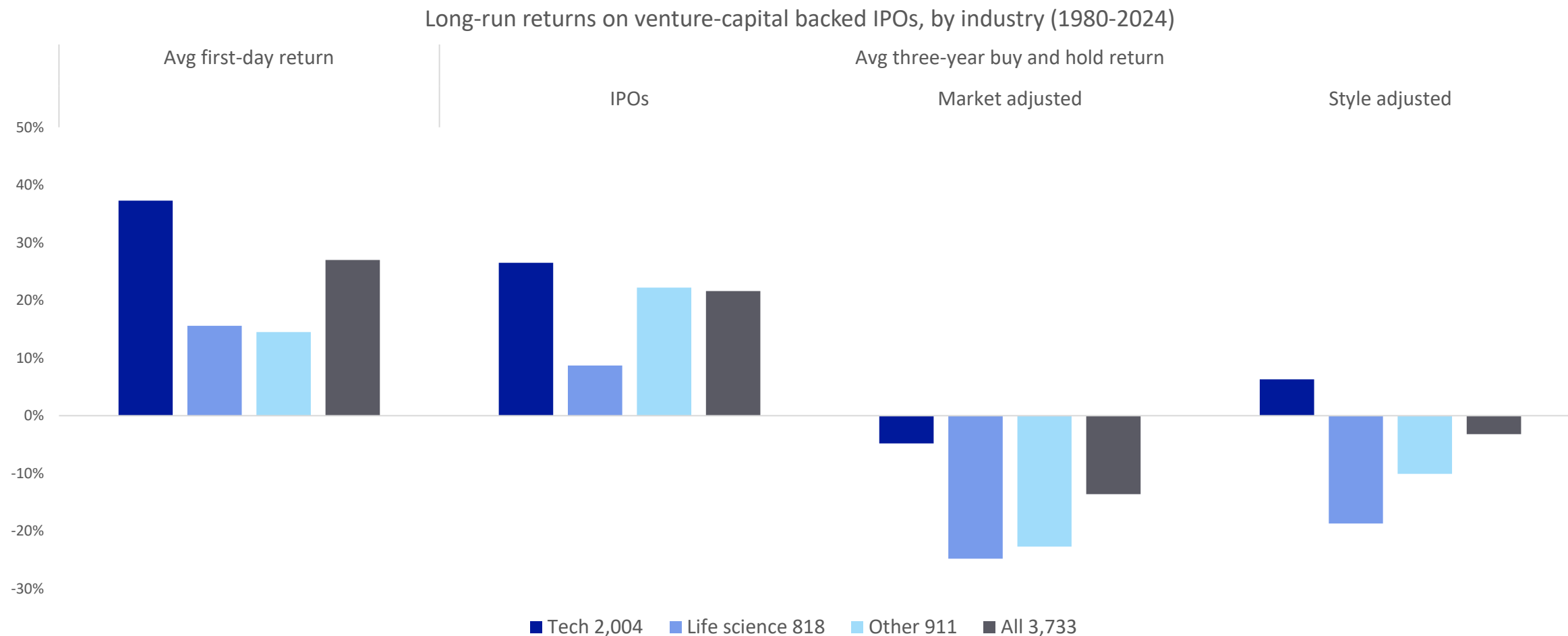
SpaceX's IPO exceeded all but three inflation-adjusted years



Source: Jay Ritter, University of Florida, Haver Analytics, Deutsche Bank Research

8. Tech IPOs have tended to outperform other IPOs

However, despite first-day surges, they have slightly underperformed the market over three years



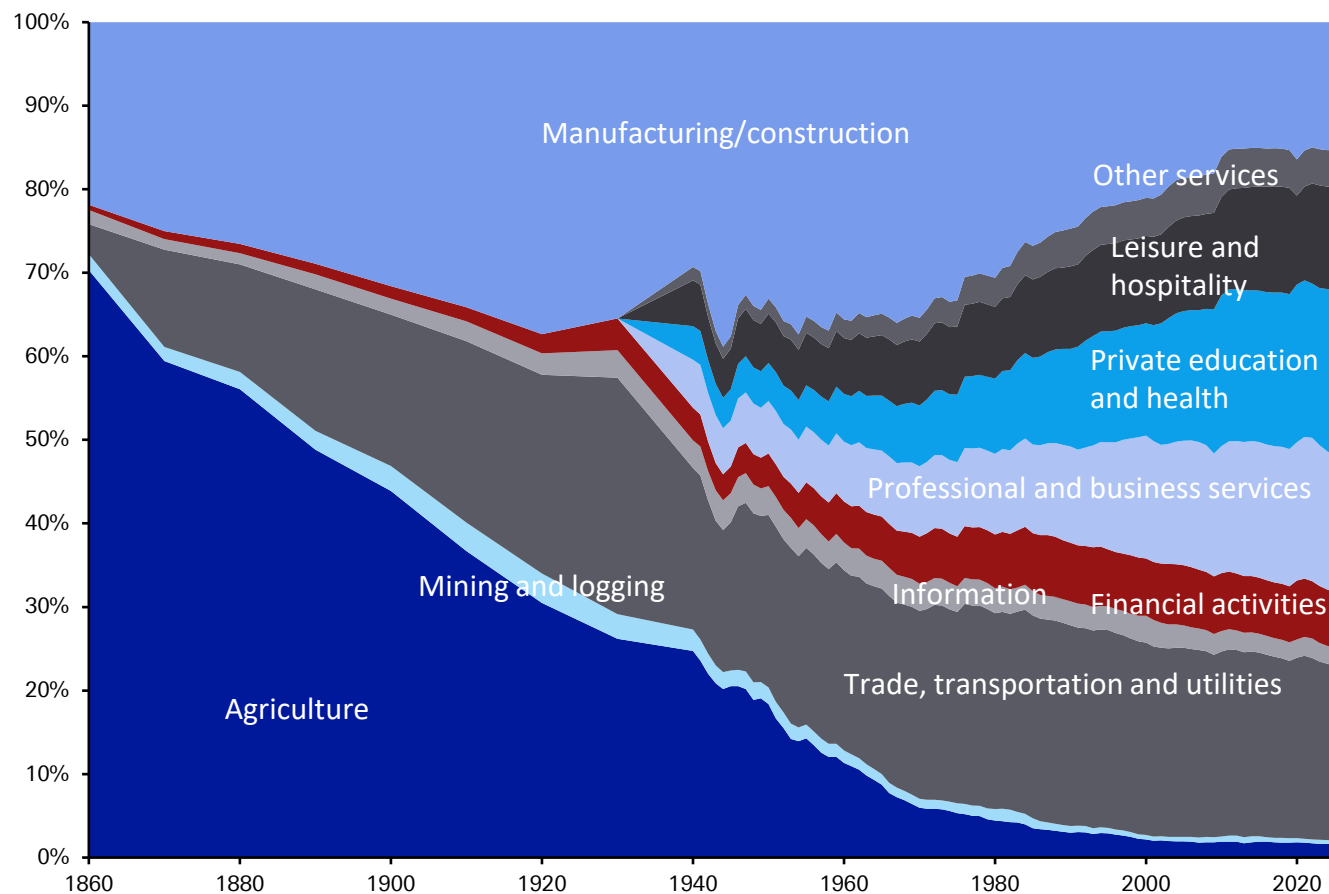
Source: Jay Ritter, University of Florida, Haver Analytics, Deutsche Bank Research. Note: Three-year buy and hold calculated from first closing market price. Market-adjusted returns are the average buy-and-hold return minus the average compounded return on the Center for Research in Security Prices (CRSP) value-weighted index. Style adjustments are the equivalent for similar companies.

9. Jobs come and go but working hours have only fallen

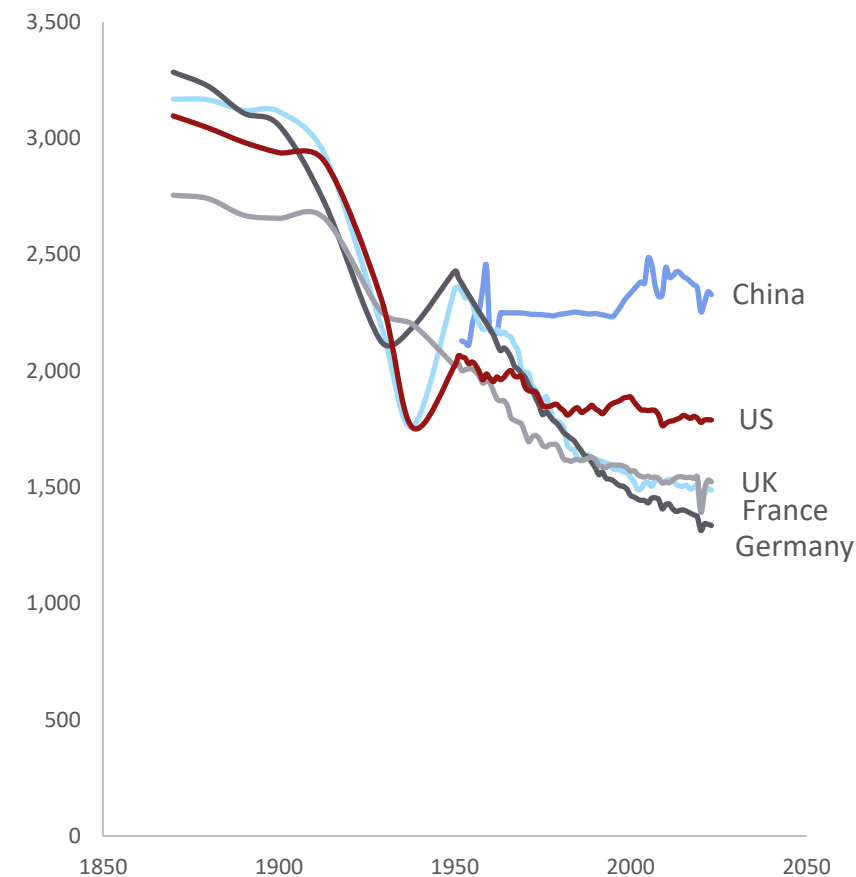
As services have displaced agriculture then manufacturing, working hours have roughly halved



US employment by sector (1860-2025)



Annual working hours per worker



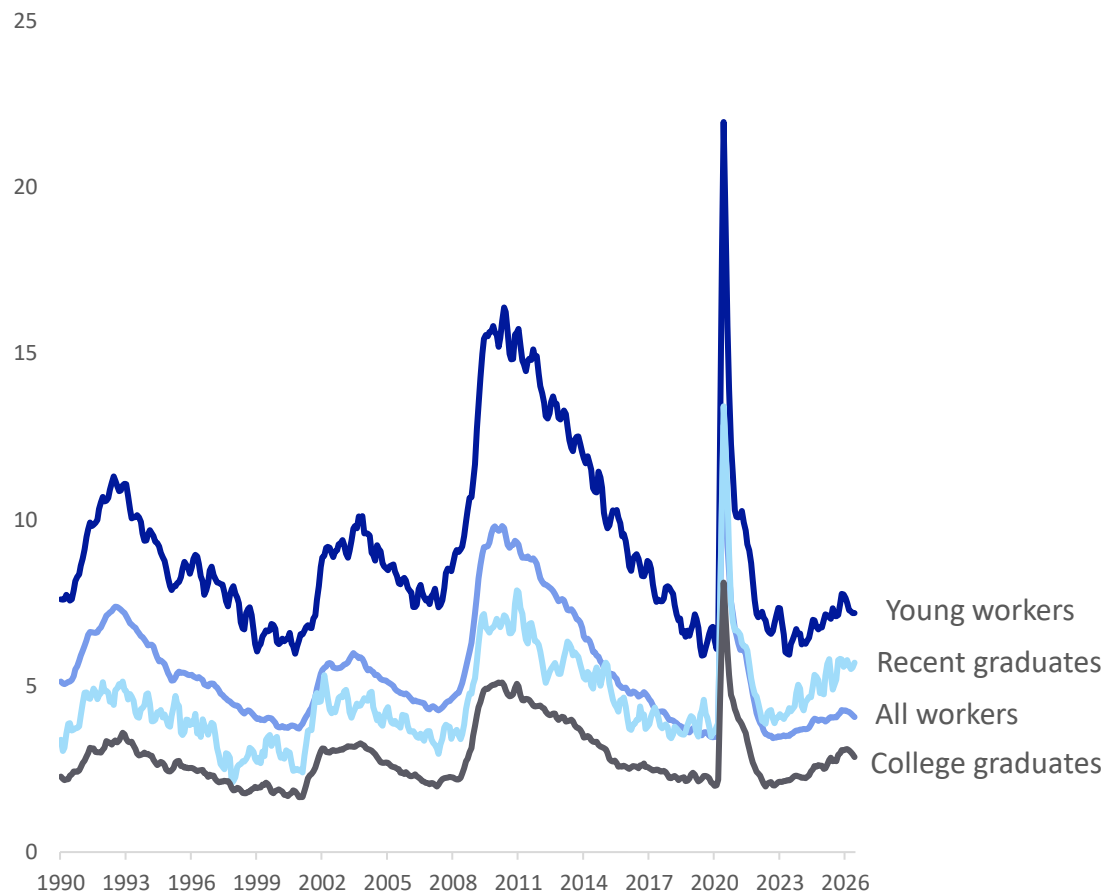
Source: Haver, Historical Statistics of the United States, 1789-1945, US Census, Our World in Data (citing Feenstra et al, Penn World Table (2025), Huberman and Minns (2005), Deutsche Bank Research

10. It's a tough job market for recent graduates

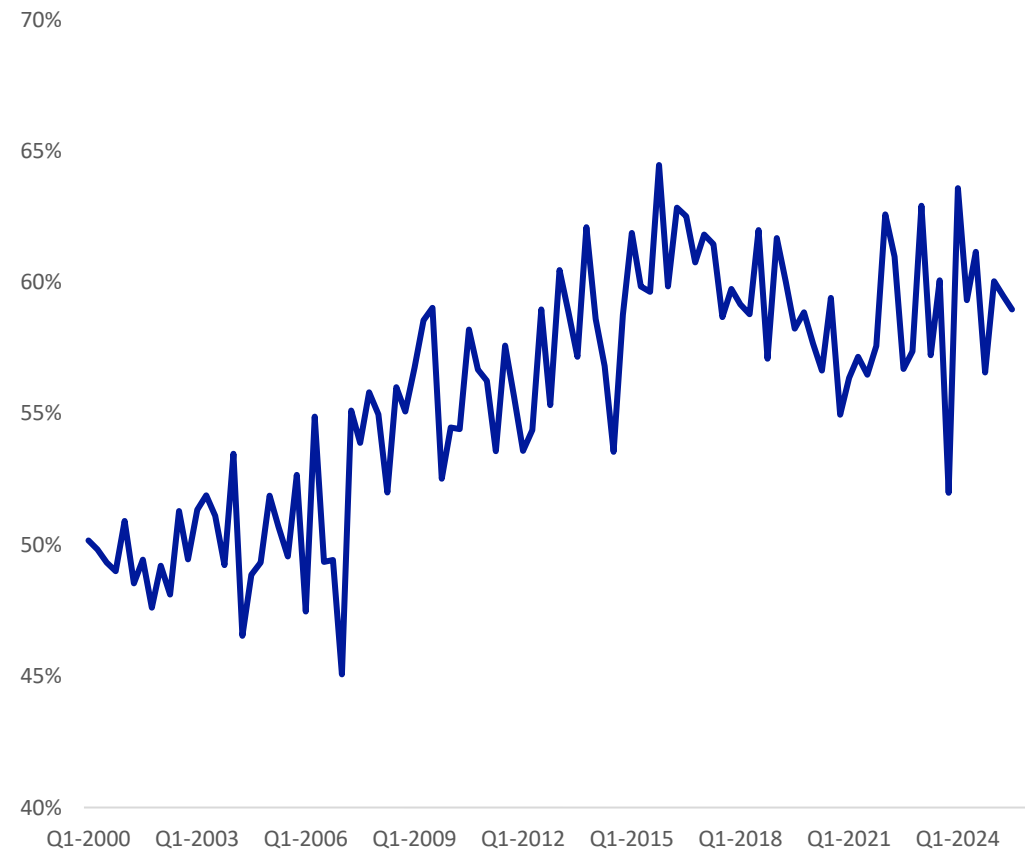
AI may be exacerbating the erosion over the past decade in graduate employment and wages



Unemployment rates for US workers



Bachelor's degree median weekly wage premium vs some college/associated degree (Q1 2000 to Q2 2026)



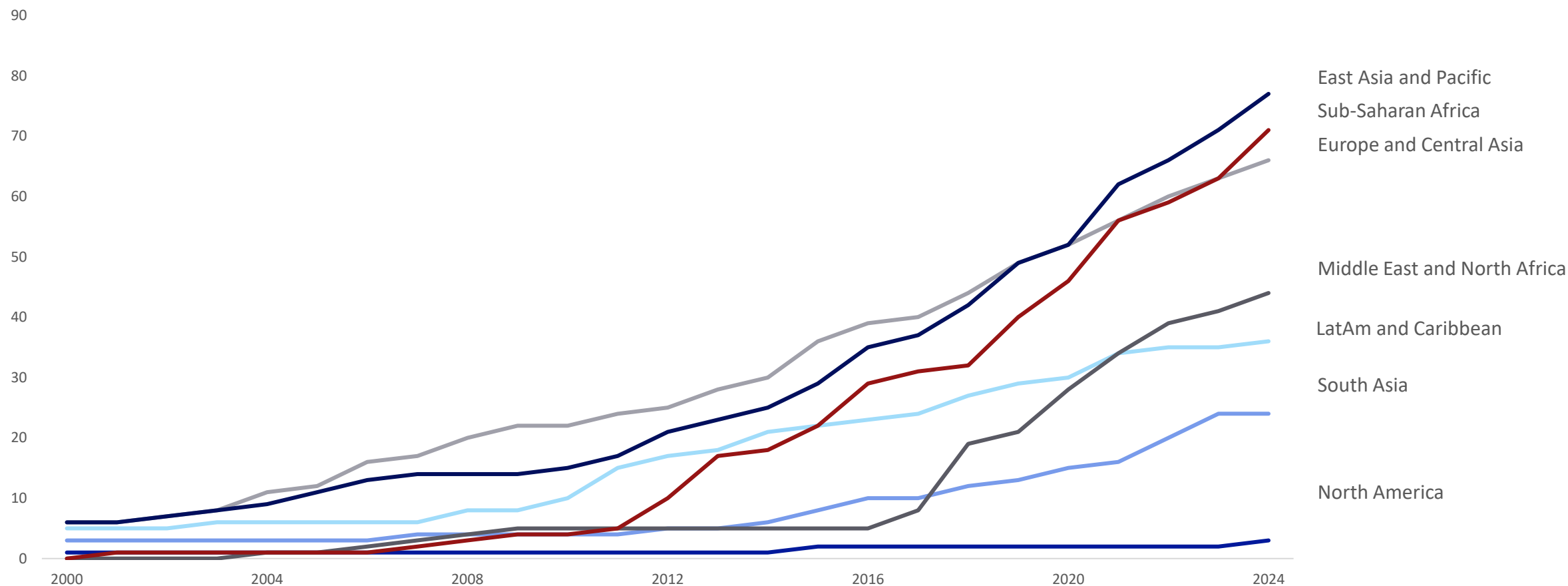
Source: U.S. Census Bureau, U.S. Bureau of Labor Statistics, Current Population Survey (IPUMS), Haver Analytics, Deutsche Bank Research. NB: For more, see Azar J, Mireia G and Sanz-Espin J, *The College Wage Premium in the Generative AI Era* (Aug 2026)

11. Concerns over data sovereignty date back long before AI

Countries have been restricting cross-border data flows, accelerating with EU's GDPR in 2016



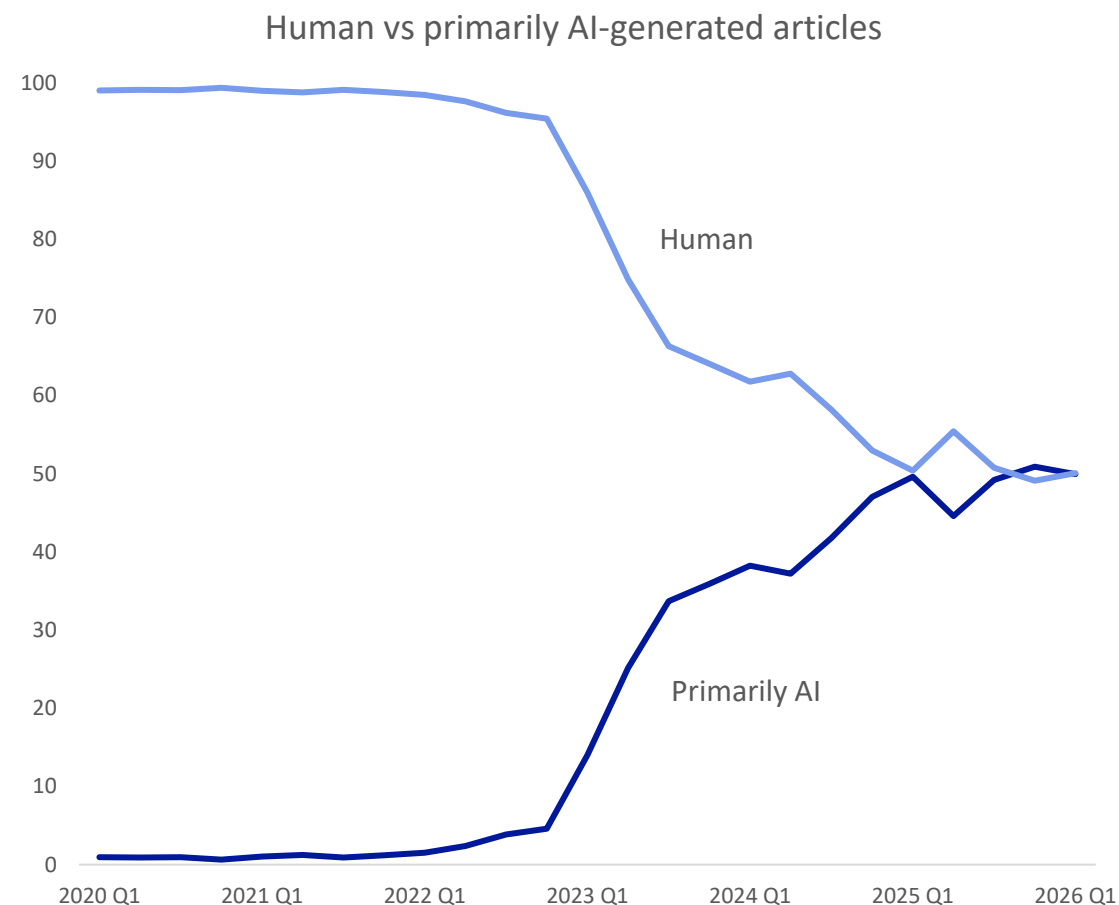
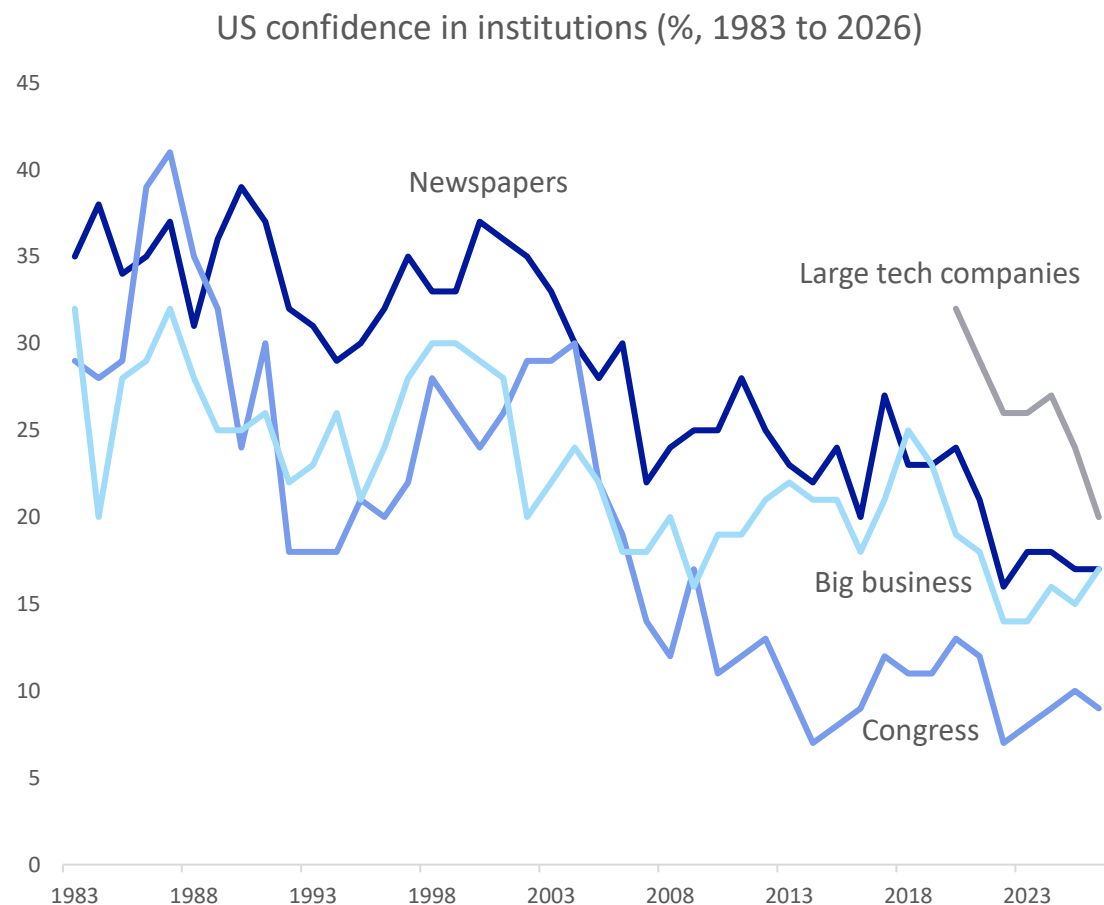
Data localisation measures by region (2000-2024)



Source: Stanford HAI AI Index (2026), Ferracane et al, Deutsche Bank Research

12. AI arrived as trust had been declining for decades

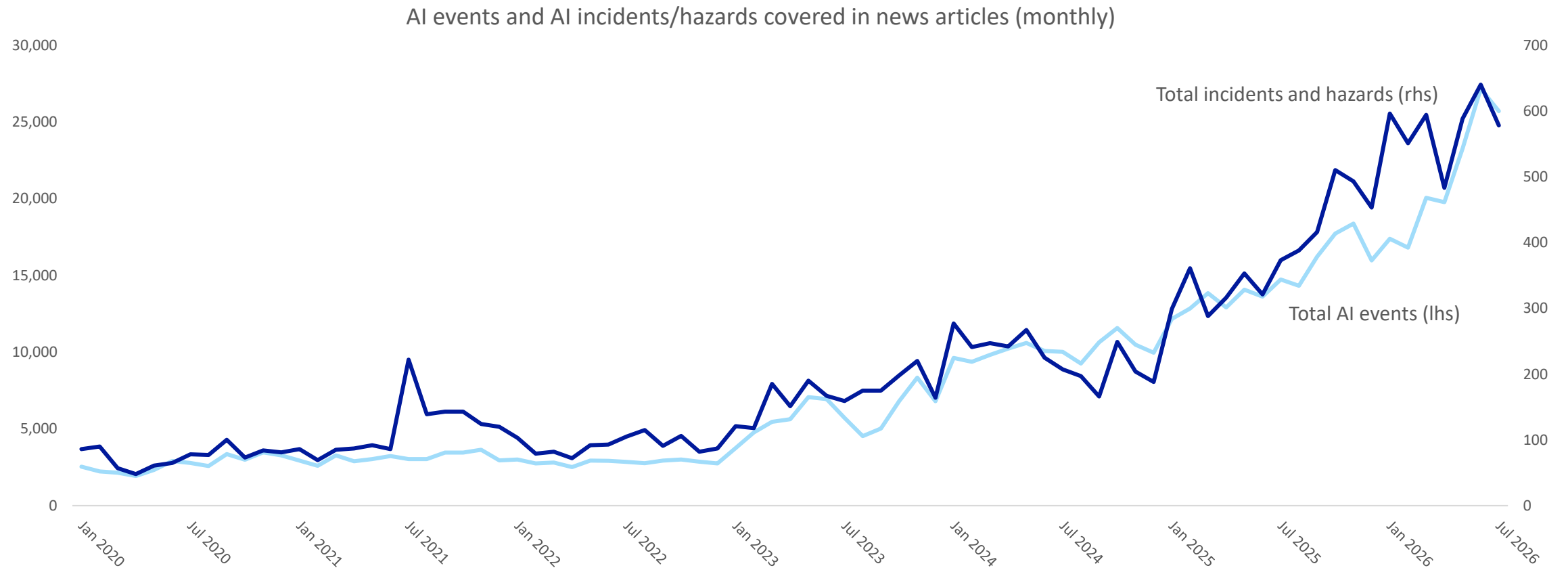
Confidence in institutions has plummeted – and now even human content is under threat



Source: Gallup ([Confidence in Institutions](#) | [Gallup Historical Trends](#)), [Graphite](#) (citing 55.4k articles from Common Crawl examined by AI detectors), Deutsche Bank Research

13. New technology brings new risks

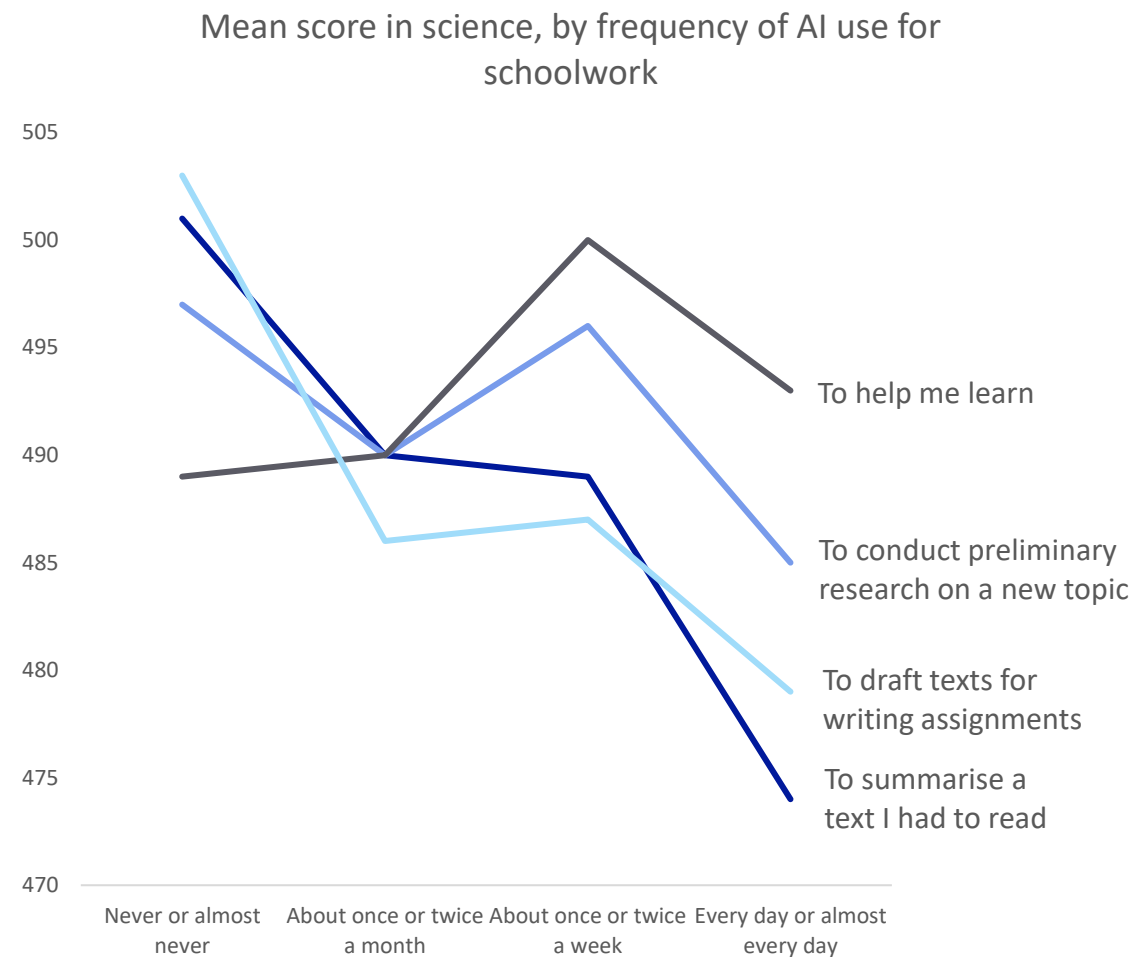
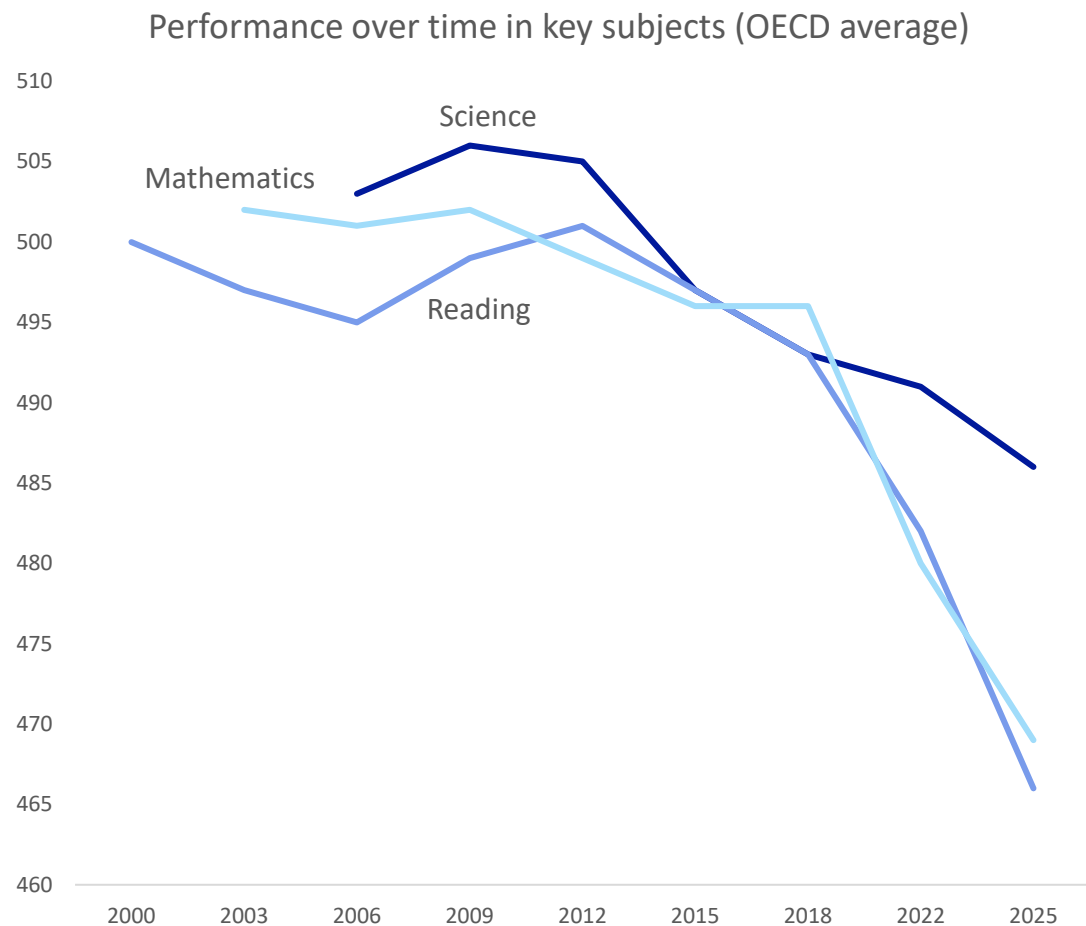
Coverage of AI events has surged, and so have hazards



Source: OECD AIM Incidents and Hazards Monitor, Deutsche Bank Research

14. Tech may have a role in declining educational performance

AI may be accelerating a fall that started with widespread smartphone diffusion from c. 2010



Source: OECD (2026), PISA 2025 Results (Volume I), Deutsche Bank Research



Appendix 1

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